SEISMIC WAVE PROPAGATION IN COMPLEX MEDIA:

SUMMING THE MULTIPLE SCATTERING SERIES

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1996

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Chapter 1

Introduction

1.1 overview

In this thesis we attempt to extend the *reflection/transmission* matrix method of *Kennett* (1974) to laterally heterogeneous media. We assume a 2−D model for the material heterogeneity of the medium, but the wavefield is allowed to spread out in 3−D, which is sometimes referred to as a $2\,{}^1/_2$−D model. We assume that the heterogeneity is an extra level of structure that overlays an otherwise horizontally stratified half space. We call the stratified component of the model, the *reference model*, and the remaining variations of material properties are relegated to the *residual model*. Wave propagation in the reference model (which will be called the *reference field*) is now well understood [e.g. *Aki & Richards* (1980); *Kennett* (1983)].

The residual structure, *scatters* the reference wavefield into the *actual wavefield*, which is the quantity of interest. The *difference between the actual wavefield and the reference wavefield is the scattered field*. This scattered field may be imagined to derive from a *secondary* or *virtual* source distribution. Consistent with *Huyghens' principle*, we assume that the secondary source distribution, may be discretized, as a *regular lattice of radiating multipoles*. Due to the *diffuse* nature of seismic scattering, the secondary source distribution *covers* the spatial support of the residual structure. Unfortunately, the secondary source field is unknown, because it depends on the actual wavefield, which is yet to be determined. In this thesis we propose to determine the secondary source field, by summing the successive interactions of the incident field with heterogeneity. We refer to this procedure as: *Summing the multiple scattering series*.

This program has its origins in the quantum mechanics formulation [e.g.

[Figure: Rdpp not found]

Figure 1.1: Multiple scattering Rpp for 6×6 point scatterers

Agassi & Gal (1973), *Keister* (1983), *Gonis* (1992)], of scalar wave scattering, and the elastic wave equation extension has been investigated in *Kennett* (1986), where it was shown how to calculate the *single scattering* approximation of the actual wavefield. In *Kennett* (1986), the superimposed heterogeneity was subdivided into *thin slabs*. In the single scattering approximation to a reflection seismogram, the reflected field is comprised of only the *primary backscatter interactions* of the incident reference field with the thin slabs of superimposed heterogeneity, while incorporating all transmission and reflections processes in the reference model.

The main focus of this thesis, is the investigation of:

- the efficient computation of multiple scattering interactions of the incident wavefield with the residual structure
- methods of summing the multiple scattering partial sums in order to secure rapid convergence of the multiple scattering series

A choice must be made concerning the representation of the superimposed heterogeneity (i.e. the residual model). The most obvious choice is to divide the residual model into thin slabs, as employed in *Kennett* (1986), and approximate the scattering properties of each slab using the *Born* or *single scattering* approximation. This model is based on the assumption that there is more energy involved in *inter-slab* scattering interactions than *intra-slab* scattering interactions. This representation ignores the possibility of *side scattering* between two point within the same thin layer.

In Fig.1.1 we depict the in-plane, downward incident $P-$wave to upward reflected $P-$wave response for 36 point lattice of scatterers arranged in a square, by summing the multiple scattering series to a convergence, for 101

[Figure: dRdpp not found]

Figure 1.2: The side scattering component of Rpp for 6×6 point scatterers

incident plane waves of varying horizontal wavenumber. In Fig.1.2 we depict the contribution in the experiment of Fig.1.1 due to side scattering between point scatterers in the same row. Certainly, for a low order approximation, we could neglect the side scatter component depicted by Fig.1.2 altogether, and we will see that there are significant computational advantages in doing just that. However, if we require a more accurate calculation of the the true response, we will need to incorporate these side scattering components. Unfortunately, there is a problem with this approach. If we employ conventional plane wave techniques, the Green's tensor which is needed to model the side scatter interactions diverges for lattice points within the same horizontal row, *especially in the limiting case where the horizontal offset between a pair of points tends to zero, so that the scattering point is interacting with itself*.

We show how to model side scattering, and particularly, self interaction, where the Green's tensor needed to transport the wave field along its scattering path is *strongly singular*. This involves finding new representations of the Green's tensor, and evaluating the self interaction terms when the source point and field point of the Green's tensor coincide. We have chosen to model the residual medium as a *tessellation* of small squares with uniform material properties, rather than thin slabs. Consistent with the Rayleigh approximation (Wu & Aki, 1985), the secondary wavefield, is modeled as a multipolar point force, situated in the middle of the respective squares. We compute the secondary source field (i.e. the multipole amplitudes) using the *pseudo-spectral method*. Indeed, our multiple scattering method shares some of the characteristics of the *elastic wave pseudo-spectral method* (Kosloff, Reshef & Loewenthal, 1984), with both methods requiring similar amounts of FFT computation.

Generally, the pseudo-spectral method requires less computation than our *multiple scattering method* for elastic media with strong superimposed heterogeneity (say 10% contrast relative to reference model), because it uses *second order time stepping*, which is very efficient to apply numerically. In the multiple scattering method of this thesis, the secondary source field is computed in *pseudo-spectral* fashion, but instead of time stepping, we have to compute a *monochromatic* multi-source/receiver seismogram, which is considerably more expensive than the analogous time stepping operation of the pseudo-spectral method. However, there are generally more time steps in the pseudo-spectral method than there are complex frequencies in the multiple scattering method.

When using second order time stepping, we must ensure that numerical errors are not amplified from one time step to the next. The time step Δt is constrained by the numerical stability requirement that $\Delta t \leq 0.26\Delta x_{min}/c_{max}$ (Daudt, Braile,Nowack, & Chiang, 1989), where Δx_{min} is the minimum spatial grid spacing and c_{max} is the maximum phase speed of the medium. The pseudo-spectral method is unique among the discrete grid methods in that the sampling requirements are optimal, so that the minimum grid spacing is chosen according to $\Delta x_{min} \leq \lambda_{min}/2 \leq c_{min}/(2\nu_{max})$ where λ_{min} is the shortest spatial wavelength, c_{min} is the minimum phase speed and ν_{max} is the maximum temporal frequency of the seismic source, consequently the time step is constrained by

$$\Delta t \leq 0.13 \frac{c_{min}}{c_{max}\nu_{max}}$$

If the temporal length of the seismic record is T, then the number of pseudo-spectral time steps N_t is given by

$$N_t = \frac{T}{\Delta t} \geq \frac{T\nu_{max}c_{max}}{0.13c_{min}}.$$

This should be compared with our multiple scattering method, where most of the computation takes place in the *complex frequency domain* followed by an inverse discrete Fourier transform to synthesize the temporal response. The number of frequencies N_f at the *Nyquist* sampling rate is then

$$N_f = T\nu_{max}.$$

Note that, although the frequency range is really $-\nu_{max} \leq \nu \leq \nu_{max}$, we need only compute those frequencies in the range $0 \leq \nu \leq \nu_{max}$, because the seismic response at a negative frequency is the complex conjugate of the

response at the corresponding positive frequency. Then, the number of time samples N_t and the number of frequency samples N_f are related by

$$N_t = \frac{c_{max}}{0.13 c_{min}} N_f.$$

For *gradient zones*, $c_{max}/0.13c_{min}$ will be appreciable, so that $N_f \ll N_t$. Also, *some degree of over sampling is commonly employed in the pseudo-spectral method, because even though the wavefield may only require two grid points per wavelength, at such coarse sampling rates the propagation environment will be poorly resolved.* If a double Nyquist sampling rate is employed, for a medium with strong velocity gradients, then there may be 100 times more time steps required than the number complex frequency points in the multiple scattering method. Also, the artificial means employed to approximate free surface and radiation conditions in the pseudo-spectral method requires that the computational domain be enlarged beyond that needed to represent the region of interest.

In *Marfurt* (1984), the relative merits of finite difference algorithms for the solution of the elastic wave equations in the time domain and frequency domain were compared. It was concluded that frequency domain (i.e. implicit) finite difference (FD) methods are often preferable to (explicit) time domain FD methods. Of course, our multiple scattering method is inherently a frequency domain method, indeed–multiple scattering does not make sense in the time domain, because causality requires that the scattering interactions at two distinct points in space-time are dependent. Consequently, both the multiple scattering (MS) and the implicit FD methods require the solution of large linear systems at each frequency. This linear system is sparse and banded in the implicit FD method, but it is fully populated in the MS method. The size of the linear system to be solved is similar for both methods. Consequently, it would appear that our MS method would not be computationally competitive with implicit FD methods. However, this comparison overlooks the relative conditioning of the linear systems. We will take great pains to formulate the MS problem as a *compact linear integral operator equation*, which means that it is relatively easy to find numerical discretizations of arbitrarily low truncation error (Milne, 1980). In particular, compact operators enjoy the favorable characteristic that a numerical representation of the operator remains well conditioned as the numerical mesh becomes finer. This is certainly not the case with the implicit finite difference approximation to the elastic wave equations. The elastic wave equations are differential operators, and consequently, they are unbounded. The unbounded nature of the problem is manifest by the deterioration in the

conditioning of the numerical discretization of the differential formulation, when the grid size becomes finer. Hence, the linear systems that result from the implicit finite difference discretization of the elastic wave equations are more poorly conditioned than the equations that arise from the numerical discretization of the integral formulation of the problem.

We recall that the conditioning of a linear system is a statement about the complex plane distribution of the eigenvalues of that system. In poorly conditioned problems, the eigenvalues are spread out over the plane, while they are bunched together in well conditioned problems (Watkins, 1994). The distribution of the eigenvalues directly impacts on the efficiency of iterative schemes for the solution of linear systems. Due to its poor conditioning, numerical practitioners often employ *preconditioning operators* to the FD equations to modify its spectrum (Bruaset, 1995). We recall that a preconditioning operator is an approximate inverse of the linear system whose solution is required. Indeed, if we employed the exact inverse as the preconditioning operator, the preconditioned system would then require the solution of a linear system with an identity matrix (all eigenvalues λ are bunched at $\lambda = 1$), and any consistent iterative scheme would converge in a single iteration.

The great advantage of the integral (i.e. multiple scattering) formulation of the elastic wave problem is that preconditioning is built into the equations from the outset. In the MS formulation, we employ a Green's tensor for the reference medium, to propagate the wave field along its multiple scattering path. But the reference Green's tensor is the *inverse* of the elastic wave equations that model wave propagation in the reference medium, and consequently, it is a viable preconditioning operator. This is the main reason that the multiple scattering equations are better conditioned than the corresponding equations that result from an implicit finite difference discretization of the elastic wave equation.

However, the apparent gains won by employing our (integral) multiple scattering method should be tempered by the realization that each successive term of the scattering series requires a single frequency, multi-source receiver seismogram. The number of terms in the series is usually about 20 for a model with 10% heterogeneity spanning the spatial region $10\lambda_{eff} \times 10\lambda_{eff}$, where λ_{eff} is the effective wavelength at the dominant frequency. Like the other discrete grid methods based on finite differences, and finite-elements, our method becomes very expensive at high temporal frequencies.

Calculating one frequency with our method, is considerably more expensive than calculating a single time step with the pseudo-spectral method. This is exacerbated by the more stringent spatial sampling requirements of

our method. In particular, the lattice spacing Δ of the discretized secondary source field, needed to model the scattering from the residual structure, must be fine on the scale λ_{eff}, say $\lambda_{eff}/16 \leq \Delta \leq \lambda_{eff}/8$. A wavefield is sensitive to variations in its propagation environment at all scale lengths, not just the scale lengths that can be resolved at the Nyquist sampling rate of the wavefield. However, for practical purposes, we must place a resolution limit on our representation of the propagation environment.

If the *spatial sampling rate* Δ is chosen as $\Delta = \lambda_{eff}/8$, the number of lattice points N_{scat}, needed to model a region encompassing $10\lambda_{eff} \times 10\lambda_{eff}$, is $N_{\text{scat}} \geq 6400$. We will soon see, that each such scattering point has 8 degrees of freedom, to give at total of at least $N = 8 \times N_{\text{scat}} = 51200$ unknowns. Since each scattering point, can interact with every other scattering point, (including itself), we get at least $N^2 \approx 2.6 \times 10^9$ possible interactions at each successive multiple scattering interaction of the incident wavefield. To directly account for each of these interactions in turn is impractical, and this is certainly, the main reason that simple multiple scattering technologies described here have not been used in computational seismology to date.

We avoid the enormous computational cost of calculating the N^2 interactions directly, by *factoring the wavefield*. We propose two factorizations that are complementary in their range of optimum utility:

- we propose an *alternating direction* sweep method based on a Riccati factorization of the wavefield, that accounts for all possible interactions between points on a regular 2−D lattice (over depth and offset), in less than $O(N^2)$ floating point operations. Using this wavefield factorization, the cost of calculating the 2.6×10^9 interactions is significantly reduced. For bigger problems, the reduction in computation is even larger. *Riccati factorization* schemes for discretely layered media are well known [*Chin, Hedstrom & Thigpen* (1984), *Ha* (1985)], and are implicit in the *invariant embedding schemes* of *Kennett* (1984), *Tromp & Snieder* (1988).

- a *cumulative forward scattering* factorization, that generalizes the method of *Wu* (1994) to a weakly stratified reference halfspace. In this method we march the forward scattered field.

The Riccati sweep method, is a two way marching scheme that factorizes the reference medium Green's tensor, but does not include any of the scattered field in the reference solution. Hence, a multiple scattering series based on this scheme converges rapidly, even when there are strong gradients in the reference model, provided that the superimposed heterogeneity is not too

strong. The complementary problem, where the reference model is rather smooth, but the superimposed heterogeneity may be strong is best handled by the cumulative forward scattering factorization, as the forward scattered energy is incorporated directly into the reference solution, and we sum a multiple scattering series for the neglected back scattered field.

A *fast multipole method* has recently been proposed [*Rokhlin* (1985), *Rokhlin* (1990)] that is based on *classical potential theory* for the *Laplace equation*. This method can compute an approximation to the N^2 gravitational interactions between N point masses, in very much less than N^2 floating point operations. In the problem addressed in this thesis, we generate the wave field by spatial differentiation of solutions of the *Helmholtz equation*, so that much of the theory proposed in *Rokhlin* (1985) and *Rokhlin* (1990) can be adapted to our problem. The main problem with the existing fast multipole theory, is that it is based on the *spatial domain free space Green's function*, so that it is not readily extensible to plane stratified media. A second, but less severe problem, is the very sophisticated mathematics, and computer programming needed to implement the method.

The main advantages of the multiple scattering approach adopted here, over competing technologies, based directly on discretization of the elastic wave equations,[e.g. *Kosloff et al.* (1984), *Reshef & Kosloff* (1985), *Vireaux* (1986), *Levander* (1988)] are:

- *free-surface, and radiation conditions* are built into the development from the outset

- it is straightforward to introduce a point source

- attenuation (both causal and acausal) are readily accommodated

- the components of the model have a direct and clear interpretation in terms of the physics of the underlying wave propagation processes, allowing the interpreter great flexibility in his/her choice of model e.g. it is straightforward to isolate single scattered arrivals from multi-scattered arrivals, or separate body wave from surface waves, etc.

- the numerical discretization of our problem has spectral properties that are favourable for numerical solution via iterative methods, whereas discretizations based directly on the differential formulation are notoriously difficult to solve using iterative solution methods

1.2 the basic formalism

Our method is but one of a whole class of methods, that attempt to construct the solution of the given elastic wave equations in terms of a soluble allied problem. This method is rather flexible, as it applies to the entire class of *linear field equations*, although it assumes a multitude of manifestations, depending upon the precise nature of the relationship between the given problem and its allied counterpart. The success of the method hinges, on the ability solve an allied problem, whose solution reflects the actual physics of the given problem.

A recent application of this concept in the context of high frequency body wave propagation was proposed by *Coates & Chapman* (1991), where the allied problem was chosen as the high frequency asymptotic approximation of the elastic wave equation. These authors went on to develop a *representation theorem*, that expressed the elastic wavefield in terms of the solution of the asymptotic allied problem. The advantage of their approach is that the allied solution can be expected to be a fairly good approximation to the required elastic wavefield, at least for high frequencies and smooth variations of material properties, so that the equations that result from their representation theorem need only be solved to within the Born approximation. A potential disadvantage is that the equation error of the asymptotic allied problem, does not have compact support, and this complicates the representation theorem through the need to incorporate complicated boundary integrals. Also, it would appear rather difficult to solve the integral equations beyond the level of the first Born approximation.

For the bulk of this thesis we follow the more traditional route, where the allied problem is chosen as the elastic wave equation, but assume that the material variations are simple enough that the solution of the allied equation may be explicitly constructed. The added material variations, not already included in the reference model, then act as a *secondary or virtual source field* to be superimposed on the reference field. As pointed out by *Coates & Charrette* (1993), this approach has the significant disadvantage that the incident field at each scattering interaction is *habitually mis-timed*. For the most part, we have chosen to live with this deficiency and focus attention instead, on the development of *iterative schemes* that are robust enough to converge the multiple scattering series representation of the secondary source field. This task is aided by the development of Riccati factorization schemes that allow us the propagate the wavefield between successive scattering interactions in an efficient and stable manner.

A recent publication (Wu, 1994), concerning *the elastic wave phase screen*

method has addressed some of the very issues of concern in our method. The method of *Wu* (1994) is basically a single scattering method, where the *cumulative forward scattered field* is marched forward. In the most basic version of our multiple scattering method, we sum the multiple scattering series as a Born series, which is equivalent to solving the matrix representation of the scattered field using a *Jacobi iteration* scheme. Instead, *Wu* (1994) employs what is equivalent, to a single iteration of the *Gauss-Seidel* scheme. In the Gauss-Seidel version, the forward scattered reference field has the correct phase for direct interaction paths between successive scattering points, which corrects for the *habitual mis-timing* in our basic (Born series) method. This is at the cost of employing a reference solution that ignores both the reflected field in the reference model, as well as the backscattered field from the residual model. The consequence of this alternate choice of reference field is that the multiple scattering series for the secondary source field converges much faster, than the corresponding series for a reference model that does not include any forward scattered energy, in those problems where reflected phases in the reference model are unimportant (i.e. in smooth reference models). We employ a set of *wavefield balances* to find a representation of the backscattered field, in so doing, we have extended the method of *Wu* (1994) into a complete wave theory, in the same way that *Coates & Chapman* (1991), extend ray theory.

However, for the bulk of the analysis and computation, we have concentrated on the two-way Riccati factorization approach, where the reference Green's tensor is the exact solution for a plane stratified halfspace, and *none* of the scattered field has been incorporated within it. The burden of convergence of the multiple scattering series is then borne by the method used to sum the multiple scattering partial sums. Herein, lies the principal contribution of this research. We have employed two summation schemes, that exploit complementary properties of the wavefield.

The first summation method is a generalization of the *Schwinger-Levine* variational method introduced into seismology by *Lapwood & Hudson* (1975). A Schwinger-Levine variational principle is an expression of the symmetry of a scattered wavefield when the source point and the field point are interchanged, i.e. *a statement of scattered wavefield reciprocity*. Mathematically, the primal and reciprocal scattering experiments are *dual* statements of the same physical problem. More explicitly, if we introduce a transition operator $\boldsymbol{\mathcal{T}}$, then the *scattering transition* from the reference medium incident state $\boldsymbol{b}_{inc}$ to reference medium the scattered state $\boldsymbol{b}_{sc}$ is modelled as

$$\langle \boldsymbol{b}_{sc}, \boldsymbol{\mathcal{T}} \boldsymbol{b}_{inc} \rangle$$

while the *reciprocal transition* from $\boldsymbol{b}_{sc}$ to $\boldsymbol{b}_{inc}$ is modelled as

$$\left\langle \boldsymbol{\mathcal{T}}^{\dagger}\boldsymbol{b}_{sc}, \boldsymbol{b}_{inc} \right\rangle$$

where $\boldsymbol{\mathcal{T}}^{\dagger}$ is the *adjoint transition operator* relative to the bilinear form $\langle \cdot, \cdot \rangle$ that expresses the scattering transitions induced by the lateral heterogeneities superimposed on the reference medium.

In *Lapwood & Hudson* (1975), a bilinear form was introduced for the plane stratified problem. We have generalized this bilinear form to the heterogeneous problem. The second extension of research of *Lapwood & Hudson* (1975) concerns extending the accuracy of their method beyond second order Born theory. The key step in this extension is to regard their variational expression as a low order *Padé* approximation to the true multiple scattering series representation of the scattered field. The higher order corrections to the leading order expression of *Lapwood & Hudson* (1975) may then be derived by writing down the relevant higher order *Padé* approximation to the true multiple scattering series. More importantly, these high order *Padé* approximations can be calculated by exploiting their well known relationship to the *bi-conjugate gradient method*. Using this line of reasoning, we have extended the *Lapwood & Hudson* variational method to arbitrarily high orders of perturbation theory. However, this method is deficient in two respects:

- due to the dependence of the method on seismic reciprocity, each source/receiver combination requires the solution of a separate problem
- the bilinear form that expresses the scattering transition is indefinite. Consequently, convergence of the method is not guaranteed

To avoid these deficiencies, we have also generalized the *Generalized Minimal Residual Method* (GMRES) of *Saad & Schultz* (1986) which is based on solving a set of *linear least-squares minimization problems*, optimized over a trial space spanned by the multiple scattering series partial sums. In each iteration of this method, the trial space is augmented by an extra multiple scattering interaction. The advantage of the GMRES method is that the residual of the current solution may not increase from one iteration to the next, so that the convergence of the method is very smooth. Also, the method calculates the response at all receivers from a given source point, simultaneously. Our generalization of the basic method involves:

- the extension to complex arithmetic

- the use of a recursive factorization scheme based on *unitary Householder reflections*, to solve the linear least squares problem at each iteration of the GMRES method

1.3 notation

In this thesis, we make use of both the *Kronecka product* between matrices, and the *tensor product* between vectors. Although both algebraic operators are represented by the binary operator "$\otimes$", they should considered as distinct mathematical operations. The Kronecka product $\boldsymbol{A} \otimes \boldsymbol{B}$ of an $m \times n$ matrix $\boldsymbol{A}$ and a $p \times q$ matrix $\boldsymbol{B}$ is the $(m+p) \times (n+q)$ matrix

$$\boldsymbol{A} \otimes \boldsymbol{B} \equiv \begin{bmatrix} A_{11}\boldsymbol{B} & \cdots & A_{1n}\boldsymbol{B} \\ \vdots & \cdots & \vdots \\ A_{m1}\boldsymbol{B} & \cdots & A_{mn}\boldsymbol{B} \end{bmatrix}$$

whereas the *tensor* product $\boldsymbol{a} \otimes \boldsymbol{b}$ of an $m-$vector $\boldsymbol{a}$ and a $p-$vector $\boldsymbol{b}$ is the $m \times p$ matrix

$$\boldsymbol{a} \otimes \boldsymbol{b} \equiv \begin{bmatrix} a_1\boldsymbol{b}^T \\ \vdots \\ a_m\boldsymbol{b}^T \end{bmatrix}$$

so that *there is an implied transposition in the tensor product between vectors, but there is no implied transposition in the Kronecka product between matrices*. The mixed cases $\boldsymbol{A} \otimes \boldsymbol{b}$ and $\boldsymbol{b} \otimes \boldsymbol{A}$ will always be interpreted as *Kronecka* products, so there is no implied transposition in these cases.

Chapter 2

Representation Theory

In this chapter we introduce the integral equations that govern the time harmonic elastic wavefield in an arbitrary 3−D isotropic elastic halfspace with superimposed 2−D space filling heterogeneity. The formulation is abstract, because we do not provide *constructive representations* for the various operators, introduced here. Computational representations for these operators will be developed incrementally in the later chapters of this thesis. The present chapter serves as a prelude to those developments.

Our presentation differs from previous work, in that it is based directly on the first order displacement-traction equations rather than the more common coupled second order displacement field equations. We derive a *displacement-traction representation theorem* in §2.2 that plays a role analogous to the *Rayleigh-Betti representation theorem* of the corresponding coupled second order theory. We then employ the new representation theorem to derive the time harmonic *elastic wave Lippmann-Schwinger type integral equation* in §2.4.3.

2.1 The multiple scattering problem

2.1.1 The multiple scattering problem geometry

We partition the plane $\boldsymbol{x} = (z, x) \in R^2$ into the disjoint union of the *scattering zone* Ω of compact support and its complement Ω^e which we will refer to as the *external zone* (Fig.2.1). The oriented boundary of Ω is denoted $\partial\Omega^+$, while $\partial\Omega^\infty$ is the *virtual semi-circular boundary* at infinity, with $\partial\Omega^f$ the *free surface* component of the external boundary. We cut Ω^e along ϑ, to ensure that Ω^e is *simply connected.* The contribution from the cut ϑ cancels

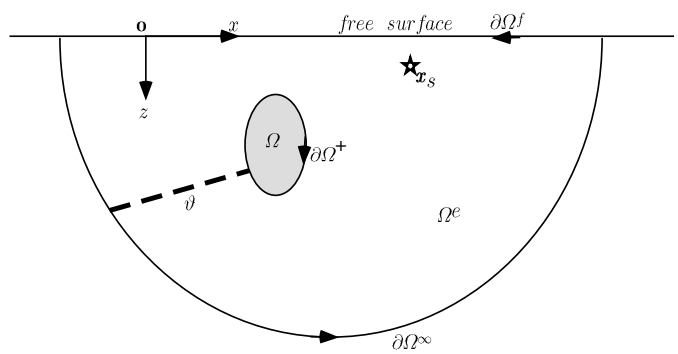

Figure 2.1: The Representation Theorem Geometry

because the wavefield is continuous across the cut. The cuts ensure that the boundary contour $\partial\Omega^+$ of the scattering zone Ω has the proper sense.

Let χ_Ω denote the *characteristic function* on Ω, defined by

$$\chi_\Omega = \begin{cases} 1 & : \quad \boldsymbol{x} \in \Omega \\ 0 & : \quad \boldsymbol{x} \notin \Omega \end{cases}, \tag{2.1}$$

so that χ_Ω takes unit value in the scattering zone but identically vanishes in the external zone. The isotropic material properties $\boldsymbol{m}(z,x) = (\rho, \alpha, \beta)(z,x)$, which denote the density, compressional and shear-wavespeeds respectively, are defined as the sum of two parts: a *reference* component $\mathring{\boldsymbol{m}} = (\mathring{\rho}, \mathring{\alpha}, \mathring{\beta})$, defined throughout the halfspace; and a *scattering* component $\Delta\boldsymbol{m}(z,x) = (\Delta\rho, \Delta\alpha, \Delta\beta)$ confined to the scattering zone. Hence,

$$\boldsymbol{m}(z,x) = \mathring{\boldsymbol{m}} + \chi_\Omega \Delta\boldsymbol{m}(z,x). \tag{2.2}$$

2.1.2 The Governing equations

Assuming that the depth below the free surface is measured by the $z-$coordinate, the forced elastic displacement $\boldsymbol{u} = (u_z, u_x, u_y)^T$ and traction vector $\boldsymbol{t} = (\tau_{zz}, \tau_{xz}, \tau_{yz})^T$ on a fixed depth plane (i.e. $z = const.$) with unit normal $\boldsymbol{e}_z$, are grouped into the 6-vector

$$\boldsymbol{b}(z,x,y) = \left[\ u_z(z,x,y), \quad u_x(z,x,y), \quad u_y(z,x,y), \quad \tau_{zz}(z,x,y), \quad \tau_{xz}(z,x,y), \quad \tau_{yz}(z,x,y)\ \right]^T \tag{2.3}$$

The governing equation for time harmonic elastic wave motion in an *isotropic medium* is then a set of 6 coupled partial differential equations (first order

parameter	(ρ,α,β)-definition	(ρ,λ,μ)-definition
P	α	$\sqrt{\frac{\lambda+2\mu}{\rho}}$
S	β	$\sqrt{\frac{\mu}{\rho}}$
H	β	$\sqrt{\frac{\mu}{\rho}}$
γ	$\left(1-2\frac{\beta^2}{\alpha^2}\right)$	$\frac{\lambda}{\lambda+2\mu}$
ζ	$4\rho\beta^2\left(1-\frac{\beta^2}{\alpha^2}\right)$	$\frac{4\mu(\lambda+\mu)}{\lambda+2\mu}$
χ	$2\rho\beta^2\left(1-2\frac{\beta^2}{\alpha^2}\right)$	$\frac{2\lambda\mu}{\lambda+2\mu}$
a	$\frac{1}{\rho\alpha^2}$	$\frac{1}{\lambda+2\mu}$
b	$\frac{1}{\rho\beta^2}$	$\frac{1}{\mu}$

Table 2.1: The material parameter definitions

with respect to the depth coordinate) which may be written in the operator notation

$$\boldsymbol{\mathcal{L}}(\boldsymbol{x};\partial_z,\partial_x,\partial_y)\cdot\boldsymbol{b}\equiv[\partial_z-\boldsymbol{\mathcal{A}}(\boldsymbol{x};\partial_x,\partial_y)]\cdot\boldsymbol{b}=\boldsymbol{f} \tag{2.4}$$

where $\boldsymbol{b}$ is given in (2.3) and $\boldsymbol{\mathcal{A}}$ is defined as

$$\boldsymbol{\mathcal{A}}(\boldsymbol{x};\partial_x,\partial_y)=\begin{bmatrix} 0 & -\gamma\partial_x & -\gamma\partial_y & a & 0 & 0\\ -\partial_x & 0 & 0 & 0 & b & 0\\ -\partial_y & 0 & 0 & 0 & 0 & b\\ -\rho\omega^2 & 0 & 0 & 0 & -\partial_x & -\partial_y\\ 0 & -\rho\omega^2-\partial_x\zeta\partial_x-\partial_y\mu\partial_y & -\partial_x\chi\partial_y-\partial_y\mu\partial_x & -\partial_x\gamma & 0 & 0\\ 0 & -\partial_y\chi\partial_x-\partial_x\mu\partial_y & -\rho\omega^2-\partial_y\zeta\partial_y-\partial_x\mu\partial_x & -\partial_y\gamma & 0 & 0 \end{bmatrix} \tag{2.5}$$

and the material parameters in (2.5) are defined in Table 2.1.

This thesis is concerned with finding solutions of (2.4) when the material parameters in Table 2.1 are functions of (z,x) but are independent of the $y-$coordinate. This problem is sometimes called the $2\,{}^1\!/_2-$D problem because the wavefield may spread in 3−D, but the heterogeneity is confined to 2−D. We exploit the translational invariance of the medium along the $y-$coordinate by assuming that the displacement/traction field $\boldsymbol{b}(z,x,y)$ may be *synthesized* from its k_y plane wave components as

$$\boldsymbol{b}(z,x,y)=\frac{1}{2\pi}\int_R \mathrm{d}k_y\,\boldsymbol{b}(z,x,k_y)\,\mathrm{e}^{ik_y y} \tag{2.6}$$

where $\boldsymbol{b}(z, x, k_y)$ is now governed by

$$\boldsymbol{\mathcal{L}}(z, x; \partial_z, \partial_x, ik_y) \cdot \boldsymbol{b}(z, x, k_y; \omega) \equiv [\partial_z - \boldsymbol{\mathcal{A}}(z, x; \partial_x, ik_y)] \cdot \boldsymbol{b}(z, x, k_y; \omega) = \tilde{\boldsymbol{f}} \tag{2.7}$$

with

$$\boldsymbol{\mathcal{A}}(z, x; \partial_x, ik_y) = \begin{bmatrix} 0 & -\gamma\partial_x & -i\gamma k_y & a & 0 & 0 \\ -\partial_x & 0 & 0 & 0 & b & 0 \\ -ik_y & 0 & 0 & 0 & 0 & b \\ -\rho\omega^2 & 0 & 0 & 0 & -\partial_x & -ik_y \\ 0 & -\rho\omega^2 - \partial_x\zeta\partial_x + \mu k_y^2 & -ik_y\partial_x\chi - ik_y\mu\partial_x & -\partial_x\gamma & 0 & 0 \\ 0 & -ik_y\chi\partial_x - ik_y\partial_x\mu & -\rho\omega^2 + \zeta k_y^2 - \partial_x\mu\partial_x & -ik_y\gamma & 0 & 0 \end{bmatrix} \tag{2.8}$$

and

$$\tilde{\boldsymbol{f}} = \int_R \mathrm{d}k_y \, \boldsymbol{f}(z, x, y) \, \mathrm{e}^{-ik_y y}. \tag{2.9}$$

In this work, we assume the the motion is driven by a point source in the $y = 0$ plane, i.e. $\boldsymbol{f}(z, x, y) = \boldsymbol{F}(z, x)\delta(y)$, so that from (2.9) we have

$$\tilde{\boldsymbol{f}} = \int_R \mathrm{d}k_y \, \boldsymbol{F}(z, x)\delta(y) \, \mathrm{e}^{-ik_y y} = \boldsymbol{F}(z, x). \tag{2.10}$$

Due to the translational invariance of the medium in the $y-$coordinate, if the source plane is translated to $y = y_1$, then the spatial domain solution $\boldsymbol{b}(z, x, y; \omega)$ is translated to $\boldsymbol{b}(z, x, y - y_1; \omega)$. Consequently, it is straight forward to derive the solution appropriate to general source conditions, by translating the field for a point source in the $y = 0$ plane, accordingly.

In (2.7) both ω and k_y are *passive parameters*. The space/time solution is constructed from a superposition of solutions with different values of ω and k_y as prescribed by the *inverse Fourier transforms* (3.2) on frequency and (2.6) on k_y. The main task remaining, is the construction the transformed displacement/traction vector $\boldsymbol{b}(z, x, k_y; \omega)$ in (2.7). Often, we will not indicate the explicit dependence of $\boldsymbol{b}(z, x, k_y; \omega)$ on k_y and ω but write $\boldsymbol{b}(z, x)$ instead.

The representation (2.7) is most convenient when the dominant material variation is in the z coordinate direction, as we have singled out this coordinate for special treatment, allowing easy application of boundary and interface conditions on fixed depth planes. Also, the displacement $\boldsymbol{u}$ and depth

plane traction $\boldsymbol{t}$ are accorded equal status. In the more traditional displacement formulation, the offset x and depth z coordinates appear within the equations in a symmetric form (due to isotropy) whereas the traction field must be derived from the displacement via the elastic analogue of *Hooke's law* which involves further spatial differentiations of $\boldsymbol{u}$.

We note in passing that (2.7) simplifies considerably in the special case where the motion is driven by a *line source*, along the $y-$axis so that we may set $k_y \equiv 0$ in (2.8) which then reduces to

$$\boldsymbol{\mathcal{A}}(z,x;\partial_x,0) = \begin{bmatrix} 0 & -\gamma\partial_x & 0 & a & 0 & 0 \\ -\partial_x & 0 & 0 & 0 & b & 0 \\ 0 & 0 & 0 & 0 & 0 & b \\ -\rho\omega^2 & 0 & 0 & 0 & -\partial_x & 0 \\ 0 & -\rho\omega^2 - \partial_x\zeta\partial_x & 0 & -\partial_x\gamma & 0 & 0 \\ 0 & 0 & -\rho\omega^2 - \partial_x\mu\partial_x & 0 & 0 & 0 \end{bmatrix}.$$

In this instance, (2.7) may be decomposed into its *in plane* equation

$$\frac{\partial}{\partial z}\begin{bmatrix} u_z \\ u_x \\ \tau_{zz} \\ \tau_{xz} \end{bmatrix} = \begin{bmatrix} 0 & -\gamma\partial_x & a & 0 \\ -\partial_x & 0 & 0 & b \\ -\rho\omega^2 & 0 & 0 & -\partial_x \\ 0 & -\rho\omega^2 - \partial_x\zeta\partial_x & -\partial_x\gamma & 0 \end{bmatrix}\begin{bmatrix} u_z \\ u_x \\ \tau_{zz} \\ \tau_{xz} \end{bmatrix} \tag{2.11}$$

for the P/SV field and *out of plane* equation

$$\frac{\partial}{\partial z}\begin{bmatrix} u_y \\ \tau_{yz} \end{bmatrix} = \begin{bmatrix} 0 & b \\ -\rho\omega^2 - \partial_x\mu\partial_x & 0 \end{bmatrix}\begin{bmatrix} u_y \\ \tau_{yz} \end{bmatrix} \tag{2.12}$$

for the SH field. For the most part, we will concentrate on the solution of (2.7) because the source conditions needed for the applicability (2.11) and (2.12) are too restrictive.

Consistent with the decomposition of the material properties in (2.2), we *split* (2.8) into

$$\boldsymbol{\mathcal{A}}(\boldsymbol{x};\partial_x,\partial_y;k_y) = \mathring{\boldsymbol{A}}(\boldsymbol{x};\partial_x,\partial_y) + \Delta\boldsymbol{A}(\boldsymbol{x};\partial_x,\partial_y) \tag{2.13}$$

contrast	definition
$\Delta\rho$	$\rho - \mathring{\rho}$
$\Delta\mu$	$\mu - \mathring{\mu}$
$\Delta\gamma$	$2\frac{\mathring{\beta}^2}{\mathring{\alpha}^2} - 2\frac{\beta^2}{\alpha^2}$
$\Delta\zeta$	$4\rho\beta^2\left[1 - \frac{\beta^2}{\alpha^2}\right] - 4\mathring{\rho}\mathring{\beta}^2\left[1 - \frac{\mathring{\beta}^2}{\mathring{\alpha}^2}\right]$
$\Delta\chi$	$2\rho\beta^2\left[1 - 2\frac{\beta^2}{\alpha^2}\right] - 2\mathring{\rho}\mathring{\beta}^2\left[1 - 2\frac{\mathring{\beta}^2}{\mathring{\alpha}^2}\right]$
Δa	$\frac{1}{\rho\alpha^2} - \frac{1}{\mathring{\rho}\mathring{\alpha}^2}$
Δb	$\frac{1}{\rho\beta^2} - \frac{1}{\mathring{\rho}\mathring{\beta}^2}$

Table 2.2: The material contrast parameters

where

$$\mathring{\boldsymbol{A}}(z,x;\partial_x,k_y) = \begin{bmatrix} 0 & -\mathring{\gamma}\partial_x & -i\mathring{\gamma}k_y & \mathring{a} & 0 & 0 \\ -\partial_x & 0 & 0 & 0 & \mathring{b} & 0 \\ -ik_y & 0 & 0 & 0 & 0 & \mathring{b} \\ -\mathring{\rho}\omega^2 & 0 & 0 & 0 & -\partial_x & -ik_y \\ 0 & -\mathring{\rho}\omega^2 - \partial_x\mathring{\zeta}\partial_x + \mathring{\mu}k_y^2 & -ik_y\partial_x\mathring{\chi} - ik_y\mathring{\mu}\partial_x & -\partial_x\mathring{\gamma} & 0 & 0 \\ 0 & -ik_y\mathring{\chi}\partial_x - ik_y\partial_x\mathring{\mu} & -\mathring{\rho}\omega^2 + \mathring{\zeta}k_y^2 - \partial_x\mathring{\mu}\partial_x & -ik_y\mathring{\gamma} & 0 & 0 \end{bmatrix} \tag{2.14}$$

where $\mathring{\gamma}$ is the *reference medium* value of γ from Table 2.1 and similar definitions hold for the remaining *reference* material parameters in (2.14). Also,

$$\Delta\boldsymbol{A}(z,x;\partial_x,k_y) = \begin{bmatrix} 0 & -\Delta\gamma\partial_x & -i\Delta\gamma k_y & \Delta a & 0 & 0 \\ 0 & 0 & 0 & 0 & \Delta b & 0 \\ 0 & 0 & 0 & 0 & 0 & \Delta b \\ -\Delta\rho\omega^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\Delta\rho\omega^2 - \partial_x\Delta\zeta\partial_x + \Delta\mu k_y^2 & -ik_y\partial_x\Delta\chi - ik_y\Delta\mu\partial_x & -\partial_x\Delta\gamma & 0 & 0 \\ 0 & -ik_y\Delta\chi\partial_x - ik_y\partial_x\Delta\mu & -\Delta\rho\omega^2 + \Delta\zeta k_y^2 - \partial_x\Delta\mu\partial_x & -ik_y\Delta\gamma & 0 & 0 \end{bmatrix} \tag{2.15}$$

with the material contrast parameters in (2.15) defined in Table 2.2.

Using the decomposition of (2.13) in the governing equation (2.7) gives

$$\boldsymbol{L}(z,x;\partial_z,\partial_x,k_y)\cdot\boldsymbol{b}(z,x,k_y;\omega) = \tilde{\boldsymbol{f}} + \Delta\boldsymbol{A}(z,x;\partial_x,k_y)\cdot\boldsymbol{b}(z,x,k_y;\omega) \tag{2.16}$$

where the *reference medium* operator $\boldsymbol{L}$ is defined as

$$\boldsymbol{L}(z,x;\partial_z,\partial_x,k_y) \equiv \left[\partial_z - \mathring{\boldsymbol{A}}\,(z,x;\partial_x,k_y)\right] \tag{2.17}$$

with $\mathring{\boldsymbol{A}}$ given in (2.14). Scrupulously indicating all the arguments of the various operators introduced in this thesis can be very cumbersome, so for example, we will often write $\boldsymbol{L}$ to mean $\boldsymbol{L}(z,x;\partial_z,\partial_x,k_y)$ when no misunderstanding is likely.

2.2 The 2−D Elastic Representation Theorem

Given the linear differential operator $\boldsymbol{L}$ of (2.16), it is possible to derive an *adjoint* operator $\boldsymbol{L}^\dagger$ relative to an arbitrary *bilinear form* $(\cdot,\cdot)$, so that (Fig.2.2)

$$\begin{aligned}(\boldsymbol{b}_2,\boldsymbol{L}\boldsymbol{b}_1) - (\boldsymbol{L}^\dagger\boldsymbol{b}_2,\boldsymbol{b}_1) &= \int (\partial_z\boldsymbol{\Phi}_z + \partial_x\boldsymbol{\Phi}_x)\,\mathrm{d}z\mathrm{d}x \\ &= \oint (\boldsymbol{\Phi}_z\,\mathrm{d}x - \boldsymbol{\Phi}_x\,\mathrm{d}z)\end{aligned} \tag{2.18}$$

for any pair of sufficiently differentiable, 6−component vector fields $\boldsymbol{b}_1(z,x)$ and $\boldsymbol{b}_2(z,x)$, not necessarily solutions of $\boldsymbol{L}$ or its adjoint $\boldsymbol{L}^\dagger$ and $\boldsymbol{\Phi} = (\boldsymbol{\Phi}_z,\boldsymbol{\Phi}_x)^T$ is a yet to be determined 2−component *flux* field. Here we adopt the specific bilinear form $(,)_\Omega$, defined over the scattering region Ω according to

$$(\boldsymbol{b}_1,\boldsymbol{b}_2)_\Omega \equiv \int_\Omega \mathrm{d}\boldsymbol{x}\,\langle\boldsymbol{b}_1(\boldsymbol{x}),\boldsymbol{b}_2(\boldsymbol{x})\rangle \tag{2.19}$$

where $\langle\boldsymbol{b}_1(\boldsymbol{x}),\boldsymbol{b}_2(\boldsymbol{x})\rangle$ is defined as

$$\langle\boldsymbol{b}_1(\boldsymbol{x}),\boldsymbol{b}_2(\boldsymbol{x})\rangle \equiv \boldsymbol{b}_1^T\boldsymbol{J}_6\boldsymbol{b}_2(\boldsymbol{x}) \tag{2.20}$$

and $\boldsymbol{J}_6$ is given by

$$\boldsymbol{J}_6 = \begin{bmatrix} \boldsymbol{0}_3 & \boldsymbol{I}_3 \\ -\boldsymbol{I}_3 & \boldsymbol{0}_3 \end{bmatrix}. \tag{2.21}$$

Using the *product rule for partial differentiation*,

$$\frac{\partial\; a(x)\; b(x)}{\partial\, x} \;=\; b(x)\;\frac{\partial\; a(x)}{\partial\, x} \;+\; a(x)\;\frac{\partial\; b(x)}{\partial\, x}$$

[Figure: G2DThm not found]

Figure 2.2: The 2−D Green's Theorem

of differentiable functions $a(x)$ and $b(x)$, it can be verified that

$$\langle \boldsymbol{b}_1, \boldsymbol{L}\boldsymbol{b}_2 \rangle - \left\langle \boldsymbol{L}^\dagger \boldsymbol{b}_1, \boldsymbol{b}_2 \right\rangle = \partial_z \langle \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle + \partial_x \langle \boldsymbol{T}_x(-k_y)\boldsymbol{b}_1, \boldsymbol{T}_x(k_y)\boldsymbol{b}_2 \rangle \quad (2.22)$$

where $\boldsymbol{L}^\dagger$ is the operator adjoint of $\boldsymbol{L}$, defined as

$$\boldsymbol{L}^\dagger(\boldsymbol{x}; \partial_z, \partial_x, k_y) \equiv -\boldsymbol{L}(\boldsymbol{x}; \partial_z, \partial_x, -k_y) \quad (2.23)$$

and

$$\boldsymbol{T}_x(k_y) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & \mathring{\zeta}\partial_x & i\,k_y\mathring{\chi} & \mathring{\gamma} & 0 & 0 \\ 0 & i\,k_y\mathring{\mu} & \mathring{\mu}\partial_x & 0 & 0 & 0 \end{bmatrix}. \quad (2.24)$$

Integrating both sides of (2.22) over a simply connected spatial region Ω, gives

$$\begin{aligned} &(\boldsymbol{b}_1, \boldsymbol{L}\cdot\boldsymbol{b}_2)_\Omega - \left(\boldsymbol{L}^\dagger \cdot \boldsymbol{b}_1, \boldsymbol{b}_2\right)_\Omega \\ &\quad = \int_\Omega \left[\langle \boldsymbol{b}_1, \boldsymbol{L}(\boldsymbol{x}; \partial_z, \partial_x, k_y)\cdot\boldsymbol{b}_2\rangle + \langle \boldsymbol{L}(\boldsymbol{x}; \partial_z, \partial_x, -k_y)\cdot\boldsymbol{b}_1, \boldsymbol{b}_2\rangle\right] \mathrm{d}\boldsymbol{x} \\ &\quad = \int_\Omega \left[\partial_z \langle \boldsymbol{b}_1, \boldsymbol{b}_2\rangle + \partial_x \langle \boldsymbol{T}_x(-k_y)\boldsymbol{b}_1, \boldsymbol{T}_x(k_y)\boldsymbol{b}_2\rangle\right] \mathrm{d}\boldsymbol{x} \\ &\quad = \oint_{\partial\Omega} \mathrm{d}x \langle \boldsymbol{b}_1, \boldsymbol{b}_2\rangle - \mathrm{d}z \langle \boldsymbol{T}_x(-k_y)\boldsymbol{b}_1, \boldsymbol{T}_x(k_y)\boldsymbol{b}_2\rangle \end{aligned} \quad (2.25)$$

where the last equation in (2.25) follows from the 2−D version of Green's theorem. The integral expression (2.25) is a *balance law* for the two component *flux field*

$$(\boldsymbol{\Phi}_z, \boldsymbol{\Phi}_x) \equiv (\langle \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle, \langle \boldsymbol{T}_x(-k_y)\boldsymbol{b}_1, \boldsymbol{T}_x(k_y)\boldsymbol{b}_2 \rangle)$$

generated by the *virtual source density*

$$\langle \boldsymbol{b}_1, \boldsymbol{L}(\boldsymbol{x}; \partial_z, \partial_x, k_y) \cdot \boldsymbol{b}_2 \rangle + \langle \boldsymbol{L}(\boldsymbol{x}; \partial_z, \partial_x, -k_y) \cdot \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle \equiv$$
$$\overbrace{\langle \boldsymbol{b}_1, \boldsymbol{L}(\boldsymbol{x}; \partial_z, \partial_x, k_y) \cdot \boldsymbol{b}_2 \rangle}^{\text{source density}} - \overbrace{\left\langle \boldsymbol{L}^\dagger(\boldsymbol{x}; \partial_z, \partial_x, k_y) \cdot \boldsymbol{b}_1, \boldsymbol{b}_2 \right\rangle}^{\text{sink density}} \tag{2.26}$$

(*see* Fig.2.2).

Equation (2.25) will be used extensively in the development of this chapter, and to save space, we rewrite it in the following compact notation

$$\boxed{(\boldsymbol{b}_1, \boldsymbol{L}\boldsymbol{b}_2)_\Omega - \left(\boldsymbol{L}^\dagger \boldsymbol{b}_1, \boldsymbol{b}_2\right)_\Omega = \langle \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle_{\partial\Omega}} \tag{2.27}$$

where the boundary integral $\langle \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle_{\partial\Omega}$ is given by the cartesian form

$$\langle \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle_{\partial\Omega} = \oint_{\partial\Omega} \left[\boldsymbol{b}_1^T(z,x)\boldsymbol{J}_6\boldsymbol{b}_2(z,x)\ \mathrm{d}x - (\boldsymbol{T}_x(-k_y)\boldsymbol{b}_1)^T(z,x)\boldsymbol{J}_6\boldsymbol{T}_x(k_y)\boldsymbol{b}_2(z,x)\ \mathrm{d}z \right] \tag{2.28}$$

or the *parametric* form

$$\langle \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle_{\partial\Omega} = \oint_{\partial\Omega} \boldsymbol{b}_1^T(z(s), x(s); -k_y, \hat{\boldsymbol{n}})\boldsymbol{J}_6\boldsymbol{b}_2(z(s), x(s); k_y, \hat{\boldsymbol{n}})\mathrm{d}s \tag{2.29}$$

where $\mathrm{d}s$ is the arc length of the boundary $\partial\Omega$,

$$\hat{\boldsymbol{n}} \equiv \boldsymbol{e}_z \frac{\mathrm{d}\,x}{\mathrm{d}\,s} - \boldsymbol{e}_x \frac{\mathrm{d}\,z}{\mathrm{d}\,s}$$

is the outward unit normal (Fig.2.2) and

$$\boldsymbol{b}_{1,2}(z, x; \hat{\boldsymbol{n}}) = \boldsymbol{N}(\hat{\boldsymbol{n}})\boldsymbol{b}_{1,2}(z, x), \tag{2.30}$$

where

$$\boldsymbol{N}(k_y, \hat{\boldsymbol{n}}) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{dx}{ds} & -\frac{dz}{ds} & 0 \\ 0 & -\frac{dz}{ds}\mathring{\zeta}\partial_x & -ik_y\frac{dz}{ds}\mathring{\chi} & -\frac{dz}{ds}\mathring{\gamma} & \frac{dx}{ds} & 0 \\ 0 & -ik_y\frac{dz}{ds}\mathring{\mu} & -\frac{dz}{ds}\mathring{\mu}\partial_x & 0 & 0 & \frac{dx}{ds} \end{bmatrix}. \tag{2.31}$$

2.3 Reciprocity

The great generality of (2.27) derives from its abstract nature, as the *test* fields $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$ in (2.27) have been arbitrary, subject only to sufficient differentiability to ensure the validity of the various integration by parts operations needed to derive (2.27). We now impose further conditions on these fields. Specifically let us assume they are point source solutions in a halfspace with no inclusions, and continuous elastic properties, so that

$$\left.\begin{array}{ll} \boldsymbol{L}^{\dagger}\cdot\mathring{\boldsymbol{G}}_1(\boldsymbol{x},\boldsymbol{x}_1,k_y)=-\boldsymbol{I}_6\delta(\boldsymbol{x}-\boldsymbol{x}_1), & \boldsymbol{x}_1\in R^2 \\ \boldsymbol{L}\cdot\mathring{\boldsymbol{G}}_2(\boldsymbol{x},\boldsymbol{x}_2,k_y)=\boldsymbol{I}_6\delta(\boldsymbol{x}-\boldsymbol{x}_2), & \boldsymbol{x}_2\in R^2 \end{array}\right\}, \tag{2.32}$$

subject to the traction free condition at the free surface, and the radiation condition at infinity. The differential operator $\boldsymbol{L}$ was given in (2.17) and the corresponding adjoint operator $\boldsymbol{L}^{\dagger}$ was given in (2.23). Note that the *point source solutions* $\mathring{\boldsymbol{G}}_1(\boldsymbol{x},\boldsymbol{x}_1,k_y)$ and $\mathring{\boldsymbol{G}}_2(\boldsymbol{x},\boldsymbol{x}_2,k_y)$ both have two arguments. The right most argument is the *source* coordinate, and the left argument is the *field* coordinate, so that $\mathring{\boldsymbol{G}}_1(\boldsymbol{x},\boldsymbol{x}_1,k_y)$ $\left(\mathring{\boldsymbol{G}}_2(\boldsymbol{x},\boldsymbol{x}_2,k_y)\right)$ has a source at $\boldsymbol{x}_1(\boldsymbol{x}_2)$ and a field coordinate at $\boldsymbol{x}$.

Then (2.27) gives

$$\begin{aligned}&\left(\mathring{\boldsymbol{G}}_1(\boldsymbol{x},\boldsymbol{x}_1,k_y),\boldsymbol{L}\mathring{\boldsymbol{G}}_2(\boldsymbol{x},\boldsymbol{x}_2,k_y)\right)_{\Omega}-\left(\boldsymbol{L}^{\dagger}\mathring{\boldsymbol{G}}_1(\boldsymbol{x},\boldsymbol{x}_1,k_y),\mathring{\boldsymbol{G}}_2(\boldsymbol{x},\boldsymbol{x}_2,k_y)\right)_{\Omega}\\ &\quad\equiv\left\langle\mathring{\boldsymbol{G}}_1(\boldsymbol{x},\boldsymbol{x}_1,k_y),\mathring{\boldsymbol{G}}_2(\boldsymbol{x},\boldsymbol{x}_2,k_y)\right\rangle_{\partial\Omega^{\infty+f}}\equiv\boldsymbol{0}\end{aligned} \tag{2.33}$$

where the boundary integral $\left\langle\mathring{\boldsymbol{G}}_1(\boldsymbol{x},\boldsymbol{x}_1,k_y),\mathring{\boldsymbol{G}}_2(\boldsymbol{x},\boldsymbol{x}_2,k_y)\right\rangle_{\partial\Omega^{\infty+f}}$ is over a surface $\partial\Omega^{\infty}$ at infinity and over the free surface $\partial\Omega^{f}$ and consequently, it vanishes in accordance with the *radiation condition* and the *free surface condition*. Indeed the annihilation of

$$\left\langle\mathring{\boldsymbol{G}}_1(\boldsymbol{x},\boldsymbol{x}_1,k_y),\mathring{\boldsymbol{G}}_2(\boldsymbol{x},\boldsymbol{x}_2,k_y)\right\rangle_{\partial\Omega^{\infty}}$$

may be taken as the definition of the radiation condition. Using (2.32) the left hand side of (2.33) is

$$\begin{aligned}&\left(\mathring{\boldsymbol{G}}_1(\boldsymbol{x},\boldsymbol{x}_1,k_y),\boldsymbol{I}_6\delta(\boldsymbol{x}-\boldsymbol{x}_2)\right)_{\Omega}+\left(\boldsymbol{I}_6\delta(\boldsymbol{x}-\boldsymbol{x}_1),\mathring{\boldsymbol{G}}_2(\boldsymbol{x},\boldsymbol{x}_2,k_y)\right)_{\Omega}=\\ &\quad\mathring{\boldsymbol{G}}_1^{\,T}(\boldsymbol{x}_2;\boldsymbol{x}_1,k_y)\boldsymbol{J}_6+\boldsymbol{J}_6\mathring{\boldsymbol{G}}_2(\boldsymbol{x}_1;\boldsymbol{x}_2,k_y)\equiv\boldsymbol{0}\end{aligned} \tag{2.34}$$

or upon rearranging we get

$$\mathring{\boldsymbol{G}}_1(\boldsymbol{x}_2;\boldsymbol{x}_1,k_y)\equiv\boldsymbol{J}_6\cdot\mathring{\boldsymbol{G}}_2^{\,T}(\boldsymbol{x}_1;\boldsymbol{x}_2,k_y)\cdot\boldsymbol{J}_6 \tag{2.35}$$

From the governing equations (2.32) and the definition of the adjoint operator $\boldsymbol{L}^\dagger$ (2.23)

$$\overset{\circ}{\boldsymbol{G}}_1(\boldsymbol{x}_2;\boldsymbol{x}_1;k_y) \equiv \overset{\circ}{\boldsymbol{G}}_2(\boldsymbol{x}_2;\boldsymbol{x}_1;-k_y)$$

so that

$$\boxed{\overset{\circ}{\boldsymbol{G}}_2(\boldsymbol{x}_2;\boldsymbol{x}_1;k_y) \equiv \boldsymbol{J}_6 \cdot \overset{\circ}{\boldsymbol{G}}{}_2^T(\boldsymbol{x}_1;\boldsymbol{x}_2;-k_y) \cdot \boldsymbol{J}_6} \tag{2.36}$$

which is the *reciprocity* relation for point source solutions in free space.

Very often, we split the Green's tensor $\overset{\circ}{\boldsymbol{G}}(\boldsymbol{x}_2;\boldsymbol{x}_1;k_y)$ into the product

$$\overset{\circ}{\boldsymbol{G}}(\boldsymbol{x}_2;\boldsymbol{x}_1;k_y) \equiv \overset{\circ}{\boldsymbol{P}}(\boldsymbol{x}_2;\boldsymbol{x}_1;k_y)\boldsymbol{J}_6$$

where $\overset{\circ}{\boldsymbol{P}}(\boldsymbol{x}_2;\boldsymbol{x}_1;k_y)$ is defined entirely by its relation to $\overset{\circ}{\boldsymbol{G}}(\boldsymbol{x}_2;\boldsymbol{x}_1;k_y)$. We refer to $\overset{\circ}{\boldsymbol{P}}(\boldsymbol{x}_2;\boldsymbol{x}_1;k_y)$ as the *propagator*. It follows directly from (2.36) that the propagator obeys the simple reciprocity relation

$$\boxed{\overset{\circ}{\boldsymbol{P}}(\boldsymbol{x}_2;\boldsymbol{x}_1;k_y) \equiv \overset{\circ}{\boldsymbol{P}}{}^T(\boldsymbol{x}_1;\boldsymbol{x}_2;-k_y).} \tag{2.37}$$

In the sections that follow, we encounter numerous instances of

$$\boldsymbol{J}_6 \left[\boldsymbol{T}_x(-k_y)\, \overset{\circ}{\boldsymbol{G}}\, (\boldsymbol{x};\boldsymbol{x}';-k_y)\right]^T \boldsymbol{J}_6$$

for various (receiver, source) combinations $(\boldsymbol{x};\boldsymbol{x}')$, where the operator $\boldsymbol{T}_x$ was given in (2.24) and acts on the field (or receiver) coordinate $\boldsymbol{x}$ in $\overset{\circ}{\boldsymbol{G}}$ $(\boldsymbol{x};\boldsymbol{x}';-k_y)$. It follows from (2.36) that:

$$\boxed{\overset{s}{\boldsymbol{T}}_x(-k_y)\, \overset{\circ}{\boldsymbol{G}}\, (\boldsymbol{x}_2;\boldsymbol{x}_1;k_y) \equiv \boldsymbol{J}_6 \left[\overset{r}{\boldsymbol{T}}_x(-k_y)\, \overset{\circ}{\boldsymbol{G}}\, (\boldsymbol{x}_1;\boldsymbol{x}_2;-k_y)\right]^T \boldsymbol{J}_6} \tag{2.38}$$

where $\overset{s}{\boldsymbol{T}}_x$ operates on the source vector $\boldsymbol{x}_1$ vector in $\overset{\circ}{\boldsymbol{G}}$ $(\boldsymbol{x}_2;\boldsymbol{x}_1)$, and we have written $\overset{r}{\boldsymbol{T}}_x\, \overset{\circ}{\boldsymbol{G}}$ $(\boldsymbol{x}_1;\boldsymbol{x}_2)$ in place of the previously employed notation $\boldsymbol{T}_x\, \overset{\circ}{\boldsymbol{G}}$ $(\boldsymbol{x}_1;\boldsymbol{x}_2)$ to indicate an operation on the field (or receiver) coordinate $\boldsymbol{x}_1$.

2.4 The Representation Theorems

2.4.1 Exterior Representation Theorem

Theorem 1 Given that

$$\boldsymbol{L}(\boldsymbol{x},\partial_z,\partial_x;k_y)\boldsymbol{G}(\boldsymbol{x},\boldsymbol{x}_S;k_y) = \\ \boldsymbol{I}_6\delta(\boldsymbol{x}-\boldsymbol{x}_S)+\chi_\Omega\Delta\boldsymbol{A}(\boldsymbol{x},\partial_x;k_y)\boldsymbol{G}(\boldsymbol{x},\boldsymbol{x}_S;k_y), \qquad \boldsymbol{x}_S\in\Omega^e$$

and

$$\boldsymbol{L}^\dagger(\boldsymbol{x},\partial_z,\partial_x;k_y)\,\mathring{\boldsymbol{G}}\,(\boldsymbol{x},\boldsymbol{x}_R;-k_y) = -\boldsymbol{I}_6\delta(\boldsymbol{x}-\boldsymbol{x}_R), \qquad \boldsymbol{x}_R\in R^2$$

then

$$\left.\begin{array}{ll}\boldsymbol{G}(\boldsymbol{x}_R;\boldsymbol{x}_S;k_y) & \boldsymbol{x}_R\in\Omega^e\\ \boldsymbol{0} & \boldsymbol{x}_R\in\Omega\end{array}\right\} = \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x}_S;k_y) \\ - \oint_{\partial\Omega^+}\Big[\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;z,x;k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\,\mathrm{d}x \\ - \overset{s}{\boldsymbol{T}}_x(-k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;z,x;k_y)\,\overset{r}{\boldsymbol{T}}_x(k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\,\mathrm{d}z\Big]$$

Proof It follows from (2.27) with Ω chosen as Ω^e in that equation, that

$$\Big(\mathring{\boldsymbol{G}}(\boldsymbol{x};\boldsymbol{x}_R;-k_y),\boldsymbol{L}\cdot\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\Big)_{\Omega^e} - \Big(\boldsymbol{L}^\dagger\cdot\mathring{\boldsymbol{G}}(\boldsymbol{x};\boldsymbol{x}_1;-k_y),\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_2;k_y)\Big)_{\Omega^e} = \\ \Big\langle\mathring{\boldsymbol{G}}(\boldsymbol{x};\boldsymbol{x}_1;-k_y),\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_2;k_y)\Big\rangle_{\partial\Omega^+} + \overbrace{\Big\langle\mathring{\boldsymbol{G}}(\boldsymbol{x};\boldsymbol{x}_1;-k_y),\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_2;k_y)\Big\rangle_{\partial\Omega^{\infty+f}}}^{\equiv\boldsymbol{0}\text{ free-surface/radiation conditions}} \quad (2.39)$$

We can write (2.39) more explicitly as

$$\int_{\Omega^e}\mathring{\boldsymbol{G}}^T(\boldsymbol{x};\boldsymbol{x}_R;-k_y)\boldsymbol{J}_6\boldsymbol{L}\cdot\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x}- \\ \int_{\Omega^e}(\boldsymbol{L}^\dagger\cdot\mathring{\boldsymbol{G}})^T(\boldsymbol{x};\boldsymbol{x}_R;-k_y)\boldsymbol{J}_6\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x} = \\ \oint_{\partial\Omega^+}\Big[\mathring{\boldsymbol{G}}^T(z,x;\boldsymbol{x}_R;-k_y)\boldsymbol{J}_6\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\,\mathrm{d}x \\ -(\boldsymbol{T}_x(-k_y)\mathring{\boldsymbol{G}}(z,x;\boldsymbol{x}_R;-k_y))^T\boldsymbol{J}_6\boldsymbol{T}_x(k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\,\mathrm{d}z\Big] \quad (2.40)$$

But multiplying (2.40) through by $\boldsymbol{J}_6$ and using the reciprocity relations

(2.36) and (2.38) we get

$$\int_{\Omega^e} \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\boldsymbol{L}\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x} - \int_{\Omega^e} \boldsymbol{J}_6(\boldsymbol{L}^\dagger\mathring{\boldsymbol{G}})^T(\boldsymbol{x};\boldsymbol{x}_R;-k_y)\boldsymbol{J}_6\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x}$$
$$= \oint_{\partial\Omega^+} \Big[\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;z,x;k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\ \mathrm{d}x$$
$$- \overset{s}{\boldsymbol{T}}_x(-k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;z,x;k_y)\ \overset{r}{\boldsymbol{T}}_x(k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\mathrm{d}z\Big] \tag{2.41}$$

Using

$$\int_{\Omega^e} \overbrace{\boldsymbol{J}_6(\boldsymbol{L}^\dagger\mathring{\boldsymbol{G}})^T(\boldsymbol{x};\boldsymbol{x}_R;-k_y)\boldsymbol{J}_6}^{\boldsymbol{I}_6\delta(\boldsymbol{x}-\boldsymbol{x}_R)}\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x} = \begin{cases} \boldsymbol{G}(\boldsymbol{x}_R;\boldsymbol{x}_S;k_y) & \boldsymbol{x}_R \in \Omega^e \\ \boldsymbol{0} & \boldsymbol{x}_R \in \Omega \end{cases}$$

and

$$\int_{\Omega^e} \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\overbrace{\boldsymbol{L}\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)}^{\boldsymbol{I}_6\delta(\boldsymbol{x}-\boldsymbol{x}_S)}\mathrm{d}\boldsymbol{x} = \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x}_S;k_y)$$

in (2.41) and rearranging proves Thm.1. ■

2.4.2 Interior Representation Theorem

Theorem 2 Given that

$$\boldsymbol{L}(\boldsymbol{x},\partial_z,\partial_x;k_y)\boldsymbol{G}(\boldsymbol{x},\boldsymbol{x}_S;k_y) = \boldsymbol{I}_6\delta(\boldsymbol{x}-\boldsymbol{x}_S) + \chi_\Omega\Delta\boldsymbol{A}(\boldsymbol{x},\partial_x;k_y)\boldsymbol{G}(\boldsymbol{x},\boldsymbol{x}_S;k_y), \qquad \boldsymbol{x}_S \in \Omega^e$$

and

$$\boldsymbol{L}^\dagger(\boldsymbol{x},\partial_z,\partial_x;k_y)\ \mathring{\boldsymbol{G}}\ (\boldsymbol{x},\boldsymbol{x}_R;-k_y) = -\boldsymbol{I}_6\delta(\boldsymbol{x}-\boldsymbol{x}_R), \qquad \boldsymbol{x}_R \in R^2$$

then

$$\left.\begin{matrix} \boldsymbol{G}(\boldsymbol{x}_R;\boldsymbol{x}_S;k_y) & \boldsymbol{x}_R \in \Omega \\ \boldsymbol{0} & \boldsymbol{x}_R \in \Omega^e \end{matrix}\right\} = \int_\Omega \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\Delta\boldsymbol{A}(\boldsymbol{x},\partial_x;k_y)\boldsymbol{G}(\boldsymbol{x},\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x}$$
$$- \oint_{\partial\Omega^-} \Big[\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;z,x;k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\ \mathrm{d}x$$
$$- \overset{s}{\boldsymbol{T}}_x(-k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;z,x;k_y)\ \overset{r}{\boldsymbol{T}}_x(k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\ \mathrm{d}z\Big]$$

Proof It follows from (2.27) that

$$\Big(\mathring{\boldsymbol{G}}(\boldsymbol{x};\boldsymbol{x}_R;-k_y),\boldsymbol{L}\cdot\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\Big)_\Omega - \Big(\boldsymbol{L}^\dagger\cdot\mathring{\boldsymbol{G}}(\boldsymbol{x};\boldsymbol{x}_1;-k_y),\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_2;k_y)\Big)_\Omega =$$
$$\Big\langle\mathring{\boldsymbol{G}}(\boldsymbol{x};\boldsymbol{x}_1;-k_y),\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_2;k_y)\Big\rangle_{\partial\Omega^-}. \tag{2.42}$$

We can write (2.42) more explicitly as

$$\begin{aligned}
&\int_\Omega \mathring{\boldsymbol{G}}^T(\boldsymbol{x};\boldsymbol{x}_R;-k_y)\boldsymbol{J}_6\boldsymbol{L}\cdot\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x}- \\
&\qquad \int_\Omega (\boldsymbol{L}^\dagger\cdot\mathring{\boldsymbol{G}})^T(\boldsymbol{x};\boldsymbol{x}_R;-k_y)\boldsymbol{J}_6\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x} = \\
&\qquad \oint_{\partial\Omega^-}\Big[\mathring{\boldsymbol{G}}^T(z,x;\boldsymbol{x}_R;-k_y)\boldsymbol{J}_6\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\ \mathrm{d}x \\
&\qquad -(\boldsymbol{T}_x(-k_y)\mathring{\boldsymbol{G}}(z,x;\boldsymbol{x}_R;-k_y))^T\boldsymbol{J}_6\boldsymbol{T}_x(k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\ \mathrm{d}z\Big] \quad (2.43)
\end{aligned}$$

But multiplying (2.42) through by $\boldsymbol{J}_6$ and using the reciprocity relations (2.36) and (2.38) we get

$$\begin{aligned}
&\int_\Omega \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\boldsymbol{L}\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x} - \int_\Omega \boldsymbol{J}_6(\boldsymbol{L}^\dagger\mathring{\boldsymbol{G}})^T(\boldsymbol{x};\boldsymbol{x}_R;-k_y)\boldsymbol{J}_6\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x} \\
&\quad = \oint_{\partial\Omega^-}\Big[\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;z,x;k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\ \mathrm{d}x \\
&\qquad\qquad - \overset{s}{\boldsymbol{T}}_x(-k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;z,x;k_y)\ \overset{r}{\boldsymbol{T}}_x(k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\Big] \quad (2.44)
\end{aligned}$$

Using

$$\int_\Omega \overbrace{\boldsymbol{J}_6(\boldsymbol{L}^\dagger\mathring{\boldsymbol{G}})^T(\boldsymbol{x};\boldsymbol{x}_R;-k_y)\boldsymbol{J}_6}^{\boldsymbol{I}_6\delta(\boldsymbol{x}-\boldsymbol{x}_R)}\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x} = \begin{cases} \boldsymbol{G}(\boldsymbol{x}_R;\boldsymbol{x}_S;k_y) & \boldsymbol{x}_R\in\Omega \\ \boldsymbol{0} & \boldsymbol{x}_R\in\Omega^e \end{cases}$$

and

$$\int_\Omega \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\ \overbrace{\boldsymbol{L}\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)}^{\Delta\boldsymbol{A}(\boldsymbol{x},\partial_x;k_y)\boldsymbol{G}(\boldsymbol{x},\boldsymbol{x}_S;k_y)}\ \mathrm{d}\boldsymbol{x} = \int_\Omega \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\Delta\boldsymbol{A}(\boldsymbol{x},\partial_x;k_y)\boldsymbol{G}(\boldsymbol{x},\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x}$$

in (2.44) and rearranging proves Thm.2. ■

2.4.3 The Lippmann-Schwinger type integral equation

We can derive a number of useful identities using Thm.1 and Thm.2 by employing the *boundary integral* identity

$$\Big\langle \mathring{\boldsymbol{G}}(\boldsymbol{x};\boldsymbol{x}_1;-k_y),\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_2;k_y)\Big\rangle_{\partial\Omega^-} \equiv -\Big\langle \mathring{\boldsymbol{G}}(\boldsymbol{x};\boldsymbol{x}_1;-k_y),\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_2;k_y)\Big\rangle_{\partial\Omega^+}$$

or equivalently

$$\oint_{\partial\Omega^-} \Big[\mathring{\boldsymbol{G}}(\boldsymbol{x}_R; z, x; k_y)\boldsymbol{G}(z, x; \boldsymbol{x}_S; k_y)\ \mathrm{d}x \\ - \overset{s}{\boldsymbol{T}}_x(-k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x}_R; z, x; k_y)\ \overset{r}{\boldsymbol{T}}_x(k_y)\boldsymbol{G}(z, x; \boldsymbol{x}_S; k_y)\ \mathrm{d}z\Big] \equiv \\ -\oint_{\partial\Omega^+} \Big[\mathring{\boldsymbol{G}}(\boldsymbol{x}_R; z, x; k_y)\boldsymbol{G}(z, x; \boldsymbol{x}_S; k_y)\ \mathrm{d}x \\ - \overset{s}{\boldsymbol{T}}_x(-k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x}_R; z, x; k_y)\ \overset{r}{\boldsymbol{T}}_x(k_y)\boldsymbol{G}(z, x; \boldsymbol{x}_S; k_y)\ \mathrm{d}z\Big] \tag{2.45}$$

because the *sense of the contour integration has been reversed in the two integrals*.

First, let us assume that $\boldsymbol{x}_R \in \Omega$ in Thm.1 and Thm.2. From Thm.1 it follows that

$$\mathring{\boldsymbol{G}}(\boldsymbol{x}_R; \boldsymbol{x}_S; k_y) = \oint_{\partial\Omega^+} \Big[\mathring{\boldsymbol{G}}(\boldsymbol{x}_R; z, x; k_y)\boldsymbol{G}(z, x; \boldsymbol{x}_S; k_y)\ \mathrm{d}x \\ - \overset{s}{\boldsymbol{T}}_x(-k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x}_R; z, x; k_y)\ \overset{r}{\boldsymbol{T}}_x(k_y)\boldsymbol{G}(z, x; \boldsymbol{x}_S; k_y)\ \mathrm{d}z\Big], \quad \boldsymbol{x}_R \in \Omega \\ \oint_{\partial\Omega^+} \Big[\mathring{\boldsymbol{G}}(\boldsymbol{x}_R; z, x; k_y)\boldsymbol{G}(z, x; \boldsymbol{x}_S; k_y)\ \mathrm{d}x \\ - \overset{s}{\boldsymbol{T}}_x(-k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x}_R; z, x; k_y)\ \overset{r}{\boldsymbol{T}}_x(k_y)\boldsymbol{G}(z, x; \boldsymbol{x}_S; k_y)\ \mathrm{d}z\Big], \quad \boldsymbol{x}_R \in \Omega \tag{2.46}$$

where the last equation follows from (2.45). Similarly, setting $\boldsymbol{x}_R \in \Omega$ in Thm.2., we get

$$\boldsymbol{G}(\boldsymbol{x}_R; \boldsymbol{x}_S; k_y) = \int_\Omega \mathring{\boldsymbol{G}}(\boldsymbol{x}_R; \boldsymbol{x}; k_y)\Delta\boldsymbol{A}(\boldsymbol{x}, \partial_x; k_y)\boldsymbol{G}(\boldsymbol{x}, \boldsymbol{x}_S; k_y)\mathrm{d}\boldsymbol{x} \\ - \oint_{\partial\Omega^-} \Big[\mathring{\boldsymbol{G}}(\boldsymbol{x}_R; z, x; k_y)\boldsymbol{G}(z, x; \boldsymbol{x}_S; k_y)\ \mathrm{d}x \\ - \overset{s}{\boldsymbol{T}}_x(-k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x}_R; z, x; k_y)\ \overset{r}{\boldsymbol{T}}_x(k_y)\boldsymbol{G}(z, x; \boldsymbol{x}_S; k_y)\ \mathrm{d}z\Big] \tag{2.47}$$

Now, using (2.46) in (2.47), we get the *interior Lippmann-Schwinger type integral equation*

$$\begin{aligned}\boldsymbol{G}(\boldsymbol{x}_R;\boldsymbol{x}_S;k_y) &= \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x}_S;k_y)\\ &+ \int_{\Omega}\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\Delta\boldsymbol{A}(\boldsymbol{x},\partial_x;k_y)\boldsymbol{G}(\boldsymbol{x},\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x}, \quad \boldsymbol{x}_R\in\Omega^e\end{aligned} \tag{2.48}$$

Next, we set $\boldsymbol{x}_R \in \Omega^e$ in Thm1. to get

$$\begin{aligned}&\boldsymbol{G}(\boldsymbol{x}_R;\boldsymbol{x}_S;k_y) - \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x}_S;k_y) = \\ &\quad - \oint_{\partial\Omega^+}\Big[\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;z,x;k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\ \mathrm{d}x\\ &\qquad - \overset{s}{\boldsymbol{T}}_x(-k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;z,x;k_y)\ \overset{r}{\boldsymbol{T}}_x(k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\ \mathrm{d}z\Big]\\ &\quad = \oint_{\partial\Omega^-}\Big[\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;z,x;k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\ \mathrm{d}x\\ &\quad - \overset{s}{\boldsymbol{T}}_x(-k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;z,x;k_y)\ \overset{r}{\boldsymbol{T}}_x(k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\ \mathrm{d}z\Big], \quad \boldsymbol{x}_R\in\Omega^e\end{aligned} \tag{2.49}$$

where the last equation in (2.49) follows from (2.45). Similarly, we set $\boldsymbol{x}_R \in \Omega^e$ in Thm2. to get

$$\begin{aligned}&\int_{\Omega}\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\Delta\boldsymbol{A}(\boldsymbol{x},\partial_x;k_y)\boldsymbol{G}(\boldsymbol{x},\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x} = \\ &\quad \oint_{\partial\Omega^-}\Big[\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;z,x;k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\ \mathrm{d}x\\ &\qquad - \overset{s}{\boldsymbol{T}}_x(-k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;z,x;k_y)\ \overset{r}{\boldsymbol{T}}_x(k_y)\boldsymbol{G}(z,x;\boldsymbol{x}_S;k_y)\ \mathrm{d}z\Big], \quad \boldsymbol{x}_R\in\Omega^e\end{aligned} \tag{2.50}$$

Using the domain integral on the left hand side of (2.50) in place of the boundary integral on the right hand side of (2.49) we get the *exterior Lippmann-Schwinger type integral equation*

$$\begin{aligned}\boldsymbol{G}(\boldsymbol{x}_R;\boldsymbol{x}_S;k_y) &= \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x}_S;k_y)\\ &+ \int_{\Omega}\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\Delta\boldsymbol{A}(\boldsymbol{x},\partial_x;k_y)\boldsymbol{G}(\boldsymbol{x},\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x}, \quad \boldsymbol{x}_R\in\Omega^e\end{aligned} \tag{2.51}$$

Corollary 1: *The Domain Integral Representation*
Given that

$$\begin{aligned}&\boldsymbol{L}(\boldsymbol{x},\partial_z,\partial_x;k_y)\boldsymbol{G}(\boldsymbol{x},\boldsymbol{x}_S;k_y) = \\ &\qquad \boldsymbol{I}_6\delta(\boldsymbol{x}-\boldsymbol{x}_S) + \chi_{\Omega}\Delta\boldsymbol{A}(\boldsymbol{x},\partial_x;k_y)\boldsymbol{G}(\boldsymbol{x},\boldsymbol{x}_S;k_y), \qquad \boldsymbol{x}_S\in\Omega^e\end{aligned}$$

and

$$\boldsymbol{L}^{\dagger}(\boldsymbol{x}, \partial_z, \partial_x; k_y)\, \overset{\circ}{\boldsymbol{G}}\,(\boldsymbol{x}, \boldsymbol{x}_R; -k_y) = -\boldsymbol{I}_6 \delta(\boldsymbol{x} - \boldsymbol{x}_R), \qquad \boldsymbol{x}_R \in R^2$$

then

$$\begin{aligned} \boldsymbol{G}(\boldsymbol{x}_R; \boldsymbol{x}_S; k_y) &= \overset{\circ}{\boldsymbol{G}}(\boldsymbol{x}_R; \boldsymbol{x}_S; k_y) \\ &\quad + \int_{\Omega} \overset{\circ}{\boldsymbol{G}}(\boldsymbol{x}_R; \boldsymbol{x}; k_y) \Delta \boldsymbol{A}(\boldsymbol{x}, \partial_x; k_y) \boldsymbol{G}(\boldsymbol{x}, \boldsymbol{x}_S; k_y) \mathrm{d}\boldsymbol{x}, \quad \boldsymbol{x}_R \in R^2 \end{aligned}$$

Proof Follows directly from (2.48) and (2.51). ■

In words, **Corollary 1** is a statement that: the 6×6 Green's tensor $\boldsymbol{G}(\boldsymbol{x}_R; \boldsymbol{x}_S; k_y)$ for a heterogeneous medium is the sum of the incident Green's tensor $\overset{\circ}{\boldsymbol{G}}(\boldsymbol{x}_R; \boldsymbol{x}_S; k_y)$ propagating through the reference medium and the radiated (or scattered) field $\int_{\Omega} \overset{\circ}{\boldsymbol{G}}(\boldsymbol{x}_R; \boldsymbol{x}; k_y) \Delta \boldsymbol{A}(\boldsymbol{x}, \partial_x; k_y) \boldsymbol{G}(\boldsymbol{x}, \boldsymbol{x}_S; k_y) \mathrm{d}\boldsymbol{x}$ from the *secondary source distribution* $\Delta \boldsymbol{A}(\boldsymbol{x}, \partial_x; k_y) \boldsymbol{G}(\boldsymbol{x}, \boldsymbol{x}_S; k_y)$, supported on Ω. In general, the larger the *contrast* $\Delta \boldsymbol{A}(\boldsymbol{x}, \partial_x; k_y)$ between the actual medium, and the reference medium, the more difficult it will be to find the solution $\boldsymbol{G}(\boldsymbol{x}_R; \boldsymbol{x}_S; k_y)$. The rest of the thesis will be concerned with solving this Lipmann-Schwinger type operator equation.

Chapter 3

Representations of the time harmonic elastic wave Green's tensor

We recall from chapter 2 that the material properties of the propagation environment have been defined as the sum of a reference and residual component, i.e. $(\rho, \alpha, \beta) = (\mathring{\rho} + \Delta\rho, \mathring{\alpha} + \Delta\alpha, \mathring{\beta} + \Delta\beta)$. In this chapter, we will be concerned exclusively with uniform media so that residual properties $(\Delta\rho, \Delta\alpha, \Delta\beta)$ vanish and we have $(\rho, \alpha, \beta) = (\mathring{\rho}, \mathring{\alpha}, \mathring{\beta})$. To simplify notation, we discontinue the practice of writing $(\mathring{\rho}, \mathring{\alpha}, \mathring{\beta})$ to denote reference medium parameters and write (ρ, α, β) instead.

This chapter is concerned with the construction of the time harmonic elastic wave motion driven by a *lattice of multipolar point forces* $\boldsymbol{f}$ *superimposed on a homogeneous medium*. The homogeneous medium is characterized by the elastic parameters (ρ, α, β), which denote the density, compressional and shear speeds respectively, in an isotropic medium.

Each point force $\boldsymbol{f}$, has an assumed time dependence of the form $\exp(\epsilon t - i2\pi\nu t)$ where ν is the frequency and ϵ is a non-negative *growth* parameter. The *adiabatic growth term* $\mathrm{e}^{\epsilon t}$, evolves the wavefield from a state of quiescence in the infinite past, to its present value, following a path that ensures that the *wavefield is always in steady state equilibrium with the driving force* $\boldsymbol{f}$. This device has been employed by *Lighthill* (1978) to enforce the *radiation boundary conditions*. For convenience, we introduce the *complex angular frequency*

$$\omega \equiv 2\pi\nu + i\epsilon. \tag{3.1}$$

We can then synthesize *transient motion* from a discrete set of *forced motions*, sampled from the *Bromwich contour* in the complex frequency domain [e.g.*Chapman* (1978)] as

$$[\cdot]\,(t) = 2\,\mathrm{e}^{\epsilon t}\,\Re \int_0^{\nu_{max}} \mathrm{d}\nu\ \mathrm{e}^{-i2\pi\nu t}\,[\cdot]\,(2\pi\nu + i\epsilon) \tag{3.2}$$

where ν_{max} is the maximum frequency of the source field. Usually the growth parameter ϵ is chosen so that $5 \leq \epsilon T \leq 10$ (Banerjee, Ahmad & Manolis, 1987), where T is the length of the required time series (which is usually determined by trial and error). Using the complex angular frequency ω smoothes the frequency domain response, which enhances the conditioning of the problem, causing the complex frequency multiple scattering series to converge faster than the real frequency version. However, we must be aware that the process is equivalent to an analytic continuation, and is inherently unstable. The instability, is manifested by the sensitivity of (3.2) to errors in the frequency domain response. This is a particularly troubling issue, as the multiple scattering series is usually only summed to a relative error of 10^{-6} and it remains an open question whether we have secured sufficient accuracy to avoid the deleterious effects of the numerical instability inherent in (3.2). An excellent survey of numerical methods for the numerical evaluation of (3.2) is presented in *Davies & Martin* (1979).

3.1 The free space, time harmonic, elastic wave Green's tensor

Given the 2-D vectors

$$\boldsymbol{x} \equiv (z, x), \quad \boldsymbol{x}' \equiv (z', x')$$

a particularly useful solution $\overset{\circ}{\boldsymbol{G}}\,(\boldsymbol{x}; \boldsymbol{x}'; k_y)$, satisfies (2.7) when the the material properties in Table 2.1 are homogeneous and the $y-$*Fourier transformed body force* $\boldsymbol{F}(\boldsymbol{x}; \boldsymbol{x}')$ assumes the special form

$$\boldsymbol{F}(\boldsymbol{x}; \boldsymbol{x}') = \boldsymbol{I}_6 \delta(\boldsymbol{x} - \boldsymbol{x}'),$$

with $\boldsymbol{I}_6$ the 6×6 *identity matrix*, *viz*

$$\boldsymbol{\mathcal{L}} \cdot \overset{\circ}{\boldsymbol{G}}\,(\boldsymbol{x}; \boldsymbol{x}'; k_y) = \boldsymbol{I}_6 \delta(z - z')\delta(x - x'). \tag{3.3}$$

We can then use the linearity of (2.7) to pose the *particular solution* $\boldsymbol{b}_P$ of (2.7) as the *superposition integral*

$$\boldsymbol{b}_P = \int \mathrm{d}\boldsymbol{x}' \; \mathring{\boldsymbol{G}}\left(\boldsymbol{x};\boldsymbol{x}';k_y\right)\boldsymbol{f}(\boldsymbol{x}') \tag{3.4}$$

for an arbitrary body force $\boldsymbol{f}$. It will be convenient for the purpose of subsequent constructions, to interpret $\mathring{\boldsymbol{G}}\left(\boldsymbol{x};\boldsymbol{x}';k_y\right)$ of (3.4), as the *integral kernel* of the formal inverse

$$\left[\boldsymbol{\mathcal{L}}^{-1}\boldsymbol{w}\right](\boldsymbol{x}) \equiv \int \mathrm{d}\boldsymbol{x}' \; \mathring{\boldsymbol{G}}\left(\boldsymbol{x};\boldsymbol{x}';k_y\right)\cdot\boldsymbol{w}(\boldsymbol{x}') \tag{3.5}$$

for an arbitrary *test* vector-function $\boldsymbol{w}$, where $\boldsymbol{\mathcal{L}}$ is the governing operator in (2.7). Equation (3.5) is strictly formal, as the *null space* of $\boldsymbol{\mathcal{L}}$ is spanned by the set of *homogeneous* solutions and consequently, $\mathring{\boldsymbol{G}}$ is not unique. To get a *well posed* inverse, we invoke the *radiation condition* that *there are no homogeneous solutions of* (2.7) *due to sources at infinity*.

To solve for $\mathring{\boldsymbol{G}}$ in (3.3), or equivalently, to construct the inverse $\boldsymbol{\mathcal{L}}^{-1}$ in (3.5), we follow the classical *Fourier Transform* route. To begin, we transform the $x-$ dependence via $\mathcal{F}_x\left[\cdot\right](k_x) \equiv \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} dx \;\; \exp\left(-i\,k_x x\right)\left[\cdot\right](x)$, where k_x is the horizontal component of the wavevector, so that the transverse *Fourier* transform of the governing equation (3.3) may be written as

$$\begin{aligned}
&\langle k_x|\boldsymbol{\mathcal{L}}|k_x{}'\rangle\langle k_x{}'|x\rangle\langle x| \;\; \mathring{\boldsymbol{G}}\left(\boldsymbol{x};\boldsymbol{x}';k_y\right)\\
&\quad\equiv \boldsymbol{L}(z,k_x,k_y) \;\; \mathring{\boldsymbol{G}}\left(z,k_x;z',x';k_y\right)\\
&\quad\equiv \left[\partial_z - \boldsymbol{A}(k_x,k_y)\right] \;\; \mathring{\boldsymbol{G}}\left(z,k_x;z',x';k_y\right) = \boldsymbol{I}_6\delta(z-z')\,\frac{\exp(-ik_x x')}{\sqrt{2\pi}}
\end{aligned} \tag{3.6}$$

where the *Fourier domain system matrix* $\langle k_x|\boldsymbol{\mathcal{A}}|k_x{}'\rangle = \delta(k_x - k_x{}')\boldsymbol{A}(k_x,k_y)$ is defined

$$\boldsymbol{A}(k_x,k_y) = \begin{bmatrix}
0 & -i\gamma k_x & -\gamma i\; k_y & a & 0 & 0\\
-ik_x & 0 & 0 & 0 & b & 0\\
-ik_y & 0 & 0 & 0 & 0 & b\\
-\rho\omega^2 & 0 & 0 & 0 & -ik_x & -ik_y\\
0 & -\rho\omega^2 + \zeta k_x^2 + \mu k_y^2 & k_x k_y\left(\chi+\mu\right) & -ik_x\gamma & 0 & 0\\
0 & k_x k_y\left(\chi+\mu\right) & -\rho\omega^2 + \zeta k_y^2 + \mu k_x^2 & -ik_y\gamma & 0 & 0
\end{bmatrix} \tag{3.7}$$

due to the *translational invariance of* $\boldsymbol{\mathcal{A}}$. The system operator $\boldsymbol{A}(k_x,k_y)$ of (3.7) is *quasi-Hamiltonian* since

$$\boldsymbol{A}^T(-k_x,-k_y)\boldsymbol{J}_6 + \boldsymbol{J}_6\boldsymbol{A}(k_x,k_y) \equiv \boldsymbol{0}_6 \tag{3.8}$$

where $\boldsymbol{J}_6$ is the *canonically symplectic* array of (2.21). The *diagonal array* $\boldsymbol{\Lambda}$ *of eigenvalues* of $\boldsymbol{A}(k_x, k_y)$ is defined as

$$\boldsymbol{\Lambda}(k_x, k_y) = \tag{3.9}$$
$$i\ \mathrm{diag}\,[k_{z,P}(k_x,k_y), k_{z,S}(k_x,k_y), k_{z,H}(k_x,k_y), -k_{z,P}(k_x,k_y), -k_{z,S}(k_x,k_y), -k_{z,H}(k_x,k_y)]$$

where

$$k_{z,c}(k_x,k_y) = \begin{cases} \sqrt{\hat{K}_c^2(k_y) \ - \ k_x^2} & |k_x| \ < \ |\hat{K}_c(k_y)| \\ i\sqrt{k_x^2 \ - \ \hat{K}_c^2(k_y)} & |k_x| \ \geq \ |\hat{K}_c(k_y)| \end{cases}, \quad c \in \{P, S, H\}, \tag{3.10}$$

with

$$\hat{K}_c(k_y) = \begin{cases} \sqrt{(\omega/c)^2 \ - \ k_y^2} & |k_y| \ < \ |\omega/c| \\ i\sqrt{k_y^2 \ - \ (\omega/c)^2} & |k_y| \ \geq \ |\omega/c| \end{cases}, \quad c \in \{P, S, H\}. \tag{3.11}$$

We note that $k_{z,S} = k_{z,H}$ because $S = H = \beta$ from Table 2.1. We distinguish between $k_{z,S}$ and $k_{z,H}$, only to maintain a self-consistent notational convention. The system matrix $\boldsymbol{A}(k_x, k_y)$ of (3.7) has an equivalent *spectral representation*:

$$\boldsymbol{A}(k_x,k_y) \equiv \boldsymbol{D}_z(k_x,k_y)\boldsymbol{\Lambda}(k_x,k_y)\boldsymbol{D}_z^{-1}(k_x,k_y); \tag{3.12}$$

where the *eigen matrix* $\boldsymbol{D}_z(k_x, k_y)$ is defined as

$$\boldsymbol{D}_z(k_x,k_y) = \tag{3.13}$$
$$\left[\ |+,P,k_x,k_y\rangle \quad |+,S,k_x,k_y\rangle \quad |+,H,k_x,k_y\rangle \quad |-,P,k_x,k_y\rangle \quad |-,S,k_x,k_y\rangle \quad |-,H,k_x,k_y\rangle\ \right]$$

with:

- the $|\pm, P, k_x, k_y\rangle$ eigen-vectors are the $\begin{smallmatrix}\text{downgoing}\\ \text{upgoing}\end{smallmatrix}$ $P-$waves

$$|\pm,P,k_x,k_y\rangle = \begin{bmatrix} \pm i\, k_{z,P}(k_x,k_y) \\ i\, k_x \\ i\, k_y \\ \rho\ \left(2\,\beta^2\,k_x^2 + 2\,\beta^2\,k_y^2 - \omega^2\right) \\ \mp 2\,\rho\,\beta^2\,k_x\,k_{z,P}(k_x,k_y) \\ \mp 2\,\rho\,\beta^2\,k_y\,k_{z,P}(k_x,k_y) \end{bmatrix} \epsilon_P \tag{3.14}$$

with the normalization factor ϵ_P

- the $|\pm, S, k_x, k_y\rangle$ eigen-vectors are the $\begin{smallmatrix}\text{downgoing}\\ \text{upgoing}\end{smallmatrix}$ quasi-$SV-$waves

$$|\pm, S, k_x, k_y\rangle = \begin{bmatrix} i\,k_x \\ \mp i\,k_{z,S}(k_x,k_y) \\ 0 \\ \mp 2\,\rho\,\beta^2\,k_x\,k_{z,S}(k_x,k_y) \\ \rho\,\left(\omega^2 - 2\,\beta^2\,k_x^2 - \beta^2\,k_y^2\right) \\ -\rho\,\beta^2\,k_x\,k_y \end{bmatrix} \epsilon_S \tag{3.15}$$

with the normalization factor ϵ_S

- the $|\pm, H, k_x, k_y\rangle$ eigen-vectors are the $\begin{smallmatrix}\text{downgoing}\\ \text{upgoing}\end{smallmatrix}$ quasi-$SH-$waves

$$|\pm, H, k_x, k_y\rangle = \begin{bmatrix} \mp k_y\,k_{z,H}(k_x,k_y) \\ -k_x\,k_y \\ \hat{K}_H^2 \\ 2\,i\,k_y\,\rho\,\left(\beta^2\,k_x^2 + \beta^2\,k_y^2 - \omega^2\right) \\ \mp 2\,i\,\rho\,\beta^2\,k_x\,k_y\,k_{z,H}(k_x,k_y) \\ \pm i\,k_{z,H}(k_x,k_y)\,\rho\,\left(\omega^2 - 2\,\beta^2\,k_y^2\right) \end{bmatrix} \epsilon_H \tag{3.16}$$

with the normalization factor ϵ_H

We have called the eigen-vectors in (3.15) and (3.16) *quasi-SV* and *quasi-SH* waves respectively, because the corresponding vector fields are *solenoidal* and are *in the* (z,x) *plane* and *out of the* (z,x) *plane* respectively, when $k_y = 0$. However, our choice of the shear wavefield decomposition into quasi-inplane and quasi-out of plane components differs from the choice commonly used in seismology [e.g. *Prange & Toksöz* (1990)]. Unlike the conventional decomposition, the basis vectors (3.15) and (3.16) are *non-degenerate* when $k_x = 0$ and $k_y = 0$. The normalization factors ϵ_P, ϵ_S and ϵ_H in Eqns.(3.14)-(3.16) are chosen so that $\boldsymbol{D}_z^{-1}(k_x,k_y)$ takes a simple form.

Using the standard result that eigenspaces of $\boldsymbol{A}^T(k_x,k_y)$ annihilate eigenspaces of $\boldsymbol{A}(k_x,k_y)$ (and vice versa) [e.g. *Milne* (1980), *p.89*], we can avoid constructing $\boldsymbol{D}_z^{-1}(k_x,k_y)$ in (3.12) by explicit matrix inversion, since we use the *Hamiltonian symmetry* (3.8), to relate $\boldsymbol{D}_z^{-1}(k_x,k_y)$ to the eigen matrix of the *transposed* or *dual* system matrix $\boldsymbol{A}^T(k_x,k_y)$ defined as

$$\boldsymbol{A}^T(k_x,k_y) \equiv \boldsymbol{J}_6\boldsymbol{A}(-k_x,-k_y)\boldsymbol{J}_6$$

To begin, it follows from (3.9) and (3.10) that $\boldsymbol{\Lambda}(-k_x,-k_y)=\boldsymbol{\Lambda}(k_x,k_y)$, so that

$$\begin{aligned}\boldsymbol{D}_z(-k_x,-k_y)\boldsymbol{\Lambda}(k_x,k_y) &= \boldsymbol{A}(-k_x,-k_y)\boldsymbol{D}_z(-k_x,-k_y)\\ &= \boldsymbol{J}_6\boldsymbol{A}^T(k_x,k_y)\boldsymbol{J}_6\boldsymbol{D}_z(-k_x,-k_y)\end{aligned}$$

which implies upon rearrangement that

$$\boldsymbol{A}^T(k_x,k_y)\boldsymbol{J}_6\boldsymbol{D}_z(-k_x,-k_y)=-\boldsymbol{J}_6\boldsymbol{D}_z(-k_x,-k_y)\boldsymbol{\Lambda}(k_x,k_y).$$

Consequently, $\boldsymbol{J}_6\boldsymbol{D}_z(-k_x,-k_y)$ is an eigenbasis of the dual operator $\boldsymbol{A}^T(k_x,k_y)$, with $-\boldsymbol{\Lambda}(k_x,k_y)$ the corresponding eigenvalue array. From direct evaluation it follows that

$$(\boldsymbol{J}_6\boldsymbol{D}_z(-k_x,-k_y))^T\boldsymbol{D}_z(k_x,k_y)=i\begin{bmatrix}\boldsymbol{0}_3 & \boldsymbol{V}\\ -\boldsymbol{V} & \boldsymbol{0}_3\end{bmatrix} \tag{3.17}$$

where

$$\boldsymbol{V}=\mathrm{diag}\left[\ 2\rho\omega^2k_{z,P}(k_x)\epsilon_P^2,\quad 2\mu k_{z,S}(k_x)\hat{K}_S^2\epsilon_S^2,\quad 2\rho\omega^2k_{z,H}(k_x)\hat{K}_H^2\epsilon_H^2\ \right]$$

and $\hat{K}_c, c\in\{P,S,H\}$ was given by (3.11), so that if we choose the normalization factors ϵ_P, ϵ_S and ϵ_H in Eqns.(3.14)-(3.16) according to

$$\epsilon_P=\frac{1}{\sqrt{2\rho\omega^2k_{z,P}}},\quad \epsilon_S=\frac{\omega}{\beta\,\hat{K}_S\,\sqrt{2\rho\omega^2k_{z,S}}},\quad \text{and}\quad \epsilon_H=\frac{1}{\hat{K}_H\,\sqrt{2\rho\omega^2k_{z,H}}} \tag{3.18}$$

then (3.17) simplifies to

$$(\boldsymbol{J}_6\boldsymbol{D}_z(-k_x,-k_y))^T\boldsymbol{D}_z(k_x,k_y)=i\boldsymbol{J}_6 \tag{3.19}$$

so that upon rearrangement of (3.19) we get

$$\boldsymbol{D}_z^{-1}(k_x,k_y)=-i\ \boldsymbol{J}_6\boldsymbol{D}_z^T(-k_x,-k_y)\boldsymbol{J}_6 \tag{3.20}$$

Equation (3.20) for the inverse is reminiscent of the *symplectic* structures found in *Hamiltonian dynamical systems* [e.g. *Goldstein* (1980)] (*see also* Box **3.1**).

We now transform the $z-$dependence of (3.6) using the *Fourier transform*

$$\mathcal{F}_z\left[\cdot\right](k_z)\equiv\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}dz\ \exp\left(-i\ k_z\ z\right)\ [\cdot](z),$$

Box **3.1**

A *fundamental* investigation of the seismic wave equations as a *Hamiltonian dynamical system* may be found in *Kennett* (1974b), *Chapman & Woodhouse* (1981), where it is shown that the coupled second order elastic wave equations are the *Euler* equations for the stationary value of a *Lagrangian variational principle*. The "Hamiltonian" first order system employed in this thesis is then obtained via a *Legendre transformation* [e.g. *Goldstein* (1980) Ch. 12]. Unlike, the equations of *static elastic equilibrium*, variational principles for the time harmonic elastic wave equations are not *extremal*, due to the complex nature of the *radiation conditions*, which siphon energy out of the system, as well as the *D'Alembert forces*, which render the wave equations *indefinite*. The lack of an extremum, vitiates some of the numerical procedures that have been developed by mathematicians and engineers to handle the static problem. None the less, identifying the time harmonic wave elastic wave equation as a Hamiltonian dynamical system suggests a very useful *propagation invariant* that follows from the *incompressibility of a Hamiltonian flow in displacement-traction phase space* [e.g. *Schultz* (1980)], so that $\boldsymbol{b}_1 \wedge \boldsymbol{b}_2|_{z_1} = \boldsymbol{b}_1 \wedge \boldsymbol{b}_2|_{z_2}$. In theoretical seismology, the exterior product "$\wedge$" is expressed in terms of the canonically symplectic array $\boldsymbol{J}_6$ of (2.21), so that

$$\boldsymbol{b}_1^T(z_1, -k_x, -k_y)\boldsymbol{J}_6\boldsymbol{b}_2(z_1, k_x, k_y) = \boldsymbol{b}_1^T(z_2, -k_x, -k_y)\boldsymbol{J}_6\boldsymbol{b}_2(z_2, k_x, k_y) \tag{1}$$

for a pair of wavefields $\boldsymbol{b}_1, \boldsymbol{b}_2$ in a *source free* depth interval $z_1 \leq \zeta \leq z_2$. An appreciable portion of the *matrix mechanics* for seismic wave propagation in stratified media, may be developed, with (1) as the starting point (Kennett, 1983). The identity (1) plays a role analogous to the *Rayleigh-Betti* representation of the conventional coupled second order theory.

to get

$$\tilde{\boldsymbol{L}}(\boldsymbol{k})\ \mathring{\boldsymbol{G}}\ (\boldsymbol{k};\boldsymbol{x}';k_y) \equiv [i\boldsymbol{I}_6 k_z - \boldsymbol{A}(k_x,k_y)\,]\ \mathring{\boldsymbol{G}}\ (\boldsymbol{k};\boldsymbol{x}';k_y) = \boldsymbol{I}_6 \frac{\exp(-i\boldsymbol{k}\cdot\boldsymbol{x}')}{2\pi}, \tag{3.21}$$

where $\boldsymbol{x}' = (z', x')$ and $\boldsymbol{k} = (k_z, k_x)$, so that

$$\mathring{\boldsymbol{G}}\ (\boldsymbol{k};\boldsymbol{x}';k_y) \equiv [i\boldsymbol{I}_6 k_z - \boldsymbol{A}(k_x,k_y)]^{-1}\,\frac{\exp(-i\boldsymbol{k}\cdot\boldsymbol{x}')}{2\pi}. \tag{3.22}$$

Using the spectral representation (3.12) in (3.22), we get

$$\begin{aligned}
\mathring{\boldsymbol{G}}\ (\boldsymbol{k};\boldsymbol{x}';k_y) &= \boldsymbol{D}_z(k_x,k_y)[i\,\boldsymbol{I}_6\,k_z\ -\ i\,\boldsymbol{\Lambda}]^{-1}\boldsymbol{D}_z^{-1}(k_x,k_y)\frac{\exp(-i\boldsymbol{k}\cdot\boldsymbol{x}')}{2\pi} \\
&= \boldsymbol{D}_z(k_x,k_y)[\boldsymbol{J}_6\,k_z\ -\ \boldsymbol{J}_6\,\boldsymbol{\Lambda}]^{-1}\boldsymbol{D}_z^T(-k_x,-k_y)\,\boldsymbol{J}_6\frac{\exp(-i\boldsymbol{k}\cdot\boldsymbol{x}')}{2\pi} \\
&= \boldsymbol{D}_z(k_x,k_y)\begin{bmatrix} \boldsymbol{0}_3 & -\boldsymbol{W}^+ \\ \boldsymbol{W}^- & \boldsymbol{0}_3 \end{bmatrix}\boldsymbol{D}_z^T(-k_x,-k_y)\,\boldsymbol{J}_6\frac{\exp(-i\boldsymbol{k}\cdot\boldsymbol{x}')}{2\pi}
\end{aligned} \tag{3.23}$$

where

$$\boldsymbol{W}^{\pm} = \mathrm{diag}\left[\ \tfrac{1}{(k_z\mp k_{z,P})},\quad \tfrac{1}{(k_z\mp k_{z,S})},\quad \tfrac{1}{(k_z\mp k_{z,S})}\ \right]$$

Applying the inverse Fourier transform $\boldsymbol{\mathcal{F}}^{-1}\,[\cdot](\boldsymbol{x}) \equiv \frac{1}{2\pi}\int_{R^2}\mathrm{d}\boldsymbol{k}\exp\left(i\boldsymbol{k}\cdot\boldsymbol{x}\right)\,[\cdot](\boldsymbol{k})$, it follows from (3.23), that $\mathring{\boldsymbol{G}}\ (\boldsymbol{x};\boldsymbol{x}';k_y)$ may be written

$$\begin{aligned}
\mathring{\boldsymbol{G}}\ (\boldsymbol{x};\boldsymbol{x}';k_y) &= \boldsymbol{\mathcal{F}}^{-1}\left[\mathring{\boldsymbol{G}}\ (\boldsymbol{k};\boldsymbol{x}';k_y)\right](\boldsymbol{x}) = \frac{1}{4\pi^2}\int_{R^2}\mathrm{d}\boldsymbol{k}\exp\left[i\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')\right] \\
&\quad\times\boldsymbol{D}_z(k_x,k_y)\begin{bmatrix} \boldsymbol{0}_3 & -\boldsymbol{W}^+ \\ \boldsymbol{W}^- & \boldsymbol{0}_3 \end{bmatrix}\boldsymbol{D}_z^T(-k_x,-k_y)\,\boldsymbol{J}_6
\end{aligned} \tag{3.24}$$

3.1.1 The classical up/down, plane wave Green's tensor

We now evaluate the k_z integral in (3.24) using the calculus of residues [e.g. *Aki & Richards* (1980)], which is possible provided the z component of $(\boldsymbol{x}-\boldsymbol{x}')$ in (3.24) is non-zero, i.e. $(z-z')\neq 0$.

We close the k_z integration contour in the $\substack{\text{upper}\\\text{lower}}$ half of the complex k_z-plane when $(z-z')\gtrless 0$, in so doing we pick up the

$$\Re(\partial\omega/\partial k_{z,c}) \gtrless 0$$

[Figure: Poles not found]

Figure 3.1: The movement of poles in the complex k_z plane consistent with $2\pi\nu \to 2\pi\nu + i\epsilon$

residue contributions (Fig.3.1), associated with $\begin{smallmatrix}\text{downgoing}\\ \text{upgoing}\end{smallmatrix}$ group velocity, to get

$$\mathring{\boldsymbol{G}}\left(\boldsymbol{x};\boldsymbol{x}';k_y\right) = \frac{-i}{2\pi}\int_R \mathrm{d}k_x \exp\left[ik_x(x-x')\right]\boldsymbol{D}_z(k_x,k_y)\boldsymbol{Q}^z(z;z')\boldsymbol{D}_z^T(-k_x,-k_y)\,\boldsymbol{J}_6 \tag{3.25}$$

where $\boldsymbol{Q}^z(z;z')$ in (3.25), is defined as

$$\boldsymbol{Q}^z(z;z') = \begin{bmatrix} \mathbf{0}_3 & H(z-z')\boldsymbol{E}(k_x;z;z') \\ H(z'-z)\boldsymbol{E}(k_x;z;z') & \mathbf{0}_3 \end{bmatrix} \tag{3.26}$$

with

$$\boldsymbol{E}(k_x;z;z') = \begin{bmatrix} \mathrm{e}^{ik_{z,P}(k_x)|z-z'|} & 0 & 0 \\ 0 & \mathrm{e}^{ik_{z,S}(k_x)|z-z'|} & 0 \\ 0 & 0 & \mathrm{e}^{ik_{z,H}(k_x)|z-z'|} \end{bmatrix}, \tag{3.27}$$

and $H(\cdot)$ is the *Heaviside step function*. The imposition of the *radiation boundary conditions* employed here, follows *Lighthill* (1978). Alternatively, we may introduce *attenuation* to move the poles off the integration contour [e.g. *Aki & Richards* (1980)].

3.2 Discrete plane wave representations of the free space elastodynamic Green's tensor

3.2.1 The z splitting

Basic results concerning the discrete Fourier (DFT) approximation to the inverse Fourier transform over horizontal wavenumber of a scalar function,

Box 3.2 The discrete Fourier transform

Given the Fourier integral

$$\phi(x) = \frac{1}{2\pi}\int_{-\Omega}^{\Omega} \mathrm{d}k_x \exp\left(ik_x x\right)\tilde{\phi}(k_x). \tag{3.28}$$

we will now show how to approximate it using quadrature. To begin, (3.28) may be approximated by the $2M$-panel *trapezoidal rule* as

$$\begin{aligned}\phi(x) \quad &\approx \quad \frac{\Delta k_x}{2\pi}\left[\frac{\exp\left(-i\Omega x\right)\tilde{\phi}(-\Omega) + \exp\left(i\Omega x\right)\tilde{\phi}(\Omega)}{2}\right] \\ &\quad + \sum_{j=-M+1}^{M-1} \exp\left(ij\Delta k_x x\right)\tilde{\phi}(j\Delta k_x)\end{aligned} \tag{3.29}$$

where the wavenumber sampling interval Δk_x is defined as

$$\Delta k_x = \frac{2\Omega}{2M} = \frac{\Omega}{M}$$

For computational purposes, we *alias* the endpoints in (3.29), i.e.

$$\frac{\exp\left(-i\Omega x\right)\tilde{\phi}(-\Omega) + \exp\left(i\Omega x\right)\tilde{\phi}(\Omega)}{2} \equiv \exp\left(-i\Omega x\right)\tilde{\phi}(-\Omega) \tag{3.30}$$

so that (3.29) becomes

$$\phi(x) \approx \frac{\Delta k_x}{2\pi}\sum_{j=-M}^{M-1} \exp\left(ij\Delta k_x x\right)\tilde{\phi}(j\Delta k_x) \tag{3.31}$$

The approximation (3.30) is feasible when

$$\tilde{\phi}(-\Omega) = \tilde{\phi}(\Omega) = 0 \tag{3.32}$$

which is the approximation adopted here.

We will now proceed to manipulate (3.31) into a *discrete Fourier transform* (DFT). To begin, we rewrite the sum in (3.31) as

$$\begin{aligned}\phi(x) \quad &\approx \quad \frac{\Delta k_x}{2\pi}\left[\sum_{j=0}^{M-1} \exp\left(ij\Delta k_x x\right)\tilde{\phi}(j\Delta k_x) + \sum_{j=-M}^{-1} \exp\left(ij\Delta k_x x\right)\tilde{\phi}(j\Delta k_x)\right] \\ &\approx \quad \frac{\Delta k_x}{2\pi}\left[\sum_{j=0}^{M-1} \exp\left(ij\Delta k_x x\right)\tilde{\phi}(j\Delta k_x)\right. \\ &\qquad\qquad \left. + \sum_{j=M}^{2M-1} \exp\left(i(j-2M)\Delta k_x x\right)\tilde{\phi}((j-2M)\Delta k_x)\right]\end{aligned} \tag{3.33}$$

We now discretize x in (3.33) so that

$$x_l = l\Delta x, \qquad 0 \le l \le 2M-1. \tag{3.34}$$

Box **3.2***continued*

Finally, we impose the fundamental condition in Discrete Fourier analysis that

$$\Delta k_x \Delta x = \frac{2\pi}{2M} \tag{3.36}$$

and employing this constraint in (3.35) we get

$$\phi(x_l) \approx \frac{\Delta k_x}{2\pi} \sum_{j=0}^{2M-1} \exp\left(\frac{2\pi i j l}{2M}\right) \tilde{\varphi}(j\Delta k_x), \qquad 0 \le l \le 2M-1 \tag{3.37}$$

where we have introduced the *DFT-ordered vector* $\tilde{\varphi}$ defined as

$$\tilde{\varphi}(j\Delta k_x) = \begin{cases} \tilde{\phi}(j\Delta k_x) & 0 \le j \le \mathrm{M}-1 \\ \tilde{\phi}([j-2M]\Delta k_x) & M \le j \le 2M-1 \end{cases} \tag{3.38}$$

and used the identity

$$\exp\left(-i2Ml\Delta k_x \Delta x\right) = \exp\left(-i2Ml\frac{2\pi}{2M}\right) = 1$$

which follows directly from the constraint (3.36). To evaluate the DFT (3.37) using the *fast Fourier transform*(FFT) algorithm, we impose the further condition

$$2M = 2^m, \quad m \in \{1, 2, 3, \cdots\}$$

in which case the floating point operation count of (3.37) is reduced from $O(4M^2)$ to $O(2mM)$.
Finally, an efficient method to incorporate *Filon quadrature* into the DFT approximation is discussed in Box **3.3**.

are summarized in Box **3.2** and a simple extension of the basic theory to incorporate Filon quadrature, while exploiting the speed of the FFT algorithm is presented in Box **3.3**. We will employ these results to develop a DFT approximation to (3.25). We assume the $N_k = 2M$ element *Fourier lattice*

$$\{k_j = j\Delta k_x \mid j = -M, \cdots, M-1\} \tag{3.39}$$

where $\Delta k_x = 2\pi j/L_x$ and L_x defines the *fundamental spatial period* of the representation. We may then approximate (3.25) by the *discrete plane wave sum*

$$\begin{aligned}\mathring{\boldsymbol{G}}^z(\boldsymbol{x};\boldsymbol{x}';k_y) &= \frac{-i\,\Delta k_x}{2\pi}\sum_{j=-M}^{j=M-1} \\ &\quad\times \exp\left[ik_j(x-x')\right]\boldsymbol{D}_z(k_j,k_y)\boldsymbol{Q}^z(k_j;z;z')\boldsymbol{D}_z^T(-k_j,-k_y)\end{aligned} \tag{3.40}$$

The discrete representation (3.40) is equivalent to assuming that the transverse dependence of the wavefield is both *bandlimited* (since N_k is finite) and periodically repeated (since Δk_x is finite).

The next step will be to rewrite the discrete sum (3.40) as an equivalent matrix operation. To this end, we define the N_k-element diagonal, horizontal-wavevector array:

$$\mathbf{k}_x = \mathrm{diag}\,[0, \Delta k_x, \cdots, (M-1)\Delta k_x, -M\Delta k_x, \cdots, -\Delta k_x,]\,; \tag{3.44}$$

the N_k-element diagonal, vertical-wavevector array

$$\mathbf{k}_{z,c} = \mathrm{diag}\,[k_{z,c}(0), k_{z,c}(\Delta k_x), \cdots, k_{z,c}((M-1)\Delta k_x), k_{z,c}(-M\Delta k_x), \cdots, k_{z,c}(-\Delta k_x)] \qquad c \in \{P, S\} \tag{3.45}$$

and the N_k-element *Fourier vector*

$$\boldsymbol{F}(x;\mathbf{k}_x) = \left[\begin{array}{ccc} 1 & ,\cdots, & 1 \end{array}\right]\cdot \mathrm{e}^{i\,\mathbf{k}_x\,x}. \tag{3.46}$$

It is important to note that $\boldsymbol{F}(x;\mathbf{k}_x)$ as defined in (3.46) is an N_k- row vector, even though $\mathbf{k}_x$ is an N_k-diagonal matrix. Also, the elements of (3.44), (3.45) and (3.46) have the DFT ordering of (3.38) for the reasons indicated in Box **3.2**.

Using the newly defined arrays (3.44)–(3.46), we can rewrite the *discrete plane wave* approximation (3.40) to the 6×6 Green's tensor in the equivalent matrix form

$$\begin{aligned}\mathring{\boldsymbol{G}}^z(\boldsymbol{x};\boldsymbol{x}';k_y) &= -i\,\left(\boldsymbol{I}_6\otimes\boldsymbol{F}(x;\mathbf{k}_x)\right)\boldsymbol{D}_z(\mathbf{k}_x,k_y) \\ &\quad\times\boldsymbol{Q}^z(\mathbf{k}_x;z;z')\boldsymbol{D}_z^T(-\mathbf{k}_x,-k_y)\,\left(\boldsymbol{I}_6\otimes\boldsymbol{F}^T(x';-\mathbf{k}_x)\right)\end{aligned} \tag{3.47}$$

where:

BOX **3.3** AN APPLICATION OF SYSTEMATIC FILON QUADRATURE TO THE DISCRETE FOURIER TRANSFORM

An interesting method for *attenuating the effects of aliasing* in the discrete Fourier transform of BOX **3.2** is to iterpolate the spectrum before transforming: a technique commonly employed in the Filon method [e.g. *Frazer & Gettrust* (1984), *Press & Teukolsky* (1989)]. We will demonstrate this technique with a specific example.
We interpolate the digitized spectrum

$$\{\tilde{\phi}(j\Delta k_x) : -M \leq j \leq M\}$$

with the *chapeau shape function* $\tilde{X}(k_x)$ (of finite element fame), defined as

$$\tilde{X}(k_x) = \begin{cases} 1 - |k_x/\Delta k_x| & -\Delta k_x \leq k_x \leq \Delta k_x \\ 0 & \text{otherwise} \end{cases}$$

so that the spectrum may be approximated as

$$\tilde{\Phi}(k_x) = \sum_{j=-M}^{M} \tilde{X}(k_x - j\Delta k_x)\tilde{\phi}(j\Delta k_x) = \tilde{X}(k_x) * \sum_{j=-M}^{M} \delta(k_x - j\Delta k_x)\tilde{\phi}(j\Delta k_x) \quad (3.41)$$

where the binary operator "$*$" denotes a *convolution* and δ is the Dirac delta function. Applying the inverse Fourier transform to (3.41) gives

$$\begin{aligned} \frac{1}{2\pi}\int \mathrm{d}\, k_x \mathrm{e}^{ik_x x}\tilde{\Phi}(k_x) &= \frac{1}{2\pi}\int \mathrm{d}k_x \mathrm{e}^{ik_x x} \sum_{j=-M}^{M} \tilde{X}(k_x - j\Delta k_x)\tilde{\phi}(j\Delta k_x) \\ &= \left[\frac{1}{2\pi}\int \mathrm{d}(k_x - j\Delta k_x)\, \mathrm{e}^{i(k_x - j\Delta k_x)x}\tilde{X}(k_x - j\Delta k_x)\right] \sum_{j=-M}^{M} \mathrm{e}^{ij\Delta k_x x}\, \tilde{\phi}(j\Delta k_x) \\ &= \left[\frac{\Delta k_x}{2\pi}\mathrm{sinc}^2\left(\frac{\Delta k_x\, x}{2}\right)\right] \sum_{j=-M}^{M} \mathrm{e}^{ij\Delta k_x x}\, \tilde{\phi}(j\Delta k_x) \end{aligned} \quad (3.42)$$

The extra *multiplicative factor*

$$A(x) = \mathrm{sinc}^2\left(\frac{\Delta k_x\, x}{2}\right)$$

in (3.42) *attenuates* the basic DFT sum

$$\frac{\Delta k_x}{2\pi} \sum_{j=-M}^{M} \mathrm{e}^{ij\Delta k_x x}\, \tilde{\phi}(j\Delta k_x) \quad (3.43)$$

that we met previously in BOX 3.2. If the fundamental period of the representation is L_x, then the wavenumber sampling interval Δk_x is chosen according to

$$\Delta k_x = \frac{2\pi}{L_x}$$

in which case (3.43) is a *monotone decreasing* function of x for $0 \leq x \leq L_x$. In particular

the *Kronecka product* $(\boldsymbol{I}_6 \otimes \boldsymbol{F}(x';\mathbf{k}_x))$ is a $6 \times 6N_k$ array, $\boldsymbol{F}^H(x;\mathbf{k}_x)$ is the *Hermitian transpose* of the Fourier vector $\boldsymbol{F}(x;\mathbf{k}_x)$ defined in (3.46) (so that $\boldsymbol{F}^H(x;\mathbf{k}_x) \equiv \boldsymbol{F}^T(x;-\mathbf{k}_x)$);
the $(6N_k) \times (6N_k)$ matrix analogue of (3.13) is defined as

$$\boldsymbol{D}_z(\mathbf{k}_x, k_y) = \tag{3.48}$$
$$\left[\ |+,P,\mathbf{k}_x,k_y\rangle \quad |+,S,\mathbf{k}_x,k_y\rangle \quad |+,H,\mathbf{k}_x,k_y\rangle \quad |-,P,\mathbf{k}_x,k_y\rangle \quad |-,S,\mathbf{k}_x,k_y\rangle \quad |-,H,\mathbf{k}_x,k_y\rangle\ \right]$$

in terms of

- the $6N_k \times N_k$ P-wave partition

$$|\pm,P,\mathbf{k}_x,k_y\rangle = \sqrt{\frac{\Delta k_x}{4\pi\rho\omega^2\mathbf{k}_{z,P}}} \begin{bmatrix} \pm i\,\mathbf{k}_{z,P} \\ i\,\mathbf{k}_x \\ i\,k_y\,\boldsymbol{I} \\ \rho\,\left(2\,\beta^2\,\mathbf{k}_x^2 + (2\,\beta^2\,k_y^2 - \omega^2)\boldsymbol{I}\right) \\ \mp 2\,\rho\,\beta^2\,\mathbf{k}_x\,\mathbf{k}_{z,P} \\ \mp 2\,\rho\,\beta^2\,k_y\,\mathbf{k}_{z,P} \end{bmatrix} \tag{3.49}$$

- the $6N_k \times N_k$ SV-wave partition

$$|\pm,S,\mathbf{k}_x,k_y\rangle = \frac{\omega}{\beta\hat{K}_S}\sqrt{\frac{\Delta k_x}{4\pi\rho\omega^2\mathbf{k}_{z,S}}} \begin{bmatrix} i\,\mathbf{k}_x \\ \mp i\,\mathbf{k}_{z,S} \\ \mathbf{0} \\ \mp 2\,\rho\,\beta^2\,\mathbf{k}_x\,\mathbf{k}_{z,S} \\ \rho\,\left((\omega^2 - \beta^2\,k_y^2)\boldsymbol{I} - 2\,\beta^2\,\mathbf{k}_x^2\right) \\ -\rho\,\beta^2\,k_y\,\mathbf{k}_x \end{bmatrix} \tag{3.50}$$

- the $6N_k \times N_k$ SH-wave partition

$$|\pm,H,\mathbf{k}_x,k_y\rangle = \frac{1}{\hat{K}_H}\sqrt{\frac{\Delta k_x}{4\pi\rho\omega^2\mathbf{k}_{z,H}}} \begin{bmatrix} \mp k_y\,\mathbf{k}_{z,H} \\ -k_y\,\mathbf{k}_x \\ \hat{K}_H^2\boldsymbol{I} \\ -2\,i\,\rho\,\beta^2\,k_y\,\mathbf{k}_{z,H}^2 \\ \mp 2\,i\,\rho\,\beta^2\,k_y\,\mathbf{k}_x\,\mathbf{k}_{z,H} \\ \pm i\,\rho\,\left(\omega^2 - 2\,\beta^2\,k_y^2\right)\,\mathbf{k}_{z,H} \end{bmatrix} \tag{3.51}$$

The $(6N_k) \times (6N_k)$ *plane wave propagator* $\boldsymbol{Q}^z(\mathbf{k}_x; z; z')$ in (3.47), is defined as

$$\boldsymbol{Q}^z(\mathbf{k}_x; z; z') = \begin{bmatrix} \mathbf{0}_{3N_k} & H(z-z')\boldsymbol{E}(\mathbf{k}_x; z; z') \\ H(z'-z)\boldsymbol{E}(\mathbf{k}_x; z; z') & \mathbf{0}_{3N_k} \end{bmatrix} \quad (3.52)$$

where the $(3N_k) \times (3N_k)$ *vertical phase shift array* is defined as

$$\boldsymbol{E}(\mathbf{k}_x; z; z') = \begin{bmatrix} \mathrm{e}^{i\mathbf{k}_{z,P}|z-z'|} & \mathbf{0}_{N_k} & \mathbf{0}_{N_k} \\ \mathbf{0}_{N_k} & \mathrm{e}^{i\mathbf{k}_{z,S}|z-z'|} & \mathbf{0}_{N_k} \\ \mathbf{0}_{N_k} & \mathbf{0}_{N_k} & \mathrm{e}^{i\mathbf{k}_{z,H}|z-z'|} \end{bmatrix},$$

in terms of the diagonal wavevector array $\mathbf{k}_{z,c}, c \in \{P, S, H\}$ of (3.45).

It will be convenient in later applications, to split (3.47) into the product

$$\mathring{\boldsymbol{G}}^z(\boldsymbol{x}; \boldsymbol{x}'; k_y) = \mathring{\boldsymbol{P}}^z(\boldsymbol{x}; \boldsymbol{x}'; k_y) \cdot \boldsymbol{J}_6$$

where $\boldsymbol{J}_6$ was given by (2.21) and

$$\begin{aligned} \mathring{\boldsymbol{P}}^z(\boldsymbol{x}; \boldsymbol{x}'; k_y) &= -i\ (\boldsymbol{I}_6 \otimes \boldsymbol{F}(x; \mathbf{k}_x))\, \boldsymbol{D}_z(\mathbf{k}_x, k_y) \\ &\quad \times \boldsymbol{Q}^z(\mathbf{k}_x; z; z') \boldsymbol{D}_z^T(-\mathbf{k}_x, -k_y)\ \left(\boldsymbol{I}_6 \otimes \boldsymbol{F}^T(x'; -\mathbf{k}_x)\right) \end{aligned} \quad (3.53)$$

might be called the *propagator*, for the want of a better name. The propagator is an entity in itself for most numerical applications considered here.

The representation (3.47) (or equivalently, (3.53)) is readily extensible to the plane stratified problem and we investigate this extension in chapter 5.

3.2.2 The x splitting

Recalling the definition (3.13) for the columns of the eigen-matrix $\boldsymbol{D}_z$, (3.24) may be rewritten as

$$\begin{aligned} \mathring{\boldsymbol{G}}\ (\boldsymbol{x}; \boldsymbol{x}'; k_y) &= \sum_{c \in \{P,S,H\}} \frac{1}{4\pi^2} \int_{R^2} \mathrm{d}\boldsymbol{k} \exp\left[i\boldsymbol{k} \cdot (\boldsymbol{x} - \boldsymbol{x}')\right] \\ &\quad \times \left[\frac{|-, c, k_x, k_y\rangle \otimes |+, c, -k_x, -k_y\rangle}{k_z + k_{z,c}(k_x)} - \frac{|+, c, k_x, k_y\rangle \otimes |-, c, -k_x, -k_y\rangle}{k_z - k_{z,c}(k_x)} \right] \boldsymbol{J}_6 \\ &= \mathring{\boldsymbol{G}}_P\ (\boldsymbol{x}; \boldsymbol{x}'; k_y) + \mathring{\boldsymbol{G}}_S\ (\boldsymbol{x}; \boldsymbol{x}'; k_y) + \mathring{\boldsymbol{G}}_H\ (\boldsymbol{x}; \boldsymbol{x}'; k_y) \end{aligned} \quad (3.54)$$

where $\otimes$ is the *tensor product*. We will ultimately evaluate the integrals in (3.54) using the residues at the simple poles, where

$$k_z^2 + k_x^2 + k_y^2 = \frac{\omega^2}{c^2}, \quad \text{for} \quad c \in \{P, S, H\}. \quad (3.55)$$

Hence, (3.54) may be replaced by an expression that is *equivalent at these poles*. Introducing the partial differential operators

$$\boldsymbol{L}_P = \begin{bmatrix} \partial_z \\ \partial_x \\ \partial_y \\ 2\mu\partial_{zz} - \frac{\lambda\omega^2}{\alpha^2} \\ 2\mu\partial_{xz} \\ 2\mu\partial_{yz} \end{bmatrix} \quad \boldsymbol{L}_S = \frac{\omega}{\beta\hat{K}_S}\begin{bmatrix} \partial_x \\ -\partial_z \\ 0 \\ 2\mu\partial_{xz} \\ \mu(-\partial_{zz}+\partial_{xx}) \\ \mu\partial_{xy} \end{bmatrix} \quad \boldsymbol{L}_H = \frac{1}{\hat{K}_H}\begin{bmatrix} \partial_{yz} \\ \partial_{xy} \\ \frac{\omega^2}{\beta^2}+\partial_{yy} \\ 2\mu\partial_{yzz} \\ 2\mu\partial_{xyz} \\ \mu(\frac{\omega^2}{\beta^2}\partial_z + 2\partial_{yyz}) \end{bmatrix} \tag{3.56}$$

$\mathring{\boldsymbol{G}}_c\,(\boldsymbol{x};\boldsymbol{x}';k_y)$ in (3.54) may be replaced with the *equivalent* expression

$$\begin{aligned} \mathring{\boldsymbol{G}}_c\,(\boldsymbol{x};\boldsymbol{x}';k_y) &= \frac{1}{8\pi^2\rho\omega^2}\int_{R^2}\mathrm{d}\boldsymbol{k}\,\frac{\boldsymbol{L}_c\otimes\boldsymbol{L}_c'\,(\exp\left[i\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')\right])}{k_{z,c}(k_x)\left[k_z+k_{z,c}(k_x)\right]}\boldsymbol{J}_6 \\ &\quad -\frac{1}{8\pi^2\rho\omega^2}\int_{R^2}\mathrm{d}\boldsymbol{k}\,\frac{\boldsymbol{L}_c\otimes\boldsymbol{L}_c'\,(\exp\left[i\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')\right])}{k_{z,c}(k_x)\left[k_z-k_{z,c}(k_x)\right]}\boldsymbol{J}_6,\ c\in\{P,S,H\}; \\ &= \frac{1}{4\pi^2\rho\omega^2}\int_{R^2}\mathrm{d}\boldsymbol{k}\,\boldsymbol{L}_c\otimes\boldsymbol{L}_c'\frac{\exp\left[i\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')\right]}{\hat{K}_c^2-k_z^2-k_x^2}\boldsymbol{J}_6 \end{aligned} \tag{3.57}$$

where we have used the algebraic relation (3.55) and the differential operator $\boldsymbol{L}_c'$ operates on the $\boldsymbol{x}'$ coordinate.

Assuming that $x \neq x'$ in (3.57) we may evaluate the k_x integral in that equation using the calculus of residues to get

$$\mathring{\boldsymbol{G}}_c\,(\boldsymbol{x};\boldsymbol{x}';k_y) = \frac{-i}{2\pi\rho\omega^2}\int_R \mathrm{d}k_z\,\boldsymbol{L}_c\otimes\boldsymbol{L}_c'\frac{\exp\left[ik_z(z-z')\right]\exp\left[ik_{x,c}(k_z,k_y)|x-x'|\right]}{2\,k_{x,c}(k_z,k_y)}\boldsymbol{J}_6, \tag{3.58}$$

with

$$k_{x,c}(k_z,k_y) = \begin{cases} \sqrt{\hat{K}_c^2(k_y)\ -\ k_z^2} & |k_x|\ <\ |\hat{K}_c(k_y)| \\ i\sqrt{k_z^2\ -\ \hat{K}_c^2(k_y)} & |k_x|\ \geq\ |\hat{K}_c(k_y)| \end{cases} \tag{3.59}$$

and $\hat{K}_c(k_y)$ was given by (3.11). Using (3.58) in (3.54) we get

$$\begin{aligned} \mathring{\boldsymbol{G}}_c\,(\boldsymbol{x};\boldsymbol{x}';k_y) &= \frac{-i}{2\pi\rho\omega^2}\sum_{c\in\{P,S,H\}} \\ &\times \int_R \mathrm{d}k_z\,\boldsymbol{L}_c\otimes\boldsymbol{L}_c'\frac{\exp\left[ik_z(z-z')\right]\exp\left[ik_{x,c}(k_z,k_y)|x-x'|\right]}{2\,k_{x,c}(k_z,k_y)}\boldsymbol{J}_6 \end{aligned} \tag{3.60}$$

Using the definition of the partial differential operators $\boldsymbol{L}_c$ from (3.56) in (3.54) and performing the relevant spatial differentiations we get

$$\mathring{\boldsymbol{G}}\left(\boldsymbol{x};\boldsymbol{x}';k_y\right) = \frac{-i}{2\pi}\int_R \mathrm{d}k_z \exp\left[ik_z(z-z')\right]\boldsymbol{D}_x(k_z,k_y)\boldsymbol{Q}^x(k_z;x;x')\boldsymbol{D}_x^T(-k_z,-k_y)\,\boldsymbol{J}_6 \tag{3.61}$$

provided $x \neq x'$, where:

$$\boldsymbol{D}_x(k_z,k_y) = \tag{3.62}$$
$$\left[\ \overline{|+,P,k_z,k_y\rangle}\quad \overline{|+,S,k_z,k_y\rangle}\quad \overline{|+,H,k_z,k_y\rangle}\quad \overline{|-,P,k_z,k_y\rangle}\quad \overline{|-,S,k_z,k_y\rangle}\quad \overline{|-,H,k_z,k_y\rangle}\ \right]$$

where:

- the $\overline{|\pm,P,k_z,k_y\rangle}$ states are the $\begin{smallmatrix}\text{rightgoing}\\ \text{leftgoing}\end{smallmatrix}$ $P-$waves

$$\overline{|\pm,P,k_{x,P},k_y\rangle} = \frac{1}{\sqrt{2\rho\omega^2 k_{x,P}}}\begin{bmatrix} i\,k_z \\ \pm i\,k_{x,P}(k_x,k_y) \\ i\,k_y \\ \rho\left(2\,\beta^2\,k_{x,P}^2(k_x,k_y)+2\,\beta^2\,k_y^2-\omega^2\right) \\ \mp 2\,\rho\,\beta^2\,k_{x,P}(k_x,k_y)\,k_z \\ -2\,\rho\,\beta^2\,k_y\,k_z \end{bmatrix} \tag{3.63}$$

- the $\overline{|\pm,S,k_z,k_y\rangle}$ states are the $\begin{smallmatrix}\text{rightgoing}\\ \text{leftgoing}\end{smallmatrix}$ quasi-$SV-$waves

$$\overline{|\pm,S,k_z,k_y\rangle} = \frac{\omega}{\beta\,\hat{K}_S\,\sqrt{2\rho\omega^2 k_{x,S}}}\begin{bmatrix} \pm i\,k_{x,S}(k_x,k_y) \\ -i\,k_z \\ 0 \\ \mp 2\,\rho\,\beta^2\,k_{x,S}(k_x,k_y)\,k_z \\ \rho\left(\omega^2-2\,\beta^2\,k_{x,S}^2(k_x,k_y)-\beta^2\,k_y^2\right) \\ \mp\rho\,\beta^2\,k_{x,S}(k_x,k_y)\,k_y \end{bmatrix} \tag{3.64}$$

- the $\overline{|\pm, H, k_z, k_y\rangle}$ states are the $\begin{smallmatrix}\text{rightgoing}\\ \text{leftgoing}\end{smallmatrix}$ quasi-$SH-$waves

$$\overline{|\pm, H, k_z, k_y\rangle} = \frac{1}{\hat{K}_H \sqrt{2\rho\omega^2 k_{x,H}}} \begin{bmatrix} -k_y\, k_z \\ \mp k_{x,H}(k_x, k_y)\, k_y \\ \hat{K}_H^2 \\ 2\, i\, k_y\, \rho\, \left(\beta^2\, k_{x,H}^2(k_x, k_y) + \beta^2\, k_y^2 - \omega^2\right) \\ \mp 2\, i\, \rho\, \beta^2\, k_{x,H}(k_x, k_y)\, k_y\, k_z \\ i\, k_z\, \rho\, \left(\omega^2 - 2\, \beta^2\, k_y^2\right) \end{bmatrix} \tag{3.65}$$

the *horizontal plane wave propagator* $\boldsymbol{Q}^x(k_z; x; x')$ in (3.61), is defined as

$$\boldsymbol{Q}^x(k_z; x; x') = \begin{bmatrix} \mathbf{0}_3 & H(x - x')\boldsymbol{E}(k_z; x; x') \\ H(x' - x)\boldsymbol{E}(k_z; x; x') & \mathbf{0}_3 \end{bmatrix} \tag{3.66}$$

with the *horizontal phase shift* array

$$\boldsymbol{E}(k_z; x; x') = \begin{bmatrix} \mathrm{e}^{ik_{x,P}(k_z)|x-x'|} & 0 & 0 \\ 0 & \mathrm{e}^{ik_{x,S}(k_z)|x-x'|} & 0 \\ 0 & 0 & \mathrm{e}^{ik_{x,H}(k_z)|x-x'|} \end{bmatrix},$$

and $H(\cdot)$ is the *Heaviside step function*.

For computational purposes, we must employ a *discrete plane wave* or *Fourier lattice* version of the wavenumber integral (3.61). The development closely parallels that of §3.2.1. One of the many problems associated with discrete Fourier representations of the Green's tensor operators commonly employed seismology is wraparound. In the standard reflectivity method, we employ a horizontal wavenumber sampling, and as a consequence we get aliased versions of the wavefield *wrapping* into the computational domain from the sides (Mallick & Frazier, 1988). This is not such a serious problem provided we extend the transverse support of the computational domain, or equivalently, decrease the horizontal wavenumber sampling interval interval Δk_x.

We will now derive a *discrete plane wave* version of (3.61), and we may expect wraparound in z. This is a far more serious problem than wraparound in x because we will eventually extend the representation to $z-$stratified media, so that the wraparound phases will interact with the stratification in a complicated manner, that may not be removed. However, we will employ (3.61) only in the special case where $z = z'$ so that $\exp\left[i\overline{k}_j(z - z')\right] \equiv 1$ and

consequently

$$\overset{\circ}{\boldsymbol{G}}\,(x;x';k_y) = \frac{-i}{2\pi}\int_R \mathrm{d}k_z \boldsymbol{D}_x(k_z,k_y)\boldsymbol{Q}^x(k_z;x;x')\boldsymbol{D}_x^T(-k_z,-k_y)\,\boldsymbol{J}_6,\ z=z',\ x\neq x' \tag{3.67}$$

so there will be no *aliasing* or *wraparound* in the $z-$coordinate. This observation is crucial to the success in the intended application of $\overset{\circ}{\boldsymbol{G}}\,(\boldsymbol{x};\boldsymbol{x'};k_y)$ from (3.67) to problems with a $z-$stratification in material properties.

Given the $\overline{N}_k = 2\overline{M}+1$ element *Fourier lattice*

$$\left\{\overline{k}_j = j\Delta k_z \mid j=-\overline{M},\cdots,\overline{M}\right\} \tag{3.68}$$

where $\Delta k_z = 2\pi j/L_z$ and L_z controls the *wavenumber sampling rate,* we may approximate (3.67) by the *discrete plane wave sum*

$$\overset{\circ}{\boldsymbol{G}}{}^x(x;x';k_y) = \frac{-i\,\Delta k_z}{2\pi}\sum_{j=-\overline{M}}^{j=\overline{M}} \boldsymbol{D}_x(\overline{k}_j,k_y)\boldsymbol{Q}^x(\overline{k}_j;x;x')\boldsymbol{D}_x^T(-\overline{k}_j,-k_y)\,\boldsymbol{J}_6 \tag{3.69}$$

The discrete representation (3.69) is equivalent to assuming that the $z-$dependence of the wavefield is *bandlimited* (since $\overline{M}$ is finite), which is certainly not the case in the limit $x \leftarrow x'$.

The next step will be to rewrite the discrete sum (3.69) as an equivalent matrix operation. To this end, we define the diagonal wave vector arrays:

$$\mathbf{k}_z = \mathrm{diag}\left[-\overline{M}\Delta k_z,\cdots,-\Delta k_z,0,\Delta k_z,\cdots,\overline{M}\Delta k_z\right]; \tag{3.70}$$

$$\mathbf{k}_{x,c} = \mathrm{diag}\left[k_{x,c}(-\overline{M}\Delta k_z),\cdots,k_{x,c}(-\Delta k_z),k_{x,c}(0),k_{z,c}(\Delta k_z),\cdots,k_{x,c}(\overline{M}\Delta k_z)\right] \tag{3.71}$$

and the $\overline{N}_k-$element *projection* vector

$$\overline{\boldsymbol{F}}(0) = \left[\begin{array}{ccccc} 1 & \cdots & 1 & \cdots & 1 \end{array}\right]. \tag{3.72}$$

We can rewrite the *discrete plane wave* approximation (3.69) to the 6×6 Green's tensor in the equivalent matrix form

$$\begin{aligned}\overset{\circ}{\boldsymbol{G}}{}^x(x;x';k_y) &= -i\,\left(\boldsymbol{I}_6\otimes\overline{\boldsymbol{F}}(0)\right)\boldsymbol{D}_x(\mathbf{k}_z,k_y)\\ &\quad\times\boldsymbol{Q}^x(\mathbf{k}_z;x;x')\boldsymbol{D}_x^T(-\mathbf{k}_z,-k_y)\,\left(\boldsymbol{I}_6\otimes\overline{\boldsymbol{F}}^T(0)\right)\boldsymbol{J}_6\end{aligned} \tag{3.73}$$

where:

the $(6\bar{N}_k) \times (6\bar{N}_k)$ matrix analogue of (3.62) is defined as

$$\boldsymbol{D}_x(\mathbf{k}_z, k_y) = \tag{3.74}$$
$$\left[\ \overline{|+, P, \mathbf{k}_z, k_y\rangle}\ \ \overline{|+, S, \mathbf{k}_z, k_y\rangle}\ \ \overline{|+, H, \mathbf{k}_z, k_y\rangle}\ \ \overline{|-, P, \mathbf{k}_z, k_y\rangle}\ \ \overline{|-, S, \mathbf{k}_z, k_y\rangle}\ \ \overline{|-, H, \mathbf{k}_z, k_y\rangle}\ \right]$$

in terms of

- the $6\overline{N}_k \times \overline{N}_k$ P-wave partition

$$\overline{|\pm, P, \mathbf{k}_z, k_y\rangle} = \sqrt{\frac{\Delta k_z}{4\pi\rho\omega^2 \mathbf{k}_{x,P}}} \begin{bmatrix} i\,\mathbf{k}_z \\ \pm i\,\mathbf{k}_{x,P} \\ i\,k_y\,\boldsymbol{I} \\ \rho\,\left(2\,\beta^2\,\mathbf{k}_{x,P}^2 + (2\,\beta^2\,k_y^2 - \omega^2)\boldsymbol{I}\right) \\ \mp 2\,\rho\,\beta^2\,\mathbf{k}_{x,P}\,\mathbf{k}_z \\ -2\,\rho\,\beta^2\,k_y\,\mathbf{k}_z \end{bmatrix} \tag{3.75}$$

- the $6\overline{N}_k \times \overline{N}_k$ SV-wave

$$\overline{|\pm, S, \mathbf{k}_z, k_y\rangle} = \frac{\omega}{\beta\hat{K}_S}\sqrt{\frac{\Delta k_z}{4\pi\rho\omega^2 \mathbf{k}_{x,S}}} \begin{bmatrix} \pm i\,\mathbf{k}_{x,S} \\ -i\,\mathbf{k}_z \\ \mathbf{0} \\ \mp 2\,\rho\,\beta^2\,\mathbf{k}_{x,S}\,\mathbf{k}_z \\ \rho\,\left((\omega^2 - \beta^2\,k_y^2)\boldsymbol{I} - 2\,\beta^2\,\mathbf{k}_{x,S}^2\right) \\ \mp\rho\,\beta^2\,k_y\,\mathbf{k}_{x,S} \end{bmatrix} \tag{3.76}$$

- the $6\overline{N}_k \times \overline{N}_k$ SH-wave partition

$$\overline{|\pm, H, \mathbf{k}_z, k_y\rangle} = \frac{1}{\hat{K}_H}\sqrt{\frac{\Delta k_z}{4\pi\rho\omega^2 \mathbf{k}_{x,H}}} \begin{bmatrix} -k_y\,\mathbf{k}_z \\ \mp k_y\,\mathbf{k}_{x,H} \\ \hat{K}_H^2\boldsymbol{I} \\ -2\,i\,\rho\,\beta^2\,k_y\,\mathbf{k}_z^2 \\ \mp 2\,i\,\rho\,\beta^2\,k_y\,\mathbf{k}_{x,H}\,\mathbf{k}_z \\ i\,\rho\,\left(\omega^2 - 2\,\beta^2\,k_y^2\right)\,\mathbf{k}_z \end{bmatrix} \tag{3.77}$$

where $\mathbf{k}_x$ and $\mathbf{k}_{z,c}$ were given in (3.71) and (3.70) respectively.

The $(6\bar{N}_k) \times (6\bar{N}_k)$ *plane wave propagator* $\boldsymbol{Q}^x(\mathbf{k}_z; x; x')$ in (3.73), is defined as

$$\boldsymbol{Q}^x(\mathbf{k}_z; x; x') = \begin{bmatrix} \mathbf{0}_{3\bar{N}_k} & H(x - x')\boldsymbol{E}(\mathbf{k}_z; x; x') \\ H(x' - x)\boldsymbol{E}(\mathbf{k}_z; x; x') & \mathbf{0}_{3\bar{N}_k} \end{bmatrix} \tag{3.78}$$

where the $(3\bar{N}_k) \times (3\bar{N}_k)$ *vertical phase shift array* is defined as

$$\boldsymbol{E}(\mathbf{k}_z; x; x') = \begin{bmatrix} \mathrm{e}^{i\mathbf{k}_{z,P}|x-x'|} & \mathbf{0}_{\overline{N}_k} & \mathbf{0}_{\overline{N}_k} \\ \mathbf{0}_{\overline{N}_k} & \mathrm{e}^{i\mathbf{k}_{z,S}|x-x'|} & \mathbf{0}_{\overline{N}_k} \\ \mathbf{0}_{\overline{N}_k} & \mathbf{0}_{\overline{N}_k} & \mathrm{e}^{i\mathbf{k}_{z,H}|x-x'|} \end{bmatrix},$$

in terms of the diagonal wavevector array $\mathbf{k}_{x,c}, c \in \{P, S, H\}$ of (3.71).

For computational efficiency, we split (3.73) into

$$\mathring{\boldsymbol{G}}^x(x; x'; k_y) = \mathring{\boldsymbol{P}}^x(x; x'; k_y)\boldsymbol{J}_6$$

where

$$\begin{aligned} \mathring{\boldsymbol{P}}^x(x; x'; k_y) &= -i\left(\boldsymbol{I}_6 \otimes \overline{\boldsymbol{F}}(0)\right)\boldsymbol{D}_x(\mathbf{k}_z, k_y) \\ &\quad \times \boldsymbol{Q}^x(\mathbf{k}_z; x; x')\boldsymbol{D}_x^T(-\mathbf{k}_z, -k_y)\left(\boldsymbol{I}_6 \otimes \overline{\boldsymbol{F}}^T(0)\right) \end{aligned} \quad (3.79)$$

3.3 Lattice propagator representations in free space

This section is central to the thesis. In this section the medium is homogeneous. The method to be expounded here is extended to a stratified halfspace in chapter 5.

To begin, we define a uniform lattice (Fig.3.2) with L rows and N_x lattice points per row. The lattice points are numbered in the same order that words are read on the written page, i.e. from left to right, from top to bottom. For many applications it is helpful to label the lattice points as if they were the elements of a 2−D array. For this purpose we introduce the matrix row index $1 \leq \rho \leq L$ and the matrix column index $1 \leq \chi \leq N_x$, so that the global index i is related to the matrix indices via

$$i = (\rho - 1)N_x + \chi, \quad 1 \leq i \leq LN_x, \quad 1 \leq \rho \leq L, \quad 1 \leq \chi \leq N_x$$

Please note that use of ρ and χ to indicate the row and column indices respectively, are confined to this section, and are not to be confused with the other applications of these symbols throughout the thesis.

We want to *communicate* the influence of the *secondary force field* 6−vector $\bar{\boldsymbol{f}}_j$, from a given lattice point to all the *other* lattice points. Using the naive matrix/vector operation

$$\mathring{\boldsymbol{\mathcal{G}}} \cdot \bar{\boldsymbol{\mathcal{F}}} \equiv \sum_{\substack{j=1 \\ j\neq i}}^{LN_x} \mathring{\boldsymbol{G}}\left(\boldsymbol{R}_i; \boldsymbol{R}_j; k_y\right) \cdot \bar{\boldsymbol{f}}_j \qquad 1 \leq i \leq LN_x \quad (3.80)$$

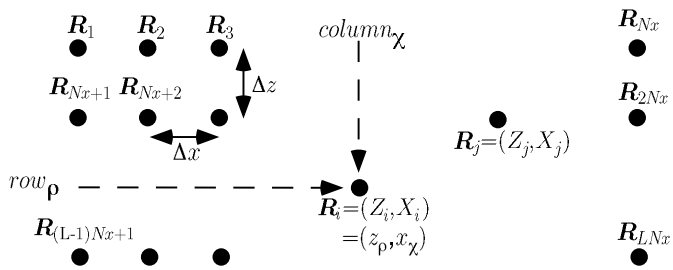

Figure 3.2: The lattice geometry

where $\mathring{\boldsymbol{G}}$ is the 6×6 Green's tensor (3.3), (3.80) may be implemented as

$$N_1 = (6 \times N_x \times L)^2 - (6 \times N_x \times L)$$

separate interactions.

We will employ the plane wave expansions (3.47), (3.73), [or equivalently, (3.53), (3.79)] , the Fast Fourier Transform and matrix factorization techniques so that (3.80) is computed using

$$\begin{aligned}
\mathring{\boldsymbol{\mathcal{G}}} \cdot \bar{\boldsymbol{\mathcal{F}}} &\equiv \sum_{\substack{j=1\\ j\neq i}}^{LN_x} \left[(1-\delta_{\rho\rho'})\, \mathring{\boldsymbol{P}}^z(\boldsymbol{R}_i;\boldsymbol{R}_j;k_y)\cdot \boldsymbol{J}_6 \bar{\boldsymbol{f}}_j + \delta_{\rho\rho'}\, \mathring{\boldsymbol{P}}^x(\boldsymbol{R}_i;\boldsymbol{R}_j;k_y)\cdot \boldsymbol{J}_6 \bar{\boldsymbol{f}}_j \right] \\
&\equiv -i \sum_{\substack{j=1\\ j\neq i}}^{LN_x} (1-\delta_{\rho\rho'})\, (\boldsymbol{I}_6 \otimes \boldsymbol{F}(X_i;\mathbf{k}_x))\, \boldsymbol{D}_z(\mathbf{k}_x,k_y) \\
&\quad \times \boldsymbol{Q}^z(\mathbf{k}_x;Z_i;Z_j)\boldsymbol{D}_z^T(-\mathbf{k}_x,-k_y)\, \left(\boldsymbol{I}_6 \otimes \boldsymbol{F}^T(x';-\mathbf{k}_x)\right) \cdot \boldsymbol{f}_j \\
&\quad -i \sum_{\substack{j=1\\ j\neq i}}^{LN_x} \delta_{\rho\rho'}\, \left(\boldsymbol{I}_6 \otimes \overline{\boldsymbol{F}}(0)\right) \boldsymbol{D}_x(\mathbf{k}_z,k_y) \\
&\quad \times \boldsymbol{Q}^x(\mathbf{k}_z;X_i;X_j)\boldsymbol{D}_x^T(-\mathbf{k}_z,-k_y)\, \left(\boldsymbol{I}_6 \otimes \overline{\boldsymbol{F}}^T(0)\right) \cdot \boldsymbol{f}_j \\
&\qquad i \in \mathrm{row}_\rho, \qquad j \in \mathrm{row}_{\rho'}, \qquad \text{for each} \quad 1 \le i \le LN_x
\end{aligned} \tag{3.81}$$

where

$$\boldsymbol{f}_j \equiv \boldsymbol{J}_6 \bar{\boldsymbol{f}}_j$$

to reduce the floating point operation count of (3.80) to

$$N_2 \approx 2\,(L-1)\,[39N_k + 6N_k \log_2 N_k] + 78\,L\,(N_x-1)\overline{N}_k$$

where N_k is the number of plane waves employed to propagate the secondary source field between rows, while $\bar{N}_k$ is the number of plane waves employed to propagate the secondary source field between lattice points within a given row. If we employ critical (i.e. Nyquist) sampling rates so that $N_k = N_x$, then $N_2 \ll N_1$. However, for the sake of accuracy, we need to employ a substantial amount of oversampling so that $N_k > N_x$, which increases the operation count N_2 of our plane wave expansion method. In the applications presented in chapter 6, $N_k = 16N_x$ or even $N_k = 32N_x$, so that the plane wave method is not substantially faster than the naive operation count N_1; at least for the scale of the problems tested, for larger problems we observe that N_2 grows much more slowly than N_1. The principal merit of the method to be presented in this section lies in its extensibility to a plane stratified medium.

Recall that we have introduced a so called *propagator*, denoted as $\overset{\circ}{\boldsymbol{P}}{}^{z,x}(\boldsymbol{R}_i;\boldsymbol{R}_j;k_y)$, and defined by reference to the Green's tensors of (3.47) and (3.73). We will denote the generic propagator as $\overset{\circ}{\boldsymbol{P}}(\boldsymbol{R}_i;\boldsymbol{R}_j;k_y)$, and will re-introduce the superscripts in $\overset{\circ}{\boldsymbol{P}}{}^{z}(\boldsymbol{R}_i;\boldsymbol{R}_j;k_y)$ and $\overset{\circ}{\boldsymbol{P}}{}^{x}(\boldsymbol{R}_i;\boldsymbol{R}_j;k_y)$, when appropriate. Let us assemble these 6×6 propagators $\overset{\circ}{\boldsymbol{P}}(\boldsymbol{R}_i;\boldsymbol{R}_j;k_y)$ for lattice points $\boldsymbol{R}_i$ and $\boldsymbol{R}_j$ into a $6N_xL \times 6N_xL$ *super-matrix* $\boldsymbol{\mathcal{P}}$ where the (i,j) partition $\boldsymbol{\mathcal{P}}_{i,j}$ of $\boldsymbol{\mathcal{P}}$ is defined as

$$\boldsymbol{\mathcal{P}}_{i,j} = (1-\delta_{ij})\,\overset{\circ}{\boldsymbol{P}}(\boldsymbol{R}_i;\boldsymbol{R}_j;k_y) \tag{3.82}$$

so that (3.81) may be expressed as the equivalent matrix vector operation

$$\boldsymbol{\mathcal{P}}\cdot\boldsymbol{\mathcal{F}} = \begin{bmatrix} \mathbf{0}_6 & \boldsymbol{\mathcal{P}}_{1,2} & \cdots & \boldsymbol{\mathcal{P}}_{1,N_xL} \\ \boldsymbol{\mathcal{P}}_{2,1} & \mathbf{0}_6 & & \vdots \\ \vdots & & \ddots & \boldsymbol{\mathcal{P}}_{N_xL-1,N_xL} \\ \boldsymbol{\mathcal{P}}_{N_xL,1} & \cdots & \boldsymbol{\mathcal{P}}_{N_xL,N_xL-1} & \mathbf{0}_6 \end{bmatrix} \cdot \begin{bmatrix} \boldsymbol{f}_1 \\ \boldsymbol{f}_2 \\ \vdots \\ \boldsymbol{f}_{LN_x} \end{bmatrix} \tag{3.83}$$

We assemble each of the $6-$vectors $\boldsymbol{f}_{N_x(\rho'-1)+\chi}$, $1 \le \chi \le N_x$ in row$_{\rho'}$ of the lattice into the $6N_x$-vector

$$\boldsymbol{\mathcal{F}}_{\rho'} = \begin{bmatrix} \boldsymbol{f}_{N_x(\rho'-1)+1} \\ \vdots \\ \boldsymbol{f}_{N_x\rho'} \end{bmatrix} \tag{3.84}$$

From (3.81), we observe that the propagator partitions $\overset{\circ}{\boldsymbol{P}}(\boldsymbol{R}_i;\boldsymbol{R}_j)$ have different plane wave expansions depending on whether the lattice points $\boldsymbol{R}_i$

and $\boldsymbol{R}_j$ are within the same or different rows of the lattice. Consequently, we label the partitions of (3.83) depending on the source and receiver row, i.e.

$$\boldsymbol{\mathcal{P}} = \begin{bmatrix} \boldsymbol{\mathcal{P}}_{1\leftarrow 1} & \boldsymbol{\mathcal{P}}_{1\leftarrow 2} & \cdots & \boldsymbol{\mathcal{P}}_{1\leftarrow L} \\ \boldsymbol{\mathcal{P}}_{2\leftarrow 1} & \boldsymbol{\mathcal{P}}_{2\leftarrow 2} & & \vdots \\ \vdots & & \ddots & \boldsymbol{\mathcal{P}}_{L-1\leftarrow L} \\ \boldsymbol{\mathcal{P}}_{L\leftarrow 1} & \cdots & \boldsymbol{\mathcal{P}}_{L\leftarrow L-1} & \boldsymbol{\mathcal{P}}_{L\leftarrow L} \end{bmatrix} \tag{3.85}$$

where the $6N_x \times 6N_x$ partition $\boldsymbol{\mathcal{P}}_{\rho\leftarrow\rho'}$ defines the *influence* of the source field $\boldsymbol{\mathcal{F}}_{\rho'}$ (3.84), from the N_x lattice points in $\text{row}_{\rho'}$ on the N_x lattice points of row_{ρ}, and is defined as

$$\boldsymbol{\mathcal{P}}_{\rho\leftarrow\rho'} = \begin{bmatrix} \boldsymbol{\mathcal{P}}_{(\rho-1)N_x+1,(\rho'-1)N_x+1} & \boldsymbol{\mathcal{P}}_{(\rho-1)N_x+1,(\rho'-1)N_x+2} & \cdots & \boldsymbol{\mathcal{P}}_{(\rho-1)N_x+1,\rho' N_x} \\ \boldsymbol{\mathcal{P}}_{(\rho-1)N_x+2,(\rho'-1)N_x+1} & \boldsymbol{\mathcal{P}}_{(\rho-1)N_x+2,(\rho'-1)N_x+2} & & \vdots \\ \vdots & & \ddots & \boldsymbol{\mathcal{P}}_{\rho N_x-1,\rho' N_x} \\ \boldsymbol{\mathcal{P}}_{\rho N_x,(\rho'-1)N_x+1} & \cdots & \boldsymbol{\mathcal{P}}_{\rho N_x,\rho' N_x-1} & \boldsymbol{\mathcal{P}}_{\rho N_x,\rho' N_x} \end{bmatrix} \tag{3.86}$$

$1 \leq \rho, \rho' \leq L$, with 6×6 component arrays $\boldsymbol{\mathcal{P}}_{i,j}$ given by (3.82). We now split (3.85) into its off-diagonal and diagonal blocks to get

$$\boldsymbol{\mathcal{P}} = \boldsymbol{\mathcal{P}}^z + \boldsymbol{\mathcal{P}}^x \tag{3.87}$$

where

$$\boldsymbol{\mathcal{P}}^z = \begin{bmatrix} \mathbf{0}_{6N_x} & \boldsymbol{\mathcal{P}}_{1\leftarrow 2} & \cdots & \boldsymbol{\mathcal{P}}_{1\leftarrow L} \\ \boldsymbol{\mathcal{P}}_{2\leftarrow 1} & \mathbf{0}_{6N_x} & & \vdots \\ \vdots & & \ddots & \boldsymbol{\mathcal{P}}_{L-1\leftarrow L} \\ \boldsymbol{\mathcal{P}}_{L\leftarrow 1} & \cdots & \boldsymbol{\mathcal{P}}_{L\leftarrow L-1} & \mathbf{0}_{6N_x} \end{bmatrix} \tag{3.88}$$

defines the interaction between *distinct* rows of the lattice and

$$\boldsymbol{\mathcal{P}}^x = \text{diag}\left[\ \boldsymbol{\mathcal{P}}_{1\leftarrow 1},\ \ \boldsymbol{\mathcal{P}}_{2\leftarrow 2},\ \ \cdots,\ \ \boldsymbol{\mathcal{P}}_{L\leftarrow L}\ \right] \tag{3.89}$$

defines the interaction between points within the *same* row of the lattice.

Our next task will be to generalize the definition of (3.47), (3.73) to find a computational construction for (3.88) and (3.89), respectively.

3.3.1 Implementing $\boldsymbol{\mathcal{P}}^z \cdot \boldsymbol{\mathcal{F}}$

Using $\mathring{\boldsymbol{P}}^z(\boldsymbol{x};\boldsymbol{x}';k_y)$ from (3.53) for the partitions of $\boldsymbol{\mathcal{P}}_{\rho\leftarrow\rho'}$ in (3.86), we get

$$\begin{aligned}\boldsymbol{\mathcal{P}}_{\rho\leftarrow\rho'} &= -i\begin{bmatrix}(\boldsymbol{I}_6\otimes\boldsymbol{F}(x_1;\mathbf{k}_x))\\ \vdots\\ (\boldsymbol{I}_6\otimes\boldsymbol{F}(x_{N_x};\mathbf{k}_x))\end{bmatrix}\boldsymbol{D}_z(\mathbf{k}_x,k_y)\boldsymbol{Q}^z_{\rho\leftarrow\rho'}(\mathbf{k}_x)\boldsymbol{D}_z^T(-\mathbf{k}_x,-k_y)\\ &\quad\times\left[\begin{array}{ccc}\left(\boldsymbol{I}_6\otimes\boldsymbol{F}^T(x_1;-\mathbf{k}_x)\right) & \cdots & \left(\boldsymbol{I}_6\otimes\boldsymbol{F}^T(x_{N_x};-\mathbf{k}_x)\right)\end{array}\right],\quad \rho\neq\rho'\end{aligned}\tag{3.90}$$

where

$$\boldsymbol{Q}^z_{\rho\leftarrow\rho'}(\mathbf{k}_x)=\begin{bmatrix}\mathbf{0}_{3N} & H(z_\rho-z_{\rho'})\mathbf{E}(\mathbf{k}_x)^{|\rho-\rho'|}\\ H(z_{\rho'}-z_\rho)\mathbf{E}(\mathbf{k}_x)^{|\rho-\rho'|} & \mathbf{0}_{3N}\end{bmatrix}\tag{3.91}$$

with

$$\mathbf{E}(\mathbf{k}_x)=\begin{bmatrix}\mathrm{e}^{i\mathbf{k}_{z,P}\,\Delta z} & \mathbf{0}_N & \mathbf{0}_N\\ \mathbf{0}_N & \mathrm{e}^{i\mathbf{k}_{z,S}\,\Delta z} & \mathbf{0}_N\\ \mathbf{0}_N & \mathbf{0}_N & \mathrm{e}^{i\mathbf{k}_{z,H}\,\Delta z}\end{bmatrix},$$

Δz the vertical separation between adjacent rows of the lattice in Fig.3.2, $H(\cdot)$ is the *Heaviside step function* and the diagonal wavevector array $\mathbf{k}_{z,c}, c\in\{P,S,H\}$ was given by (3.45) and is an implicit function of $\mathbf{k}_x$.

We will now show how to evaluate the matrix/vector product

$$\boldsymbol{\mathcal{P}}_{\rho\leftarrow\rho'}\cdot\boldsymbol{\mathcal{F}}_{\rho'}$$

using the fast Fourier transform (FFT) algorithm. To utilize the FFT algorithm, we must transform the $6N_x$-vector $\boldsymbol{\mathcal{F}}_{\rho'}$ from (3.84) into the $6N_k$-vector $\tilde{\boldsymbol{\mathcal{F}}}_{\rho'}\equiv\boldsymbol{\Phi}[\boldsymbol{\mathcal{F}}_{\rho'}]$ where

$$\boldsymbol{\Phi}\left[\boldsymbol{\mathcal{F}}_{\rho'}\right](N_k(n-1)+m\,N_\Delta) = \begin{cases}\boldsymbol{\mathcal{F}}_{\rho'}\,(6(m-1)+n) & 1\le m\le N_x,\quad 1\le n\le 6\\ 0 & \text{otherwise}\end{cases}\tag{3.92}$$

with

$$N_k=N_\Delta N_L N_x.\tag{3.93}$$

The mapping (3.92) and the corresponding inverse mapping are summarized in the pseudo-code of Alg.3.1.

We observe that (3.92) defines a dual purpose mapping:

- firstly, we re-organize the components of $\boldsymbol{\mathcal{F}}_{\rho'}$ defined in (3.84). We recall that all 6 components of the force vector $\boldsymbol{f}_{N_x(\rho'-1)+\chi}$ for each $1 \leq \chi \leq N_x$ are stored contiguously in the $6N_x$-vector $\boldsymbol{\mathcal{F}}_{\rho'}$. To employ the DFT, we need to store each fixed component, at all gridpoints in a contiguous fashion. This may be explained in another way, $\boldsymbol{\mathcal{F}}_{\rho'}$ is constructed with an outer loop over each of the N_x the lattice points and an inner loop over each of the 6-components, while the order of the inner and outer loop has been swapped in the construction of $\boldsymbol{\Phi}[\boldsymbol{\mathcal{F}}_{\rho'}]$, so that we have an outer loop over the 6-components and an inner loop over the lattice points.

- secondly, the mapping $\boldsymbol{\Phi}[\cdot]$ defines an upsampling, due to the N_Δ factor in (3.92). The reason for the upsampling will be explained presently.

Also the integer relation (3.93) requires further explanation. While the width of the lattice is $x_{N_x} \equiv N_x \, \Delta x$, the corresponding *Nyquist sampling rate* for the k_x wavenumber based on this width, i.e.

$$\Delta k_x = \frac{2\pi}{N_x \, \Delta x},$$

is usually too coarse, resulting in a significant amount aliasing. To get a finer value of Δk_x, we extend the width of the computational grid by a factor N_L (Fig.3.3) so that the actual value of Δk_x is given by

$$\Delta k_x = \frac{2\pi}{N_L \, N_x \, \Delta x}$$

Also, the wavenumber band width $-\Omega \leq k_x \leq \Omega$ is usually very wide, ($\Omega \approx 20\omega/\beta$ in the applications of chapter 6) so that the *Nyquist spatial sampling rate* value of Δx, defined as

$$\Delta x = \frac{\pi}{\Omega}$$

is much finer than the actual lattice spacing Δx in Fig.3.2 and Fig.3.3. Consequently, we have introduced the positive integer N_Δ in (3.93) so that the lattice spacing is a multiple of the corresponding Nyquist spatial sampling rate value, i.e.

$$\Delta x = \frac{N_\Delta \pi}{\Omega}$$

The positive integers N_Δ and N_L can be chosen to improve the accuracy of the plane wave representation (3.90). However, this improvement comes

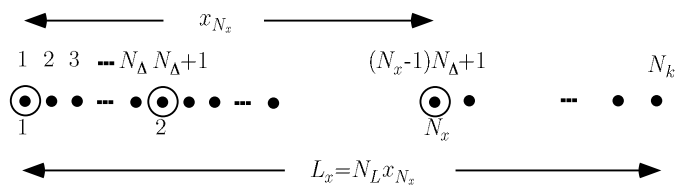

Figure 3.3: The x-lattice superimposed on the computational grid. Circles are the lattice points and bullets are the corresponding grid points at the Nyquist sampling rate. L_x is the fundamental period of the Fourier representation, while x_{N_x} is the width of the lattice

at the cost of employing $N_\Delta \times N_L$ more wavenumbers in (3.90) than are required by *Shannon's sampling theorem*.

The upsampled version of $\boldsymbol{\mathcal{P}}_{\rho\leftarrow\rho'} \cdot \boldsymbol{\mathcal{F}}_{\rho'}$ from (3.90), corresponding the oversampled grid (a single row of that grid is depicted by the "bullets" in Fig.3.3) and operating on the $6N_k-$element upsampled force vector $\boldsymbol{\Phi}\left[\boldsymbol{\mathcal{F}}_{\rho'}\right]$ of (3.92), and may be written as

$$\tilde{\boldsymbol{\mathcal{P}}}_{\rho\leftarrow\rho'} \cdot \boldsymbol{\Phi}\left[\boldsymbol{\mathcal{F}}_{\rho'}\right]$$

where the $6N_k \times 6N_k$ *upsampled Green's tensor* $\tilde{\boldsymbol{\mathcal{P}}}_{\rho\leftarrow\rho'}$ is an augmented version of $6N_x \times 6N_x$ $\boldsymbol{\mathcal{P}}_{\rho\leftarrow\rho'}$ from (3.90) and is defined as

$$\begin{aligned}\tilde{\boldsymbol{\mathcal{P}}}_{\rho\leftarrow\rho'} = & \\ & -i\left(\boldsymbol{I}_6 \otimes \mathbf{DFT}(\mathbf{k}_x)\right)\boldsymbol{D}_z(\mathbf{k}_x,k_y)\boldsymbol{Q}^z_{\rho\leftarrow\rho'}(\mathbf{k}_x)\boldsymbol{D}^T_z(-\mathbf{k}_x,-k_y)\left(\boldsymbol{I}_6 \otimes \mathbf{DFT}^T(-\mathbf{k}_x)\right)\end{aligned} \tag{3.94}$$

In (3.94), the $\mathbf{DFT}(\mathbf{k}_x)$ represents the $N_k \times N_k$ discrete Fourier transform array, with elements

$$[\mathbf{DFT}(\mathbf{k}_x)]_{l,j} = \exp\left(\frac{2\pi i(l-1)(j-1)}{N_k}\right), \qquad 1 \le l,j \le N_k \tag{3.95}$$

while the elements of *Hermitian transpose* $\mathbf{DFT}^T(-\mathbf{k}_x)$ are defined as

$$\left[\mathbf{DFT}^T(-\mathbf{k}_x)\right]_{l,j} = \left[\mathbf{DFT}^H(\mathbf{k}_x)\right]_{l,j} = \exp\left(\frac{-2\pi i(l-1)(j-1)}{N_k}\right), \quad 1 \le l,j \le N_k \tag{3.96}$$

Algorithm 3.1 THE UPSAMPLING $\boldsymbol{\Phi}[\cdot]$ AND DOWNSAMPLING $\boldsymbol{\Phi}^{-1}[\cdot]$ OPERATORS

Complex $\tilde{\boldsymbol{\mathcal{F}}}_\rho(N_k, 6), \boldsymbol{\mathcal{F}}_\rho(6, N_x)$

C0=(0,0)

- Initialize $\tilde{\boldsymbol{\mathcal{F}}}_\rho$

```
For j=1,6
    For k=1,N_k
        F̃_ρ(k,j)=C0
    End !k-loop
End !j-loop
```

- The upsampling mapping $\tilde{\boldsymbol{\mathcal{F}}}_\rho = \boldsymbol{\Phi}[\boldsymbol{\mathcal{F}}_\rho]$

```
For j=1,6
    For k=1,N_x
        F̃_ρ((k-1)N_Δ+1,j)=F_ρ(j,k)
    End !k-loop
End !j-loop
```

- The downsampling mapping $\boldsymbol{\mathcal{F}}_\rho = \boldsymbol{\Phi}^{-1}[\tilde{\boldsymbol{\mathcal{F}}}_\rho]$

```
For j=1,6
    For k=1,N_x
        F_ρ(j,k)= F̃_ρ((k-1)N_Δ+1,j)
    End !k-loop
End !j-loop
```

We recall that the Kronecka product $(\boldsymbol{I}_6 \otimes \mathbf{DFT}(\mathbf{k}_x))$ in (3.94) may be written more explicitly as the $6N_k \times 6N_k$ block diagonal array

$$(\boldsymbol{I}_6 \otimes \mathbf{DFT}(\mathbf{k}_x)) \equiv \\ \operatorname{diag}\left[\ \mathbf{DFT}(\mathbf{k}_x) \quad \mathbf{DFT}(\mathbf{k}_x) \quad \mathbf{DFT}(\mathbf{k}_x) \quad \mathbf{DFT}(\mathbf{k}_x) \quad \mathbf{DFT}(\mathbf{k}_x) \quad \mathbf{DFT}(\mathbf{k}_x)\ \right]$$

To summarize, to implement (3.90) using the DFT algorithm, we:

- *upsample* the $6N_x$-element lattice force vector $\boldsymbol{\mathcal{F}}_{\rho'}$ from (3.84) using the upsampling operator $\boldsymbol{\Phi}[\cdot]$ of (3.92) to get the $6N_k$-element computational vector $\tilde{\boldsymbol{\mathcal{F}}}_{\rho'}$

- we perform the mapping

  $$\tilde{\boldsymbol{\mathcal{P}}}_{\rho\leftarrow\rho'} \cdot \boldsymbol{\Phi}[\boldsymbol{\mathcal{F}}_{\rho'}],$$

  where the $6N_k \times 6N_k$ Green's tensor $\tilde{\boldsymbol{\mathcal{P}}}_{\rho\leftarrow\rho'}$ mapping the lattice force on $\text{row}_{\rho'}$ to the incident displacement/traction field on upsampled row_{ρ} (*cf.* Fig.3.3), was defined in (3.94).

- we apply the inverse operation $\boldsymbol{\Phi}^{-1}[\cdot]$ to the result, i.e.

  $$\boldsymbol{\Phi}^{-1}\left[\tilde{\boldsymbol{\mathcal{P}}}_{\rho\leftarrow\rho'} \cdot \boldsymbol{\Phi}[\boldsymbol{\mathcal{F}}_{\rho'}]\right]$$

  that down-samples or *decimates* the field, to the *circles* in Fig.3.3, while re-ordering the components, so that they are stored with an outer loop over offset, and an inner loop over the $6-$components of displacement an traction.

If we discount the cost of the transformations $\boldsymbol{\Phi}[\cdot]$, $\boldsymbol{\Phi}^{-1}[\cdot]$, then the floating point operation count of $\tilde{\boldsymbol{\mathcal{P}}}_{\rho\leftarrow\rho'} \cdot \boldsymbol{\Phi}[\boldsymbol{\mathcal{F}}_{\rho'}]$ is approximately

$$N_3 \approx 75\, N_k + 12\ N_k^2$$

However, provided $N_k = 2^n$ for some positive integer n, the DFT operations in (3.94) may be performed with the FFT algorithm, in which case $\tilde{\boldsymbol{\mathcal{P}}}_{\rho\leftarrow\rho'} \cdot \boldsymbol{\Phi}[\boldsymbol{\mathcal{F}}_{\rho'}]$ may be implemented in

$$N_4 \approx 75\, N_k + 12\ N_k \log_2 N_k$$

floating point operations.

Next, we consider the mapping

$$\boldsymbol{\mathcal{P}}^z \cdot \boldsymbol{\mathcal{F}} = \begin{bmatrix} \mathbf{0}_{6N_x} & \boldsymbol{\mathcal{P}}_{1\leftarrow 2} & \cdots & \boldsymbol{\mathcal{P}}_{1\leftarrow L} \\ \boldsymbol{\mathcal{P}}_{2\leftarrow 1} & \mathbf{0}_{6N_x} & & \vdots \\ \vdots & & \ddots & \boldsymbol{\mathcal{P}}_{L-1\leftarrow L} \\ \boldsymbol{\mathcal{P}}_{L\leftarrow 1} & \cdots & \boldsymbol{\mathcal{P}}_{L\leftarrow L-1} & \mathbf{0}_{6N_x} \end{bmatrix} \begin{bmatrix} \boldsymbol{\mathcal{F}}_1 \\ \boldsymbol{\mathcal{F}}_2 \\ \vdots \\ \boldsymbol{\mathcal{F}}_L \end{bmatrix} \tag{3.97}$$

where $\boldsymbol{\mathcal{P}}^z$ was introduced in (3.87), the $6N_x \times 6N_x$ partition $\boldsymbol{\mathcal{P}}_{\rho\leftarrow\rho'}$ was defined previously by (3.90) and the $6N_x$ secondary force vector $\boldsymbol{\mathcal{F}}_{\rho'}$ was introduced in (3.83). Once again, we will utilize the DFT (or more particularly, the FFT) algorithm to implement (3.97) in a 3 step process:

- we introduce the upsampled lattice force vector $\tilde{\boldsymbol{\mathcal{F}}}$, defined as

$$\tilde{\boldsymbol{\mathcal{F}}} = \begin{bmatrix} \boldsymbol{\Phi}[\boldsymbol{\mathcal{F}}_1] \\ \boldsymbol{\Phi}[\boldsymbol{\mathcal{F}}_2] \\ \vdots \\ \boldsymbol{\Phi}[\boldsymbol{\mathcal{F}}_L] \end{bmatrix} = \begin{bmatrix} \tilde{\boldsymbol{\mathcal{F}}}_1 \\ \tilde{\boldsymbol{\mathcal{F}}}_2 \\ \vdots \\ \tilde{\boldsymbol{\mathcal{F}}}_L \end{bmatrix}$$

- we replace (3.97) by the upsampled equation

$$\tilde{\boldsymbol{\mathcal{P}}}^z \cdot \tilde{\boldsymbol{\mathcal{F}}} = \begin{bmatrix} \mathbf{0}_{6N_x} & \tilde{\boldsymbol{\mathcal{P}}}_{1\leftarrow 2} & \cdots & \tilde{\boldsymbol{\mathcal{P}}}_{1\leftarrow L} \\ \tilde{\boldsymbol{\mathcal{P}}}_{2\leftarrow 1} & \mathbf{0}_{6N_x} & & \vdots \\ \vdots & & \ddots & \tilde{\boldsymbol{\mathcal{P}}}_{L-1\leftarrow L} \\ \tilde{\boldsymbol{\mathcal{P}}}_{L\leftarrow 1} & \cdots & \tilde{\boldsymbol{\mathcal{P}}}_{L\leftarrow L-1} & \mathbf{0}_{6N_x} \end{bmatrix} \begin{bmatrix} \tilde{\boldsymbol{\mathcal{F}}}_1 \\ \tilde{\boldsymbol{\mathcal{F}}}_2 \\ \vdots \\ \tilde{\boldsymbol{\mathcal{F}}}_L \end{bmatrix} \tag{3.98}$$

  where the $6N_k \times 6N_k$ partition $\tilde{\boldsymbol{\mathcal{P}}}_{\rho\leftarrow\rho'}$ was defined previously by (3.94).

- We then downsample the relevant blocks of $\tilde{\boldsymbol{\mathcal{P}}}^z \cdot \tilde{\boldsymbol{\mathcal{F}}}$ using the down sampling operator $\boldsymbol{\Phi}^{-1}[\cdot]$

However, $\tilde{\boldsymbol{\mathcal{P}}}^z$ in (3.98) may be replaced by the factored equation

$$\begin{aligned}\tilde{\boldsymbol{\mathcal{P}}}^z &= -i\boldsymbol{I}_L \otimes \left[\left(\boldsymbol{I}_6 \otimes \mathbf{DFT}(\mathbf{k}_x)\right) \boldsymbol{D}_z(\mathbf{k}_x, k_y)\right] \boldsymbol{\mathcal{Q}}^z \\ &\quad \times \boldsymbol{I}_L \otimes \left[\boldsymbol{D}_z^T(-\mathbf{k}_x, -k_y) \left(\boldsymbol{I}_6 \otimes \mathbf{DFT}^T(-\mathbf{k}_x)\right)\right] \end{aligned} \tag{3.99}$$

where

$$\boldsymbol{\mathcal{Q}}^z = \begin{bmatrix} \mathbf{0}_{6N_k} & \boldsymbol{Q}^z_{1\leftarrow 2} & \cdots & \boldsymbol{Q}^z_{1\leftarrow L} \\ \boldsymbol{Q}^z_{2\leftarrow 1} & \mathbf{0}_{6N_k} & & \vdots \\ \vdots & & \ddots & \boldsymbol{Q}^z_{L-1\leftarrow L} \\ \boldsymbol{Q}^z_{L\leftarrow 1} & \cdots & \boldsymbol{Q}^z_{L\leftarrow L-1} & \mathbf{0}_{6N_k} \end{bmatrix}. \tag{3.100}$$

We employ (3.91) for the partitions $\boldsymbol{Q}^z_{i\leftarrow j}$ in (3.100) to get

$$\boldsymbol{\mathcal{Q}}^z = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{E} & \mathbf{0} & \mathbf{E}^2 & \mathbf{0} & & \mathbf{E}^{L-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{E} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{E} & \mathbf{0} & \cdots & \mathbf{E}^{L-2} & \mathbf{0} \\ \mathbf{0} & \mathbf{E}^2 & & & & \ddots & & \vdots & \\ \mathbf{0} & \mathbf{0} & & & & & & & \\ \vdots & & & & & & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ & & & & & & \mathbf{0} & \mathbf{0} & \mathbf{E} & \mathbf{0} \\ \mathbf{0} & \mathbf{E}^{L-1} & \mathbf{0} & \mathbf{E}^{L-2} & & & \mathbf{0} & \mathbf{E} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & & \cdots & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \tag{3.101}$$

We can reduce the floating point operation count of $\tilde{\boldsymbol{\mathcal{P}}}^z \cdot \tilde{\boldsymbol{\mathcal{F}}}$, with $\tilde{\boldsymbol{\mathcal{P}}}^z$ given by (3.99), by factorizing the $\boldsymbol{\mathcal{Q}}^z$ factor, defined by (3.101).

We demonstrate the plane wave propagator factorization technique, by factoring $\boldsymbol{\mathcal{Q}}^z$ when the lattice has 3 rows, i.e. $L = 3$, so that (3.101) reduces to

$$\boldsymbol{\mathcal{Q}}^z = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{E} & \mathbf{0} & \mathbf{E}^2 & \mathbf{0} \\ \mathbf{0} & \mathbf{E} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{E} & \mathbf{0} \\ \mathbf{0} & \mathbf{E}^2 & \mathbf{0} & \mathbf{E} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \tag{3.102}$$

which may then be factored as

$$\boldsymbol{\mathcal{Q}}^z = \begin{bmatrix} \boldsymbol{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ -\mathbf{E} & \mathbf{0} & \boldsymbol{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \boldsymbol{I} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & -\mathbf{E} & \mathbf{0} & \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \boldsymbol{I} \end{bmatrix}^{-1} \begin{bmatrix} \boldsymbol{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{I} & \mathbf{0} & -\mathbf{E} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{I} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \boldsymbol{I} & \mathbf{0} & -\mathbf{E} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \boldsymbol{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \boldsymbol{I} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{E} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{E} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{E} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{E} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \tag{3.103}$$

Let

$$\begin{bmatrix} \boldsymbol{\Sigma}_{\uparrow}^{\rho'} \\ \boldsymbol{\Sigma}_{\downarrow}^{\rho'} \end{bmatrix} \equiv -i\, \boldsymbol{D}_z^T(-\mathbf{k}_x, -k_y) \left(\boldsymbol{I}_6 \otimes \mathbf{DFT}^T(-\mathbf{k}_x)\right) \cdot \tilde{\boldsymbol{\mathcal{F}}}_{\rho'} \tag{3.104}$$

denote the $\begin{bmatrix} \text{upgoing} \\ \text{downgoing} \end{bmatrix}$ planewave radiation from $\text{row}_{\rho'}$ for each $1 \le \rho' \le 3$.

Then the action

$$\boldsymbol{\mathcal{Q}}^z \boldsymbol{I}_L \otimes \left[\boldsymbol{D}_z^T(-\mathbf{k}_x, -k_y) \left(\boldsymbol{I}_6 \otimes \mathbf{DFT}^T(-\mathbf{k}_x)\right)\right] \cdot \tilde{\boldsymbol{\mathcal{F}}}$$

may be computed using the factored representation of (3.103), computing the inverses using forward and backward substitution. The result of the calculation may be summarized as:

$$\left.\begin{aligned} \Delta\boldsymbol{V}_{\uparrow}^{\rho'} &= \mathbf{E}\left(\boldsymbol{\Sigma}_{\uparrow}^{\rho'+1} + \Delta\boldsymbol{V}_{\uparrow}^{\rho'+1}\right) \quad \rho' = 1, 2 \\ \Delta\boldsymbol{V}_{\downarrow}^{\rho'} &= \mathbf{E}\left(\boldsymbol{\Sigma}_{\downarrow}^{\rho'-1} + \Delta\boldsymbol{V}_{\downarrow}^{\rho'-1}\right) \quad \rho' = 2, 3 \end{aligned}\right\} \tag{3.105}$$

where $\Delta\boldsymbol{V}_{\uparrow}^{\rho'}(\Delta\boldsymbol{V}_{\downarrow}^{\rho'})$ is the upward(downward) incident field on $\text{row}_{\rho'}$ of the lattice. It is not too difficult to generalize (3.105) to a lattice with L rows, in which case we get

$$\left.\begin{aligned} \Delta\boldsymbol{V}_{\uparrow}^{\rho'} &= \mathbf{E}\left(\boldsymbol{\Sigma}_{\uparrow}^{\rho'+1} + \Delta\boldsymbol{V}_{\uparrow}^{\rho'+1}\right) \quad \rho' = 1, L-1 \\ \Delta\boldsymbol{V}_{\downarrow}^{\rho'} &= \mathbf{E}\left(\boldsymbol{\Sigma}_{\downarrow}^{\rho'-1} + \Delta\boldsymbol{V}_{\downarrow}^{\rho'-1}\right) \quad \rho' = 2, L \end{aligned}\right\} \tag{3.106}$$

Given that

$$\begin{bmatrix} \Delta\boldsymbol{V}_{\downarrow}^{\rho} \\ \Delta\boldsymbol{V}_{\uparrow}^{\rho} \end{bmatrix}$$

denotes the $\begin{bmatrix} \text{downgoing} \\ \text{upgoing} \end{bmatrix}$ incident plane wave radiation on row_{ρ} of the lattice,

we synthesize the $6N_k$ displacement/traction vector using

$$\begin{bmatrix} \tilde{\boldsymbol{u}}_\rho \\ \tilde{\boldsymbol{t}}_\rho \end{bmatrix} = (\boldsymbol{I}_6 \otimes \mathbf{DFT}(\mathbf{k}_x))\, \boldsymbol{D}_z(\mathbf{k}_x, k_y) \begin{bmatrix} \Delta \boldsymbol{V}^\rho_\downarrow \\ \Delta \boldsymbol{V}^\rho_\uparrow \end{bmatrix} \tag{3.107}$$

operation. Finally, we downsample or decimate this result, to get

$$\begin{bmatrix} \boldsymbol{u}_\rho \\ \boldsymbol{t}_\rho \end{bmatrix} = \boldsymbol{\Phi}^{-1} \begin{bmatrix} \tilde{\boldsymbol{u}}_\rho \\ \tilde{\boldsymbol{t}}_\rho \end{bmatrix},$$

the 6 component displacement/traction vector at each of the N_x lattice points in row_ρ of the lattice.

In summary, if we implement the $6N_xL$ matrix/vector mapping $\boldsymbol{\mathcal{P}}^z \cdot \boldsymbol{\mathcal{F}}$ of (3.97) using (3.104)–(3.107), the floating point operation count is asymptotically

$$N_5 = 2\,(L-1)\,[39N_k + 6N_k \log_2 N_k] \tag{3.108}$$

This algorithm is an elastic wave generalization of the acoustic wave (ω, k) *phase shift method* of *Gazdag* (1981). However, our algorithm has been set up with the intention of extending it to a plane stratified background model.

3.3.2 Implementing $\boldsymbol{\mathcal{P}}^x \cdot \boldsymbol{\mathcal{F}}$

We now find a representation of the $6N_x \times 6N_x$ partition $\boldsymbol{\mathcal{P}}_{\rho\leftarrow\rho}$ of array $\boldsymbol{\mathcal{P}}^x$ in (3.89) using (3.79) for the 6×6 component partitions.

From (3.86) with $\rho = \rho'$, and using (3.82) for the definition of the partitions we have

$$\boldsymbol{\mathcal{P}}_{\rho\leftarrow\rho} = \begin{bmatrix} \mathbf{0}_6 & \mathring{\boldsymbol{P}}^x(x_1, x_2) & \cdots & \mathring{\boldsymbol{P}}^x(x_1, x_{N_x}) \\ \mathring{\boldsymbol{P}}^x(x_2, x_1) & \mathbf{0}_6 & & \vdots \\ \vdots & & \ddots & \mathring{\boldsymbol{P}}^x(x_{N_x-1}, x_{N_x}) \\ \mathring{\boldsymbol{P}}^x(x_{N_x}, x_1) & \cdots & \mathring{\boldsymbol{P}}^x(x_{N_x}, x_{N_x-1}) & \mathbf{0}_6 \end{bmatrix} \tag{3.109}$$

By employing the propagator of (3.79) for the partitions of (3.109), we get

$$\mathring{\boldsymbol{\mathcal{P}}}_{\rho\leftarrow\rho}\equiv \tag{3.110}$$
$$-i\boldsymbol{I}_{N_x}\otimes\left[\left(\boldsymbol{I}_6\otimes\overline{\boldsymbol{F}}(0)\right)\boldsymbol{D}_x(\mathbf{k}_z,k_y)\right]\boldsymbol{\mathcal{Q}}^x_{\rho\leftarrow\rho}\boldsymbol{I}_{N_x}\otimes\left[\boldsymbol{D}_x^T(-\mathbf{k}_z,-k_y)\left(\boldsymbol{I}_6\otimes\overline{\boldsymbol{F}}^T(0)\right)\right]$$

where:

- the $(6N_xN_k)\times(6N_xN_k)$, *transverse* plane wave propagator $\boldsymbol{\mathcal{Q}}^x_{\rho\leftarrow\rho}$ is defined as

$$\boldsymbol{\mathcal{Q}}^x_{\rho\leftarrow\rho}=\begin{bmatrix}\mathbf{0}_{6N_k} & \boldsymbol{Q}^x(x_1,x_2) & \cdots & \boldsymbol{Q}^x(x_1,x_{N_x})\\ \boldsymbol{Q}^x(x_2,x_1) & \mathbf{0}_{6N_k} & & \vdots\\ \vdots & & \ddots & \boldsymbol{Q}^x(x_{N_x-1},x_{N_x})\\ \boldsymbol{Q}^x(x_{N_x},x_1) & \cdots & \boldsymbol{Q}^x(x_{N_x},x_{N_x-1}) & \mathbf{0}_{6N_k}\end{bmatrix} \tag{3.111}$$

  where the $6N_k\times6N_k$ plane wave propagator partition, $\boldsymbol{Q}^x(x_\chi,x_{\chi'})$, $1\le\chi,\chi'\le N_x$ in (3.111) is given by

$$\boldsymbol{Q}^x(x_\chi,x_{\chi'})=\begin{bmatrix}\mathbf{0}_{3N_k} & H(\chi-\chi')\mathbf{E}_x^{|\chi-\chi'|}\\ H(\chi'-\chi)\mathbf{E}_x^{|\chi-\chi'|} & \mathbf{0}_{3N_k}\end{bmatrix} \tag{3.112}$$

  with the $3N_k$-diagonal phase shift array $\mathbf{E}_x$ defined by

$$\mathbf{E}_x=\begin{bmatrix}\mathrm{e}^{i\mathbf{k}_{x,P}\,\Delta x} & \mathbf{0}_N & \mathbf{0}_N\\ \mathbf{0}_N & \mathrm{e}^{i\mathbf{k}_{x,S}\,\Delta x} & \mathbf{0}_N\\ \mathbf{0}_N & \mathbf{0}_N & \mathrm{e}^{i\mathbf{k}_{x,H}\,\Delta x}\end{bmatrix} \tag{3.113}$$

  and $\mathbf{k}_{x,c},\ c\in\{P,S,H\}$ was given by (3.71), so that

$$\boldsymbol{\mathcal{Q}}^x_{\rho\leftarrow\rho}=\begin{bmatrix}\mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{E}_x & \mathbf{0} & \mathbf{E}_x^2 & \mathbf{0} & & \mathbf{E}_x^{N-1} & \mathbf{0}\\ \mathbf{0} & \mathbf{E}_x & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{E}_x & \mathbf{0} & \cdots & \mathbf{E}_x^{N-2} & \mathbf{0}\\ \mathbf{0} & \mathbf{E}_x^2 & & & & \ddots & & \vdots & \\ \mathbf{0} & \mathbf{0} & & & & & & & \\ & \vdots & & & & & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0}\\ & & & & & & \mathbf{0} & \mathbf{0} & \mathbf{E}_x & \mathbf{0}\\ \mathbf{0} & \mathbf{E}_x^{N-1} & \mathbf{0} & \mathbf{E}_x^{N-2} & & \cdots & \mathbf{0} & \mathbf{E}_x & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & & & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0}\end{bmatrix} \tag{3.114}$$

- the $6N_k \times 6N_k$ array $\boldsymbol{D}_x(\mathbf{k}_z, k_y)$ was defined in (3.74)
- the $6N_k$-vector $\overline{\boldsymbol{F}}(0)$ was defined in (3.72)

With $\overset{\circ}{\boldsymbol{\mathcal{P}}}_{\rho\leftarrow\rho}$ given by (3.110), and $\boldsymbol{\mathcal{F}}_\rho$ given by (3.84), we can reduce the cost of the $6N_x$ matrix/vector products

$$\overset{\circ}{\boldsymbol{\mathcal{P}}}_{\rho\leftarrow\rho} \cdot \boldsymbol{\mathcal{F}}_\rho, \qquad 1 \le \rho \le L$$

by factoring the plane wave propagator $\boldsymbol{\mathcal{Q}}^x_{\rho\leftarrow\rho}$ component of $\overset{\circ}{\boldsymbol{\mathcal{P}}}_{\rho\leftarrow\rho}$ that was defined in (3.114).

Let us denote the left(right) going plane waves from a lattice point in column$_\chi$ of row$_\rho$ by $\boldsymbol{\Sigma}^{(\rho,\chi)}_{\leftarrow}(\boldsymbol{\Sigma}^{(\rho,\chi)}_{\rightarrow})$ for $1 \le \rho \le L$, $1 \le \chi \le N_x$, then

$$\begin{aligned} \begin{bmatrix} \boldsymbol{\Sigma}^{(\rho,\chi)}_{\leftarrow} \\ \boldsymbol{\Sigma}^{(\rho,\chi)}_{\rightarrow} \end{bmatrix} &= \boldsymbol{D}_x^T(-\mathbf{k}_z, -k_y) \left(\boldsymbol{I}_6 \otimes \overline{\boldsymbol{F}}^T(0)\right) \boldsymbol{J}_6 \bar{\boldsymbol{f}}_{N_x(\rho-1)+\chi} \\ &= \boldsymbol{D}_x^T(-\mathbf{k}_z, -k_y) \left(\boldsymbol{I}_6 \otimes \overline{\boldsymbol{F}}^T(0)\right) \boldsymbol{f}_{N_x(\rho-1)+\chi} \end{aligned} \tag{3.115}$$

By analogy with (3.106) we have

$$\left.\begin{aligned} \Delta\boldsymbol{V}^{(\rho,\chi)}_{\leftarrow} &= \mathbf{E}_x \left(\boldsymbol{\Sigma}^{(\rho,\chi+1)}_{\leftarrow} + \Delta\boldsymbol{V}^{(\rho,\chi+1)}_{\leftarrow}\right) & \chi = 1, N_x - 1 \\ \Delta\boldsymbol{V}^{(\rho,\chi)}_{\rightarrow} &= \mathbf{E}_x \left(\boldsymbol{\Sigma}^{(\rho,\chi-1)}_{\rightarrow} + \Delta\boldsymbol{V}^{(\rho,\chi-1)}_{\rightarrow}\right) & \chi = 2, N_x \end{aligned}\right\} \quad \rho = 1, L \tag{3.116}$$

where $\Delta\boldsymbol{V}^{(\rho,\chi)}_{\leftarrow}(\Delta\boldsymbol{V}^{(\rho,\chi)}_{\rightarrow})$ is the leftward(rightward) incident field on (row$_\rho$, column$_\chi$) of the lattice. Finally, we synthesize the components of displacement and traction at (row$_\rho$, column$_\chi$) from the incident plane waves, using

$$\begin{bmatrix} \boldsymbol{u}^{(\rho,\chi)} \\ \boldsymbol{t}^{(\rho,\chi)} \end{bmatrix} = \left(\boldsymbol{I}_6 \otimes \overline{\boldsymbol{F}}(0)\right) \boldsymbol{D}_x(\mathbf{k}_z, k_y) \begin{bmatrix} \Delta\boldsymbol{V}^{(\rho,\chi)}_{\rightarrow} \\ \Delta\boldsymbol{V}^{(\rho,\chi)}_{\leftarrow} \end{bmatrix}, \quad \rho = 1, L, \quad \chi = 1, N_x \tag{3.117}$$

The computations (3.115)–(3.117) may be summarized by the matrix formula

$$\begin{bmatrix} \boldsymbol{u} \\ \boldsymbol{t} \end{bmatrix} = \overset{\circ}{\boldsymbol{\mathcal{P}}}{}^x \cdot \boldsymbol{\mathcal{F}}$$

with

$$\begin{aligned} \overset{\circ}{\boldsymbol{\mathcal{P}}}{}^x \equiv -i\boldsymbol{I}_L \otimes (\\ \boldsymbol{I}_{N_x} \otimes \left[\left(\boldsymbol{I}_6 \otimes \overline{\boldsymbol{F}}(0)\right) \boldsymbol{D}_x(\mathbf{k}_z, k_y)\right] \boldsymbol{\mathcal{Q}}^x_{\rho\leftarrow\rho} \boldsymbol{I}_{N_x} \otimes \left[\boldsymbol{D}_x^T(-\mathbf{k}_z, -k_y) \left(\boldsymbol{I}_6 \otimes \overline{\boldsymbol{F}}^T(0)\right)\right]) \end{aligned} \tag{3.118}$$

an explicit representation of the array $\boldsymbol{\mathcal{P}}^x$ introduced in (3.89), that follows after using (3.110) for the diagonal blocks $\boldsymbol{\mathcal{P}}^x_{\rho\leftarrow\rho}$, $1 \le \rho \le L$. The asmyptotic

floating point operation count of eqns. (3.115)–(3.117) is

$$N_6 = 78\, L\, (N_x - 1)\overline{N}_k \tag{3.119}$$

3.3.3 Implementing $\boldsymbol{\mathcal{P}} \cdot \boldsymbol{\mathcal{F}}$

Finally, in the culmination of the chapter, we use (3.99) and (3.118) in (3.87) to get the representation of the *lattice propagator* $\overset{\circ}{\boldsymbol{\mathcal{P}}}$, which may be written as

$$\begin{aligned}
\overset{\circ}{\boldsymbol{\mathcal{P}}} \;\equiv\; & \overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{z}[\cdot] + \overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{x} \\
= & -i\boldsymbol{\Phi}^{-1}\left[\boldsymbol{I}_L \otimes \left[\left(\boldsymbol{I}_6 \otimes \mathbf{DFT}(\mathbf{k}_x)\right)\boldsymbol{D}_z(\mathbf{k}_x, k_y)\right]\boldsymbol{\mathcal{Q}}^z\right. \\
& \qquad\qquad \left.\times\, \boldsymbol{I}_L \otimes \left[\boldsymbol{D}_z^T(-\mathbf{k}_x, -k_y)\left(\boldsymbol{I}_6 \otimes \mathbf{DFT}^T(-\mathbf{k}_x)\right)\boldsymbol{\Phi}[\cdot]\right]\right] \\
& -i\boldsymbol{I}_L \otimes \left\{\boldsymbol{I}_{N_x} \otimes \left[\left(\boldsymbol{I}_6 \otimes \overline{\boldsymbol{F}}(0)\right)\boldsymbol{D}_x(\mathbf{k}_z, k_y)\right]\boldsymbol{\mathcal{Q}}^x_{\rho\leftarrow\rho}\right. \\
& \qquad\qquad \left.\times \boldsymbol{I}_{N_x} \otimes \left[\boldsymbol{D}_x^T(-\mathbf{k}_z, -k_y)\left(\boldsymbol{I}_6 \otimes \overline{\boldsymbol{F}}^T(0)\right)\right]\right\}
\end{aligned} \tag{3.120}$$

(3.120)

Note that we have incorporated to upsampling and downsampling operators within (3.3.3), so that $\overset{\circ}{\boldsymbol{\mathcal{P}}}$ operates on $6LN_x$-vectors.

In (3.116), we have a left/right sweep in each of the L rows of the lattice, where as (3.106) is composed of a single up/down sweep. An accurate measure of the number of floating point operations embodied in (3.106)[(3.116)] may be ascertained by counting the number of applications of the vertical[horizontal] phase shift arrays $\mathbf{E}[\mathbf{E}_x]$. For our $L-$row, N_x-column lattice, we have $2(L-1)$ applications of $\mathbf{E}$ and $2L(N_x-1)$ applications of $\mathbf{E}_x$, where $\mathbf{E}$ is a $3N_k$-element diagonal array and $\mathbf{E}_x$ is a $3\overline{N}$-element diagonal array, with $\overline{N}_k \leq N_k$.

We generally use less plane waves in the transverse phase shifts of $\mathbf{E}_x$ than the vertical phase shifts of $\mathbf{E}$ because all points in a given row have the same depth coordinate z and consequently, there is no possibility of *aliasing* caused by undersampling of the complex exponential $\exp(ik_z\Delta z)$ because $\Delta z \equiv 0$ when phase shifting between points within the same row. However, when we compute the vertical phase shifts, we must ensure that we have employed an adequate k_x-wavenumber bandwidth 2Ω and sampling rate Δk_x so that the DFT arrays $\mathbf{DFT}$ in (3.3.3) correctly model the transverse phase shifts between lattice points at *opposite ends of adjacent rows in the*

lattice. At the Nyquist sampling rate, the number of lattice points within a given row N_x is equal to the number of plane waves N_k used to phase shift between rows. However, in numerical applications of the techniques described here, we have found it necessary to employ a considerable amount of *oversampling*, so that

$$N_k = N_\Delta N_L N_x.$$

which was defined above in (3.93). In the numerical computations of chapter 6, $N_\Delta = 8$, $N_L = 4$, so that

$$N_k = 32 N_x$$

i.e., 32 times the number of plane waves that would be necessary from Shannon's sampling theorem (Jerri, 1977; Brigham, 1988). We compute the relevant quantities at the actual grid points by *decimating* or *downsampling* the results at the fictitious oversampled lattice points (Alg. 3.1). Usually, we have taken $\overline{N}_k = N_k/2$ or $\overline{N_k} = N_k/4$, where $N_k = 2048$ or $N_k = 4096$ is the number of oversampled plane waves employed in the vertical phase shifts between rows and $\overline{N}_k$ is the corresponding number of plane waves in the transverse shifts between lattice points within the same row.

The advantages of the representation (3.3.3) are twofold:

1. the action of (3.3.3) on a secondary source field $\boldsymbol{\mathcal{F}}$ with 6-vector components $\boldsymbol{f}_j$ at each of the $N_x \times L$ lattice points of Fig3.2, may be computed in the order of

$$N_2 \approx 2\,(L-1)\,[39 N_k + 6 N_k \log_2 N_k] + 78\,L\,(N_x - 1)\overline{N}_k$$

floating point operations, which is often far less arithmetic than the ostensible

$$N_1 \approx 36 N_x^2 \times L^2 - 6 N_x \times L$$

floating point operation count. However, the oversampling requirement (3.93), erodes the numerical advantage of N_2 over N_1.

2. (3.3.3) is readily extensible to the important case where the lattice is superimposed on a plane stratified half space. We consider this problem in chapter 5.

In summary, the key steps in achieving the goal expressed by item-(**1**) above is to:

1. factor the plane wave propagators $\boldsymbol{\mathcal{Q}}^z$ and $\boldsymbol{\mathcal{Q}}^x$ in (3.3.3) into the product of *sparse* matrices

2. employ the *Fast Fourier Transform*(FFT) algorithm to compute the action of the *Discrete Fourier Transform*(DFT) matrix **DFT** from (3.95) on a field vector. The FFT algorithm factors **DFT** into a product of sparse arrays so that the cost of applying **DFT** to an N_k-vector is reduced from $O(N_k^2)$ to $O(N_k \log_2 N_k)$ floating point operations.

In closing this chapter, we note that although we have assumed that the secondary source vector $\boldsymbol{\mathcal{F}}$ in (3.80), has $6-$vector components $\boldsymbol{f}_j,\ 1 \leq j \leq LN_x$, it will be shown in chapter 4, that $\boldsymbol{f}_j,\ 1 \leq j \leq LN_x$ is most conveniently expressed as an $8-$vector. The reason we have employed $6-$vectors here is related to the fact that the partial wave displacement/traction states used to construct the Green's tensor have 6 components, i.e. 3 components of displacement (u_z, u_x, u_y), and 3 components of traction $(\tau_{zz}, \tau_{xz}, \tau_{yz})$. In chapter 4, we employ integration by parts, so that the partial wave components of the Green's tensor that propagates the field between successive scattering interactions, are augmented by the $(\partial_x u_x, \partial_x u_y)$ components. Consequently, the *effective* secondary source vector $\hat{\boldsymbol{\mathcal{F}}}$ has $8-$vector components, say, $\hat{\boldsymbol{f}}_j,\ 1 \leq j \leq LN_x$. Some of the results presented here will then require minor modification, and these modifications will be outlined in chapter 4.

Chapter 4

Representations of the coupling matrix

In this chapter we recommence the practice of writing the material properties as the sum of a reference and residual component, i.e. $(\rho, \alpha, \beta) = (\mathring{\rho} + \Delta\rho, \mathring{\alpha} + \Delta\alpha, \mathring{\beta} + \Delta\beta)$ where the uniform host material has the reference properties $(\mathring{\rho}, \mathring{\alpha}, \mathring{\beta})$.

4.1 Overview

In chapter 3, we developed various plane wave representations of the displacement/ traction elastic wave Green's tensor $\mathring{\boldsymbol{G}}$, appropriate for a lattice of source points. We were purposely vague about the nature of the source $\boldsymbol{f}_j$ at each lattice point $1 \leq j \leq LN_x$ other than to suppose that it was a $6-$vector. In this chapter we will see that some of the components of $\boldsymbol{f}_j$ have ∂_x operators. We always transfer these partial differentiations to the source and receiver coordinates of the relevant Green's tensors, using integration by parts. Consequently, it is convenient to augment the definitions of the Green's tensors to carry these partially differentiated components as well as the usual displacement and traction fields. The Green's tensors are then expressed through 8×8 matrices, rather than the 6×6 arrays that we have employed in chapter 3. The advantage of this representation is that the scattering induced by heterogeneity superimposed on the reference medium may be modelled as a *multiplicative operator* rather than the differential operator (2.15). This is the essence of the *pseudo-spectral approximation* (Orszag, 1972), where a *spectral convolution* is replaced by a *spatial multiplication*.

We recall that the convolution of two $N-$sequences, is a $2N-$sequence. The "approximation" in the term "pseudo-spectral approximation" derives from the neglect of the aliasing terms that result from sampling the $2N-$spectral convolution (of two $N-$spectral sequences) as if it was an $N-$sequence– the approximation that is commonly employed in what has become known as the pseudo-spectral method [e.g. *Fornberg* (1987)]. In our method, we are oversampling by a factor ≥ 2 (usually 16, or 32). So that the aliasing contribution we have just mentioned will not occur.

In Corollary 1, of §2.4.3 we found the singular integro/differential *Lippmann-Schwinger type operator equation*, reproduced here as

$$\begin{aligned}\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y) &= \mathring{\boldsymbol{G}}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\\ &\quad+\int_\Omega \mathring{\boldsymbol{G}}(\boldsymbol{x};\boldsymbol{x}';k_y)\Delta\boldsymbol{A}(\boldsymbol{x}',\partial_{x'};k_y)\boldsymbol{G}(\boldsymbol{x}',\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x}', \quad \boldsymbol{x}\in\mathbb{R}^2\end{aligned} \tag{4.1}$$

for the unknown Green's tensor $\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)$, defined in terms of the *known reference medium Green's tensor* $\mathring{\boldsymbol{G}}$ where the *differential coupling operator* $\Delta\boldsymbol{A}$ is defined as

$$\Delta\boldsymbol{A}(\boldsymbol{x};\partial_x,k_y) = \tag{2.15}$$
$$\begin{bmatrix} 0 & -\Delta\gamma\partial_x & -i\Delta\gamma k_y & \Delta a & 0 & 0\\ 0 & 0 & 0 & 0 & \Delta b & 0\\ 0 & 0 & 0 & 0 & 0 & \Delta b\\ -\Delta\rho\omega^2 & 0 & 0 & 0 & 0 & 0\\ 0 & -\Delta\rho\omega^2-\partial_x\Delta\zeta\partial_x+\Delta\mu k_y^2 & -ik_y\partial_x\Delta\chi-ik_y\Delta\mu\partial_x & -\partial_x\Delta\gamma & 0 & 0\\ 0 & -ik_y\Delta\chi\partial_x-ik_y\partial_x\Delta\mu & -\Delta\rho\omega^2+\Delta\zeta k_y^2-\partial_x\Delta\mu\partial_x & -ik_y\Delta\gamma & 0 & 0 \end{bmatrix}$$

with the material contrast parameters in (2.15) defined in Table 2.2. The integral $\int_\Omega\cdot$ in Corollary 1 was defined over the spatial support Ω of the inclusion.

In this chapter, we will replace the singular integro/differential *Lippmann-Schwinger type operator equation* of chapter 2 by a nonsingular, compact integral operator equation, that is much simpler to discretize and solve. To effect this transformation we employ a *Galerkin* type projection of the range space of (4.1) onto the span of

$$\int_\Omega \mathrm{d}\boldsymbol{x}\,\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\Delta\boldsymbol{A}(\boldsymbol{x},\partial_x;k_y)\cdot$$

so that (4.1) is replaced by

$$\begin{aligned}
&\int_\Omega \mathrm{d}\boldsymbol{x}\, \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\boldsymbol{\Delta A}(\boldsymbol{x},\partial_x;k_y)\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y) = \\
&\quad \int_\Omega \mathrm{d}\boldsymbol{x}\, \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\boldsymbol{\Delta A}(\boldsymbol{x},\partial_x;k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x};\boldsymbol{x}_S;k_y) \\
&\quad + \int_\Omega \mathrm{d}\boldsymbol{x}\, \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\boldsymbol{\Delta A}(\boldsymbol{x},\partial_x;k_y) \int_\Omega \mathrm{d}\boldsymbol{x}'\, \mathring{\boldsymbol{G}}(\boldsymbol{x};\boldsymbol{x}';k_y)\boldsymbol{\Delta A}(\boldsymbol{x}',\partial_{x'};k_y)\boldsymbol{G}(\boldsymbol{x}',\boldsymbol{x}_S;k_y).
\end{aligned} \tag{4.2}$$

Let us denote the reference $\mathring{\boldsymbol{G}}$ and actual $\boldsymbol{G}$ displacement/traction Green's tensors in terms of their 3×3 displacement and traction partitions as

$$\mathring{\boldsymbol{G}} = \begin{bmatrix} \mathring{\boldsymbol{P}}_{uu} & \mathring{\boldsymbol{P}}_{ut} \\ \mathring{\boldsymbol{P}}_{tu} & \mathring{\boldsymbol{P}}_{tt} \end{bmatrix} \boldsymbol{J}_6, \quad \text{and} \quad \boldsymbol{G} = \begin{bmatrix} \boldsymbol{P}_{uu} & \boldsymbol{P}_{ut} \\ \boldsymbol{P}_{tu} & \boldsymbol{P}_{tt} \end{bmatrix} \boldsymbol{J}_6$$

respectively. Next we employ integration by parts in (4.2) to transfer the ∂_x operators in $\boldsymbol{\Delta A}$ onto the field coordinate of the unknown Green's tensor $\boldsymbol{G}$ to give the augmented operator $\boldsymbol{G}_r$, and onto both the source and field coordinate of the reference Green's tensor $\mathring{\boldsymbol{G}}$, to give $\mathring{\boldsymbol{P}}_{r,s}$ where the augmented operators may be written as:

$$\boldsymbol{G}_r = \begin{bmatrix} \boldsymbol{P}_{uu} & \boldsymbol{P}_{ut} \\ \boldsymbol{P}_{tu} & \boldsymbol{P}_{tt} \\ \boldsymbol{P}_{du} & \boldsymbol{P}_{dt} \end{bmatrix} \boldsymbol{J}_6, \quad \mathring{\boldsymbol{P}}_s = \begin{bmatrix} \mathring{\boldsymbol{P}}_{uu} & \mathring{\boldsymbol{P}}_{ut} & \mathring{\boldsymbol{P}}_{ud} \\ \mathring{\boldsymbol{P}}_{tu} & \mathring{\boldsymbol{P}}_{tt} & \mathring{\boldsymbol{P}}_{td} \end{bmatrix}, \quad \mathring{\boldsymbol{P}}_{r,s} = \begin{bmatrix} \mathring{\boldsymbol{P}}_{uu} & \mathring{\boldsymbol{P}}_{ut} & \mathring{\boldsymbol{P}}_{ud} \\ \mathring{\boldsymbol{P}}_{tu} & \mathring{\boldsymbol{P}}_{tt} & \mathring{\boldsymbol{P}}_{td} \\ \mathring{\boldsymbol{P}}_{du} & \mathring{\boldsymbol{P}}_{dt} & \mathring{\boldsymbol{P}}_{dd} \end{bmatrix}$$

These integration by parts operations may be summarized in the somewhat stylized notation

$$\int_\Omega \mathrm{d}\boldsymbol{x}\, \mathring{\boldsymbol{P}}_s(\boldsymbol{x}_R;\boldsymbol{x};k_y)\boldsymbol{\Delta C}(\boldsymbol{x};k_y)\boldsymbol{G}_r(\boldsymbol{x};\boldsymbol{x}_S;k_y) =$$
$$\int_\Omega \mathrm{d}\boldsymbol{x}\, \mathring{\boldsymbol{P}}_s(\boldsymbol{x}_R;\boldsymbol{x};k_y)\boldsymbol{\Delta C}(\boldsymbol{x};k_y)\mathring{\boldsymbol{G}}_r(\boldsymbol{x};\boldsymbol{x}_S;k_y) \tag{4.3}$$
$$+\int_\Omega \mathrm{d}\boldsymbol{x}\, \mathring{\boldsymbol{P}}_s(\boldsymbol{x}_R;\boldsymbol{x};k_y)\boldsymbol{\Delta C}(\boldsymbol{x};k_y)\int_\Omega \mathrm{d}\boldsymbol{x}'\, \mathring{\boldsymbol{P}}_{r,s}(\boldsymbol{x};\boldsymbol{x}';k_y)\boldsymbol{\Delta C}(\boldsymbol{x}';k_y)\boldsymbol{G}_r(\boldsymbol{x}',\boldsymbol{x}_S;k_y).$$

where $\boldsymbol{\Delta C}$ is an 8×8 multiplicative operator defined as

$$\boldsymbol{\Delta C}(\boldsymbol{x};k_y) =$$
$$\begin{bmatrix} -\Delta\rho\omega^2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \Delta\mu k_y^2-\Delta\rho\omega^2 & 0 & 0 & 0 & 0 & 0 & -ik_y\Delta\mu \\ 0 & 0 & \Delta\zeta k_y^2-\Delta\rho\omega^2 & -ik_y\Delta\gamma & 0 & 0 & -ik_y\chi & 0 \\ 0 & 0 & i\Delta\gamma k_y & -\Delta a & 0 & 0 & \Delta\gamma & 0 \\ 0 & 0 & 0 & 0 & -\Delta b & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\Delta b & 0 & 0 \\ 0 & 0 & ik_y\Delta\chi & \Delta\gamma & 0 & 0 & \Delta\zeta & 0 \\ 0 & ik_y\Delta\mu & 0 & 0 & 0 & 0 & 0 & \Delta\mu \end{bmatrix}$$

However, (4.3) is equivalent to

$$\int_\Omega \mathrm{d}\boldsymbol{x}\, \mathring{\boldsymbol{P}}_s(\boldsymbol{x}_R;\boldsymbol{x};k_y)\boldsymbol{\Delta C}(\boldsymbol{x};k_y)\Big[\boldsymbol{G}_r(\boldsymbol{x};\boldsymbol{x}_S;k_y)-\mathring{\boldsymbol{G}}_r(\boldsymbol{x};\boldsymbol{x}_S;k_y)$$
$$-\int_\Omega \mathrm{d}\boldsymbol{x}'\, \mathring{\boldsymbol{P}}_{r,s}(\boldsymbol{x};\boldsymbol{x}';k_y)\boldsymbol{\Delta C}(\boldsymbol{x}';k_y)\boldsymbol{G}_r(\boldsymbol{x}',\boldsymbol{x}_S;k_y)\Big] = \boldsymbol{0}.$$

which is the *weak* statement of the *orthogonality* of the residual

$$\boldsymbol{R} \equiv \boldsymbol{G}_r(\boldsymbol{x};\boldsymbol{x}_S;k_y)-\mathring{\boldsymbol{G}}_r(\boldsymbol{x};\boldsymbol{x}_S;k_y)-\int_\Omega \mathrm{d}\boldsymbol{x}'\, \mathring{\boldsymbol{P}}_{r,s}(\boldsymbol{x};\boldsymbol{x}';k_y)\boldsymbol{\Delta C}(\boldsymbol{x}';k_y)\boldsymbol{G}_r(\boldsymbol{x}',\boldsymbol{x}_S;k_y)$$

to the *test* space spanned by

$$\int_\Omega \mathrm{d}\boldsymbol{x}\, \mathring{\boldsymbol{P}}_s(\boldsymbol{x}_R;\boldsymbol{x};k_y)\boldsymbol{\Delta C}(\boldsymbol{x};k_y)\,[\cdot]\,, \qquad \text{for arbitrary} \quad \boldsymbol{x}_R$$

In this chapter, we will solve for $\boldsymbol{G}_r$ from the *augmented equation*

$$\boldsymbol{G}_r(\boldsymbol{x};\boldsymbol{x}_S;k_y) = \mathring{\boldsymbol{G}}_r(\boldsymbol{x};\boldsymbol{x}_S;k_y)+\int_\Omega \mathrm{d}\boldsymbol{x}'\, \mathring{\boldsymbol{P}}_{r,s}(\boldsymbol{x};\boldsymbol{x}';k_y)\boldsymbol{\Delta C}(\boldsymbol{x}';k_y)\boldsymbol{G}_r(\boldsymbol{x}',\boldsymbol{x}_S;k_y) \tag{4.4}$$

instead of $\boldsymbol{G}$ from (4.2). The advantage of (4.4), is that it is an integral equation, while the original formulation (4.2) was an integro/differential equation.

However, (4.4) is *strongly singular* because

$$\mathring{\boldsymbol{P}}_{r,s}(\boldsymbol{x};\boldsymbol{x}';k_y) \approx O\left(\|\boldsymbol{x}-\boldsymbol{x}'\|^{-2}\right), \quad \text{in the limit} \quad \|\boldsymbol{x}-\boldsymbol{x}'\| \to 0$$

We can *subtract* the singular part of (4.4) and invert it analytically. The resulting *renormalized* equation

$$\begin{aligned}\boldsymbol{G}_r(\boldsymbol{x};\boldsymbol{x}_S;k_y) &= \mathring{\boldsymbol{G}}_r(\boldsymbol{x};\boldsymbol{x}_S;k_y) \\ &+ \int_\Omega \mathrm{d}\boldsymbol{x}'\, \mathring{\boldsymbol{P}}_{r,s}(\boldsymbol{x};\boldsymbol{x}';k_y)\Delta\boldsymbol{C}_{eff}(\boldsymbol{x}';k_y)\boldsymbol{G}_r(\boldsymbol{x}',\boldsymbol{x}_S;k_y), \quad \boldsymbol{x}\neq\boldsymbol{x}' \quad (4.5)\end{aligned}$$

is both *nonsingular* and *compact*, where we have introduced the *effective coupling matrix* $\Delta\boldsymbol{C}_{eff}$ defined later in (4.69).

4.2 The two centred, harmonic expansion of the reference Green's tensor

In this section, we derive a closed form expression for the reference medium Green's tensor $\mathring{\boldsymbol{G}}$, in a *basis of cylindrical harmonics*. The expansion is two centred, i.e. it is constructed in two separate coordinate systems, one coordinate system for the *source* point $\boldsymbol{x}'$, and one coordinate system for the *field* point $\boldsymbol{x}$. We will utilize $\mathring{\boldsymbol{G}}$ to investigate the near field singularity of $\mathring{\boldsymbol{P}}_{r,s}$ from (4.4).

We recall from §3.2.2, that $\mathring{\boldsymbol{G}}$ may be written as

$$\begin{aligned}\mathring{\boldsymbol{G}}\left(\boldsymbol{x};\boldsymbol{x}';k_y\right) &= \sum_{c\in\{P,S,H\}} \frac{1}{4\pi^2\rho\omega^2}\int_{R^2}\mathrm{d}\boldsymbol{k}\,\boldsymbol{L}_c\otimes\boldsymbol{L}_c'\frac{\exp\left[i\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')\right]}{\hat{K}_c^2-k_z^2-k_x^2}\boldsymbol{J}_6 \\ &= \mathring{\boldsymbol{G}}_P\left(\boldsymbol{x};\boldsymbol{x}';k_y\right)+\mathring{\boldsymbol{G}}_S\left(\boldsymbol{x};\boldsymbol{x}';k_y\right)+\mathring{\boldsymbol{G}}_H\left(\boldsymbol{x};\boldsymbol{x}';k_y\right) \quad (4.6)\end{aligned}$$

where $\otimes$ is the *tensor product* and the partial differential operators

$$\boldsymbol{L}_P = \begin{bmatrix}\partial_z\\ \partial_x\\ \partial_y\\ 2\,\mathring{\mu}\,\partial_{zz}-\frac{\lambda\omega^2}{\mathring{\alpha}^2}\\ 2\,\mathring{\mu}\,\partial_{xz}\\ 2\,\mathring{\mu}\,\partial_{yz}\end{bmatrix}, \quad \boldsymbol{L}_S = \frac{\omega}{\mathring{\beta}\hat{K}_S}\begin{bmatrix}\partial_x\\ -\partial_z\\ 0\\ 2\,\mathring{\mu}\,\partial_{xz}\\ \mathring{\mu}\,(-\partial_{zz}+\partial_{xx})\\ \mathring{\mu}\,\partial_{xy}\end{bmatrix}, \quad \boldsymbol{L}_H = \frac{1}{\hat{K}_H}\begin{bmatrix}\partial_{yz}\\ \partial_{xy}\\ \frac{\omega^2}{\mathring{\beta}^2}+\partial_{yy}\\ 2\,\mathring{\mu}\,\partial_{yzz}\\ 2\,\mathring{\mu}\,\partial_{xyz}\\ \mathring{\mu}\,(\frac{\omega^2}{\mathring{\beta}^2}\partial_z+2\partial_{yyz})\end{bmatrix} \quad (4.7)$$

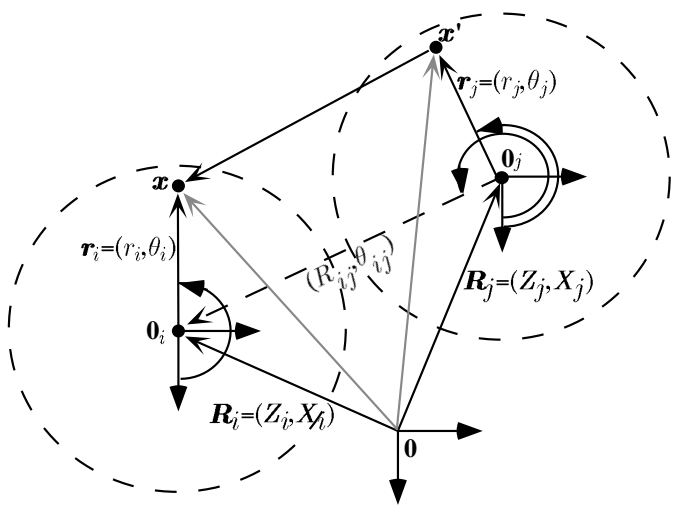

Figure 4.1: A pair of shifted cylindrical coordinate systems, with coordinate centres at $\boldsymbol{R}_i \equiv (Z_i, X_i)$ and $\boldsymbol{R}_j \equiv (Z_j, X_j)$. The source point $\boldsymbol{x}'$ and field point $\boldsymbol{x}$ are inside non-overlapping discs. This geometrical constraint is often called the *muffin tin* condition.

were defined earlier in (3.56).

Assuming that $\boldsymbol{x} \neq \boldsymbol{x}'$ in (4.6), the wavenumber integrals are *uniformly convergent* (due to the choice of complex frequency), so that the differential operators may be taken outside the respective integrals to get

$$\begin{aligned}\mathring{\boldsymbol{G}}\left(\boldsymbol{x};\boldsymbol{x}';k_y\right) &= \sum_{c\in\{P,S,H\}} \boldsymbol{L}_c \otimes \boldsymbol{L}'_c \left[\frac{1}{4\pi^2\rho\omega^2}\int_{R^2} \mathrm{d}\boldsymbol{k}\frac{\exp\left[i\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')\right]}{\hat{K}_c^2 - k_z^2 - k_x^2}\right]\boldsymbol{J}_6 \\ &= \mathring{\boldsymbol{G}}_P\left(\boldsymbol{x};\boldsymbol{x}';k_y\right) + \mathring{\boldsymbol{G}}_S\left(\boldsymbol{x};\boldsymbol{x}';k_y\right) + \mathring{\boldsymbol{G}}_H\left(\boldsymbol{x};\boldsymbol{x}';k_y\right) \qquad (4.8)\end{aligned}$$

where $\hat{K}_c^2$ was given by (3.11). Or equivalently, we have

$$\begin{aligned}\mathring{\boldsymbol{G}}\left(\boldsymbol{x};\boldsymbol{x}';k_y\right) &= \frac{1}{\rho\omega^2}\left[\boldsymbol{L}_P \otimes \boldsymbol{L}'_P\left[g_P(\boldsymbol{x}-\boldsymbol{x}')\right]\boldsymbol{J}_6 \;+\; \boldsymbol{L}_S \otimes \boldsymbol{L}'_S\left[g_S(\boldsymbol{x}-\boldsymbol{x}')\right]\boldsymbol{J}_6\right. \\ &\quad \left. + \boldsymbol{L}_H \otimes \boldsymbol{L}'_H\left[g_H(\boldsymbol{x}-\boldsymbol{x}')\right]\boldsymbol{J}_6\right], \quad \boldsymbol{x}\neq\boldsymbol{x}' \qquad (4.9)\end{aligned}$$

where the *scalar Green's function*,

$$g_c(\boldsymbol{x}-\boldsymbol{x}') = \frac{1}{4\pi^2}\int_{\mathbb{R}^2} \frac{\exp\left[i\;\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')\right]}{\hat{K}_c^2 - k_z^2 - k_x^2}\,\mathrm{d}\boldsymbol{k}, \quad c\in\{P,S,H\} \qquad (4.10)$$

is the solution of the *Helmholtz equation*

$$\Delta g_c(\boldsymbol{x}-\boldsymbol{x}') + \hat{K}_c^2 g_c(\boldsymbol{x}-\boldsymbol{x}') \;=\; \delta(\boldsymbol{x}-\boldsymbol{x}'). \qquad (4.11)$$

We have invoked the condition $\boldsymbol{x} \neq \boldsymbol{x}'$ in (4.6) in order to take the differential operators outside the integral signs, so that (4.9) is inapplicable in the limit $\|\boldsymbol{x}-\boldsymbol{x}'\| \to 0$. In this limit, g_c in (4.11) is a *distribution*, so that the ∂_z, ∂_x operators implicit in some of the components of $\boldsymbol{L}_c \otimes \boldsymbol{L}'_c g_c,\ c \in \{P, S, H\}$, will generate *delta functions*. We will re-introduce these missing delta function terms in §4.3.

Introducing the *local coordinate transformations*(*vid.* Fig.4.1)

$$\boldsymbol{x} = \boldsymbol{R}_i + \boldsymbol{r}_i \quad \text{and} \quad \boldsymbol{x}' = \boldsymbol{R}_j + \boldsymbol{r}_j$$

for *local coordinate centres* $\boldsymbol{R}_i \equiv (Z_i, X_i)$ and $\boldsymbol{R}_j \equiv (Z_j, X_j)$, (4.10) may be written as

$$\begin{aligned} & g_c(\boldsymbol{x}-\boldsymbol{x}') \\ & = \frac{1}{4\pi^2} \int \frac{\exp\left(i\ \boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{R}_i)\right) \exp\left(i\ \boldsymbol{k}\cdot(\boldsymbol{R}_i-\boldsymbol{R}_j)\right) \exp\left(-i\ \boldsymbol{k}\cdot(\boldsymbol{x}'-\boldsymbol{R}_j)\right)}{\hat{K}_c^2 - k_z^2 - k_x^2}\ \mathrm{d}\boldsymbol{k} \\ & = \frac{1}{4\pi^2} \int \frac{\exp\left(i\ \boldsymbol{k}\cdot\boldsymbol{r}_i\right) \exp\left(i\ \boldsymbol{k}\cdot(\boldsymbol{R}_i-\boldsymbol{R}_j)\right) \exp\left(-i\ \boldsymbol{k}\cdot\boldsymbol{r}_j\right)}{\hat{K}_c^2 - k^2}\ \mathrm{d}\boldsymbol{k} \end{aligned} \tag{4.12}$$

where $k^2 = k_z^2 + k_x^2$. We now introduce a set of *shifted cylindrical coordinate* systems so that

$$\begin{aligned} \boldsymbol{k} &= k(\cos\lambda\ , \sin\lambda) \\ \mathrm{d}\boldsymbol{k} &= k\mathrm{d}k\mathrm{d}\lambda \\ \boldsymbol{r}_i &= \boldsymbol{x} - \boldsymbol{R}_i = r_i(\cos\theta_i\ , \sin\theta_i) \\ \boldsymbol{R}_{ij} &= \boldsymbol{R}_i - \boldsymbol{R}_j = R_{ij}(\cos\theta_{ij}\ , \sin\theta_{ij}) \\ \boldsymbol{r}_j &= \boldsymbol{x}' - \boldsymbol{R}_j = r_j(\cos\theta_j\ , \sin\theta_j) \end{aligned} \tag{4.13}$$

hence (4.12) is transformed into

$$\begin{aligned} g_c(\boldsymbol{x}-\boldsymbol{x}') &= \frac{1}{4\pi^2} \int_0^\infty \frac{\mathrm{d}k\ k}{\hat{K}_c^2 - k^2} \times \\ & \int_0^{2\pi} \mathrm{d}\lambda \exp ik\left[r_i\cos(\theta_i-\lambda) + R_{ij}\cos(\theta_{ij}-\lambda) - r_j\cos(\theta_j-\lambda)\right] \end{aligned} \tag{4.14}$$

Using the *Jacobi-Anger* expansion [e.g *Arfken* (1985), *p*.585]

$$\exp(ir\cos s) = \sum_{n\,=\,-\infty}^{\infty} i^n\ J_n(r)\ \exp\left(i\ n\ s\right) \tag{4.15}$$

for arbitrary real numbers r and s, we transform the exponential appearing in (4.14) into

$$
\begin{aligned}
g_c(\boldsymbol{x}-\boldsymbol{x}') &= \frac{1}{4\pi^2}\sum_{m,n,p=-\infty}^{\infty}(-1)^n i^{n+m+p}\int_0^\infty \frac{\mathrm{d}k\ k}{\hat{K}_c^2-k^2}J_m(kr_i)\ J_n(kr_j)\ J_p(kR_{ij}) \\
&\times \int_0^{2\pi}\mathrm{d}\lambda \exp i\left[m(\theta_i-\lambda)+p(\theta_{ij}-\lambda)-n(\theta_j-\lambda)\right] \\
&= \frac{1}{2\pi}\sum_{m,n,p=-\infty}^{\infty}(-1)^n i^{n+m+p}\delta_{n-m-p}\exp i\left[m\theta_i+p\theta_j-n\theta_j\right] \\
&\times \int_0^\infty \frac{\mathrm{d}k\ k}{\hat{K}_c^2-k^2}J_m(kr_i)\ J_n(kr_j)\ J_p(kR_{ij}) \\
&= \frac{1}{2\pi}\sum_{m,n=-\infty}^{\infty}\mathrm{e}^{i[m\theta_i+(n-m)\theta_{ij}-n\theta_j]}\int_0^\infty \frac{\mathrm{d}k\ k}{\hat{K}_c^2-k^2}J_m(kr_i)J_n(kr_j)J_{n-m}(kR_{ij})
\end{aligned}
\tag{4.16}
$$

Assuming the geometrical (or *muffin tin*) constraint [see *Gonis* (1992)]

$$R_{ij} > r_i + r_j, \tag{4.17}$$

so that $\boldsymbol{r}_i$ and $\boldsymbol{r}_j$ are inside *non-overlapping* discs centred at $\boldsymbol{R}_i$ and $\boldsymbol{R}_j$ respectively, we can evaluate the k-integral in (4.16) using the calculus of residues. The geometrical constraint (4.17) is necessary to ensure the *uniform convergence* of subsequent contour integrals.

Given the *unit* cartesian 2−vectors $\hat{\boldsymbol{k}} = (k_z, k_x)/k$ and $\hat{\boldsymbol{r}} = (z, x)/r$ where $k^2 = k_z^2 + k_x^2$ and $r^2 = z^2 + x^2$, we define the *cylindrical harmonics*

$$Y_n(\hat{\boldsymbol{k}}) = \left(\frac{k_z + ik_x}{k}\right)^n, \quad \text{and} \quad Y_n(\hat{\boldsymbol{r}}) = \left(\frac{z+ix}{r}\right)^n \tag{4.18}$$

and their complex conjugates

$$Y_n^*(\hat{\boldsymbol{k}}) = \left(\frac{k_z - ik_x}{k}\right)^n, \quad \text{and} \quad Y_n^*(\hat{\boldsymbol{r}}) = \left(\frac{z-ix}{r}\right)^n. \tag{4.19}$$

Recalling the standard Bessel function identity $J_{n-m}(kR_{ij}) = \frac{1}{2}(H^{(1)}_{n-m}(kR_{ij}) + H^{(2)}_{n-m}(kR_{ij}))$, and evaluating the k integral in (4.16) using the residue the-

orem, we get

$$
\begin{aligned}
&g_c(\boldsymbol{x}-\boldsymbol{x}') \\
&\quad = \frac{-i}{4}\sum_{m,n=-\infty}^{\infty} \exp i\left[m\theta_i + (n-m)\theta_{ij} - n\theta_j\right] J_m(\hat{K}_c r_i) J_n(\hat{K}_c r_j)\; H^{(1)}_{n-m}(\hat{K}_c R_{ij}) \\
&\quad = \frac{-i}{4}\sum_{n,m=-\infty}^{\infty} J_m(\hat{K}_c r_i) Y_m(\hat{\boldsymbol{r}}_i) H^{(1)}_{n-m}(\hat{K}_c R_{ij}) Y_{n-m}(\hat{\boldsymbol{R}}_{ij}) J_n(\hat{K}_c r_j) Y^*_n(\hat{\boldsymbol{r}}_j) \\
&\quad = \frac{-i}{4}\sum_{n,m=-\infty}^{\infty} J_m(\hat{K}_c r_i) Y_m(\hat{\boldsymbol{r}}_i) H^{(1)}_{m-n}(\hat{K}_c R_{ij}) Y^*_{m-n}(\hat{\boldsymbol{R}}_{ji}) J_n(\hat{K}_c r_j) Y^*_n(\hat{\boldsymbol{r}}_j) \qquad (4.20)
\end{aligned}
$$

Upon using g_c of (4.20) in the expression (4.9) we find the Green's tensor in terms of its *cylindrical partial wave* components, *viz.*

$$
\begin{aligned}
&\mathring{\boldsymbol{G}}\,(\boldsymbol{x},\boldsymbol{x}';k_y) = \\
&\frac{-i}{4\rho\omega^2}\sum_{\substack{m,n\\ c\in\{P,S,H\}}} H^{(1)}_{m-n}(\hat{K}_c R_{ij}) Y^*_{m-n}(\hat{\boldsymbol{R}}_{ji}) \boldsymbol{L}_c\left[J_m(\hat{K}_c r_i) Y_m(\hat{\boldsymbol{r}}_i)\right] \otimes \boldsymbol{L}'_c\left[J_n(\hat{K}_c r_j) Y^*_n(\hat{\boldsymbol{r}}_j)\right] \boldsymbol{J}_6 \\
&= \frac{-i}{4\rho\omega^2}\sum_{\substack{m,n\\ c\in\{P,S,H\}}} H^{(1)}_{m-n}(\hat{K}_c R_{ij}) Y^*_{m-n}(\hat{\boldsymbol{R}}_{ji}) |(\underline{\mathrm{re}}, k_y, c, m), \boldsymbol{r}_i\rangle\langle(\underline{\mathrm{re}}, k_y, c, n), \boldsymbol{r}_j| \boldsymbol{J}_6 \qquad (4.21)
\end{aligned}
$$

(4.21)

where

$$
\boldsymbol{x} = \boldsymbol{R}_i + \boldsymbol{x}_i, \quad \boldsymbol{x}' = \boldsymbol{R}_j + \boldsymbol{x}_j, \quad R_{ij} = \|\boldsymbol{R}_j - \boldsymbol{R}_i\|, \quad \hat{\boldsymbol{R}}_{ji} = (\boldsymbol{R}_j - \boldsymbol{R}_i)/R_{ij},
$$

the 6−component basis states $|(\underline{\mathrm{re}}, k_y, c, m), \boldsymbol{r}_i\rangle$ are defined in Box **4.1**, with the dual states $\langle(\underline{\mathrm{re}}, k_y, c, m), \boldsymbol{r}_i|$ given by the rule

$$
\langle(\underline{\mathrm{re}}, k_y, c, m), \boldsymbol{r}_i| \equiv (-1)^m |(\underline{\mathrm{re}}, -k_y, c, -m), \boldsymbol{r}_i\rangle^T.
$$

4.3 An effective point scattering model

In this section we will take care of the nearfield singularity of the reference medium Green's tensor $\mathring{\boldsymbol{G}}$ in the limit as the source point and field point

Box 4.1 THE 6−COMPONENT CYLINDRICAL POLAR BASIS STATES
Given that

$$M = m(m+1), \qquad Y_m(\hat{\boldsymbol{r}}) = \frac{(z\ + i\,x)^m}{r^m}$$

The *regular* basis states are defined as

$$|(\underline{\text{re}}, k_y, c, m), \boldsymbol{r}\rangle \equiv \boldsymbol{L}_c \left[J_m(\hat{K}_c r) Y_m(\hat{\boldsymbol{r}}) \right], \quad c \in \{P, S, H\}$$

and the *outgoing* basis states are defined as

$$|(\underline{\text{ou}}, k_y, c, m), \boldsymbol{r}\rangle \equiv \boldsymbol{L}_c \left[H_m^{(1)}(\hat{K}_c r) Y_m(\hat{\boldsymbol{r}}) \right], \quad c \in \{P, S, H\}$$

where the differential operators $\boldsymbol{L}_c$ were defined in (3.56).
Then, assuming $C_m \in \{J_m, H_m^{(1)}\}$ we have

$$|(\{\underline{\text{re}}, \underline{\text{ou}}\}, k_y, P, m), \boldsymbol{r}\rangle \equiv \boldsymbol{L}_P \left[C_m(\hat{K}_P r) Y_m(\hat{\boldsymbol{r}}) \right] = Y_m(\hat{\boldsymbol{r}}) \times$$

$$\begin{bmatrix} \frac{C_{m-1}(\hat{K}_P r)\,\hat{K}_P\,z}{r} - \frac{C_m(\hat{K}_P r)\,m\,(z+i\,x)}{r^2} \\ \frac{C_{m-1}(\hat{K}_P r)\,\hat{K}_P\,x}{r} - \frac{C_m(\hat{K}_P r)\,m\,(x-i\,z)}{r^2} \\ i k_y C_m(\hat{K}_P r) \\ \frac{2\,\mathring{\mu}\,C_{m-1}(\hat{K}_P r)\,\hat{K}_P\left(x^2-z^2-2\,i\,m\,z\,x\right)}{r^3} - C_m(\hat{K}_P r)\left[\lambda\,K_P^2 - \frac{2\,\mathring{\mu}\left(-r^2\,z^2\,\hat{K}_P^2 + M\,(z+i\,x)^2\right)}{r^4}\right] \\ 2\,\mathring{\mu}\left[\frac{C_{m-1}(\hat{K}_P r)\,\hat{K}_P\left(i\,m\,(z^2-x^2)-2\,z\,x\right)}{r^3} - \frac{C_m(\hat{K}_P r)\left(r^2\,z\,x\,\hat{K}_P^2 + i\,M\,(z+i\,x)^2\right)}{r^4}\right] \\ 2\,\mathring{\mu}\,i\,k_y\left[\frac{C_{m-1}(\hat{K}_P r)\,\hat{K}_P\,z}{r} - \frac{C_m(\hat{K}_P r)\,m\,(z+i\,x)}{r^2}\right] \end{bmatrix} \tag{4.22}$$

$$|(\{\underline{\text{re}}, \underline{\text{ou}}\}, k_y, S, m), \boldsymbol{r}\rangle \equiv \boldsymbol{L}_S \left[C_m(\hat{K}_S r) Y_m(\hat{\boldsymbol{r}}) \right] = \frac{\omega\,Y_m(\hat{\boldsymbol{r}})}{\mathring{\beta}\,\hat{K}_S} \times$$

$$\begin{bmatrix} \frac{C_{m-1}(\hat{K}_S r)\,\hat{K}_S\,x}{r} - \frac{C_m(\hat{K}_S r)\,m\,(x-i\,z)}{r^2} \\ -\frac{C_{m-1}(\hat{K}_S r)\,\hat{K}_S\,z}{r} + \frac{C_m(\hat{K}_S r)\,m\,(z+i\,x)}{r^2} \\ 0 \\ 2\,\mathring{\mu}\left[\frac{C_{m-1}(\hat{K}_S r)\,\hat{K}_S\left(i\,m\left(z^2-x^2\right)-2\,z\,x\right)}{r^3} - \frac{C_m(\hat{K}_S r)\left(r^2\,z\,x\,\hat{K}_S^2 + i\,M\,(z+i\,x)^2\right)}{r^4}\right] \\ 2\,\mathring{\mu}\left[\frac{C_{m-1}(\hat{K}_S r)\,\hat{K}_S\left(z^2+2\,i\,m\,z\,x-x^2\right)}{r^3} + C_m(\hat{K}_S r)\left(\frac{\hat{K}_S^2(z^2-x^2)}{2r^2} - \frac{M\,(z+i\,x)^2}{r^4}\right)\right] \\ \mathring{\mu}\,i\,k_y\left[\frac{C_{m-1}(\hat{K}_S r)\,\hat{K}_S\,x}{r} - \frac{C_m(\hat{K}_S r)\,m\,(x-i\,z)}{r^2}\right] \end{bmatrix} \tag{4.23}$$

$$|(\{\underline{\text{re}}, \underline{\text{ou}}\}, k_y, H, m), \boldsymbol{r}\rangle \equiv \boldsymbol{L}_H \left[C_m(\hat{K}_H r) Y_m(\hat{\boldsymbol{r}}) \right] = \frac{Y_m(\hat{\boldsymbol{r}})}{\hat{K}_H} \times$$

$$\begin{bmatrix} i\,k_y\left[\frac{C_{m-1}(\hat{K}_H r)\,\hat{K}_H\,z}{r} - \frac{C_m(\hat{K}_H r)\,m\,(z+i\,x)}{r^2}\right] \\ i\,k_y\left[\frac{C_{m-1}(\hat{K}_H r)\,\hat{K}_H\,x}{r} - \frac{C_m(\hat{K}_H r)\,m\,(x-i\,z)}{r^2}\right] \\ \hat{K}_H^2 C_m(\hat{K}_H r) \\ 2\,\mathring{\mu}\,i\,k_y\left[\frac{C_{m-1}(\hat{K}_H r)\,\hat{K}_H\left(x^2-z^2-2\,i\,m\,z\,x\right)}{r^3} + \frac{C_m(\hat{K}_H r)\left(-r^2\,z^2\,\hat{K}_H^2 + M\,(z+i\,x)^2\right)}{r^4}\right] \end{bmatrix} \tag{4.24}$$

coincide. The plane wave representations of the Green's tensor from chapter 3 are ill suited to this task, and we will need to employ the *spatial domain* representation (4.2) of $\mathring{\boldsymbol{G}}$. Using this newly acquired Green's tensor, we will sum the multiple scattering series for the special case of a uniform medium with heterogeneity supported exclusively on the square subregion Ω_j with a midpoint at $\boldsymbol{R}_j$, and a side length $\Delta = \Delta x = \Delta z$ that is small on the scale of the incident wavelength. We let the square Ω_j shrink to point, while maintaining its square shape. All of the singular multiple scattering interactions may then be summarized by replacing the material contrasts responsible for the scattering by their *effective* values, i.e. *a single scattering interaction with the effective material contrasts on Ω_j is equivalent to the sum of all multiple scattering interactions from the actual material contrasts*. This approach has two advantages:

- we replace the multiple scattering series with a single scattering interaction, and consequently, we avoid the nearfield singularity of $\mathring{\boldsymbol{G}}$
- the effective material contrasts are generally smaller than the actual contrasts. As a result the multiple scattering series appropriate to the effective medium scattering problem is easier to sum than the corresponding series in the original medium. This is consistent with the numerical experience of *Schuster* (1985), who employed a similar renormalization strategy to solve a multi-inclusion acoustic scattering problem using a boundary integral equation method.

4.3.1 Single scattering

The *single scattering interaction* at Ω_j may be expressed as

$$\mathring{\boldsymbol{G}}\Delta\boldsymbol{A}\mathring{\boldsymbol{G}} \equiv \int_{\Omega_j} \mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\Delta\boldsymbol{A}(\boldsymbol{x},\partial_x;k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x},\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x}. \tag{4.25}$$

Using (4.2) for the components of $\mathring{\boldsymbol{G}}$ in (4.25) gives

$$\begin{aligned}\mathring{\boldsymbol{G}}\Delta\boldsymbol{A}\mathring{\boldsymbol{G}} = &\sum_{\substack{m,n\\ c\in\{P,S,H\}}}\ \sum_{\substack{m',n'\\ c'\in\{P',S',H'\}}} \frac{-1}{16\ \mathring{\rho}^{\,2}\omega^4}\\ &\times|(\underline{\mathrm{re}},k_y,c,m),\boldsymbol{r}_R\rangle H^{(1)}_{m-n}(\hat{K}_c R_{Rj})Y^*_{m-n}(\hat{\boldsymbol{R}}_{jR})\\ &\times\int_{\Omega_j}\langle(\underline{\mathrm{re}},k_y,c,n),\boldsymbol{x}-\boldsymbol{R}_j|\boldsymbol{J}_6\Delta\boldsymbol{A}(\boldsymbol{x},\partial_x;k_y)|(\underline{\mathrm{re}},k_y,c',m'),\boldsymbol{x}-\boldsymbol{R}_j\rangle\mathrm{d}\boldsymbol{x}\\ &\times H^{(1)}_{m'-n'}(\hat{K}_{c'}R_{jS})Y^*_{m'-n'}(\hat{\boldsymbol{R}}_{Sj})\langle(\underline{\mathrm{re}},k_y,c',n'),\boldsymbol{r}_S|\boldsymbol{J}_6\end{aligned} \tag{4.26}$$

where

$$\boldsymbol{x}_R = \boldsymbol{R}_R + \boldsymbol{r}_R, \qquad \boldsymbol{x}_S = \boldsymbol{R}_S + \boldsymbol{r}_S$$

in (4.25) and (4.26). The integrals

$$\Delta \boldsymbol{C}^j_{c,n;c',m'} = \int_{\Omega_j} \left\langle (\underline{\mathrm{re}}, k_y, c, n), \boldsymbol{x} - \boldsymbol{R}_j \left| \boldsymbol{J}_6 \Delta \boldsymbol{A}(\boldsymbol{x}; \partial_x, k_y) \right| (\underline{\mathrm{re}}, k_y, c', m'), \boldsymbol{x} - \boldsymbol{R}_j \right\rangle \mathrm{d}\boldsymbol{x} \tag{4.27}$$

in (4.26) represent the *single scattering approximation* to the coupling between the multipoles induced by the material heterogeneity at $\boldsymbol{R}_j$.

We evaluate the coupling integrals (4.27) in the following two step process:

- we interpret the ∂_x operators in $\Delta \boldsymbol{A}(\boldsymbol{x}; \partial_x, k_y)$ in a *weak* or *projectional* fashion (*see* Box **4.2**). Consequently we employ integration by parts to transfer the ∂_x operators from $\Delta \boldsymbol{A}(\boldsymbol{x}; \partial_x, k_y)$ onto the basis vectors $\langle (\underline{\mathrm{re}}, k_y, c, n), \boldsymbol{x} - \boldsymbol{R}_j|$ and $|(\underline{\mathrm{re}}, k_y, c', m'), \boldsymbol{x} - \boldsymbol{R}_j\rangle$. The basis vectors are *augmented* by their partially differentiated components $(\partial_x u_x, \partial_x u_y)$. These augmented basis states are denoted by $|(\underline{\mathrm{re}}, k_y, c, m), \boldsymbol{r})$, where

  $$|(\underline{\mathrm{re}}, k_y, c, m), \boldsymbol{r}) \equiv \hat{\boldsymbol{L}}_c \left[J_m(\hat{K}_c r) Y_m(\hat{\boldsymbol{r}}) \right], \ c \in \{P, S, H\}$$

  with

  $$\hat{\boldsymbol{L}}_P = \begin{bmatrix} \partial_z \\ \partial_x \\ \partial_y \\ 2\,\mathring{\mu}\,\partial_{zz} - \frac{\mathring{\lambda}\omega^2}{\mathring{\alpha}^2} \\ 2\,\mathring{\mu}\,\partial_{xz} \\ 2\,\mathring{\mu}\,\partial_{yz} \\ \partial_{xx} \\ \partial_{xy} \end{bmatrix}, \quad \hat{\boldsymbol{L}}_S = \frac{\omega}{\mathring{\beta}\hat{K}_S} \begin{bmatrix} \partial_x \\ -\partial_z \\ 0 \\ 2\,\mathring{\mu}\,\partial_{xz} \\ \mathring{\mu}\,(-\partial_{zz} + \partial_{xx}) \\ \mathring{\mu}\,\partial_{xy} \\ -\partial_{xz} \\ 0 \end{bmatrix}, \quad \hat{\boldsymbol{L}}_H = \frac{1}{\hat{K}_H} \begin{bmatrix} \partial_{yz} \\ \partial_{xy} \\ \frac{\omega^2}{\mathring{\beta}^2} + \partial_{yy} \\ 2\,\mathring{\mu}\,\partial_{yzz} \\ 2\,\mathring{\mu}\,\partial_{xyz} \\ \mathring{\mu}\,(\frac{\omega^2}{\mathring{\beta}^2}\partial_z + 2\partial_{yyz}) \\ \partial_{xxy} \\ \frac{\omega^2}{\mathring{\beta}^2}\partial_x + \partial_{xyy} \end{bmatrix} \tag{4.35}$$

  so that the 6−vectors $\langle\cdot|$ and $|\cdot\rangle$ all concatenated with two extra components. The corresponding dual states $((\underline{\mathrm{re}}, k_y, c, m), \boldsymbol{r}|$ are given by the familiar rule

  $$((\underline{\mathrm{re}}, k_y, c, m), \boldsymbol{r}| \equiv (-1)^m \, |(\underline{\mathrm{re}}, -k_y, c, -m), \boldsymbol{r})^T$$

BOX **4.2** WEAK FORMULATIONS FOR THE DISPLACEMENT TRACTION WAVEFIELD
The L_x-periodic wave field

$$\boldsymbol{b}(z,x) = \begin{pmatrix} u_z & u_x & u_y & \tau_{zz} & \tau_{xz} & \tau_{yz} \end{pmatrix}^T,$$

is continuous in a source free region, because it must satisfy the wave equation, so that the components are differentiable, and consequently, they are continuous. More explicitly,

$$u_z, u_x, u_y \in C^1[0, L_x] \quad \text{and} \quad \tau_{zz}, \tau_{xz}, \tau_{yz} \in C^0[0, L_x].$$

However, we have not chosen to include τ_{xx} in the dependent variable $\boldsymbol{b}(z,x)$. Using a momentum balance argument, we know that τ_{xx} should be continuous across transverse discontinuities of the medium properties. Yet, we have no guarantee that this condition will be fulfilled, because we do not model τ_{xx} explicitly, instead we construct it via the elastic wave stress-strain relations that may be expressed in the adopted notation as

$$\tau_{xx} = \mathring{\zeta}\partial_x u_x + i\, k_y \mathring{\chi} u_y + \mathring{\gamma}\tau_{zz}, \tag{4.28}$$

where the material parameters $\mathring{\gamma}$, $\mathring{\chi}$ and $\mathring{\zeta}$ where introduced in Table 2.1 and are re-written here as

$$\mathring{\gamma} = \left[1 - 2\frac{\mathring{\beta}^{\,2}}{\mathring{\alpha}^{\,2}}\right] \quad \mathring{\chi} = 2\,\mathring{\rho}\mathring{\beta}^{\,2}\left[1 - 2\frac{\mathring{\beta}^{\,2}}{\mathring{\alpha}^{\,2}}\right] \quad \text{and} \quad \mathring{\zeta} = 4\,\mathring{\rho}\mathring{\beta}^{\,2}\left[1 - \frac{\mathring{\beta}^{\,2}}{\mathring{\alpha}^{\,2}}\right]$$

Now, let us assume that the material parameters $\mathring{\gamma}$, $\mathring{\chi}$ and $\mathring{\zeta}$ in (4.28) have a *finite jump discontinuity* at $x = d \in (0, L_x)$, so that

$$\left[\mathring{\gamma}\right](d) \equiv \lim_{\varepsilon\to 0+} \mathring{\gamma}(d+\varepsilon) - \mathring{\gamma}(d-\varepsilon) \quad \left[\mathring{\chi}\right](d) \equiv \lim_{\varepsilon\to 0+} \mathring{\chi}(d+\varepsilon) - \mathring{\chi}(d-\varepsilon)$$

$$\text{and} \quad \left[\mathring{\zeta}\right](d) \equiv \lim_{\varepsilon\to 0+} \mathring{\zeta}(d+\varepsilon) - \mathring{\zeta}(d-\varepsilon)$$

then the traction component τ_{xx} suffers a similar saltus, since

$$\lim_{\varepsilon\to 0+} \tau_{xx}(d+\varepsilon) - \tau_{xx}(d-\varepsilon) = \left[\mathring{\zeta}\right](d)\,\partial_x u_x + i\,k_y\,\left[\mathring{\chi}\right](d)\,u_y + \left[\mathring{\gamma}\right](d)\,\tau_{zz}$$

where we have used the continuity of $\partial_x u_x$, u_y and τ_{zz}, because these are components of the solution to a source free wave equation.
Given the L_x-periodic test function $\phi(x)$, we want to find the *projection*

$$\int_0^{L_x} \phi\partial_x\tau_{xx}dx \equiv \int_0^{d-} \phi\partial_x\tau_{xx}dx + \int_{d+}^{L_x} \phi\partial_x\tau_{xx}\mathrm{d}x \tag{4.29}$$

and the right hand side of (4.29) follows from the left because the integral $\int_{d-}^{d+}(\cdot)\mathrm{d}x$ has a vanishing measure.
The projection (4.29) is mathematically equivalent to

$$\int_0^{L_x} \phi\partial_x\tau_{xx}\mathrm{d}x \equiv -\int_0^{d-} (\partial_x\phi)\tau_{xx}\mathrm{d}x - \int_{d+}^{L_x} (\partial_x\phi)\tau_{xx}\mathrm{d}x + (\phi\tau_{xx}]_0^{d-} + (\phi\tau_{xx}]_{d+}^{L_x} \tag{4.30}$$

Box 4.2 *continued*

$[\phi\tau_{xx}](d) \equiv \lim_{\varepsilon\to 0+} (\phi\tau_{xx})(d+\varepsilon) - (\phi\tau_{xx})(d-\varepsilon)$

If we now drop the jump term $[\phi\tau_{xx}](d)$ from (4.31) so that

$$\int_0^{L_x} \phi\partial_x\tau_{xx}\mathrm{d}x \equiv -\int_0^{L_x} (\partial_x\phi)\tau_{xx}\mathrm{d}x \tag{4.32}$$

we are implicitly assuming that

$$[\phi\tau_{xx}](d) \equiv \int_0^{L_x} \phi(x)\delta(x-d)\,[\tau_{xx}](d)\,\mathrm{d}x \equiv 0 \tag{4.33}$$

for all periodic test functions $\phi(x)$. The identity (4.32) is then a statement that τ_{xx} is continuous across jumps in $\mathring{\gamma}$, $\mathring{\chi}$ and $\mathring{\zeta}$ in a *projectional sense*, i.e. τ_{xx} is *weakly continuous* on the interval $x \in [-, L_x)$. To achieve a better understanding of the *weak* nature of (4.33), we should contrast it with the analogous *strong form*

$$\|\,[\tau_{xx}](d)\| \equiv 0$$

in particular, if we choose the *uniform norm* $\|\cdot\|_\infty$, then

$$\|\,[\tau_{xx}](d)\|_\infty \equiv [\tau_{xx}](d) = 0 \tag{4.34}$$

so that continuity is enforced *pointwise* at $x = d$. While (4.34) $\Rightarrow$(4.33), the converse is not generally true, i.e. (4.34)$\not\Leftarrow$ (4.33). In the method of this thesis, the wave equations are being solved in a weak sense, because the boundary conditions (7.23) on the Green's tensor, were enforced by projection onto the space of bandlimited, periodic wavefields in the reference medium. Consequently, it is logically consistent to enforce the continuity of τ_{xx} in a projectional sense also.

Finally, the weak approximation of the *coupling integral*

$$\int \mathrm{d}x \begin{bmatrix} \tilde{u}_z & \tilde{u}_x & \tilde{u}_y & \tilde{\tau}_{zz} & \tilde{\tau}_{xz} & \tilde{\tau}_{yz} \end{bmatrix} \boldsymbol{J}_6 \times$$

$$\begin{bmatrix} 0 & -\Delta\gamma\partial_x & -i\Delta\gamma k_y & \Delta a & 0 & 0 \\ 0 & 0 & 0 & 0 & \Delta b & 0 \\ 0 & 0 & 0 & 0 & 0 & \Delta b \\ -\Delta\rho\omega^2 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\Delta\rho\omega^2 - \partial_x\Delta\zeta\partial_x + \Delta\mu k_y^2 & -ik_y\partial_x\Delta\chi - ik_y\Delta\mu\partial_x & -\partial_x\Delta\gamma & 0 & 0 \\ 0 & -ik_y\Delta\chi\partial_x - ik_y\partial_x\Delta\mu & -\Delta\rho\omega^2 + \Delta\zeta k_y^2 - \partial_x\Delta\mu\partial_x & -ik_y\Delta\gamma & 0 & 0 \end{bmatrix} \begin{bmatrix} u_z \\ u_x \\ u_y \\ \tau_{zz} \\ \tau_{xz} \\ \tau_{yz} \end{bmatrix}$$

may be written as

$$\int \mathrm{d}x \begin{bmatrix} \tilde{u}_z & \tilde{u}_x & \tilde{u}_y & \tilde{\tau}_{zz} & \tilde{\tau}_{xz} & \tilde{\tau}_{yz} & \partial_x\tilde{u}_x & \partial_x\tilde{u}_y \end{bmatrix} \times$$

$$\begin{bmatrix} -\Delta\rho\omega^2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \Delta\mu k_y^2 - \Delta\rho\omega^2 & 0 & 0 & 0 & 0 & 0 & -ik_y\Delta\mu \\ 0 & 0 & \Delta\zeta k_y^2 - \Delta\rho\omega^2 & -ik_y\Delta\gamma & 0 & 0 & -ik_y\chi & 0 \\ 0 & 0 & i\Delta\gamma k_y & -\Delta a & 0 & 0 & \Delta\gamma & 0 \\ 0 & 0 & 0 & 0 & -\Delta b & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\Delta b & 0 & 0 \end{bmatrix} \begin{bmatrix} u_z \\ u_x \\ u_y \\ \tau_{zz} \\ \tau_{xz} \\ \tau_{yz} \end{bmatrix}$$

Box 4.2 *continued*
where

$$\left[\ u_z,\quad u_x,\quad u_y,\quad \tau_{zz},\quad \tau_{xz},\quad \tau_{yz}\ \right]^T \quad \text{and} \quad \left[\ \tilde{u}_z,\quad \tilde{u}_x,\quad \tilde{u}_y,\quad \tilde{\tau}_{zz},\quad \tilde{\tau}_{xz},\quad \tilde{\tau}_{yz}\ \right]^T$$

are the displacement/traction components of the incident and scattered partial wave states respectively.

The resulting *weak* approximation to the elements of the coupling matrix (4.27) may now be written as

$$\int_{\Omega_j} \left\langle (\underline{\mathrm{re}}, k_y, c, n), \boldsymbol{x} - \boldsymbol{R}_j \left| \boldsymbol{J}_6 \Delta \boldsymbol{A}(\boldsymbol{x}; \partial_x, k_y) \right| (\underline{\mathrm{re}}, k_y, c', m'), \boldsymbol{x} - \boldsymbol{R}_j \right\rangle \mathrm{d}\boldsymbol{x}$$
$$= \int_{\Omega_j} \left((\underline{\mathrm{re}}, k_y, c, n), \boldsymbol{x} - \boldsymbol{R}_j \left| \Delta \boldsymbol{C}(\boldsymbol{x}; k_y) \right| (\underline{\mathrm{re}}, k_y, c', m'), \boldsymbol{x} - \boldsymbol{R}_j \right) \mathrm{d}\boldsymbol{x} \tag{4.36}$$

where the *augmented coupling matrix* is defined as

$$\Delta \boldsymbol{C}(\boldsymbol{R}_j; k_y) = \tag{4.37}$$
$$\begin{bmatrix} -\Delta\rho\omega^2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \Delta\mu k_y^2 - \Delta\rho\omega^2 & 0 & 0 & 0 & 0 & 0 & -ik_y\Delta\mu \\ 0 & 0 & \Delta\zeta k_y^2 - \Delta\rho\omega^2 & -ik_y\Delta\gamma & 0 & 0 & -ik_y\chi & 0 \\ 0 & 0 & i\Delta\gamma k_y & -\Delta a & 0 & 0 & \Delta\gamma & 0 \\ 0 & 0 & 0 & 0 & -\Delta b & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\Delta b & 0 & 0 \\ 0 & 0 & ik_y\Delta\chi & \Delta\gamma & 0 & 0 & \Delta\zeta & 0 \\ 0 & ik_y\Delta\mu & 0 & 0 & 0 & 0 & 0 & \Delta\mu \end{bmatrix}$$

with all material contrast parameters evaluated at the mid point $\boldsymbol{R}_j$ of Ω_j. The operator is *multiplicative*, with the material contrast parameters in (4.37) defined (with reference to Table 2.1) by Table 2.2, so that $\Delta\zeta$ is the difference between ζ inside the inclusion and the value of ζ in the homogeneous background medium, with analogous definitions for the remaining contrasts.

- we evaluate the resulting integrals (4.36) using the *midpoint* or *Rayleigh* approximation, to get the *bilinear form*

$$\int_{\Omega_j} \left((\underline{\mathrm{re}}, k_y, c, n), \boldsymbol{x} - \boldsymbol{R}_j \left| \boldsymbol{J}_6 \Delta \boldsymbol{A}(\boldsymbol{x}; \partial_x, k_y) \right| (\underline{\mathrm{re}}, k_y, c', m'), \boldsymbol{x} - \boldsymbol{R}_j \right) \mathrm{d}\boldsymbol{x}$$
$$= \Delta^2 \left((\underline{\mathrm{re}}, k_y, c, n), \boldsymbol{0} + \left| \Delta \boldsymbol{C}(\boldsymbol{R}_j; k_y) \right| (\underline{\mathrm{re}}, k_y, c', m') \right) \tag{4.38}$$

where the *the augmented basis vectors* are given in Box **4.3** and we have used the Taylor series representations [e.g. *Ben-Menahem & Singh* (1981)]

$$J_n(\hat{K}_c r) \simeq \begin{cases} 1 - \frac{(\hat{K}_c r)^2}{2} & n = 0 \\ \frac{(\hat{K}_c r)^n}{2^n n!} & n > 0 \end{cases} \quad \text{and employed} \quad J_{-n}(\hat{K}_c r) = (-1)^n J_n(\hat{K}_c r)$$

where necessary.

Writing down the high order multiple scattering analogs of (4.25) in component form is very cumbersome. To simplify the notation, we introduce a matrix notation. To this end, we assemble the basis states (4.40)–(4.42) from Box **4.3** into the 8×14 array

$$|\underline{\text{re}}, k_y, \mathbf{0}+) \equiv \left[\; |(\underline{\text{re}}, k_y, P), \mathbf{0}+) \quad |(\underline{\text{re}}, k_y, S), \mathbf{0}+) \quad |(\underline{\text{re}}, k_y, H), \mathbf{0}+) \; \right] \tag{4.43}$$

Also, we introduce inter-scatterer propagator $\boldsymbol{H}(\boldsymbol{R}_i, \boldsymbol{R}_j)$ defined as

$$\boldsymbol{H}(\boldsymbol{R}_i, \boldsymbol{R}_j) = \begin{bmatrix} \boldsymbol{H}_P(\boldsymbol{R}_i; \boldsymbol{R}_j) & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{H}_S(\boldsymbol{R}_i; \boldsymbol{R}_j) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{H}_H(\boldsymbol{R}_i; \boldsymbol{R}_j) \end{bmatrix} \tag{4.44}$$

with partitions

$$\boldsymbol{H}_c(\boldsymbol{R}_i; \boldsymbol{R}_j) = \frac{1}{4\,\mathring{\rho}\,\omega^2} \times \tag{4.45}$$

$$\begin{bmatrix} H_0^{(1)}(\hat{K}_c R_{ij}) & -H_1^{(1)}(\hat{K}_c R_{ij}) Y_1(\hat{\mathbf{R}}_{ji}) & H_2^{(1)}(\hat{K}_c R_{ij}) Y_2(\hat{\mathbf{R}}_{ji}) & -H_3^{(1)}(\hat{K}_c R_{ij}) Y_3(\hat{\mathbf{R}}_{ji}) & H_4^{(1)}(\hat{K}_c R_{ij}) Y_4(\hat{\mathbf{R}}_{ji}) \\ H_1^{(1)}(\hat{K}_c R_{ij}) Y_1^*(\hat{\mathbf{R}}_{ji}) & H_0^{(1)}(\hat{K}_c R_{ij}) & -H_1^{(1)}(\hat{K}_c R_{ij}) Y_1(\hat{\mathbf{R}}_{ji}) & H_2^{(1)}(\hat{K}_c R_{ij}) Y_2(\hat{\mathbf{R}}_{ji}) & -H_3^{(1)}(\hat{K}_c R_{ij}) Y_3(\hat{\mathbf{R}}_{ji}) \\ H_2^{(1)}(\hat{K}_c R_{ij}) Y_2^*(\hat{\mathbf{R}}_{ji}) & H_1^{(1)}(\hat{K}_c R_{ij}) Y_1^*(\hat{\mathbf{R}}_{ji}) & H_0^{(1)}(\hat{K}_c R_{ij}) & -H_1^{(1)}(\hat{K}_c R_{ij}) Y_1(\hat{\mathbf{R}}_{ji}) & H_2^{(1)}(\hat{K}_c R_{ij}) Y_2(\hat{\mathbf{R}}_{ji}) \\ H_3^{(1)}(\hat{K}_c R_{ij}) Y_3^*(\hat{\mathbf{R}}_{ji}) & H_2^{(1)}(\hat{K}_c R_{ij}) Y_2^*(\hat{\mathbf{R}}_{ji}) & H_1^{(1)}(\hat{K}_c R_{ij}) Y_1^*(\hat{\mathbf{R}}_{ji}) & H_0^{(1)}(\hat{K}_c R_{ij}) & -H_1^{(1)}(\hat{K}_c R_{ij}) Y_1(\hat{\mathbf{R}}_{ji}) \\ H_4^{(1)}(\hat{K}_c R_{ij}) Y_4^*(\hat{\mathbf{R}}_{ji}) & H_3^{(1)}(\hat{K}_c R_{ij}) Y_3^*(\hat{\mathbf{R}}_{ji}) & H_2^{(1)}(\hat{K}_c R_{ij}) Y_2^*(\hat{\mathbf{R}}_{ji}) & H_1^{(1)}(\hat{K}_c R_{ij}) Y_1^*(\hat{\mathbf{R}}_{ji}) & H_0^{(1)}(\hat{K}_c R_{ij}) \end{bmatrix};$$

for $c \in \{P, H\}$, and

$$\boldsymbol{H}_S(\boldsymbol{R}_i; \boldsymbol{R}_j) = \frac{1}{4\,\mathring{\rho}\,\omega^2} \times \tag{4.46}$$

$$\begin{bmatrix} H_0^{(1)}(\hat{K}_S R_{ij}) & -H_1^{(1)}(\hat{K}_S R_{ij}) Y_1(\hat{\mathbf{R}}_{ji}) & -H_3^{(1)}(\hat{K}_S R_{ij}) Y_3(\hat{\mathbf{R}}_{ji}) & H_4^{(1)}(\hat{K}_S R_{ij}) Y_4(\hat{\mathbf{R}}_{ji}) \\ H_1^{(1)}(\hat{K}_S R_{ij}) Y_1^*(\hat{\mathbf{R}}_{ji}) & H_0^{(1)}(\hat{K}_S R_{ij}) & H_2^{(1)}(\hat{K}_S R_{ij}) Y_2(\hat{\mathbf{R}}_{ji}) & -H_3^{(1)}(\hat{K}_S R_{ij}) Y_3(\hat{\mathbf{R}}_{ji}) \\ H_3^{(1)}(\hat{K}_S R_{ij}) Y_3^*(\hat{\mathbf{R}}_{ji}) & H_2^{(1)}(\hat{K}_S R_{ij}) Y_2^*(\hat{\mathbf{R}}_{ji}) & H_0^{(1)}(\hat{K}_S R_{ij}) & -H_1^{(1)}(\hat{K}_S R_{ij}) Y_1(\hat{\mathbf{R}}_{ji}) \\ H_4^{(1)}(\hat{K}_S R_{ij}) Y_4^*(\hat{\mathbf{R}}_{ji}) & H_3^{(1)}(\hat{K}_S R_{ij}) Y_3^*(\hat{\mathbf{R}}_{ji}) & H_1^{(1)}(\hat{K}_S R_{ij}) Y_1^*(\hat{\mathbf{R}}_{ji}) & H_0^{(1)}(\hat{K}_S R_{ij}) \end{bmatrix}.$$

When $|k_y| > |\omega/c|$, we take the $+ve$ imaginary branch of the square root of $\hat{K}_c$ in (3.11), so that the Hankel functions $H_m^{(1)}(\hat{K}_c r)$ in (4.45)–(4.46) decay exponentially for large values of r.

Box 4.3 MULTIPOLE AMPLITUDES OF THE AUGMENTED CYLINDRICAL HARMONIC BASIS STATES

When we take the limits

$$\lim_{\boldsymbol{x}\to\boldsymbol{R}_j} |(\underline{\mathrm{re}}, k_y, c, m), \boldsymbol{x}-\boldsymbol{R}_j) \equiv \lim_{r\to 0+} \hat{\boldsymbol{L}}_c \left[J_m(\hat{K}_c r) Y_m(\hat{\boldsymbol{r}}) \right] \quad c \in \{P, S, H\}$$

we find that:

- there are only 5 nonvanishing harmonics $m \in \{-2, -1, 0, 1, 2\}$ for the $P-$mode
- there are only 4 nonvanishing harmonics $m \in \{-2, -1, 1, 2\}$ for the $SV-$mode
- there are only 5 nonvanishing harmonics $m \in \{-2, -1, 0, 1, 2\}$ for the $SH-$mode

We assemble these nonvanishing states $(\underline{\mathrm{re}}, k_y, c, m), c \in \{P, S, H\}$ into the 8×14 array $|\underline{\mathrm{re}}, k_y, \mathbf{0}+)$ with the ordering

$$|\underline{\mathrm{re}}, k_y, \mathbf{0}+) = \left[\; |(\underline{\mathrm{re}}, k_y, P), \mathbf{0}+) \quad |(\underline{\mathrm{re}}, k_y, S), \mathbf{0}+) \quad |(\underline{\mathrm{re}}, k_y, H), \mathbf{0}+) \; \right] \tag{4.39}$$

where

$$|(\underline{\mathrm{re}}, k_y, P), \mathbf{0}+) = \begin{array}{c} \begin{array}{ccccc} |(k_y,P,-2),\mathbf{0}+) & |(k_y,P,-1),\mathbf{0}+) & |(k_y,P,0),\mathbf{0}+) & |(k_y,P,1),\mathbf{0}+) & |(k_y,P,2),\mathbf{0}+) \end{array} \\ \left[\begin{array}{ccccc} 0 & \frac{-\hat{K}_P}{2} & 0 & \frac{K_P}{2} & 0 \\ 0 & \frac{i\,\hat{K}_P}{2} & 0 & \frac{i}{2}\,\hat{K}_P & 0 \\ 0 & 0 & i\,k_y & 0 & 0 \\ \frac{\hat{K}_P^2\,\mathring{\mu}}{2} & 0 & -\frac{\omega^2\,\lambda}{\mathring{\alpha}^2} - \hat{K}_P^2\;\mathring{\mu} & 0 & \frac{\hat{K}_P^2\,\mathring{\mu}}{2} \\ \frac{-i\,\hat{K}_P^2\,\mathring{\mu}}{2} & 0 & 0 & 0 & \frac{i\,\hat{K}_P^2\,\mathring{\mu}}{2} \\ 0 & -i\,k_y\,\hat{K}_P\;\mathring{\mu} & 0 & i\,k_y\,\hat{K}_P\;\mathring{\mu} & 0 \\ \frac{-\hat{K}_P^2}{4} & 0 & \frac{-\hat{K}_P^2}{2} & 0 & \frac{-\hat{K}_P^2}{4} \\ 0 & \frac{-k_y\,\hat{K}_P}{2} & 0 & \frac{-k_y\,\hat{K}_P}{2} & 0 \end{array} \right] \end{array} \tag{4.40}$$

$$|(\underline{\mathrm{re}}, k_y, S), \mathbf{0}+) = \frac{\omega}{\mathring{\beta}} \begin{array}{c} \begin{array}{cccc} |(k_y,S,-2),\mathbf{0}+) & |(k_y,S,-1),\mathbf{0}+) & |(k_y,S,1),\mathbf{0}+) & |(k_y,S,2),\mathbf{0}+) \end{array} \\ \left[\begin{array}{cccc} 0 & \frac{i}{2} & \frac{i}{2} & 0 \\ 0 & \frac{1}{2} & \frac{-1}{2} & 0 \\ 0 & 0 & 0 & 0 \\ \frac{-i\,\hat{K}_S\,\mathring{\mu}}{2} & 0 & 0 & \frac{i\,\hat{K}_S\,\mathring{\mu}}{2} \\ \frac{-\hat{K}_S\,\mathring{\mu}}{2} & 0 & 0 & \frac{-\hat{K}_S\,\mathring{\mu}}{2} \\ 0 & \frac{-k_y\,\mathring{\mu}}{2} & \frac{-k_y\,\mathring{\mu}}{2} & 0 \\ \frac{i\,\hat{K}_S}{4} & 0 & 0 & \frac{-i\,\hat{K}_S}{4} \\ 0 & 0 & 0 & 0 \end{array} \right] \end{array} \tag{4.41}$$

$$|(\underline{\mathrm{re}}, k_y, H), \mathbf{0}+) = \tag{4.42}$$

$$\begin{array}{c} \begin{array}{ccccc} |(k_y,H,-2),\mathbf{0}+) & |(k_y,H,-1),\mathbf{0}+) & |(k_y,H,0),\mathbf{0}+) & |(k_y,H,1),\mathbf{0}+) & |(k_y,H,2),\mathbf{0}+) \end{array} \\ \left[\begin{array}{ccccc} 0 & \frac{-i\,k_y}{2} & & \frac{i\,k_y}{2} & 0 \\ 0 & \frac{-k_y}{2} & 0 & \frac{-k_y}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \end{array}$$

Using (4.43) and (4.44), we define the further arrays

$$\mathring{\boldsymbol{G}}(\boldsymbol{x},\boldsymbol{x}';k_y) \equiv -i\,|\underline{\mathrm{re}},k_y,\boldsymbol{r}_i\rangle \boldsymbol{H}(\boldsymbol{R}_i;\boldsymbol{R}_j)\langle\underline{\mathrm{re}},k_y,\boldsymbol{r}_j|\boldsymbol{J}_6 \tag{4.47}$$

which is the matrix analogue of (4.2),

$$\mathring{\boldsymbol{G}}_r(\boldsymbol{R}_i,\boldsymbol{x}';k_y) \equiv -i\,|\underline{\mathrm{re}},k_y,\boldsymbol{0}+)\boldsymbol{H}(\boldsymbol{R}_i;\boldsymbol{R}_j)\langle\underline{\mathrm{re}},k_y,\boldsymbol{r}_j|\boldsymbol{J}_6, \tag{4.48}$$

and

$$\mathring{\boldsymbol{P}}_s(\boldsymbol{R}_i,\boldsymbol{R}_j;k_y) \equiv -i\,|\underline{\mathrm{re}},k_y,\boldsymbol{r}_i\rangle \boldsymbol{H}(\boldsymbol{R}_i;\boldsymbol{R}_j)(\underline{\mathrm{re}},k_y,\boldsymbol{0}+|. \tag{4.49}$$

The components (4.26) of the single scattering interaction, may be assembled into an equivalent matrix form

$$\begin{aligned}\mathring{\boldsymbol{G}}\boldsymbol{\Delta A}\mathring{\boldsymbol{G}} &= \mathring{\boldsymbol{P}}_s\boldsymbol{\Delta C}\mathring{\boldsymbol{G}}_r\\ &= \left(-i\,|\underline{\mathrm{re}},k_y,\boldsymbol{r}_R\rangle\boldsymbol{H}(\boldsymbol{R}_R;\boldsymbol{R}_j)\tilde{\boldsymbol{\Pi}}|\underline{\mathrm{re}},-k_y,\boldsymbol{0}+)^T\right)\\ &\times\left(\boldsymbol{\Delta C}(\boldsymbol{R}_j;k_y)\,\Delta^2\right)\left(-i\,|\underline{\mathrm{re}},k_y,\boldsymbol{0}+)\boldsymbol{H}(\boldsymbol{R}_j;\boldsymbol{R}_S)\tilde{\boldsymbol{\Pi}}|\underline{\mathrm{re}},-k_y,\boldsymbol{r}_S\rangle^T\right)\boldsymbol{J}_6\end{aligned} \tag{4.50}$$

4.3.2 Double scattering

Let us now consider the *double scattering interaction*

$$\begin{aligned}&\mathring{\boldsymbol{G}}\boldsymbol{\Delta A}\mathring{\boldsymbol{G}}\boldsymbol{\Delta A}\mathring{\boldsymbol{G}} \equiv\\ &\quad\int_{\Omega_j}\int_{\Omega_k}\mathring{\boldsymbol{G}}(\boldsymbol{x}_R;\boldsymbol{x};k_y)\boldsymbol{\Delta A}(\boldsymbol{x},\partial_x;k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x},\boldsymbol{x}';k_y)\boldsymbol{\Delta A}(\boldsymbol{x}',\partial_x;k_y)\mathring{\boldsymbol{G}}(\boldsymbol{x}',\boldsymbol{x}_S;k_y)\mathrm{d}\boldsymbol{x}\,\mathrm{d}\boldsymbol{x}'\end{aligned} \tag{4.51}$$

between two small, square subregions of heterogeneity Ω_k with centre at $\boldsymbol{R}_k$ and Ω_j with centre at $\boldsymbol{R}_j$. Using (4.2) for the components of $\mathring{\boldsymbol{G}}$ in (4.51) gives

$$\begin{aligned}\mathring{\boldsymbol{G}}\boldsymbol{\Delta A}\mathring{\boldsymbol{G}}\boldsymbol{\Delta A}\mathring{\boldsymbol{G}} = &\sum_{\substack{m,n\\ c\in\{P,S,H\}}}\;\sum_{\substack{m',n'\\ c'\in\{P,S,H\}}}\;\sum_{\substack{m'',n''\\ c''\in\{P,S,H\}}}\frac{i}{64\;\mathring{\rho}^3\,\omega^6}\\ &\times|(\underline{\mathrm{re}},k_y,c,m),\boldsymbol{r}_R\rangle H^{(1)}_{m-n}(\hat{K}_cR_{Rj})Y^*_{m-n}(\hat{\boldsymbol{R}}_{jR})\\ &\times\int_{\Omega_j}\langle(\underline{\mathrm{re}},k_y,c,n),\boldsymbol{x}-\boldsymbol{R}_j|\boldsymbol{J}_6\boldsymbol{\Delta A}(\boldsymbol{x},\partial_x;k_y)|(\underline{\mathrm{re}},k_y,c',m'),\boldsymbol{x}-\boldsymbol{R}_j\rangle\mathrm{d}\boldsymbol{x}\\ &\times H^{(1)}_{m'-n'}(\hat{K}_{c'}R_{jk})Y^*_{m'-n'}(\hat{\boldsymbol{R}}_{kj})\\ &\times\int_{\Omega_k}\langle(\underline{\mathrm{re}},k_y,c,n'),\boldsymbol{x}'-\boldsymbol{R}_k|\boldsymbol{J}_6\boldsymbol{\Delta A}(\boldsymbol{x}',\partial_x;k_y)|(\underline{\mathrm{re}},k_y,c'',m''),\boldsymbol{x}'-\boldsymbol{R}_k\rangle\mathrm{d}\boldsymbol{x}'\\ &\times H^{(1)}_{m''-n''}(\hat{K}_{c''}R_{kS})Y^*_{m''-n''}(\hat{\boldsymbol{R}}_{Sk})\langle(\underline{\mathrm{re}},k_y,c'',n''),\boldsymbol{r}_S|\boldsymbol{J}_6\end{aligned} \tag{4.52}$$

where we interpret the coupling integrals using the weak form of the Rayleigh approximation.

Once again, the components (4.52) may be assembled into an equivalent matrix equation, which may be written as

$$\begin{aligned}\mathring{\boldsymbol{G}}\Delta\boldsymbol{A}\mathring{\boldsymbol{G}}\Delta\boldsymbol{A}\mathring{\boldsymbol{G}} &= \mathring{\boldsymbol{P}}_s\Delta\boldsymbol{C}\mathring{\boldsymbol{P}}_{r,s}\Delta\boldsymbol{C}\mathring{\boldsymbol{G}}_r \\ &= (-i\,|\underline{\mathrm{re}}, k_y, \boldsymbol{r}_R\rangle\boldsymbol{H}(\boldsymbol{R}_R;\boldsymbol{R}_j)(\underline{\mathrm{re}}, -k_y, \boldsymbol{0}+|)\left(\Delta\boldsymbol{C}(\boldsymbol{R}_j;k_y)\,\Delta^2\right) \\ &\times\mathring{\boldsymbol{P}}_{r,s}(\boldsymbol{R}_j,\boldsymbol{R}_k)\left(\Delta\boldsymbol{C}(\boldsymbol{R}_j;k_y)\,\Delta^2\right)(-i\,|\underline{\mathrm{re}}, k_y, \boldsymbol{0}+)\boldsymbol{H}(\boldsymbol{R}_i;\boldsymbol{R}_j)\langle\underline{\mathrm{re}}, k_y, \boldsymbol{r}_S|\end{aligned} \tag{4.53}$$

to within the *weak Rayleigh approximation*, where the 8×8 augmented coupling matrix was given by (4.37) and the 8×8 augmented propagator array $\mathring{\boldsymbol{P}}_{r,s}$ has been written in the matrix form

$$\mathring{\boldsymbol{P}}_{r,s}(\boldsymbol{R}_i,\boldsymbol{R}_j) \equiv -i\,|\underline{\mathrm{re}}, k_y, \boldsymbol{0}+)\boldsymbol{H}(\boldsymbol{R}_i;\boldsymbol{R}_j)(\underline{\mathrm{re}}, k_y, \boldsymbol{0}+|, \tag{4.54}$$

and $\boldsymbol{H}(\boldsymbol{R}_i,\boldsymbol{R}_j)$ was given by (4.44).

4.3.3 Evaluating the near field singularity

We now consider the limiting case of (4.53) where $\Omega_j = \Omega_k = \Omega_\varepsilon$ is an *infinitesimal square* at the coordinate origin Fig.4.2. In this limiting case, the modified Green's tensor of (4.54) is strongly singular, and we will have to be very careful how we interpret integrals involving (4.54). We employ the small argument asymptotic approximations in Table 4.1, to the Hankel function entries in the diagonal blocks of (4.44) that were defined in (4.45)–(4.46). After a substantial amount of algebra, the leading order behaviour of the Green's tensor in (4.54) reduces to $\tilde{\boldsymbol{P}}_{r,s}(\boldsymbol{0},\boldsymbol{r})$ given in Box **4.4**. The $\delta-$function terms in $\tilde{\boldsymbol{P}}_{r,s}(\boldsymbol{0},\boldsymbol{r})$ are correction terms that arose when we took the partial differential operators $\boldsymbol{L}_c, c \in \{P,S,H\}$ outside the integrals in (4.6). We ignored these terms in the earlier representations of the Green's tensor because it was always assumed that the source and field point of the Green's tensor were distinct.

The integral

$$\lim_{\Omega_\varepsilon\,\to\,0}\int_{\Omega_\varepsilon}\mathring{\boldsymbol{P}}_{r,s}(\boldsymbol{0},\boldsymbol{r})\cdot\Delta\boldsymbol{C}(\boldsymbol{r})\;\mathrm{d}\boldsymbol{r}$$

has a *strong singularity* because the Green's tensor $\mathring{\boldsymbol{P}}_{r,s}(\boldsymbol{0},\boldsymbol{r})$ is $O(\|\boldsymbol{r}\|^{-2})$ in the near field. To handle this integral we employ the classical technique

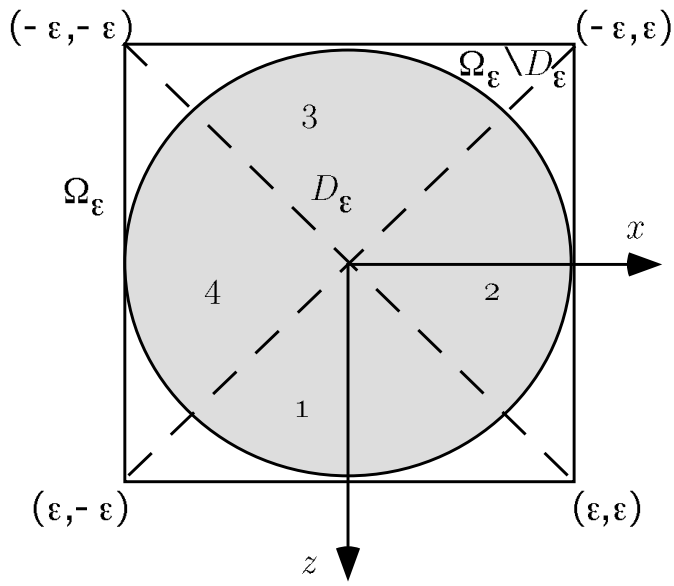

Figure 4.2: The infinitesimal square Ω with an inscribed disc D_ε

$H_m^{(1)}(r)$		leading order asymptotics
$H_0^{(1)}(r)$	$\approx$	$1 + i\,\frac{2}{\pi}\,\big(\gamma_E + \log(\frac{1}{2}) + \log(r)\big) + O(r)$
$H_1^{(1)}(r)$	$\approx$	$-\frac{2\,i}{\pi\,r} + O(r)$
$H_2^{(1)}(r)$	$\approx$	$-\frac{i}{\pi} - \frac{4\,i}{\pi\,r^2} + O(r)$
$H_3^{(1)}(r)$	$\approx$	$-\frac{2\,i}{\pi\,r} - \frac{16\,i}{\pi\,r^3} + O(r)$
$H_4^{(1)}(r)$	$\approx$	$-\frac{i}{2\,\pi} - \frac{8\,i}{\pi\,r^2} - \frac{96\,i}{\pi\,r^4} + O(r)$

Table 4.1: $\gamma_E = 0.5772156649015328606...$ *is the Euler-Mascheroni constant* (e.g. *Arfken* (1985), p.284)

of *subtracting the singularity*, i.e.

$$\lim_{\Omega_\varepsilon \to 0} \int_{\Omega_\varepsilon} \mathring{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r}) \cdot \boldsymbol{\Delta C}(\boldsymbol{r})\ \mathrm{d}\boldsymbol{r} = \lim_{\Omega_\varepsilon \to 0} \int_{\Omega_\varepsilon} \left(\mathring{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r}) - \tilde{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r})\right) \cdot \boldsymbol{\Delta C}(\boldsymbol{r})\ \mathrm{d}\boldsymbol{r}$$
$$+ \left(\lim_{\Omega_\varepsilon \to 0} \int_{\Omega_\varepsilon} \tilde{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r})\mathrm{d}\boldsymbol{r}\right) \cdot \boldsymbol{\Delta C}(\mathbf{0}+) \tag{4.55}$$

as the domain of integration Ω_ε shrinks down to a point, where the dominant near field term $\tilde{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r})$ of the Green's tensor is given in Box **4.4**. If the coupling matrix $\boldsymbol{\Delta C}(\boldsymbol{r})$ is *Holder continuous* (i.e. it is smoother than just continuous , but not necessarily so smooth as to be differentiable) in an open neighbourhood of the point $\boldsymbol{r} = \mathbf{0}$, then

$$\lim_{\Omega_\varepsilon \to 0} \int_{\Omega_\varepsilon} \left(\mathring{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r}) - \tilde{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r})\right) \cdot \boldsymbol{\Delta C}(\boldsymbol{r})\ \mathrm{d}\boldsymbol{r}\ \to\ \mathbf{0}_8$$

in (4.55) so that we are left with

$$\lim_{\Omega_\varepsilon \to 0} \int_{\Omega_\varepsilon} \mathring{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r}) \cdot \boldsymbol{\Delta C}(\boldsymbol{r})\ \mathrm{d}\boldsymbol{r} = \left(\lim_{\Omega_\varepsilon \to 0} \int_{\Omega_\varepsilon} \tilde{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r})\mathrm{d}\boldsymbol{r}\right) \cdot \boldsymbol{\Delta C}(\mathbf{0}+) \tag{4.56}$$

We divide the infinitesimal square $\Omega_\varepsilon = \{(z,x) : -\varepsilon \le z, x \le \varepsilon\}$ in (4.56) into the sum of an inscribed *disc* $D_\varepsilon = \{(z,x) : \sqrt{z^2+x^2} \le \varepsilon\}$ and the remainder $\Omega_\varepsilon \backslash D_\varepsilon$ (Fig.4.2), so that (4.56) may be written as

$$\lim_{\Omega_\varepsilon \to 0} \int_{\Omega_\varepsilon} \mathring{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r}) \cdot \boldsymbol{\Delta C}(\boldsymbol{r})\ \mathrm{d}\boldsymbol{r} = \tag{4.57}$$
$$\left(\lim_{D_\varepsilon \to 0} \int_{D_\varepsilon} \tilde{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r})\mathrm{d}\boldsymbol{r}\right) \cdot \boldsymbol{\Delta C}(\mathbf{0}+) + \left(\lim_{\Omega_\varepsilon \to 0} \int_{\Omega_\varepsilon \backslash D_\varepsilon} \tilde{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r})\mathrm{d}\boldsymbol{r}\right) \cdot \boldsymbol{\Delta C}(\mathbf{0}+)$$

Once again, we employ classical techniques to evaluate the two *strongly singular* integrals on the right hand side of (4.57). First, we consider the integral

$$\lim_{D_\varepsilon \to 0} \int_{D_\varepsilon} \tilde{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r})\mathrm{d}\boldsymbol{r}$$

The $\delta-$function contributions to this integral that were indicated in Box **4.4** are trivial to integrate. To evaluate the $\int_{\Omega_\varepsilon} \frac{\cos m\,\theta}{r^2}$ and $\int_{\Omega_\varepsilon} \frac{\sin m\,\theta}{r^2}, m \in \{2,4\}$ integrals, we employ the 2−D *divergence theorem* in cylindrical coordinates, i.e. given the 2−D vector field $\boldsymbol{v} = (v_r, v_\theta)$ in cylindrical coordinates, then

$$\int_{D_\varepsilon} \nabla \cdot \boldsymbol{v}\mathrm{d}\boldsymbol{r} \equiv \int_0^{2\pi} \mathrm{d}\theta \int_0^{\varepsilon} r\ \mathrm{d}r \left[\frac{1}{r}\frac{\partial\, v_\theta}{\partial\,\theta} + \frac{1}{r}\frac{\partial\, r v_r}{\partial\, r}\right] \equiv \int_0^{2\pi} r v_r \mathrm{d}\theta\Big|_{r=\varepsilon} \tag{4.58}$$

Box 4.4 The leading order terms of the modified Green's tensor (4.54) in the near field

$$\tilde{\boldsymbol{P}}_{r,s}(\boldsymbol{0},\boldsymbol{r}) =$$

$$-\delta(r)\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathring{\lambda}+2\,\mathring{\mu} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathring{\mu} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathring{\mu} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} + \frac{\left(\mathring{\alpha}^2-\mathring{\beta}^2\right)\cos 4\,\theta}{\mathring{\alpha}^2\,\pi\,r^2}\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathring{\mu} & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & -\mathring{\mu} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & \frac{1}{4\,\mathring{\mu}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$+\frac{\left(\mathring{\alpha}^2-\mathring{\beta}^2\right)\sin 4\,\theta}{\mathring{\alpha}^2\,\pi\,r^2}\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathring{\mu} & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathring{\mu} & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$+\frac{\cos 2\,\theta}{r^2}\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\,\mathring{\beta}^2\left(\mathring{\alpha}^2-\mathring{\beta}^2\right)\mathring{\rho}}{\mathring{\alpha}^2\,\pi} & 0 & 0 & \frac{-\mathring{\alpha}^2+2\,\mathring{\beta}^2}{2\,\mathring{\alpha}^2\,\pi} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\mathring{\beta}^2\,\mathring{\rho}}{2\,\pi} & 0 & 0 \\ 0 & 0 & 0 & \frac{-\mathring{\alpha}^2+2\,\mathring{\beta}^2}{2\,\mathring{\alpha}^2\,\pi} & 0 & 0 & \frac{-1}{2\,\mathring{\alpha}^2\,\pi\,\mathring{\rho}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{-1}{2\,\mathring{\beta}^2\,\pi\,\mathring{\rho}} \end{bmatrix}$$

$$+\frac{\sin 2\,\theta}{r^2}\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\mathring{\beta}^2\left(\mathring{\alpha}^2-\mathring{\beta}^2\right)\mathring{\rho}}{\mathring{\alpha}^2\,\pi} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\mathring{\beta}^2}{2\,\mathring{\alpha}^2\,\pi} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2\,\pi} \\ 0 & 0 & 0 & 0 & \frac{\mathring{\beta}^2}{2\,\mathring{\alpha}^2\,\pi} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2\,\pi} & 0 & 0 \end{bmatrix}$$

Using

$$\frac{\cos\ (m\theta)}{r^2} = \frac{1}{r}\frac{\partial}{\partial\,\theta}\frac{\sin\ (m\theta)}{m\ r} \qquad \text{and} \qquad \frac{\sin\ (m\theta)}{r^2} = -\frac{1}{r}\frac{\partial}{\partial\,\theta}\frac{\cos\ (m\theta)}{m\ r}$$

it follows from (4.58) that

$$\int_{D_\varepsilon} \frac{\cos m\,\theta}{r^2}\mathrm{d}\boldsymbol{r} \equiv 0, \quad \text{and} \quad \int_{D_\varepsilon} \frac{\sin m\,\theta}{r^2}\mathrm{d}\boldsymbol{r} \equiv 0, \quad m \in \{2,4\}$$

because the v_r components in (4.58) vanish. Consequently, we are left with the $\delta-$function terms, i.e.

$$\lim_{D_\varepsilon\ \to\ 0}\int_{D_\varepsilon} \tilde{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r})\mathrm{d}\boldsymbol{r} \equiv -\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathring{\lambda} + 2\,\mathring{\mu} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathring{\mu} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathring{\mu} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (4.59)$$

Next we consider the integral

$$\lim_{\Omega_\varepsilon\ \to\ 0}\int_{\Omega_\varepsilon\setminus D_\varepsilon} \tilde{\boldsymbol{P}}_{r,s}(\mathbf{0},\boldsymbol{r})\mathrm{d}\boldsymbol{r} \tag{4.60}$$

from (4.57). In this case the $\delta-$function terms in $\tilde{\boldsymbol{P}}_{r,s}$ are excluded from the domain of integration, and we need only consider the integrals

$$\int_{\Omega_\varepsilon\setminus D_\varepsilon} \frac{\cos m\,\theta}{r^2}\mathrm{d}\boldsymbol{r}, \quad \text{and} \quad \int_{\Omega_\varepsilon\setminus D_\varepsilon} \frac{\sin m\,\theta}{r^2}\mathrm{d}\boldsymbol{r}, \quad m \in \{2,4\}.$$

But $\int_{\Omega_\varepsilon\setminus D_\varepsilon}$ may be split into the sum of the *quadrant* integrals

$$\begin{aligned}\int_{\Omega_\varepsilon\setminus D_\varepsilon} [\cdot]\,\mathrm{d}\boldsymbol{r} \quad \equiv \quad & \int_{-\pi/4}^{\pi/4}\mathrm{d}\theta\int_{\varepsilon}^{\varepsilon/\cos\ (\theta)}[\cdot]\,r\ \mathrm{d}r \ + \int_{3\pi/4}^{5\pi/4}\mathrm{d}\theta\int_{\varepsilon}^{-\varepsilon/\cos\ (\theta)}[\cdot]\,r\ \mathrm{d}r \\ & + \int_{\pi/4}^{3\pi/4}\mathrm{d}\theta\int_{\varepsilon}^{\varepsilon/\sin\ (\theta)}[\cdot]\,r\ \mathrm{d}r \ + \int_{5\pi/4}^{7\pi/4}\mathrm{d}\theta\int_{\varepsilon}^{-\varepsilon/\sin\ (\theta)}[\cdot]\,r\ \mathrm{d}r\end{aligned} \tag{4.61}$$

Using the quadrant integrals (and integration by parts where necessary) we get

$$\int_{\Omega_\varepsilon\setminus D_\varepsilon} \frac{\cos 4\,\theta}{r^2}\mathrm{d}\boldsymbol{r} = \frac{\pi-4}{2}$$

and

$$\int_{\Omega_\varepsilon \setminus D_\varepsilon} \frac{\cos 2\,\theta}{r^2} \mathrm{d}\boldsymbol{r} = \int_{\Omega_\varepsilon \setminus D_\varepsilon} \frac{\sin 4\,\theta}{r^2} \mathrm{d}\boldsymbol{r} = \int_{\Omega_\varepsilon \setminus D_\varepsilon} \frac{\sin 2\,\theta}{r^2} \mathrm{d}\boldsymbol{r} = 0.$$

so that (4.60) becomes

$$\lim_{\Omega_\varepsilon \to 0} \int_{\Omega_\varepsilon \setminus D_\varepsilon} \tilde{\boldsymbol{P}}_{r,s}(\boldsymbol{0}, \boldsymbol{r}) \mathrm{d}\boldsymbol{r} = \frac{(\pi - 4)(\mathring{\alpha}^2 - \mathring{\beta}^2)}{2\,\pi\,\mathring{\alpha}^2} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathring{\mu} & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & -\mathring{\mu} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & \frac{1}{4\,\mathring{\mu}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{4.62}$$

Using (4.59) and (4.62) in the right hand side of (4.57) recovers (4.56) with

$$\lim_{\Omega_\varepsilon \to 0} \int_{\Omega_\varepsilon} \tilde{\boldsymbol{P}}_{r,s}(\boldsymbol{0}, \boldsymbol{r}) \mathrm{d}\boldsymbol{r} = \tag{4.63}$$

$$- \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathring{\lambda} + 2\,\mathring{\mu} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \mathring{\mu} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \mathring{\mu} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} + \frac{(\pi - 4)\,(\mathring{\alpha}^2 - \mathring{\beta}^2)}{2\,\mathring{\alpha}^2\,\pi} \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathring{\mu} & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & -\mathring{\mu} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & \frac{1}{4\,\mathring{\mu}} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

4.3.4 Summing the multiple scattering series at a point

We now return to the *double scattering integral* of (4.53), and considering the limiting case where $\Omega_k \to \Omega_j$, so that the two small subregions coincide. To within the *weak Rayleigh approximation* of (4.53) , we have

$$\begin{aligned} \mathring{\boldsymbol{G}} \Delta \boldsymbol{A} \mathring{\boldsymbol{G}} \Delta \boldsymbol{A} \mathring{\boldsymbol{G}} \;=\; & \left(-i\,|\underline{\mathrm{re}}, k_y, \boldsymbol{0}+\rangle \boldsymbol{P}(\boldsymbol{R}_R; \boldsymbol{R}_j) \tilde{\boldsymbol{\Pi}} |\underline{\mathrm{re}}, -k_y, \boldsymbol{0}+)^T \right) \cdot \left(\Delta \boldsymbol{C}(\boldsymbol{R}_j; k_y)\, \Delta^2 \right) \\ & \times \int_{\Omega_j} \tilde{\boldsymbol{G}}(\boldsymbol{R}_j, \boldsymbol{R}_j + \boldsymbol{r}_j) \mathrm{d}\boldsymbol{r}_j \cdot \Delta \boldsymbol{C}(\boldsymbol{R}_j; k_y) \\ & \times \left(-i\,|\underline{\mathrm{re}}, k_y, \boldsymbol{0}+) \boldsymbol{P}(\boldsymbol{R}_i; \boldsymbol{R}_j) \tilde{\boldsymbol{\Pi}} |\underline{\mathrm{re}}, -k_y, \boldsymbol{0}+\rangle^T \right) \boldsymbol{J}_6 \end{aligned} \tag{4.64}$$

Similarly, the *order-n multiple scattering interaction*

$$\mathring{\boldsymbol{G}}\Delta\boldsymbol{A}\left(\mathring{\boldsymbol{G}}\Delta\boldsymbol{A}\right)^n\mathring{\boldsymbol{G}}$$

at Ω_j, may be written as

$$\begin{aligned}\mathring{\boldsymbol{G}}\Delta\boldsymbol{A}\left(\mathring{\boldsymbol{G}}\Delta\boldsymbol{A}\right)^n\mathring{\boldsymbol{G}} &= \left(-i\,|\underline{\mathrm{re}},k_y,\boldsymbol{0}+\rangle\boldsymbol{P}(\boldsymbol{R}_R;\boldsymbol{R}_j)\tilde{\boldsymbol{\Pi}}|\underline{\mathrm{re}},-k_y,\boldsymbol{0}+)^T\right)\\ &\times\left(\Delta\boldsymbol{C}(\boldsymbol{R}_j;k_y)\,\Delta^2\right)\left(\int_{\Omega_j}\tilde{\boldsymbol{G}}(\boldsymbol{R}_j,\boldsymbol{R}_j+\boldsymbol{r}_j)\mathrm{d}\boldsymbol{r}_j\cdot\Delta\boldsymbol{C}(\boldsymbol{R}_j;k_y)\right)^n\\ &\times\left(-i\,|\underline{\mathrm{re}},k_y,\boldsymbol{0}+)\boldsymbol{P}(\boldsymbol{R}_i;\boldsymbol{R}_j)\tilde{\boldsymbol{\Pi}}|\underline{\mathrm{re}},-k_y,\boldsymbol{0}+\rangle^T\right)\boldsymbol{J}_6\end{aligned}\tag{4.65}$$

If we sum all the multiple scattering interactions within Ω_j we get

$$\begin{aligned}\sum_{n\,=\,0}^{\infty}\mathring{\boldsymbol{G}}\Delta\boldsymbol{A}\left(\mathring{\boldsymbol{G}}\Delta\boldsymbol{A}\right)^n\mathring{\boldsymbol{G}} &= \left(-i\,|\underline{\mathrm{re}},k_y,\boldsymbol{0}+\rangle\boldsymbol{P}(\boldsymbol{R}_R;\boldsymbol{R}_j)\tilde{\boldsymbol{\Pi}}|\underline{\mathrm{re}},-k_y,\boldsymbol{0}+)^T\right)\\ &\times\left(\Delta\boldsymbol{C}(\boldsymbol{R}_j;k_y)\,\Delta^2\right)\sum_{n\,=\,0}^{\infty}\left(\int_{\Omega_j}\tilde{\boldsymbol{G}}(\boldsymbol{R}_j,\boldsymbol{R}_j+\boldsymbol{r}_j)\mathrm{d}\boldsymbol{r}_j\cdot\Delta\boldsymbol{C}(\boldsymbol{R}_j;k_y)\right)^n\\ &\times\left(-i\,|\underline{\mathrm{re}},k_y,\boldsymbol{0}+)\boldsymbol{P}(\boldsymbol{R}_i;\boldsymbol{R}_j)\tilde{\boldsymbol{\Pi}}|\underline{\mathrm{re}},-k_y,\boldsymbol{0}+\rangle^T\right)\boldsymbol{J}_6\end{aligned}\tag{4.66}$$

4.3.5 The effective scattering parameters

By comparing the multiple scattering expression (4.66) with the single scattering expression (4.50), we observe the correspondence

$$\Delta\boldsymbol{C}(\boldsymbol{R}_j;k_y)\Leftrightarrow\Delta\boldsymbol{C}(\boldsymbol{R}_j;k_y)\sum_{n\,=\,0}^{\infty}\left(\int_{\Omega_j}\tilde{\boldsymbol{G}}(\boldsymbol{R}_j,\boldsymbol{R}_j+\boldsymbol{r}_j)\mathrm{d}\boldsymbol{r}_j\cdot\Delta\boldsymbol{C}(\boldsymbol{R}_j;k_y)\right)^n$$

We introduce an *effective coupling matrix* $\Delta\boldsymbol{C}_{eff}$ defined by

$$\begin{aligned}&\Delta\boldsymbol{C}_{eff}(\boldsymbol{R}_j;k_y)\equiv\\ &\qquad\lim_{\Omega_j\to 0}\Delta\boldsymbol{C}(\boldsymbol{R}_j;k_y)\sum_{n\,=\,0}^{\infty}\left(\int_{\Omega_j}\tilde{\boldsymbol{G}}(\boldsymbol{R}_j,\boldsymbol{R}_j+\boldsymbol{r}_j)\mathrm{d}\boldsymbol{r}_j\cdot\Delta\boldsymbol{C}(\boldsymbol{R}_j;k_y)\right)^n\end{aligned}\tag{4.67}$$

and using $\Delta\boldsymbol{C}_{eff}(\boldsymbol{R}_j;k_y)$ in place of the coupling matrix $\Delta\boldsymbol{C}(\boldsymbol{R}_j;k_y)$ in (4.50), the single scattering interaction

$$\begin{aligned}\mathring{\boldsymbol{G}}\Delta\boldsymbol{A}_{eff}\mathring{\boldsymbol{G}} &= \left(-i\,|\underline{\mathrm{re}},k_y,\boldsymbol{r}_R\rangle\boldsymbol{P}(\boldsymbol{R}_R;\boldsymbol{R}_j)\tilde{\boldsymbol{\Pi}}|\underline{\mathrm{re}},-k_y,\boldsymbol{0}+)^T\right)\\ &\times\left(\Delta\tilde{\boldsymbol{C}}(\boldsymbol{R}_j;k_y)\,\Delta^2\right)\left(-i\,|\underline{\mathrm{re}},k_y,\boldsymbol{0}+)\boldsymbol{P}(\boldsymbol{R}_j;\boldsymbol{R}_S)\tilde{\boldsymbol{\Pi}}|\underline{\mathrm{re}},-k_y,\boldsymbol{r}_S\rangle^T\right)\boldsymbol{J}_6\end{aligned}\tag{4.68}$$

$$
\begin{aligned}
\Delta\tilde{\boldsymbol{C}}_{33} &= \Delta\zeta\, k_y^2 - \Delta\rho\,\omega^2 + \tfrac{k_y^2}{D}\Big(\Delta\chi^2\,\tilde{\zeta} - 4\,\Delta\chi\,\Delta\gamma\,\tilde{\zeta}\;\mathring{\mu} + 4\,\Delta\gamma^2\,\tilde{\zeta}\,\mathring{\mu}^2 \\
&\quad -\mathring{\alpha}^2\,\Delta a\,\Delta\chi^2\,\tilde{\zeta}\;\mathring{\rho} - 2\,\mathring{\alpha}^2\,\Delta\chi\,\Delta\gamma^2\,\tilde{\zeta}\;\mathring{\rho} + \mathring{\alpha}^2\,\Delta\gamma^2\,\Delta\zeta\,\tilde{\zeta}\;\mathring{\rho} - 4\,\mathring{\alpha}^2\,\Delta\gamma^2\,\mathring{\mu}^2\;\mathring{\rho}\Big) \\
\Delta\tilde{\boldsymbol{C}}_{43} &= -\tfrac{i\,k_y}{D}\Big(-\Delta\chi\,\Delta\gamma\,\tilde{\zeta} + \Delta\gamma\,\Delta\zeta\,\tilde{\zeta} - 2\,\Delta a\,\Delta\chi\,\tilde{\zeta}\;\mathring{\mu} - 2\,\Delta\gamma^2\,\tilde{\zeta}\;\mathring{\mu} - 4\,\Delta\gamma\,\mathring{\mu}^2\;\Big) \\
\Delta\tilde{\boldsymbol{C}}_{73} &= -\tfrac{2\,i\,k_y\,\mathring{\mu}}{D}\Big(-\Delta\chi\,\Delta\gamma\,\tilde{\zeta} + \Delta\gamma\,\Delta\zeta\,\tilde{\zeta} - 2\,\Delta\chi\;\mathring{\mu} - 2\,\Delta a\,\Delta\chi\,\tilde{\zeta}\;\mathring{\mu} \\
&\quad -2\,\Delta\gamma^2\,\tilde{\zeta}\;\mathring{\mu} + 2\,\mathring{\alpha}^2\,\Delta a\,\Delta\chi\;\mathring{\mu}\,\mathring{\rho} + 2\,\mathring{\alpha}^2\,\Delta\gamma^2\;\mathring{\mu}\,\mathring{\rho}\Big) \\
\Delta\tilde{\boldsymbol{C}}_{44} &= \tfrac{\Delta\gamma^2\,\tilde{\zeta} + \Delta a\,\Delta\zeta\,\tilde{\zeta} - 4\,\Delta a\,\mathring{\mu}^2}{D} \\
\Delta\tilde{\boldsymbol{C}}_{74} &= \frac{2\,\mathring{\mu}\,\left(\Delta\gamma^2\,\tilde{\zeta} + \Delta a\,\Delta\zeta\,\tilde{\zeta} + 2\,\Delta\gamma\,\mathring{\mu}\right)}{D} \\
\Delta\tilde{\boldsymbol{C}}_{55} &= \frac{\Delta b}{\Delta b\,\tilde{\zeta} + \mathring{\mu}\;\Delta b - 1} \\
\Delta\tilde{\boldsymbol{C}}_{66} &= \frac{\Delta b}{\mathring{\mu}\;\Delta b - 1} \\
\Delta\tilde{\boldsymbol{C}}_{77} &= \frac{4\,\mathring{\mu}^2\,\left(\Delta\zeta + \Delta\gamma^2\,\tilde{\zeta} + \Delta a\,\Delta\zeta\,\tilde{\zeta} - \mathring{\alpha}^2\,\Delta\gamma^2\;\mathring{\rho} - \mathring{\alpha}^2\,\Delta a\,\Delta\zeta\;\mathring{\rho}\right)}{D} \\
D &= 4\,\Delta\gamma\,\tilde{\zeta}\;\mathring{\mu} + 4\,\mathring{\mu}^2 + 4\,\Delta a\,\tilde{\zeta}\,\mathring{\mu}^2 + \mathring{\alpha}^2\,\Delta\gamma^2\,\tilde{\zeta}\;\mathring{\rho} + \\
&\quad \mathring{\alpha}^2\,\Delta a\,\Delta\zeta\,\tilde{\zeta}\;\mathring{\rho} - 4\,\mathring{\alpha}^2\,\Delta a\,\mathring{\mu}^2\;\mathring{\rho} - \Delta\zeta\,\tilde{\zeta} \\
\tilde{\zeta} &= \tfrac{(\pi-4)\,\mathring{\zeta}}{8\,\pi}
\end{aligned}
$$

Table 4.2: The elements of the effective coupling matrix $\Delta\boldsymbol{C}_{eff}$

where $\Delta\boldsymbol{A}_{eff}$ is $\Delta\boldsymbol{A}$ from (2.15) but constructed using the *effective parameter contrasts* from Table 4.2.

Eqn.(4.68) *includes all multiple scattering interactions* within Ω_j to within the *weak Rayleigh approximation*. This technique is sometimes called *renormalization*. The technique is sometimes referred to as a *self avoiding renormalization* in quantum field theory, and is also implicit in the *redefinition of a scalar potential as an equivalent mass operator* in various applications of stochastic wave theory [e.g. *Sobczyk* (1985)]. The self avoiding renormalization has important implications for the convergence of the *multiple scattering series*. This later point has been noted in *Schuster* (1985), where an analogous strategy was employed to solve the *boundary integral equations* for a multiple inclusion problem.

In practice the renormalization (4.67) is effected using the matrix inversion

$$\Delta\boldsymbol{C}_{eff}(\boldsymbol{R}_j;k_y) = \tag{4.69}$$

$$\Delta\boldsymbol{C}(\boldsymbol{R}_j;k_y)\cdot\left[\boldsymbol{I}_8-\left(\lim_{\Omega_j\to 0}\int_{\Omega_j}\tilde{\boldsymbol{P}}_{r,s}(\boldsymbol{R}_j,\boldsymbol{R}_j+\boldsymbol{r}_j)\mathrm{d}\boldsymbol{r}_j\right)\cdot\Delta\boldsymbol{C}(\boldsymbol{R}_j;k_y)\right]^{-1}$$

$$=\begin{bmatrix} -\Delta\rho\,\omega^2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \Delta\mu\,k_y^2-\Delta\rho\,\omega^2 & 0 & 0 & 0 & 0 & 0 & -i\,k_y\,\Delta\mu \\ 0 & 0 & \Delta\tilde{\boldsymbol{C}}_{33} & -\Delta\tilde{\boldsymbol{C}}_{43} & 0 & 0 & -\Delta\tilde{\boldsymbol{C}}_{73} & 0 \\ 0 & 0 & \Delta\tilde{\boldsymbol{C}}_{43} & \Delta\tilde{\boldsymbol{C}}_{44} & 0 & 0 & \Delta\tilde{\boldsymbol{C}}_{74} & 0 \\ 0 & 0 & 0 & 0 & \Delta\tilde{\boldsymbol{C}}_{55} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \Delta\tilde{\boldsymbol{C}}_{66} & 0 & 0 \\ 0 & 0 & \Delta\tilde{\boldsymbol{C}}_{73} & \Delta\tilde{\boldsymbol{C}}_{74} & 0 & 0 & \Delta\tilde{\boldsymbol{C}}_{77} & 0 \\ 0 & i\,k_y\,\Delta\mu & 0 & 0 & 0 & 0 & 0 & \Delta\mu \end{bmatrix}$$

with the components given in Table 2.2 and Table 4.2. All of the elements of $\Delta\boldsymbol{C}_{eff}(\boldsymbol{R}_j;k_y)$ in (4.69) are evaluated at $\boldsymbol{R}_j$.

4.4 The matrix Lippmann-Schwinger equation

In this final section, we will derive a matrix analogue of the augmented Lippmann-Schwinger type integral equation (4.5). To discretize (4.5) into a matrix equation, we replace the integral term with a matrix multiplication. The matrix analogue translates directly into a computer code, provided the reference medium is homogeneous. The more useful case where the reference medium is a stratified half-space is considered at length in chapter 5.

First, we will arrange the effective coupling matrix $\boldsymbol{\Delta C}_{eff}(\boldsymbol{R}_j; k_y)$ of (4.69) into a larger matrix structure. This matrix is block-diagonal because the coupling operator is *local* in the spatial domain.

The second extension concerns the plane wave expansion of the propagators. We have already developed matrix representations of the propagators appropriate to a lattice of source and receiver points, in §3.3. However, these representations will require minor modification because the partial wave states used in the construction of the lattice propagators in §3.3 were 6−component displacement/traction vectors. In this chapter, we have augmented those states by the additional components $\partial_x u_x, \partial_x u_y$, for the purposes expounded above. Consequently, we will need to augment the plane wave lattice propagators of §3.3 in a similar way. It is important to note that the number of states, and the manner in which they evolve over space remains unchanged. The only change is that each state carries some extra information in the two additional components. Consequently, it is straightforward to find the augmented propagators, and we simply state the extensions.

4.4.1 The lattice representation of the effective, weak-Rayleigh coupling matrix

We recall the definition of the *secondary source field lattice* introduced in §3.3 and depicted in Fig.3.2. This lattice was comprised of L rows of lattice points, with N_x lattice points per row, to give a total of $L \times N_x$ points.

We now define a *lattice expansion* of the *effective coupling array* $\boldsymbol{\Delta C}_{eff}(\boldsymbol{R}_j; k_y)$ that was defined in (4.69). We recall from §3.3 that the lattice points are numbered in the same order as words to be read on the written page, i.e. top to bottom, left to right. Once again, we may also label the lattice points as if they were the elements of a 2−D array. For this purpose we re-introduce the matrix row index $1 \leq \rho \leq L$ and the matrix column index $1 \leq \chi \leq N_x$, so that the global index i is related to the matrix indices via

$$i = (\rho - 1)N_x + \chi, \quad 1 \leq i \leq LN_x, \quad 1 \leq \rho \leq L, \quad 1 \leq \chi \leq N_x$$

The same *caveat* regarding the use of ρ and χ to indicate the row and column indices respectively applies here, i.e. the ρ and χ in this section, are not to confused with the material parameters given by the same symbols.

If we denote the coupling matrix for lattice points in row$_\rho$, as $\boldsymbol{\Delta C}^{\rho}_{eff}(k_y)$, this array may be defined explicitly as the N_x element block diagonal array

$$\boldsymbol{\Delta C}^{\rho}_{eff}(k_y) = \Delta^2 \times \tag{4.70}$$
$$\text{diag}\left[\ \boldsymbol{\Delta C}_{eff}(\boldsymbol{R}_{N_x(\rho-1)+1}; k_y) \quad \boldsymbol{\Delta C}_{eff}(\boldsymbol{R}_{N_x(\rho-1)+2}; k_y) \quad \cdots \quad \boldsymbol{\Delta C}_{eff}(\boldsymbol{R}_{\rho N_x}; k_y)\ \right]$$

where Δ^2 is the area of the small square subregion, with the 8×8 diagonal block $\Delta \boldsymbol{C}_{eff}$ given by (4.69) at the respective lattice point.

Finally, we assemble the block diagonal arrays $\Delta \boldsymbol{C}^{\rho}_{eff}(k_y)$ from (4.70) for each row of the lattice into the $L-$element, block diagonal array $\Delta \boldsymbol{\mathcal{C}}_{eff}(k_y)$ defined as

$$\boxed{\Delta \boldsymbol{\mathcal{C}}_{eff}(k_y) = \mathrm{diag}\left[\ \Delta \boldsymbol{\mathcal{C}}^{1}_{eff}(k_y) \quad \Delta \boldsymbol{\mathcal{C}}^{1}_{eff}(k_y) \quad \cdots \quad \Delta \boldsymbol{\mathcal{C}}^{1}_{eff}(k_y)\ \right] \tag{4.71}}$$

4.4.2 The augmented propagators

In §3.3, we wrote down explicit matrix representations for the propagators appropriate for a regular lattice expansion of some virtual or secondary source distribution. We assumed that this secondary source was represented by a $6-$vector denoted by $\boldsymbol{f}_j$ at each of the $L \times N_x$ lattice points. However, in §4.3 of the present chapter, we have employed integration by parts to derive the effective coupling matrix $\Delta \boldsymbol{C}_{eff}(\boldsymbol{R}_j; k_y)$ that was defined in (4.69). This later operation entailed transferring the ∂_x operators in $\Delta \boldsymbol{A}(\boldsymbol{x}; \partial_x, k_y)$ of (4.38) onto the source and receiver coordinates of the propagator. Due to the *bilinear nature* of the propagator, the integration by parts was entirely equivalent to augmenting the propagator as well as the coupling matrix. The augmented form was very convenient for numerical applications, because the 6×6 *differential* coupling operator $\Delta \boldsymbol{A}(\boldsymbol{x}; \partial_x, k_y)$ defined in (2.15) was replaced by the 8×8 *multiplicative* coupling matrix $\Delta \boldsymbol{C}_{eff}(\boldsymbol{R}_j; k_y)$ that was defined in (4.69). Mathematically, we are approximating a singular integro/differential operator equation, by a non-singular compact integral equation.

We now present an augmented version of the plane wave expansion of the *lattice propagator* from (3.3.3) of §3.3. Certainly, we will not duplicate the manifold steps that lead us to the original representation (3.3.3). Instead,

we merely point out that (3.3.3) should be replaced by

$$\begin{aligned}
\mathring{\boldsymbol{\mathcal{P}}}_{r,s} &\equiv \mathring{\boldsymbol{\mathcal{P}}}^{z}_{r,s}[\cdot] + \mathring{\boldsymbol{\mathcal{P}}}^{x}_{r,s} \\
&= -i\hat{\boldsymbol{\Phi}}^{-1}\Big[\boldsymbol{I}_L \otimes \Big\{(\boldsymbol{I}_8 \otimes \mathbf{DFT}(\mathbf{k}_x))\,\hat{\boldsymbol{D}}_z(\mathbf{k}_x, k_y)\Big\}\,\boldsymbol{\mathcal{Q}}^z \\
&\qquad\qquad \times \boldsymbol{I}_L \otimes \Big\{\hat{\boldsymbol{D}}_z^T(-\mathbf{k}_x, -k_y)\,\big(\boldsymbol{I}_8 \otimes \mathbf{DFT}^T(-\mathbf{k}_x)\big)\,\hat{\boldsymbol{\Phi}}[\cdot]\Big\}\Big] \\
&\quad -i\boldsymbol{I}_L \otimes \Big(\boldsymbol{I}_{N_x} \otimes \Big\{\big(\boldsymbol{I}_8 \otimes \overline{\boldsymbol{F}}(0)\big)\,\hat{\boldsymbol{D}}_x(\mathbf{k}_z, k_y)\Big\}\,\boldsymbol{\mathcal{Q}}^x_{\rho\leftarrow\rho} \\
&\qquad\qquad \times \boldsymbol{I}_{N_x} \otimes \Big\{\hat{\boldsymbol{D}}_x^T(-\mathbf{k}_z, -k_y)\,\Big(\boldsymbol{I}_8 \otimes \overline{\boldsymbol{F}}^T(0)\Big)\Big\}\Big)
\end{aligned} \tag{4.72}$$

(4.72)

where the $8N_k \times 6N_k$ array $\hat{\boldsymbol{D}}_z$ is an augmented version of the $6N_k \times 6N_k$ array $\boldsymbol{D}_z$ (3.48) and is given in Box **4.5**. Similarly the $8\bar{N}_k \times 6\bar{N}_k$ array $\hat{\boldsymbol{D}}_x$ is an augmented version of the $6\bar{N}_k \times 6\bar{N}_k$ array $\boldsymbol{D}_x$ (3.48) and is given in Box **4.6**. The plane wave propagators $\boldsymbol{\mathcal{Q}}^z$ and $\boldsymbol{\mathcal{Q}}^x_{\rho\leftarrow\rho}$ were given by (3.100) and (3.111) respectively. The DFT matrix $\mathbf{DFT}$ was given by (3.95), and the projection matrix $\overline{\boldsymbol{F}}(0)$ was given by (3.72). The augmented upsampling and downsampling transformations $\hat{\boldsymbol{\Phi}}[\cdot]$ and $\hat{\boldsymbol{\Phi}}^{-1}[\cdot]$ respectively, are defined in Alg.4.1. By incorporating the upsampling and down sampling operators within the definition of $\mathring{\boldsymbol{\mathcal{P}}}^z_{r,s}[\cdot]$ in (4.4.2), $\mathring{\boldsymbol{\mathcal{P}}}_{r,s}$ operates directly on $8LN_x-$vectors.

4.4.3 The matrix formulation of the governing equation

Using the propagator $\mathring{\boldsymbol{\mathcal{P}}}_{r,s}$ from (4.4.2) and the coupling matrix $\Delta\boldsymbol{\mathcal{C}}_{eff}$ from (4.71), the matrix analogue of the governing equation (4.5) may be written as

$$\boldsymbol{\mathcal{G}}_r = \mathring{\boldsymbol{\mathcal{G}}}_r + \mathring{\boldsymbol{\mathcal{P}}}_{r,s}\Delta\boldsymbol{\mathcal{C}}_{eff}\boldsymbol{\mathcal{G}}_r \tag{4.81}$$

where $\mathring{\boldsymbol{\mathcal{G}}}_r$ is the augmented reference Green's tensor whose explicit form depends on the location and nature of the seismic source.

Algorithm 4.1 THE AUGMENTED ANALOGUES $\hat{\boldsymbol{\Phi}}[\cdot]$ AND $\hat{\boldsymbol{\Phi}}^{-1}[\cdot]$ OF THE SAMPLING OPERATORS FROM **Algorithm 3.1**

Complex $\tilde{\boldsymbol{\mathcal{F}}}_\rho(N_k, 8), \boldsymbol{\mathcal{F}}_\rho(8, N_x)$
C0=(0,0)

- Initialize $\tilde{\boldsymbol{\mathcal{F}}}_\rho$

```
For j=1,8
    For k=1,N_k
        F̃_ρ(k,j)=C0
    End !k-loop
End !j-loop
```

- The upsampling mapping $\tilde{\boldsymbol{\mathcal{F}}}_\rho = \hat{\boldsymbol{\Phi}}[\boldsymbol{\mathcal{F}}_\rho]$

```
For j=1,8
    For k=1,N_x
        F̃_ρ((k-1)N_Δ+1,j)=F_ρ(j,k)
    End !k-loop
End !j-loop
```

- The downsampling mapping $\boldsymbol{\mathcal{F}}_\rho = \hat{\boldsymbol{\Phi}}^{-1}[\tilde{\boldsymbol{\mathcal{F}}}_\rho]$

```
For j=1,8
    For k=1,N_x
        F_ρ(j,k)= F̃_ρ((k-1)N_Δ+1,j)
    End !k-loop
End !j-loop
```

BOX **4.5** THE AUGMENTED UP/DOWN PLANE WAVE STATES

$$\hat{\boldsymbol{D}}_z(\mathbf{k}_x, k_y) = \tag{4.73}$$
$$\left[\ |+,P,\mathbf{k}_x,k_y)\quad |+,S,\mathbf{k}_x,k_y)\quad |+,H,\mathbf{k}_x,k_y)\quad |-,P,\mathbf{k}_x,k_y)\quad |-,S,\mathbf{k}_x,k_y)\quad |-,H,\mathbf{k}_x,k_y)\ \right]$$

is defined in terms of the partial wave states

- the $|\pm,P,\mathbf{k}_x,k_y)$ partial wave-vectors are the $\substack{\text{downgoing}\\ \text{upgoing}}$ $P-$waves

$$|\pm,P,\mathbf{k}_x,k_y) = \sqrt{\frac{\Delta k_x}{4\pi\mathring{\rho}\omega^2\mathbf{k}_{z,c}}}\begin{bmatrix} \pm i\,\mathbf{k}_{z,P} \\ i\,\mathbf{k}_x \\ i\,k_y\,\boldsymbol{I} \\ \mathring{\rho}\,\left(2\,\mathring{\beta}^2\,\mathbf{k}_x^2 + (2\,\mathring{\beta}^2\,k_y^2 - \omega^2)\boldsymbol{I}\right) \\ \mp 2\,\mathring{\rho}\,\mathring{\beta}^2\,\mathbf{k}_x\,\mathbf{k}_{z,P} \\ \mp 2\,\mathring{\rho}\,\mathring{\beta}^2\,k_y\,\mathbf{k}_{z,P} \\ -\mathbf{k}_x^2 \\ -k_y\,\mathbf{k}_x \end{bmatrix} \tag{4.74}$$

- the $|\pm,S,\mathbf{k}_x,k_y)$ partial wave-vector are the $\substack{\text{downgoing}\\ \text{upgoing}}$ quasi-$SV-$waves

$$|\pm,S,\mathbf{k}_x,k_y) = \frac{\omega}{\mathring{\beta}\,\hat{K}_S}\sqrt{\frac{\Delta k_x}{4\pi\mathring{\rho}\omega^2\mathbf{k}_{z,c}}}\begin{bmatrix} i\,\mathbf{k}_x \\ \mp i\,\mathbf{k}_{z,S} \\ \mathbf{0} \\ \mp 2\,\mathring{\rho}\,\mathring{\beta}^2\,\mathbf{k}_x\,\mathbf{k}_{z,S} \\ \mathring{\rho}\,\left((\omega^2 - \mathring{\beta}^2\,k_y^2)\boldsymbol{I} - 2\,\mathring{\beta}^2\,\mathbf{k}_x^2\right) \\ -\mathring{\rho}\,\mathring{\beta}^2\,k_y\,\mathbf{k}_x \\ \pm\mathbf{k}_x\,\mathbf{k}_{z,S} \\ \mathbf{0} \end{bmatrix} \tag{4.75}$$

- the $|\pm,H,\mathbf{k}_x,k_y)$ partial wave-vectors are the $\substack{\text{downgoing}\\ \text{upgoing}}$ quasi-$SH-$waves

$$|\pm,H,\mathbf{k}_x,k_y) = \frac{1}{\hat{K}_H}\sqrt{\frac{\Delta k_x}{4\pi\mathring{\rho}\omega^2\mathbf{k}_{z,c}}}\begin{bmatrix} \mp k_y\,\mathbf{k}_{z,H} \\ -k_y\,\mathbf{k}_x \\ \hat{K}_H^2\boldsymbol{I} \\ -2\,i\,\mathring{\rho}\,\mathring{\beta}^2\,k_y\,\mathbf{k}_{z,H}^2 \\ \mp 2\,i\,\mathring{\rho}\,\mathring{\beta}^2\,k_y\,\mathbf{k}_x\,\mathbf{k}_{z,H} \\ \pm i\,\mathring{\rho}\,\left(\omega^2 - 2\,\mathring{\beta}^2\,k_y^2\right)\,\mathbf{k}_{z,H} \\ -i\,k_y\,\mathbf{k}_x^2 \\ i\,\hat{K}_H^2\,\mathbf{k}_x \end{bmatrix} \tag{4.76}$$

where $\mathbf{k}_x$ and $\mathbf{k}_{z,c}$ were given in (3.44) and (3.45) respectively.

Box 4.6 The augmented left/right plane wave states

$$\hat{\boldsymbol{D}}_x(\mathbf{k}_z, k_y) = \tag{4.77}$$
$$\left[\ \overline{|+,P,\mathbf{k}_z,k_y)}\ \ \overline{|+,S,\mathbf{k}_z,k_y)}\ \ \overline{|+,H,\mathbf{k}_z,k_y)}\ \ \overline{|-,P,\mathbf{k}_z,k_y)}\ \ \overline{|-,S,\mathbf{k}_z,k_y)}\ \ \overline{|-,H,\mathbf{k}_z,k_y)}\ \right]$$

is defined in terms of the partial wave states

- the $8\overline{N}_k \times \overline{N}_k$ P-wave partition

$$\overline{|\pm,P,\mathbf{k}_z,k_y)} = \sqrt{\frac{\Delta k_z}{4\pi\mathring{\rho}\omega^2\mathbf{k}_{x,P}}}\begin{bmatrix} i\,\mathbf{k}_z \\ \pm i\,\mathbf{k}_{x,P} \\ i\,k_y\,\boldsymbol{I} \\ \mathring{\rho}\,\left(2\,\mathring{\beta}^2\,\mathbf{k}_{x,P}^2 + (2\,\mathring{\beta}^2\,k_y^2 - \omega^2)\boldsymbol{I}\right) \\ \mp 2\,\mathring{\rho}\,\mathring{\beta}^2\,\mathbf{k}_{x,P}\,\mathbf{k}_z \\ -2\,\mathring{\rho}\,\mathring{\beta}^2\,k_y\,\mathbf{k}_z \\ -\,\mathbf{k}_{x,P}^2 \\ \mp\,k_y\,\mathbf{k}_{x,P} \end{bmatrix} \tag{4.78}$$

- the $8\overline{N}_k \times \overline{N}_k$ SV-wave partition

$$\overline{|\pm,S,\mathbf{k}_z,k_y)} = \frac{\omega}{\mathring{\beta}\,\hat{K}_S}\sqrt{\frac{\Delta k_z}{4\pi\mathring{\rho}\omega^2\mathbf{k}_{x,S}}}\begin{bmatrix} \pm i\,\mathbf{k}_{x,S} \\ -i\,\mathbf{k}_z \\ \mathbf{0} \\ \mp 2\,\mathring{\rho}\,\mathring{\beta}^2\,\mathbf{k}_{x,S}\,\mathbf{k}_z \\ \mathring{\rho}\,\left((\omega^2 - \mathring{\beta}^2\,k_y^2)\boldsymbol{I} - 2\,\mathring{\beta}^2\,\mathbf{k}_{x,S}^2\right) \\ \mp\mathring{\rho}\,\mathring{\beta}^2\,k_y\,\mathbf{k}_{x,S} \\ \pm\,\mathbf{k}_z\mathbf{k}_{x,S} \\ \mathbf{0} \end{bmatrix} \tag{4.79}$$

- the $8\overline{N}_k \times \overline{N}_k$ SH-wave partition

$$\overline{|\pm,H,\mathbf{k}_z,k_y)} = \frac{1}{\hat{K}_H}\sqrt{\frac{\Delta k_z}{4\pi\mathring{\rho}\omega^2\mathbf{k}_{x,H}}}\begin{bmatrix} -k_y\,\mathbf{k}_z \\ \mp k_y\,\mathbf{k}_{x,H} \\ \hat{K}_H^2\boldsymbol{I} \\ -2\,i\,\mathring{\rho}\,\mathring{\beta}^2\,k_y\,\mathbf{k}_z^2 \\ \mp 2\,i\,\mathring{\rho}\,\mathring{\beta}^2\,k_y\,\mathbf{k}_{x,H}\,\mathbf{k}_z \\ i\,\mathring{\rho}\,\left(\omega^2 - 2\,\mathring{\beta}^2\,k_y^2\right)\,\mathbf{k}_z \\ -i\,k_y\,\mathbf{k}_{x,H}^2 \\ \pm i\,\hat{K}_H^2\mathbf{k}_{x,H} \end{bmatrix} \tag{4.80}$$

where $\mathbf{k}_x$ and $\mathbf{k}_{z,c}$ were given in (3.71) and (3.70) respectively.

Chapter 5

Multiple scattering formulations for a stratified half-space with superimposed heterogeneity

In this chapter, we derive representations of the *elastodynamic Lippmann-Schwinger* equation (Lippmann, & Schwinger, 1950), appropriate to a stratified halfspace with superimposed lateral heterogeneity. When we attempt to solve this equation in terms of the solutions to the component subregions (usually, either thin slabs, or small squares) of the layer stack, we are effectively developing a *multiple scattering theory*. Multiple scattering theory goes back at least as far as *Rayleigh* (1892) and has had numerous contributors since that time. Mathematically rigorous formulations were given by *Foldy* (1945) and *Lax* (1951), and their development proceeds from a pair of axioms, referred to here as the *Multiple Scattering Axioms*. Although the axioms are physically plausible, they are not self-evident.

Multiple scattering theory for a stratified half-space may be approached via two complementary routes:

- a generalization of the *numerical analysis* approach of *Chin, Hedstrom & Thigpen* (1984). While the standard method is based on a set of displacement-traction balances for the interfaces between each plane layer, our method is derived from a set of *wavefield balances*. The wavefield balance for a subregion is a statement that the scattered field is related to the incident field through the agency of the subregion

scattering operator

- the *formal scattering theory* approach, [e.g. *Kennett* (1981), *Newton* (1982)]. This approach is more abstract (and consequently, more general) than the numerical analysis route, but this is not necessarily a hindrance, as it allows us to to translate some of the theory developed in quantum mechanics to the present elastic wave theory. This approach is particularly useful for the investigation of surface waves.

We will have cause to investigate both routes to multiple scattering theory. In §5.1 we investigate the numerical analysis method, while the formal scattering method is presented in §7.2.

In chapter 3, we introduced a *secondary source lattice* comprised of $L-$layers, with N_x source points per layer. The entire lattice was embedded in free space. In this chapter, *each row of the lattice, is embedded in a separate plane layer.* To map the wavefield between the lattice points, we generalize the planewave propagator representations of chapter 3 to a stratified halfspace.

Let us assume that we have a plane-stratified halfspace spanning the depth interval $0 \leq z \leq \infty$ (Fig.5.1), with superimposed lateral heterogeneity supported on the compact depth interval $0 < z_0 \leq z \leq z_L < \infty$. For each of the plane layers in the depth interval $z_0 \leq z \leq z_L$, the material parameters $\boldsymbol{m} = (\rho, \alpha, \beta)$ are split into the sum of a homogeneous component

$$\boldsymbol{m}_i \equiv (\rho_i, \alpha_i, \beta_i) : i = 0, 1, \cdots, L+1$$

and a laterally heterogeneous component

$$\Delta\boldsymbol{m}_i(x) \equiv (\Delta\rho_i(x), \Delta\alpha_i(x), \Delta\beta_i(x)) : i = 1, \cdots, L$$

(*cf.*Fig.5.1). The laterally heterogeneous component $\Delta\boldsymbol{m}_i$ in layer-i is split into the sum of N_x small squares, i.e.

$$\Delta\boldsymbol{m}_i(x) \equiv \{\Delta\boldsymbol{m}_{i,j} : 1 \leq j \leq N_x\}$$

Note that the heterogeneity is confined to layers $i = 1, \cdots, L$, while the free surface, zone $0 = z_f \leq z \leq z_0$ and the lower halfspace $z > z_L$ remain unperturbed in their plane stratified state. Also, we have assumed that the thickness Δz of each reference layer $i = 1, \cdots, L$, is small on the scale of the incident wavelength, so that vertical variations of the material heterogeneity may be ignored. Consequently, the perturbations in the material parameters $\Delta\boldsymbol{m}_i$ superimposed on each reference layer, are functions of the offset x only,

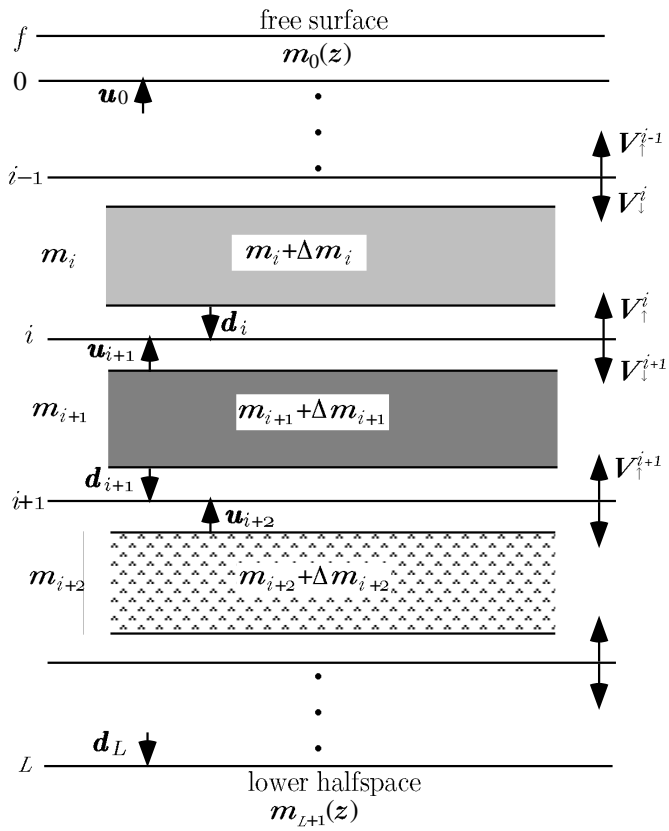

Figure 5.1: The labelling convention employed to model a stratified halfspace with superimposed heterogeneity

i.e. $\boldsymbol{\Delta m}_i(x)$. This assumption is not essential for the validity of the theory, but simplifies the development.

We are just about ready to launch into the central part of the thesis, were we attempt to *build the reflection and transmission response of a stratified medium with superimposed heterogeneity, directly in terms of the reflection and transmission operators of the component subregions*. At this juncture, we pause to define our reflection/transmission operator notation [which follows *Bostrom & Karlson* (1984)]. This notation is slightly amended from that adopted in *Kennett* (1983). Throughout this work, superscripts $(\cdot,\cdot)$ on scattering operators indicate their spatial domains of dependence. The subscript arrows, indicate the sense of the incident and scattered radiation *read right to left, in typical matrix fashion*, i.e. the right arrow indicates the sense of the incident field, and the left arrow, the sense of the scattered field. This notation is best explained by a specific example. Recalling the mathematical notation

$$(z_1, z_2) \equiv z_1 < z < z_2, \quad \text{and} \quad [z_1, z_2] \equiv z_1 \leq z \leq z_2$$

and with reference to Fig.5.2, we write:

- $\boldsymbol{S}_{\uparrow\downarrow}^{(1,2)}\left(\boldsymbol{S}_{\uparrow\downarrow}^{[1,2]}\right)$ for the reference medium downward incidence, upward reflection operator for the *open*(*closed*) depth interval $z_1 < z < z_2\,(z_1 \leq z \leq z_2)$, which is written as $\boldsymbol{R}_D^{12}$ in the notation of *Kennett* (1983)
- $\boldsymbol{S}_{\uparrow\uparrow}^{(1,2)}\left(\boldsymbol{S}_{\uparrow\uparrow}^{[1,2]}\right)$ for the reference medium upward incidence, upward transmission operator for the *open*(*closed*) depth interval $z_1 < z < z_2\,(z_1 \leq z \leq z_2)$, which is written as $\boldsymbol{T}_U^{12}$ in the notation of *Kennett* (1983)

Hence, $\boldsymbol{S}_{\uparrow\downarrow}^{(1,2)}$ maps $\downarrow-$*going* P,SV and SH plane wave components at depth z_1 to the $\uparrow-$*going* P,SV and SH plane wave components at the same level. Similarly, $\boldsymbol{S}_{\uparrow\downarrow}^{[1,2]}$ is the reference medium downward reflection operator for the *closed* depth interval $z_1 \leq z \leq z_2$. The distinction between *open* and *closed* depth intervals is important when the material properties are discontinuous at the end points of the depth interval. While the open interval excludes the discontinuous change in the material properties, the closed interval includes them. Recalling the *adopted ordering* of the wavestates in $6N_k \times 6N_k$ array $\boldsymbol{D}_z(\mathbf{k}_z, k_y)$ in (3.48),the $\boldsymbol{S}_{\uparrow\downarrow}^{(1,2)}$ matrix has the component representation

$$\boldsymbol{S}_{\uparrow\downarrow}^{(1,2)} \equiv \begin{pmatrix} (\boldsymbol{S}_{\uparrow\downarrow}^{(1,2)})_{PP} & (\boldsymbol{S}_{\uparrow\downarrow}^{(1,2)})_{PS} & (\boldsymbol{S}_{\uparrow\downarrow}^{(1,2)})_{PH} \\ (\boldsymbol{S}_{\uparrow\downarrow}^{(1,2)})_{SP} & (\boldsymbol{S}_{\uparrow\downarrow}^{(1,2)})_{SS} & (\boldsymbol{S}_{\uparrow\downarrow}^{(1,2)})_{SH} \\ (\boldsymbol{S}_{\uparrow\downarrow}^{(1,2)})_{HP} & (\boldsymbol{S}_{\uparrow\downarrow}^{(1,2)})_{HS} & (\boldsymbol{S}_{\uparrow\downarrow}^{(1,2)})_{HH} \end{pmatrix} \tag{5.1}$$

where the $N_k \times N_k$ downward incident c-wave to upward reflected c'-wave array $\left(\boldsymbol{S}^{(1,2)}_{\uparrow\downarrow}\right)_{c',c}$ is defined as

$$\left(\boldsymbol{S}^{(1,2)}_{\uparrow\downarrow}\right)_{c',c} = \mathrm{diag}\Big[\ {\scriptstyle (-,c',0)\leftarrow(+,c,0),}\quad {\scriptstyle (-,c',1)\leftarrow(+,c,1),}\quad \cdots,\quad {\scriptstyle (-,c',M-1)\leftarrow(+,c,M-1),}$$
$$\qquad {\scriptstyle (-,c',-M)\leftarrow(+,c,-M),}\quad \cdots,\quad {\scriptstyle (-,c',-1)\leftarrow(+,c,-1)}\Big] \tag{5.2}$$

The label $(+,c,j)$ indicates a downgoing (indicated by $+$) c-wave (indicated by c) with horizontal wavenumber $k_x = 2\pi j/L_x$, while $(-,c',j)$ is a upgoing (indicated by $-$) c'-wave (indicated by c') with horizontal wavenumber $k_x = 2\pi j/L_x$. The diagonal elements of (5.2) are individual plane wave components of the downward incident c-wave to the upward reflected c'-wave transition. The array is diagonal, because the medium is laterally invariant on the depth interval $z_1 < z < z_2$ so that there is no coupling between the various plane wave components. A similar interpretation may be ascribed to the other diagonal partitions of (5.1). Of course, when the depth interval $z_1 < z < z_2$ is laterally heterogeneous, due to *plane wave coupling*, each of the indicated partitions in (5.1) is generally fully populated. An explicit construction for the plane interface reflection/transmission arrays is detailed in **Box 5.1**.

There are two common exceptions to this nomenclature:

- for the interface between layers, (say interface-i) we write $\boldsymbol{S}^{i}_{\uparrow\downarrow} \equiv \boldsymbol{S}^{(i-,i+)}_{\uparrow\downarrow}$ (*see* Fig.5.2) The appended $\pm$ in $i\pm$ indicates

  $$i\pm \equiv \lim_{\epsilon\to 0+} z_i \pm \epsilon$$

- for the open depth interval $z_{i-1} < z < z_i$ spanning a single layer, we write

  $$\boldsymbol{S}^{(i-1,i)}_{\uparrow\uparrow} = \boldsymbol{S}^{(i-1,i)}_{\downarrow\downarrow} = \boldsymbol{E}_i$$

  The $3N_k \times 3N_k$ diagonal, plane-layer phase shift array is defined as

  $$\boldsymbol{E}_i = \mathrm{diag}\left[\mathrm{e}^{i\mathbf{k}_{z,P_i}(z_i-z_{i-1})},\ \mathrm{e}^{i\mathbf{k}_{z,S_i}(z_i-z_{i-1})},\ \mathrm{e}^{i\mathbf{k}_{z,H_i}(z_i-z_{i-1})}\right], \tag{5.9}$$

  with the $N_k \times N_k$ vertical wavenumber array $\mathbf{k}_{z,c_i}$ given by

  $$\mathbf{k}_{z,c_i} \equiv \mathrm{diag}\,[k_{z,c_i}(0), k_{z,c_i}(\Delta k_x),\ \cdots,$$
  $$k_{z,c_i}((M-1)\Delta k_x),\ k_{z,c_i}(-M\Delta k_x),\ \cdots,\ k_{z,c_i}(-\Delta k_x)] \tag{5.10}$$

Box 5.1 THE THE PLANE INTERFACE REFLECTION/ TRANSMISSION ARRAYS

Let us partition the $6N_k \times 6N_k$ array $\boldsymbol{D}_z(\mathbf{k}_z, k_y)$ from (3.48) into the $6N_k \times 3N_k$ array

$$\boldsymbol{B}_{\downarrow}(\mathbf{k}_x, k_y) = \left[\begin{array}{ccc} |+, P, \mathbf{k}_x, k_y\rangle & |+, S, \mathbf{k}_x, k_y\rangle & |+, H, \mathbf{k}_x, k_y\rangle \end{array}\right] \tag{5.3}$$

of downgoing partial wave states, and the $6N_k \times 3N_k$ array

$$\boldsymbol{B}_{\uparrow}(\mathbf{k}_x, k_y) = \left[\begin{array}{ccc} |-, P, \mathbf{k}_x, k_y\rangle & |-, S, \mathbf{k}_x, k_y\rangle & |-, H, \mathbf{k}_x, k_y\rangle \end{array}\right] \tag{5.4}$$

of upgoing partial wave states. The component states in (5.3) and (5.4) were defined in eqns.(3.49)–(3.51).

Let us assume that we have a plane interface problem, where the upper and lower half spaces have uniform material properties $(\rho^1, \alpha^1, \beta^1)$ and $(\rho^2, \alpha^2, \beta^2)$, respectively. We append the superscripts 1 and 2 to the arrays in (5.3) and (5.4) to indicate that (ρ, α, β) in the definitions (3.49)–(3.51) are to be replaced by $(\rho^1, \alpha^1, \beta^1)$ and $(\rho^2, \alpha^2, \beta^2)$, respectively.

The method we employ to construct the plane interface reflection/ transmission arrays is straightforward generalization of *Kennett* (1983), §**5.1**. The canonical displacement/traction balance

$$\left[\begin{array}{cc} \boldsymbol{B}^1_{\downarrow}(\mathbf{k}_x, k_y) & \boldsymbol{B}^1_{\uparrow}(\mathbf{k}_x, k_y) \end{array}\right] \left[\begin{array}{cc} \boldsymbol{I}_{3N_k} & \mathbf{0}_{3N_k} \\ \boldsymbol{S}^1_{\uparrow\downarrow} & \boldsymbol{S}^1_{\uparrow\uparrow} \end{array}\right] = \left[\begin{array}{cc} \boldsymbol{B}^2_{\downarrow}(\mathbf{k}_x, k_y) & \boldsymbol{B}^2_{\uparrow}(\mathbf{k}_x, k_y) \end{array}\right] \left[\begin{array}{cc} \boldsymbol{S}^1_{\downarrow\downarrow} & \boldsymbol{S}^1_{\downarrow\uparrow} \\ \mathbf{0}_{3N_k} & \boldsymbol{I}_{3N_k} \end{array}\right] \tag{5.5}$$

expresses the continuity of the 3− components of displacement (u_z, u_x, u_y) and 3− components of traction $(\tau_{zz}, \tau_{xz}, \tau_{yz})$ across the discontinuity of the material parameters at the plane interface. We have $3N_k$ unit amplitude downgoing planewaves states incident on the interface from the upper half space, and a further $3N_k$ unit amplitude upgoing planewaves states incident on the interface from the lower half space. The $3N_k \times 3N_k$ arrays $\boldsymbol{S}^1_{\uparrow\downarrow}$, $\boldsymbol{S}^1_{\downarrow\downarrow}$, $\boldsymbol{S}^1_{\uparrow\uparrow}$ and $\boldsymbol{S}^1_{\downarrow\uparrow}$ are the undetermined reflection and transmission arrays, whose entries are to be adjusted to balance the displacement/traction field on the two sides of the boundary.

By rearranging (5.5) so that the unknown reflection and transmission arrays are on the left side of the rearranged equations, we get

$$\left[\begin{array}{cc} \boldsymbol{B}^1_{\uparrow}(\mathbf{k}_x, k_y) & -\boldsymbol{B}^2_{\downarrow}(\mathbf{k}_x, k_y) \end{array}\right] \left[\begin{array}{cc} \boldsymbol{S}^1_{\uparrow\downarrow} & \boldsymbol{S}^1_{\uparrow\uparrow} \\ \boldsymbol{S}^1_{\downarrow\downarrow} & \boldsymbol{S}^1_{\downarrow\uparrow} \end{array}\right] = \left[\begin{array}{cc} -\boldsymbol{B}^1_{\downarrow}(\mathbf{k}_x, k_y) & \boldsymbol{B}^2_{\uparrow}(\mathbf{k}_x, k_y) \end{array}\right] \tag{5.6}$$

If we introduce the bilinear form $\langle \cdot, \cdot \rangle$ defined by

$$\langle \boldsymbol{a}, \boldsymbol{b} \rangle = \boldsymbol{a}^T \left(\boldsymbol{J}_6 \otimes \boldsymbol{I}_{N_k}\right) \boldsymbol{b}$$

for arbitrary $6N_k-$vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ then

$$\begin{aligned} &\left[\begin{array}{cc} \mathbf{0}_{3N_k} & -\left\langle \boldsymbol{B}^1_{\uparrow}(-\mathbf{k}_x, -k_y), \boldsymbol{B}^2_{\downarrow}(\mathbf{k}_x, k_y) \right\rangle \\ \left\langle \boldsymbol{B}^2_{\downarrow}(-\mathbf{k}_x, -k_y), \boldsymbol{B}^1_{\uparrow}(\mathbf{k}_x, k_y) \right\rangle & \mathbf{0}_{3N_k} \end{array}\right] \left[\begin{array}{cc} \boldsymbol{S}^1_{\uparrow\downarrow} & \boldsymbol{S}^1_{\uparrow\uparrow} \\ \boldsymbol{S}^1_{\downarrow\downarrow} & \boldsymbol{S}^1_{\downarrow\uparrow} \end{array}\right] = \\ &\left[\begin{array}{cc} -i\,\frac{\Delta k_x}{2\pi}\,\boldsymbol{I}_{3N_k} & \left\langle \boldsymbol{B}^1_{\uparrow}(-\mathbf{k}_x, -k_y), \boldsymbol{B}^2_{\uparrow}(\mathbf{k}_x, k_y) \right\rangle \\ -\left\langle \boldsymbol{B}^2_{\downarrow}(-\mathbf{k}_x, -k_y), \boldsymbol{B}^1_{\downarrow}(\mathbf{k}_x, k_y) \right\rangle & -i\,\frac{\Delta k_x}{2\pi}\,\boldsymbol{I}_{3N_k} \end{array}\right] \end{aligned} \tag{5.7}$$

Box 5.1 *continued*

$$\left\langle \boldsymbol{B}^1_{\uparrow}(-\mathbf{k}_x,-k_y), \boldsymbol{B}^1_{\downarrow}(\mathbf{k}_x,k_y)\right\rangle = i\,\frac{\Delta k_x}{2\pi}\boldsymbol{I}_{3N_K}$$

along with

$$\left\langle \boldsymbol{B}^2_{\downarrow}(-\mathbf{k}_x,-k_y), \boldsymbol{B}^2_{\uparrow}(\mathbf{k}_x,k_y)\right\rangle = -i\,\frac{\Delta k_x}{2\pi}\boldsymbol{I}_{3N_K}$$

to get the right side of (5.7).
Finally, by multiplying both sides of (5.7) by

$$\begin{bmatrix} \mathbf{0}_{3N_k} & -\left\langle \boldsymbol{B}^1_{\uparrow}(-\mathbf{k}_x,-k_y), \boldsymbol{B}^2_{\downarrow}(\mathbf{k}_x,k_y)\right\rangle \\ \left\langle \boldsymbol{B}^2_{\downarrow}(-\mathbf{k}_x,-k_y), \boldsymbol{B}^1_{\uparrow}(\mathbf{k}_x,k_y)\right\rangle & \mathbf{0}_{3N_k} \end{bmatrix}^{-1} = \begin{bmatrix} \mathbf{0}_{3N_k} & \left\langle \boldsymbol{B}^2_{\downarrow}(-\mathbf{k}_x,-k_y), \boldsymbol{B}^1_{\uparrow}(\mathbf{k}_x,k_y)\right\rangle^{-1} \\ -\left\langle \boldsymbol{B}^1_{\uparrow}(-\mathbf{k}_x,-k_y), \boldsymbol{B}^2_{\downarrow}(\mathbf{k}_x,k_y)\right\rangle^{-1} & \mathbf{0}_{3N_k} \end{bmatrix}$$

we get

$$\begin{bmatrix} \boldsymbol{S}^1_{\uparrow\downarrow} & \boldsymbol{S}^1_{\uparrow\uparrow} \\ \boldsymbol{S}^1_{\downarrow\downarrow} & \boldsymbol{S}^1_{\downarrow\uparrow} \end{bmatrix} = \begin{bmatrix} \mathbf{0}_{3N_k} & \left\langle \boldsymbol{B}^2_{\downarrow}(-\mathbf{k}_x,-k_y), \boldsymbol{B}^1_{\uparrow}(\mathbf{k}_x,k_y)\right\rangle^{-1} \\ -\left\langle \boldsymbol{B}^1_{\uparrow}(-\mathbf{k}_x,-k_y), \boldsymbol{B}^2_{\downarrow}(\mathbf{k}_x,k_y)\right\rangle^{-1} & \mathbf{0}_{3N_k} \end{bmatrix}$$
$$\times \begin{bmatrix} -i\,\frac{\Delta k_x}{2\pi}\boldsymbol{I}_{3N_k} & \left\langle \boldsymbol{B}^1_{\uparrow}(-\mathbf{k}_x,-k_y), \boldsymbol{B}^2_{\uparrow}(\mathbf{k}_x,k_y)\right\rangle \\ -\left\langle \boldsymbol{B}^2_{\downarrow}(-\mathbf{k}_x,-k_y), \boldsymbol{B}^1_{\downarrow}(\mathbf{k}_x,k_y)\right\rangle & -i\,\frac{\Delta k_x}{2\pi}\boldsymbol{I}_{3N_k} \end{bmatrix} \quad (5.8)$$

On a serial machine, (5.8) requires the evaluation of a large number of 3×3 matrix inversions. However, on a vector machine it is preferable to evaluate (5.8) in terms of its N_k-element diagonal partitions, because this approach is highly vectorizable.

[Figure: Sfig not found]

Figure 5.2: The reflection/transmission operator notation for regions and interfaces

for $N_k = 2M$ and $c_i \in \{P_i, S_i, H_i\}$, where

$$k_{z,c_i}(k_x) = \begin{cases} \sqrt{\hat{K}_{c_i}^2 \; - \; k_x^2} & |k_x| \; < \; |\hat{K}_{c_i}| \\ i\sqrt{k_x^2 \; - \; \hat{K}_{c_i}^2} & |k_x| \; \geq \; |\hat{K}_{c_i}| \end{cases} , \quad c_i \in \{P_i, S_i, H_i\}, \tag{5.11}$$

and

$$\hat{K}_{c_i} = \begin{cases} \sqrt{(\omega_+/c_i)^2 \; - \; k_y^2} & |k_y| \; < \; |\omega_+/c_i| \\ i\sqrt{k_y^2 \; - \; (\omega_+/c_i)^2} & |k_y| \; \geq \; |\omega_+/c_i| \end{cases} , \quad c_i \in \{P_i, S_i, H_i\}, \tag{5.12}$$

an implicit function of k_y.

The essence of the *standard multiple scattering formalism* [e.g. *Kennett* (1986)] is to assume that *all propagation takes place within a reference medium, with the laterally invariant properties* $\boldsymbol{m}_i$. The material variations $\boldsymbol{\Delta m}_i$ of the superimposed heterogeneity then induce scattering of the reference medium wave field. There is an important variation on the standard theory [e.g. *Wu* (1994)] that incorporates the forward transmission through the reference model and the forward scattered field from the superimposed heterogeneity directly within in the reference solution, while the reflected phases in the reference model and the back scattered field from the heterogeneity are governed by a modified Lippmann-Schwinger equation. This approach is best suited to problems where forward scattering and transmission dominates back scattering and reflection [e.g. for smooth reference models at high frequencies]. We concentrate on the standard multiple scattering model in this thesis, with details of the forward scattering model restricted to §5.1.2.

5.1 The multiple scattering equations for a stratified halfspace

Following the *numerical analysis route* of *Chin, Hedstrom & Thigpen* (1984), propagation through the plane stratified component of Fig.5.1, may be modeled via the solution of a linear system of equations that results from the enforcement of *displacement-traction* continuity across the interfacial boundaries between each plane layer. However, it is not clear how the superimposed heterogeneity would be included into the analysis, using the method

described in *Chin et al.* (1984), as the influence of the heterogeneity extends over space, and is not confined to interfacial boundaries. Instead, we will adopt a *multiple scattering* formalism, where the wavefield is governed by a linear algebraic system that results from a set of *wavefield balances*, rather then the traditional *displacement-traction* balances.

With reference to Fig.5.1, the standard multiple scattering model is composed from two distinct physical processes:

1. *interfacial wavefield balances* (Fig.5.3)

$$\begin{cases} \boldsymbol{V}^1_{\downarrow} = \boldsymbol{S}^{[f,0]}_{\downarrow\uparrow}\boldsymbol{u}_1 & \text{upward incident reflection from the free surface zone} \\ \begin{pmatrix} \boldsymbol{V}^i_{\uparrow} \\ \boldsymbol{V}^i_{\downarrow} \end{pmatrix} = \begin{pmatrix} \boldsymbol{S}^i_{\uparrow\downarrow} & \boldsymbol{S}^i_{\uparrow\uparrow} \\ \boldsymbol{S}^i_{\downarrow\downarrow} & \boldsymbol{S}^i_{\downarrow\uparrow} \end{pmatrix} \begin{pmatrix} \boldsymbol{d}_i \\ \boldsymbol{u}_i \end{pmatrix} & 1 \le i \le L-1 \\ \boldsymbol{V}^L_{\uparrow} = \boldsymbol{S}^{[L,\infty)}_{\downarrow\uparrow}\boldsymbol{d}_L & \text{downward incident reflection from the lower halfspace} \end{cases} \tag{5.13}$$

to model the transition of the wavefield between adjacent plane layers of the reference medium. An explicit construction for the interfacial reflection/transmission arrays is given in **Box 5.1**.

2. *layer wavefield balances* (Fig.5.4)

$$\begin{pmatrix} \boldsymbol{u}_i \\ \boldsymbol{d}_i \end{pmatrix} = \begin{pmatrix} \Delta\boldsymbol{S}^i_{\uparrow\downarrow} & \boldsymbol{E}_i + \Delta\boldsymbol{S}^i_{\uparrow\uparrow} \\ \boldsymbol{E}_i + \Delta\boldsymbol{S}^i_{\downarrow\downarrow} & \Delta\boldsymbol{S}^i_{\downarrow\uparrow} \end{pmatrix} \begin{pmatrix} \boldsymbol{V}^i_{\downarrow} \\ \boldsymbol{V}^i_{\uparrow} \end{pmatrix} \tag{5.14}$$

to model the plane layer phase shift due to propagation across the homogeneous reference layer, as well as the scattered field due to the superimposed heterogeneity. An explicit construction for the layer scattering $\Delta\boldsymbol{S}^i$ arrays is given in Box **5.2**.

In particular, we note that $\boldsymbol{S}^i$ effects the reflection and transmission transitions of the field for propagation between adjacent layers of the reference stratification, while $\Delta\boldsymbol{S}^i$ effects the additional shift in phase and distortion of amplitude due to scattering from the superimposed heterogeneities, *above and beyond* the phase shift (modeled by $\boldsymbol{E}_i$) for propagation across the homogeneous background. For this reason, $\Delta\boldsymbol{S}^i$ is referred to as a *transition matrix* in formal scattering theory, and not as a *scattering matrix*, although the term *residual scattering matrix* might be more informative, as it models the scattering from the material variations that are left over, after we have

[Figure: IntScat not found]

Figure 5.3: The wavefield balance for plane interface scattering.

[Figure: LayerScat not found]

Figure 5.4: The wavefield balance for a single heterogeneous layer superimposed on a homogeneous background medium. The balance involves both background propagation, effected by the phase shift $\boldsymbol{E}_i$, and the scattering processes modeled by $\boldsymbol{\Delta S}^i$.

BOX **5.2** THE LAYERSTACK SCATTERING MATRIX

We define the sideways propagator

$$\mathring{\boldsymbol{\mathcal{P}}}^{x}_{r,s,\rho} = -i\boldsymbol{I}_{N_x} \otimes \left\{ \left(\boldsymbol{I}_8 \otimes \overline{\boldsymbol{F}}(0)\right) \hat{\boldsymbol{D}}_{x,\rho}(\mathbf{k}_z, k_y) \right\} \boldsymbol{\mathcal{Q}}^{x}_{\rho\leftarrow\rho} \boldsymbol{I}_{N_x} \otimes \left\{ \hat{\boldsymbol{D}}^{T}_{x,\rho}(-\mathbf{k}_z, -k_y) \left(\boldsymbol{I}_8 \otimes \overline{\boldsymbol{F}}^{T}(0)\right) \right\} \tag{5.15}$$

for row$_\rho$ of the heterogeneity lattice, where the $8\bar{N}_k \times 6\bar{N}_k$ array $\hat{\boldsymbol{D}}_{x,\rho}$ is an augmented version of the $6\bar{N}_k \times 6\bar{N}_k$ array $\boldsymbol{D}_x$ (3.48) and is given in BOX **4.6**. The appended ρ in $\hat{\boldsymbol{D}}_{x,\rho}$ indicates that it should be constructed using the reference material parameters appropriate to row$_\rho$. The planewave propagator $\boldsymbol{\mathcal{Q}}^{x}_{\rho\leftarrow\rho}$ was given by (3.111). The projection matrix $\overline{\boldsymbol{F}}(0)$ was given by (3.72).

Using (5.15) we can sum all possible sideways multiple scattering interactions between the N_x squares of heterogeneity superimposed on layer-ρ, by introducing the $6N_x \times 6N_x$ layer transition matrix, defined as

$$\Delta\boldsymbol{\mathcal{T}}^{\rho}(k_y) = \Delta\boldsymbol{\mathcal{C}}^{\rho}_{eff}(k_y) \left[\boldsymbol{I} - \mathring{\boldsymbol{\mathcal{P}}}^{x}_{r,s,\rho} \cdot \Delta\boldsymbol{\mathcal{C}}^{\rho}_{eff}(k_y)\right]^{-1}, \quad 1 \le \rho \le L, \tag{5.16}$$

where the *effective coupling matrix* $\Delta\boldsymbol{\mathcal{C}}^{\rho}_{eff}(k_y)$ for row$_\rho$ was defined by (4.70).
Using (5.16) , we define the $6N_k \times 6N_k$ array of plane wave reflection/transmission transitions from row$_\rho$ of the heterogeneity by

$$\Delta\boldsymbol{\mathcal{S}}^{\rho}(k_y) = \begin{pmatrix} \Delta\boldsymbol{S}^{\rho}_{\uparrow\downarrow} & \Delta\boldsymbol{S}^{\rho}_{\uparrow\uparrow} \\ \Delta\boldsymbol{S}^{\rho}_{\downarrow\downarrow} & \Delta\boldsymbol{S}^{\rho}_{\downarrow\uparrow} \end{pmatrix} = -i \begin{pmatrix} \sqrt{\boldsymbol{E}_\rho} & \mathbf{0} \\ \mathbf{0} & \sqrt{\boldsymbol{E}_\rho} \end{pmatrix} \hat{\boldsymbol{D}}^{T}_{z,\rho}(-\mathbf{k}_x, -k_y) \tag{5.17}$$
$$\times \left(\boldsymbol{I}_8 \otimes \mathbf{DFT}^{T}(-\mathbf{k}_x)\right) \hat{\boldsymbol{\Phi}} \Delta\boldsymbol{T}^{\rho}(k_y) \hat{\boldsymbol{\Phi}}^{-1} \left(\boldsymbol{I}_8 \otimes \mathbf{DFT}(\mathbf{k}_x)\right) \hat{\boldsymbol{D}}_{z,\rho}(\mathbf{k}_x, k_y) \begin{pmatrix} \sqrt{\boldsymbol{E}_\rho} & \mathbf{0} \\ \mathbf{0} & \sqrt{\boldsymbol{E}_\rho} \end{pmatrix}$$

which incorporates the upsampling $\hat{\boldsymbol{\Phi}}$ and downsampling $\hat{\boldsymbol{\Phi}}^{-1}$ operators that were defined in Alg 4.1. The appended ρ in $\hat{\boldsymbol{D}}_{z,\rho}$ indicates that it should be constructed using the reference material parameters appropriate to row$_\rho$ according to its definition BOX **4.5**. Also,

$$\begin{pmatrix} \sqrt{\boldsymbol{E}_\rho} & \mathbf{0} \\ \mathbf{0} & \sqrt{\boldsymbol{E}_\rho} \end{pmatrix}$$

implements the *phase shift* of the interface variables $\boldsymbol{\mathcal{V}}_{int}$ of (5.24) from their assumed phase reference values at the edges of layer-ρ, to the centre of that layer. We recall that $\boldsymbol{E}_\rho$ was defined in (5.9).
We now assemble the blocks (5.17) for each $1 \le \rho \le L$ into the block diagonal *layer stack scattering array* $\Delta\boldsymbol{\mathcal{S}}(k_y)$ defined as

$$\Delta\boldsymbol{\mathcal{S}}(k_y) \equiv \begin{pmatrix} \Delta\boldsymbol{S}^{1}_{\uparrow\downarrow} & \Delta\boldsymbol{S}^{1}_{\uparrow\uparrow} & & \\ \Delta\boldsymbol{S}^{1}_{\downarrow\downarrow} & \Delta\boldsymbol{S}^{1}_{\downarrow\uparrow} & & \\ & & \ddots & \\ & & & \Delta\boldsymbol{S}^{L}_{\uparrow\downarrow} \quad \Delta\boldsymbol{S}^{L}_{\uparrow\uparrow} \\ & & & \Delta\boldsymbol{S}^{L}_{\downarrow\downarrow} \quad \Delta\boldsymbol{S}^{L}_{\downarrow\uparrow} \end{pmatrix} \tag{5.18}$$

Box **5.2** *continued*

For computational purposes, it will be convenient to factor the block diagonal array $\boldsymbol{\Delta\mathcal{S}}(k_y)$ of (5.18) into the product of separate block diagonal arrays. To this end, we define the block diagonal arrays:

$$\boldsymbol{\Theta}^z_{ou} \equiv \mathrm{diag}\left[\begin{pmatrix} \sqrt{\boldsymbol{E}_1} & \mathbf{0} \\ \mathbf{0} & \sqrt{\boldsymbol{E}_1} \end{pmatrix} \hat{\boldsymbol{D}}^T_{z,1}(-\mathbf{k}_x, -k_y)\left(\boldsymbol{I}_8 \otimes \mathbf{DFT}^T(-\mathbf{k}_x)\right)\hat{\boldsymbol{\Phi}}, \right.$$
$$\left. \cdots, \begin{pmatrix} \sqrt{\boldsymbol{E}_L} & \mathbf{0} \\ \mathbf{0} & \sqrt{\boldsymbol{E}_L} \end{pmatrix} \hat{\boldsymbol{D}}^T_{z,L}(-\mathbf{k}_x, -k_y)\left(\boldsymbol{I}_8 \otimes \mathbf{DFT}^T(-\mathbf{k}_x)\right)\hat{\boldsymbol{\Phi}}\right]; \tag{5.19}$$

$$\boldsymbol{\Delta\mathcal{T}}(k_y) \equiv \mathrm{diag}\left[\boldsymbol{\Delta\mathcal{T}}^1(k_y), \cdots, \boldsymbol{\Delta\mathcal{T}}^L(k_y)\right]; \tag{5.20}$$

$$\boldsymbol{\Theta}^z_{in} \equiv \mathrm{diag}\left[\hat{\boldsymbol{\Phi}}^{-1}\left(\boldsymbol{I}_8 \otimes \mathbf{DFT}(\mathbf{k}_x)\right)\hat{\boldsymbol{D}}_{z,1}(\mathbf{k}_x, k_y)\begin{pmatrix} \sqrt{\boldsymbol{E}_1} & \mathbf{0} \\ \mathbf{0} & \sqrt{\boldsymbol{E}_1} \end{pmatrix}, \right.$$
$$\left. \cdots, \hat{\boldsymbol{\Phi}}^{-1}\left(\boldsymbol{I}_8 \otimes \mathbf{DFT}(\mathbf{k}_x)\right)\hat{\boldsymbol{D}}_{z,L}(\mathbf{k}_x, k_y)\begin{pmatrix} \sqrt{\boldsymbol{E}_L} & \mathbf{0} \\ \mathbf{0} & \sqrt{\boldsymbol{E}_L} \end{pmatrix}\right]. \tag{5.21}$$

Using (5.19)–(5.21), the layer stack scattering array $\boldsymbol{\Delta\mathcal{S}}$ of (5.18) may be written as

$$\boldsymbol{\Delta\mathcal{S}}(k_y) \equiv \begin{pmatrix} \boldsymbol{\Delta S}^1_{\uparrow\downarrow} & \boldsymbol{\Delta S}^1_{\uparrow\uparrow} & & \\ \boldsymbol{\Delta S}^1_{\downarrow\downarrow} & \boldsymbol{\Delta S}^1_{\downarrow\uparrow} & & \\ & & \ddots & \\ & & \boldsymbol{\Delta S}^L_{\uparrow\downarrow} & \boldsymbol{\Delta S}^L_{\uparrow\uparrow} \\ & & \boldsymbol{\Delta S}^L_{\downarrow\downarrow} & \boldsymbol{\Delta S}^L_{\downarrow\uparrow} \end{pmatrix} \equiv -i\,\boldsymbol{\Theta}^z_{ou}\boldsymbol{\Delta\mathcal{T}}(k_y)\boldsymbol{\Theta}^z_{in} \tag{5.22}$$

removed the effect of the background model. The Δ is used to remind us, that $\Delta \boldsymbol{S}^i \to \boldsymbol{0}$ as $\Delta \boldsymbol{m}_i \to \boldsymbol{0}$.

Let us assume the following ordering of the field variables $\boldsymbol{\mathcal{V}}$:

$$\boldsymbol{\mathcal{V}} = \begin{pmatrix} \boldsymbol{\mathcal{V}}_{int} \\ \boldsymbol{\mathcal{V}}_{layer} \end{pmatrix} \tag{5.23}$$

where (Fig.5.1)

$$\boldsymbol{\mathcal{V}}_{int} = \begin{pmatrix} \boldsymbol{V}^1_{\downarrow} \\ \boldsymbol{V}^1_{\uparrow} \\ \boldsymbol{V}^2_{\downarrow} \\ \vdots \\ \boldsymbol{V}^L_{\uparrow} \end{pmatrix}, \qquad \boldsymbol{\mathcal{V}}_{layer} = \begin{pmatrix} \boldsymbol{u}_1 \\ \boldsymbol{d}_1 \\ \boldsymbol{u}_2 \\ \vdots \\ \boldsymbol{d}_L \end{pmatrix}. \tag{5.24}$$

When the balances (5.13) and (5.14) are assembled into a matrix consistent with the adopted ordering (5.23) of the field variables, we get

$$\begin{pmatrix} \boldsymbol{I}_{6N_kL} & -\boldsymbol{\mathcal{S}}_{int} \\ -\boldsymbol{\mathcal{S}}_{layer} & \boldsymbol{I}_{6N_kL} \end{pmatrix} \begin{pmatrix} \boldsymbol{\mathcal{V}}_{int} \\ \boldsymbol{\mathcal{V}}_{layer} \end{pmatrix} = \begin{pmatrix} \boldsymbol{b}_{int} \\ \boldsymbol{0}_{6N_kL\times 1} \end{pmatrix} \tag{5.25}$$

where the $(6N_kL) \times (6N_kL)$ partition $\boldsymbol{\mathcal{S}}_{int}$ is written:

$$\boldsymbol{\mathcal{S}}_{int} = \begin{pmatrix} \boldsymbol{S}^{[f,0]}_{\downarrow\uparrow} & & & & & \\ & \boldsymbol{S}^1_{\uparrow\downarrow} & \boldsymbol{S}^1_{\uparrow\uparrow} & & & \\ & \boldsymbol{S}^1_{\downarrow\downarrow} & \boldsymbol{S}^1_{\downarrow\uparrow} & & & \\ & & & \ddots & & \\ & & & \boldsymbol{S}^{L-1}_{\uparrow\downarrow} & \boldsymbol{S}^{L-1}_{\uparrow\uparrow} & \\ & & & \boldsymbol{S}^{L-1}_{\downarrow\downarrow} & \boldsymbol{S}^{L-1}_{\downarrow\uparrow} & \\ & & & & & \boldsymbol{S}^{[L,\infty)}_{\uparrow\downarrow} \end{pmatrix}; \tag{5.26}$$

and the $(6N_kL) \times (6N_kL)$ partition $\boldsymbol{\mathcal{S}}_{layer}$ is defined

$$\boldsymbol{\mathcal{S}}_{layer} = \begin{pmatrix} \Delta\boldsymbol{S}^1_{\uparrow\downarrow} & \boldsymbol{E}_1 + \Delta\boldsymbol{S}^1_{\uparrow\uparrow} & & & \\ \boldsymbol{E}_1 + \Delta\boldsymbol{S}^1_{\downarrow\downarrow} & \Delta\boldsymbol{S}^1_{\downarrow\uparrow} & & & \\ & & \ddots & & \\ & & & \Delta\boldsymbol{S}^L_{\uparrow\downarrow} & \boldsymbol{E}_L + \Delta\boldsymbol{S}^L_{\uparrow\uparrow} \\ & & & \boldsymbol{E}_L + \Delta\boldsymbol{S}^L_{\downarrow\downarrow} & \Delta\boldsymbol{S}^L_{\downarrow\uparrow} \end{pmatrix} \tag{5.27}$$

with $\boldsymbol{E}_j$ given by (5.9), for the *phase shift* across layer-j of the reference medium.

Finally, the source vector $\boldsymbol{b}_{int}$ in (5.25) depends on the location and nature of the source point. For simplicity, we restrict ourselves to problems with the source in the free surface layer $f \leq z \leq z_0$ or the lower halfspace $z_L \leq z \leq \infty$. For a source in the surface zone (Fig.5.5), we get

$$\boldsymbol{b}_{int} = \begin{pmatrix} \boldsymbol{S}_{\downarrow\downarrow}^{(s,0]} \left(\boldsymbol{I} - \boldsymbol{S}_{\downarrow\uparrow}^{[f,s)} \boldsymbol{S}_{\downarrow\uparrow}^{(s,0]} \right)^{-1} \left(\boldsymbol{\Sigma}_{\downarrow} + \boldsymbol{S}_{\downarrow\uparrow}^{[f,s)} \boldsymbol{\Sigma}_{\uparrow} \right) \\ \boldsymbol{0} \\ \vdots \end{pmatrix} \tag{5.28}$$

and for a source in the lower halfspace (Fig.5.6), we have

$$\boldsymbol{b}_{int} = \begin{pmatrix} \vdots \\ \boldsymbol{0} \\ \boldsymbol{S}_{\uparrow\uparrow}^{[L,s)} \left(\boldsymbol{I} - \boldsymbol{S}_{\uparrow\downarrow}^{(s,\infty)} \boldsymbol{S}_{\downarrow\uparrow}^{[L,s)} \right)^{-1} \left(\boldsymbol{\Sigma}_{\uparrow} + \boldsymbol{S}_{\uparrow\downarrow}^{(s,\infty)} \boldsymbol{\Sigma}_{\downarrow} \right) \end{pmatrix} \tag{5.29}$$

where the upgoing source radiation is given by $\boldsymbol{\Sigma}_{\uparrow}$, and the downgoing source radiation is given by $\boldsymbol{\Sigma}_{\downarrow}$ (*see* Box **5.3**).

Before we attempt to invert the left hand side of (5.25), we eliminate the layer variables $\boldsymbol{\mathcal{V}}_{layer}$, to get the following *Schur complement system* for the interface variables $\boldsymbol{\mathcal{V}}_{int}$

$$(\boldsymbol{I} - \boldsymbol{\mathcal{S}}_{int}\boldsymbol{\mathcal{S}}_{layer})\boldsymbol{\mathcal{V}}_{int} = \boldsymbol{b}_{int} \tag{5.31}$$

The linear system (5.31) is much too large to be to solved via Gaussian based matrix factorization because the diagonal blocks of $\boldsymbol{\mathcal{S}}_{layer}$ are fully populated $(6N_k) \times (6N_k)$ complex arrays (for the problems tested in Chapter 6, $6N_k \approx 24 \times 10^3$). In this thesis we propose to *split* the left hand side array in (5.31) into the sum of two parts. One part is then transferred to the right hand side of the equation, while the other part is numerically inverted.

5.1.1 The standard splitting and the recursive Riccati algorithm

We split the *layer* term $\boldsymbol{\mathcal{S}}_{layer}$ of (5.27) into the sum

$$\boldsymbol{\mathcal{S}}_{layer} = \boldsymbol{\mathcal{E}} + \Delta\boldsymbol{\mathcal{S}} \tag{5.32}$$

Box 5.3 Calculating the plane wave source radiation $\boldsymbol{\Sigma}_\uparrow$, $\boldsymbol{\Sigma}_\downarrow$ given a dipolar body $\boldsymbol{F}$

We assume the general form

$$\boldsymbol{F}(k_x, k_y; z) = \boldsymbol{F}_1(k_x, k_y; z)\delta(z - z_S) + \boldsymbol{F}_2(k_x, k_y; z)\delta'(z - z_S)$$

for a *dipolar point force* in the intersection of the $y = 0$, $z = z_S$ planes, which is essentially *Kennett* (1983), Eqn.(2.78).

Introducing the bilinear form $\langle \cdot, \cdot \rangle$ defined as (see also (7.11))

$$\langle \boldsymbol{b}_1, \boldsymbol{b}_2 \rangle \stackrel{\text{def}}{=} (\boldsymbol{b}_1)^T \boldsymbol{J}_6 \boldsymbol{b}_2$$

for arbitrary 6−vector fields $\boldsymbol{b}_1$ and $\boldsymbol{b}_2$, the $\substack{\text{upgoing}\\ \text{downgoing}}$ source radiation $\begin{bmatrix} \boldsymbol{\Sigma}_\uparrow(k_x, k_y) \\ \boldsymbol{\Sigma}_\downarrow(k_x, k_y) \end{bmatrix}$ at fixed horizontal wavenumber (k_x, k_y), may be witten as

$$\begin{bmatrix} \boldsymbol{\Sigma}_\uparrow(k_x, k_y) \\ \boldsymbol{\Sigma}_\downarrow(k_x, k_y) \end{bmatrix} = -i \left\langle \boldsymbol{D}_{z,S}(-k_x, -k_y), \boldsymbol{F}_1(k_x, k_y; z) + \mathring{\boldsymbol{A}}_S\,(k_x, k_y)\boldsymbol{F}_2(k_x, k_y; z) \right\rangle \tag{5.30}$$

where x_S is the horizontal offset of the source point, $\boldsymbol{D}_{z,S}(k_x, k_y)$ and $\mathring{\boldsymbol{A}}_S\,(k_x, k_y)$ were given by (3.13) and (3.7) respectively, and the appended "S" subscript indicates that they should be constructed from the reference material properties $(\mathring{\rho}, \mathring{\alpha}, \mathring{\beta})$ on the source plane $z = z_S$.

To demonstrate the application of (5.30) we find the source radiation generated by a point explosion at $(z, x, y) = (z_S, x_S, 0)$. The *volume force equivalent* of a point explosion in the (k_x, k_y, ω) domain may be written as

$$\boldsymbol{F}(k_x, k_y; z) = M(\omega) \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ ik_x \exp(-ik_x x_S) \\ ik_y \exp(-ik_x x_S) \end{bmatrix} \delta(z - z_S) + M(\omega) \begin{bmatrix} 0 \\ 0 \\ 0 \\ \exp(-ik_x x_S) \\ 0 \\ 0 \end{bmatrix} \delta'(z - z_S)$$

where $M(\omega)$ is the source spectrum. Consequently,

$$\begin{aligned} \boldsymbol{F}_1(k_x, k_y; z) + \mathring{\boldsymbol{A}}_S\,(k_x, k_y)\boldsymbol{F}_2(k_x, k_y; z) &= M(\omega) \exp(-ik_x x_S)\, \delta(z - z_S) \\ &\quad \times \left[\frac{1}{\mathring{\rho}\,\mathring{\alpha}^2},\ 0,\ 0,\ 0,\ \frac{2\,i\,k_x\,\mathring{\beta}^2}{\mathring{\alpha}^2},\ \frac{2\,i\,k_y\,\mathring{\beta}^2}{\mathring{\alpha}^2} \right]^T \end{aligned}$$

so that

$$\begin{bmatrix} \boldsymbol{\Sigma}_\uparrow(k_x, k_y) \\ \boldsymbol{\Sigma}_\downarrow(k_x, k_y) \end{bmatrix} = \frac{-i\,\omega\,M(\omega)\,\exp(-ik_x x_S)}{\mathring{\alpha}^2 \sqrt{2\,\mathring{\rho}\,k_{z,P_S}(k_x, k_y)}} \begin{bmatrix} 1, & 0, & 0, & 1, & 0, & 0 \end{bmatrix}^T$$

where $k_{z,P_S}(k_x, k_y)$ was given by (5.11).

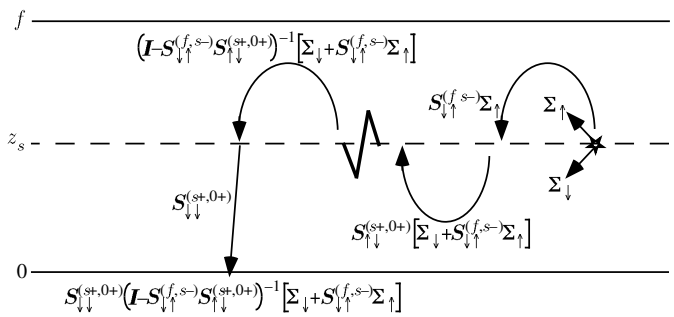

Figure 5.5: The incident field for a point source in the surface zone $f \leq z_s \leq z_0$

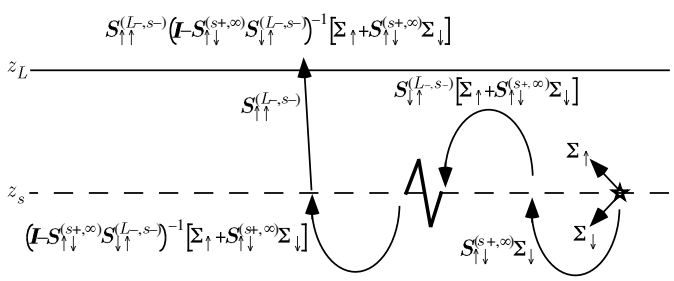

Figure 5.6: The incident field for a point source in the lower halfspace $z_s \geq z_L$

where

$$\boldsymbol{\mathcal{E}} = \begin{pmatrix} \mathbf{0} & \boldsymbol{E}_1 & & & \\ \boldsymbol{E}_1 & \mathbf{0} & & & \\ & & \ddots & & \\ & & & \mathbf{0} & \boldsymbol{E}_L \\ & & & \boldsymbol{E}_L & \mathbf{0} \end{pmatrix}; \tag{5.33}$$

is *very sparse*, but the indicated partitions of $\boldsymbol{\Delta\mathcal{S}}$ defined in (5.18) of Box **5.2** are *dense*. We now transfer the $\boldsymbol{\Delta\mathcal{S}}$ terms to the right hand side of (5.31) to get

$$(\boldsymbol{I} - \boldsymbol{\mathcal{S}}_{int}\boldsymbol{\mathcal{E}})\boldsymbol{\mathcal{V}}_{int} = \boldsymbol{b}_{int} + \boldsymbol{\mathcal{S}}_{int}\boldsymbol{\Delta\mathcal{S}}\boldsymbol{\mathcal{V}}_{int} \tag{5.34}$$

where

$$(\boldsymbol{I} - \boldsymbol{\mathcal{S}}_{int}\boldsymbol{\mathcal{E}}) = \tag{5.35}$$

$$\begin{bmatrix} \boldsymbol{I} & -\boldsymbol{S}^{[f,0]}_{\downarrow\uparrow}\boldsymbol{E}_1 & & & & & & & \\ -\boldsymbol{S}^{1}_{\uparrow\downarrow}\boldsymbol{E}_1 & \boldsymbol{I} & \mathbf{0} & -\boldsymbol{S}^{1}_{\uparrow\uparrow}\boldsymbol{E}_2 & & & & & \\ -\boldsymbol{S}^{1}_{\downarrow\downarrow}\boldsymbol{E}_1 & \mathbf{0} & \boldsymbol{I} & -\boldsymbol{S}^{1}_{\downarrow\uparrow}\boldsymbol{E}_2 & & & & & \\ & & \ddots & & & & & & \\ & & -\boldsymbol{S}^{i}_{\uparrow\downarrow}\boldsymbol{E}_i & \boldsymbol{I} & \mathbf{0} & -\boldsymbol{S}^{i}_{\uparrow\uparrow}\boldsymbol{E}_{i+1} & & & \\ & & -\boldsymbol{S}^{i}_{\downarrow\downarrow}\boldsymbol{E}_i & \mathbf{0} & \boldsymbol{I} & -\boldsymbol{S}^{i}_{\downarrow\uparrow}\boldsymbol{E}_{i+1} & & & \\ & & & & \ddots & & & & \\ & & & & -\boldsymbol{S}^{L-1}_{\uparrow\downarrow}\boldsymbol{E}_{L-1} & \boldsymbol{I} & \mathbf{0} & -\boldsymbol{S}^{L-1}_{\uparrow\uparrow}\boldsymbol{E}_L \\ & & & & -\boldsymbol{S}^{L-1}_{\downarrow\downarrow}\boldsymbol{E}_{L-1} & \mathbf{0} & \boldsymbol{I} & -\boldsymbol{S}^{L-1}_{\downarrow\uparrow}\boldsymbol{E}_L \\ & & & & & & -\boldsymbol{S}^{[L,\infty)}_{\uparrow\downarrow}\boldsymbol{E}_L & \boldsymbol{I} \end{bmatrix}$$

This splitting is best suited to a problems with weak heterogeneity superimposed on a plane stratified reference model that may change rapidly on the scale of the incident wavelength.

The left hand side array in (5.34) governs the wavefield in the plane stratified reference model, and it has a sparsity pattern that affords an efficient matrix inversion. Consequently, we invert the left hand side of (5.34) to get the *matrix analog of the elastodynamic Lippmann-Schwinger equation*

$$\boxed{\boldsymbol{\mathcal{V}}_{int} = (\boldsymbol{I} - \boldsymbol{\mathcal{S}}_{int}\boldsymbol{\mathcal{E}})^{-1}\boldsymbol{b}_{int} + (\boldsymbol{I} - \boldsymbol{\mathcal{S}}_{int}\boldsymbol{\mathcal{E}})^{-1}\boldsymbol{\mathcal{S}}_{int}\boldsymbol{\Delta\mathcal{S}}\boldsymbol{\mathcal{V}}_{int}} \tag{5.36}$$

or

$$\boldsymbol{\mathcal{V}}_{int} = \boldsymbol{\mathcal{V}}_{inc} + \boldsymbol{\mathcal{Q}}^{\partial} \Delta \boldsymbol{\mathcal{S}} \boldsymbol{\mathcal{V}}_{int}$$

where

$$\boldsymbol{\mathcal{Q}}^{\partial} \equiv (\boldsymbol{I} - \boldsymbol{\mathcal{S}}_{int} \boldsymbol{\mathcal{E}})^{-1} \boldsymbol{\mathcal{S}}_{int} \tag{5.37}$$

is the *required extension of the plane wave propagator (3.100) to a plane stratified medium*, and

$$\boldsymbol{\mathcal{V}}_{inc} \equiv (\boldsymbol{I} - \boldsymbol{\mathcal{S}}_{int} \boldsymbol{\mathcal{E}})^{-1} \boldsymbol{b}_{int} \tag{5.38}$$

is the *incident field* propagating through the reference medium from the source point.The explicit form of $\boldsymbol{\mathcal{V}}_{inc}$ depends on the source vector $\boldsymbol{b}_{int}$ of (5.28) or (5.29) depending on whether the source point is in the surface zone, or the lower halfspace respectively. For a source in the surface zone, the incident field in layer-i is given by

$$\boldsymbol{V}^i_{inc} = \begin{bmatrix} \boldsymbol{V}^i_{inc,\downarrow} \\ \boldsymbol{V}^i_{inc,\uparrow} \end{bmatrix} = \begin{bmatrix} \boldsymbol{I}_{3N_k} \\ \boldsymbol{S}^{[i,\infty)}_{\uparrow\downarrow} \boldsymbol{E}_i \end{bmatrix} \left(\boldsymbol{I}_{3N_k} - \boldsymbol{S}^{(s,i-1]}_{\downarrow\uparrow} \boldsymbol{S}^{(i-1,\infty)}_{\uparrow\downarrow} \right)^{-1} \boldsymbol{S}^{(s,i-1]}_{\downarrow\downarrow}$$
$$\times \left(\boldsymbol{I}_{3N_k} - \boldsymbol{S}^{[f,s)}_{\downarrow\uparrow} \boldsymbol{S}^{(s,\infty)}_{\uparrow\downarrow} \right)^{-1} \left[\boldsymbol{S}^{[f,s)}_{\downarrow\uparrow} \boldsymbol{\Sigma}_{\uparrow} + \boldsymbol{\Sigma}_{\downarrow} \right] \tag{5.39}$$

Alternatively, for a source in the lower halfspace, the incident field in layer-i is given by

$$\boldsymbol{V}^i_{inc} = \begin{bmatrix} \boldsymbol{V}^i_{inc,\downarrow} \\ \boldsymbol{V}^i_{inc,\uparrow} \end{bmatrix} = \begin{bmatrix} \boldsymbol{S}^{[f,i-1]}_{\downarrow\uparrow} \boldsymbol{E}_i \\ \boldsymbol{I}_{3N_k} \end{bmatrix} \left(\boldsymbol{I}_{3N_k} - \boldsymbol{S}^{[i,s)}_{\uparrow\downarrow} \boldsymbol{S}^{[f,i)}_{\downarrow\uparrow} \right)^{-1} \boldsymbol{S}^{[i,z_s)}_{\uparrow\uparrow}$$
$$\times \left(\boldsymbol{I}_{3N_k} - \boldsymbol{S}^{(s,\infty)}_{\uparrow\downarrow} \boldsymbol{S}^{[f,s)}_{\downarrow\uparrow} \right)^{-1} \left[\boldsymbol{\Sigma}_{\uparrow} + \boldsymbol{S}^{(s,\infty)}_{\uparrow\downarrow} \boldsymbol{\Sigma}_{\downarrow} \right]. \tag{5.40}$$

If we now model

$$\boldsymbol{\Sigma} \equiv \Delta \boldsymbol{\mathcal{S}} \boldsymbol{\mathcal{V}}_{int}$$

in (5.36) as a *virtual* or *secondary* source field, then the the *scattered field* $\Delta \boldsymbol{\mathcal{V}}$ is given by

$$\Delta \boldsymbol{\mathcal{V}} = (\boldsymbol{I} - \boldsymbol{\mathcal{S}}_{int} \boldsymbol{\mathcal{E}})^{-1} \boldsymbol{\mathcal{S}}_{int} \Delta \boldsymbol{\mathcal{S}} \boldsymbol{\mathcal{V}}_{int} = \boldsymbol{\mathcal{Q}}^{\partial} \boldsymbol{\Sigma}. \tag{5.41}$$

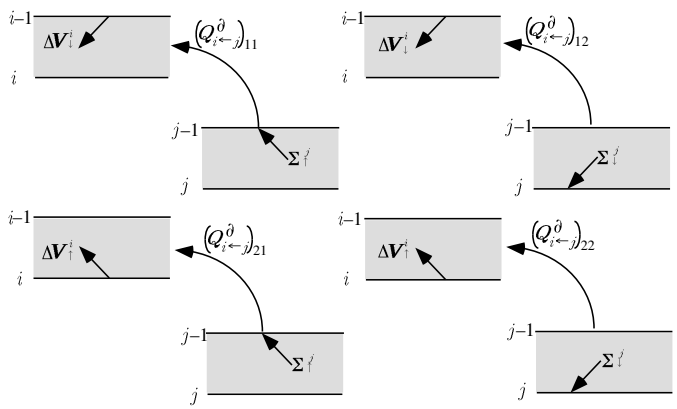

Figure 5.7: The *plane wave propagator* $\boldsymbol{Q}^{\partial}_{i\leftarrow j}$ maps the *virtual source field* $\boldsymbol{\Sigma}^j \equiv (\boldsymbol{\Sigma}^j_\uparrow, \boldsymbol{\Sigma}^j_\downarrow)^T$ from layer-j to the *scattered field* $\boldsymbol{\Delta V^i} \equiv (\boldsymbol{\Delta V^i_\downarrow}, \boldsymbol{\Delta V^i_\uparrow})^T$ in layer-i.

Partitioning the scattered field $\boldsymbol{\Delta\mathcal{V}}$ and secondary source field $\boldsymbol{\Sigma}$ in (5.41) according to

$$\boldsymbol{\Delta\mathcal{V}} = \begin{bmatrix} \begin{bmatrix} \boldsymbol{\Delta V}^1_\downarrow \\ \boldsymbol{\Delta V}^1_\uparrow \end{bmatrix} \\ \vdots \\ \begin{bmatrix} \boldsymbol{\Delta V}^L_\downarrow \\ \boldsymbol{\Delta V}^L_\uparrow \end{bmatrix} \end{bmatrix}, \quad \text{and} \quad \boldsymbol{\Sigma} = \begin{bmatrix} \begin{bmatrix} \boldsymbol{\Sigma}^1_\uparrow \\ \boldsymbol{\Sigma}^1_\downarrow \end{bmatrix} \\ \vdots \\ \begin{bmatrix} \boldsymbol{\Sigma}^L_\uparrow \\ \boldsymbol{\Sigma}^L_\downarrow \end{bmatrix} \end{bmatrix} \tag{5.42}$$

then, (5.41) may be written as (Fig.5.7)

$$\begin{bmatrix} \begin{bmatrix} \boldsymbol{\Delta V}^1_\downarrow \\ \boldsymbol{\Delta V}^1_\uparrow \end{bmatrix} \\ \vdots \\ \begin{bmatrix} \boldsymbol{\Delta V}^L_\downarrow \\ \boldsymbol{\Delta V}^L_\uparrow \end{bmatrix} \end{bmatrix} = \begin{pmatrix} \begin{bmatrix} (\boldsymbol{Q}^\partial_{1\leftarrow 1})_{11} & (\boldsymbol{Q}^\partial_{1\leftarrow 1})_{12} \\ (\boldsymbol{Q}^\partial_{1\leftarrow 1})_{21} & (\boldsymbol{Q}^\partial_{1\leftarrow 1})_{22} \end{bmatrix} & \cdots & \begin{bmatrix} (\boldsymbol{Q}^\partial_{1\leftarrow L})_{11} & (\boldsymbol{Q}^\partial_{1\leftarrow L})_{12} \\ (\boldsymbol{Q}^\partial_{1\leftarrow L})_{21} & (\boldsymbol{Q}^\partial_{1\leftarrow L})_{22} \end{bmatrix} \\ \vdots & \cdots & \vdots \\ \begin{bmatrix} (\boldsymbol{Q}^\partial_{L\leftarrow 1})_{11} & (\boldsymbol{Q}^\partial_{L\leftarrow 1})_{12} \\ (\boldsymbol{Q}^\partial_{L\leftarrow 1})_{21} & (\boldsymbol{Q}^\partial_{L\leftarrow 1})_{22} \end{bmatrix} & \cdots & \begin{bmatrix} (\boldsymbol{Q}^\partial_{L\leftarrow L})_{11} & (\boldsymbol{Q}^\partial_{L\leftarrow L})_{12} \\ (\boldsymbol{Q}^\partial_{L\leftarrow L})_{21} & (\boldsymbol{Q}^\partial_{L\leftarrow L})_{22} \end{bmatrix} \end{pmatrix} \begin{bmatrix} \begin{bmatrix} \boldsymbol{\Sigma}^1_\uparrow \\ \boldsymbol{\Sigma}^1_\downarrow \end{bmatrix} \\ \vdots \\ \begin{bmatrix} \boldsymbol{\Sigma}^L_\uparrow \\ \boldsymbol{\Sigma}^L_\downarrow \end{bmatrix} \end{bmatrix} \tag{5.43}$$

where an explicit construction for $\boldsymbol{Q}^\partial$ is presented in Box **5.4**

The *matrix Lippmann-Schwinger equation* (5.36) may now be written in the slightly more abstract form

$$\boldsymbol{\mathcal{V}}_{int} = \boldsymbol{\mathcal{V}}_{inc} + \boldsymbol{\Delta\mathcal{V}} \tag{5.44}$$

Box 5.4 An explicit representation of the $\boldsymbol{Q}^{\partial}$ propagator

$$\boldsymbol{Q}^{\partial}_{m\leftarrow l} = \left[(\boldsymbol{I} - \boldsymbol{\mathcal{S}}_{int}\boldsymbol{\mathcal{E}})^{-1}\boldsymbol{\mathcal{S}}_{int}\right]_{m,l}$$

$$\begin{cases} \begin{bmatrix} \boldsymbol{I} \\ \boldsymbol{S}^{(m-,L)}_{\uparrow\downarrow}\boldsymbol{E}_m \end{bmatrix} \left(\boldsymbol{I}_{3N} - \boldsymbol{S}^{(l-,m-1+)}_{\downarrow\uparrow}\boldsymbol{S}^{(m-1+,L)}_{\uparrow\downarrow}\right)^{-1} \boldsymbol{S}^{(l-,m-1+)}_{\downarrow\downarrow} \\ \qquad\qquad \times\left(\boldsymbol{I}_{3N} - \boldsymbol{S}^{(f,l-)}_{\downarrow\uparrow}\boldsymbol{S}^{(l-,L)}_{\uparrow\downarrow}\right)^{-1} \begin{bmatrix} \boldsymbol{E}_l\boldsymbol{S}^{(f,l-1+)}_{\downarrow\uparrow} & \boldsymbol{I} \end{bmatrix} & : m > l \\ \begin{bmatrix} \left(\boldsymbol{I} - \boldsymbol{S}^{(f,m-1)}_{\downarrow\uparrow}\boldsymbol{S}^{(m-1,L)}_{\uparrow\downarrow}\right)^{-1}\boldsymbol{S}^{(f,m-1)}_{\downarrow\uparrow} & \boldsymbol{S}^{(f,m-1)}_{\downarrow\uparrow}\left(\boldsymbol{I} - \boldsymbol{S}^{(m-1,L)}_{\uparrow\downarrow}\boldsymbol{S}^{(f,m-1)}_{\downarrow\uparrow}\right)^{-1}\boldsymbol{E}_m\boldsymbol{S}^{(m,L)}_{\uparrow\downarrow} \\ \boldsymbol{S}^{(m,L)}_{\uparrow\downarrow}\boldsymbol{E}_m\left(\boldsymbol{I} - \boldsymbol{S}^{(f,m-1)}_{\downarrow\uparrow}\boldsymbol{S}^{(m-1,L)}_{\uparrow\downarrow}\right)^{-1}\boldsymbol{S}^{(f,m-1)}_{\downarrow\uparrow} & \boldsymbol{S}^{(m,L)}_{\uparrow\downarrow}\left(\boldsymbol{I} - \boldsymbol{S}^{(f,m)}_{\downarrow\uparrow}\boldsymbol{S}^{(m,L)}_{\uparrow\downarrow}\right)^{-1} \end{bmatrix} & : m = l \\ \begin{bmatrix} \boldsymbol{S}^{(f,m-1+)}_{\downarrow\uparrow}\boldsymbol{E}_m \\ \boldsymbol{I} \end{bmatrix} \left(\boldsymbol{I}_{3N} - \boldsymbol{S}^{(m-,L+)}_{\uparrow\downarrow}\boldsymbol{S}^{(f,m-)}_{\downarrow\uparrow}\right)^{-1} \boldsymbol{S}^{(m-,l-1+)}_{\uparrow\uparrow} \\ \qquad\qquad \times\left(\boldsymbol{I}_{3N} - \boldsymbol{S}^{(l-1+,L+)}_{\uparrow\downarrow}\boldsymbol{S}^{(m-,l-1+)}_{\downarrow\uparrow}\right)^{-1} \begin{bmatrix} \boldsymbol{I} & \boldsymbol{E}_l\boldsymbol{S}^{(l-,L+)}_{\uparrow\downarrow} \end{bmatrix} & : m < l \end{cases}$$

(5.43)

so that the *total field* $\boldsymbol{\mathcal{V}}_{int}$, *phased relative to the reference medium interfaces*, is the sum of the *incident field* $\boldsymbol{\mathcal{V}}_{inc}$ and the *scattered field* $\Delta\boldsymbol{\mathcal{V}}$ from the superimposed heterogeneity.

Due to the enormous matrix dimension, we must be very careful to optimize the efficiency of the matrix operation (5.41). We now propose a *Riccati matrix factorization scheme* based on *reflection/transmission matrix recursions* that are quite similar to those of *Kennett* (1983), Chapter 7 (indeed, we make use of the Kennett's recursions in our derivation), but are more convenient for the *multi-source, multi-receiver* application, currently under investigation. To begin the derivation of the Riccati factorization, we observe that the array (5.35) is *block tri-diagonal*, i.e.

$$\boldsymbol{\mathcal{A}} \equiv (\boldsymbol{I} - \boldsymbol{\mathcal{S}}_{int}\boldsymbol{\mathcal{E}}) = \text{tridiag}\begin{cases} \boldsymbol{A}_{0,0} & \boldsymbol{A}_{0,1} & & i=0 \\ \boldsymbol{A}_{i,i-1} & \boldsymbol{A}_{i,i} & \boldsymbol{A}_{i,i+1} & i=1,\cdots,L-1 \\ & \boldsymbol{A}_{L,L-1} & \boldsymbol{A}_{L,L} & i=L \end{cases} ; \tag{5.45}$$

where the matrix partitions of (5.45) are defined:

$$\boldsymbol{A}_{0,0} = \boldsymbol{A}_{L,L} = \boldsymbol{I}_{3N_k},\ \boldsymbol{A}_{0,1} = \begin{pmatrix} -\boldsymbol{S}^{[f,0]}_{\downarrow\uparrow}\boldsymbol{E}_1 \end{pmatrix},\ \boldsymbol{A}_{L,L-1} = \begin{pmatrix} -\boldsymbol{S}^{[L,\infty)}_{\uparrow\downarrow}\boldsymbol{E}_L \end{pmatrix} \tag{5.46}$$

and the remaining elements of $\boldsymbol{\mathcal{A}}$ are $6N_k \times 6N_k$ arrays

$$\boldsymbol{A}_{i,i} = \boldsymbol{I}_{6N_k}, \boldsymbol{A}_{i,i-1} = \begin{pmatrix} \boldsymbol{0}_{3N_k} & -\boldsymbol{S}^{i}_{\uparrow\downarrow}\boldsymbol{E}_i \\ \boldsymbol{0}_{3N_k} & -\boldsymbol{S}^{i}_{\downarrow\downarrow}\boldsymbol{E}_i \end{pmatrix},\ \boldsymbol{A}_{i,i+1} = \begin{pmatrix} -\boldsymbol{S}^{i}_{\uparrow\uparrow}\boldsymbol{E}_{i+1} & \boldsymbol{0}_{3N_k} \\ -\boldsymbol{S}^{i}_{\downarrow\uparrow}\boldsymbol{E}_{i+1} & \boldsymbol{0}_{3N_k} \end{pmatrix} \tag{5.47}$$

except for the $6N_k \times 3N_k$ arrays

$$\boldsymbol{A}_{1,0} = \begin{pmatrix} -\boldsymbol{S}^{1}_{\uparrow\downarrow}\boldsymbol{E}_1 \\ -\boldsymbol{S}^{1}_{\downarrow\downarrow}\boldsymbol{E}_1 \end{pmatrix},\ \boldsymbol{A}_{L-1,L} = \begin{pmatrix} -\boldsymbol{S}^{L-1}_{\uparrow\uparrow}\boldsymbol{E}_L \\ -\boldsymbol{S}^{L-1}_{\downarrow\uparrow}\boldsymbol{E}_L \end{pmatrix}. \tag{5.48}$$

Although the matrix dimension of the reference medium reflection/transmission arrays such as $\boldsymbol{S}^{i}_{\uparrow\downarrow}$ in (5.45) is very large in general ($\approx 10^4$ in the applications considered in this work), *due to the lateral invariance of the reference medium, they are quasi-diagonal cf.*(5.1). We can exploit the block tridiagonal structure of $\boldsymbol{\mathcal{A}}$ and the *sparsity* (5.1) of the components, to calculate

the action of $\boldsymbol{\mathcal{A}}^{-1}$, on a field vector, via the factorization

$$\boldsymbol{\mathcal{A}} = \boldsymbol{\mathcal{U}}\boldsymbol{\mathcal{L}} = \begin{bmatrix} \boldsymbol{D}_0^{-1} & \boldsymbol{A}_{01} & & & \\ & \boldsymbol{D}_1^{-1} & \boldsymbol{A}_{12} & & \\ & & \ddots & & \\ & & & \boldsymbol{D}_{L-1}^{-1} & \boldsymbol{A}_{L-1,L} \\ & & & & \boldsymbol{D}_L^{-1} \end{bmatrix} \begin{bmatrix} \boldsymbol{I}_{3N_k} & & & & \\ \boldsymbol{D}_1\boldsymbol{A}_{10} & \boldsymbol{I}_{6N_k} & & & \\ & & \ddots & & \\ & & & \boldsymbol{I}_{6N_k} & \\ & & & \boldsymbol{D}_L\boldsymbol{A}_{L,L-1} & \boldsymbol{I}_{3N_k} \end{bmatrix} \tag{5.49}$$

where the diagonal blocks $\boldsymbol{D}_i$ are constructed according to

$$\begin{aligned} \boldsymbol{D}_i &= (\boldsymbol{A}_{i,i} - \boldsymbol{A}_{i,i+1}\boldsymbol{D}_{i+1}\boldsymbol{A}_{i+1,i})^{-1} \\ &= \begin{cases} \left(\boldsymbol{I}_{3N_k} - \boldsymbol{S}_{\downarrow\uparrow}^{[f,0]}\boldsymbol{S}_{\uparrow\downarrow}^{(0,\infty)}\right)^{-1} & : \quad i = 0 \\ \begin{pmatrix} \boldsymbol{I}_{3N_k} & \boldsymbol{S}_{\uparrow\uparrow}^{i}\boldsymbol{S}_{\uparrow\downarrow}^{(i,\infty)}\left(\boldsymbol{I}_{3N_k} - \boldsymbol{S}_{\downarrow\uparrow}^{i}\boldsymbol{S}_{\uparrow\downarrow}^{(i,\infty)}\right)^{-1} \\ \boldsymbol{0}_{3N_k} & \left(\boldsymbol{I}_{3N_k} - \boldsymbol{S}_{\downarrow\uparrow}^{i}\boldsymbol{S}_{\uparrow\downarrow}^{(i,\infty)}\right)^{-1} \end{pmatrix} & : \quad i = 1, \cdots, L-1 \\ \boldsymbol{I}_{3N_k} & : \quad i = L \end{cases} \end{aligned} \tag{5.50}$$

where $\boldsymbol{S}_{\uparrow\downarrow}^{(i,\infty)}$ in (5.50) is the *net downward reflection operator* for the region of the layer stack below interface-i (i.e. $z_i < z < \infty$), and is defined recursively in terms of $\boldsymbol{S}_{\uparrow\downarrow}^{(i+1,\infty)}$ as

$$\boldsymbol{S}_{\uparrow\downarrow}^{(i,\infty)} = \boldsymbol{E}_{i+1}\left[\boldsymbol{S}_{\uparrow\downarrow}^{i+1} + \boldsymbol{S}_{\uparrow\uparrow}^{i+1}\boldsymbol{S}_{\uparrow\downarrow}^{(i+1,\infty)}\left(\boldsymbol{I} - \boldsymbol{S}_{\downarrow\uparrow}^{i+1}\boldsymbol{S}_{\uparrow\downarrow}^{(i+1,\infty)}\right)\boldsymbol{S}_{\downarrow\downarrow}^{i+1}\right]\boldsymbol{E}_{i+1}. \tag{5.51}$$

Eqn. (5.51) is essentially the recursion for the net downward reflection operator [*Kennett* (1983) Eqn. (6.3)].

We now construct the scattered field $\Delta\boldsymbol{\mathcal{V}}$ in (5.41) by solving the triangular systems

$$\begin{aligned} \boldsymbol{\mathcal{U}}\boldsymbol{t} &= \boldsymbol{\mathcal{S}}_{int}\boldsymbol{\Sigma} \\ \boldsymbol{\mathcal{L}}\Delta\boldsymbol{\mathcal{V}} &= \boldsymbol{t} \end{aligned} \tag{5.52}$$

where:

Algorithm 5.1 A Riccati sweep implementation of $\Delta\boldsymbol{\mathcal{V}} = (\boldsymbol{I} - \boldsymbol{\mathcal{S}}_{int}\boldsymbol{\mathcal{E}})^{-1}\boldsymbol{\mathcal{S}}_{int}\boldsymbol{\Sigma} = \boldsymbol{\mathcal{Q}}^{\partial}\boldsymbol{\Sigma}$

- given the *virtual source field* $\boldsymbol{\Sigma}$ compute the *scattered field* $\Delta\boldsymbol{\mathcal{V}} = \boldsymbol{\mathcal{Q}}^{\partial}\boldsymbol{\Sigma}$

  (**bottom of layer stack* *)

- $\boldsymbol{R}^L_{\uparrow\downarrow} \equiv \boldsymbol{S}^{[L,\infty)}_{\uparrow\downarrow}$
- $\boldsymbol{t}^L_{\uparrow} = \boldsymbol{R}^L_{\uparrow\downarrow}\boldsymbol{\Sigma}^L_{\downarrow}$

  (**upsweep* *)

- **For** $i = L-1, \cdots, 1$

$$\boldsymbol{R}^i_{\uparrow\downarrow} = \boldsymbol{S}^i_{\uparrow\downarrow} + \boldsymbol{S}^i_{\uparrow\uparrow}\boldsymbol{E}_{i+1}\boldsymbol{R}^{i+1}_{\uparrow\downarrow}\boldsymbol{E}_{i+1}\Big(\boldsymbol{I} - \boldsymbol{S}^i_{\downarrow\uparrow}\boldsymbol{E}_{i+1}\boldsymbol{R}^{i+1}_{\uparrow\downarrow}\boldsymbol{E}_{i+1}\Big)^{-1}\boldsymbol{S}^i_{\downarrow\downarrow}$$
$$\boldsymbol{R}^i_{\uparrow\uparrow} = \boldsymbol{S}^i_{\uparrow\uparrow}\Big(\boldsymbol{I} - \boldsymbol{E}_{i+1}\boldsymbol{R}^{i+1}_{\uparrow\downarrow}\boldsymbol{E}_{i+1}\boldsymbol{S}^i_{\downarrow\uparrow}\Big)^{-1}$$
$$\boldsymbol{R}^i_{\downarrow\downarrow} = \Big(\boldsymbol{I} - \boldsymbol{S}^i_{\downarrow\uparrow}\boldsymbol{E}_{i+1}\boldsymbol{R}^{i+1}_{\uparrow\downarrow}\boldsymbol{E}_{i+1}\Big)^{-1}\boldsymbol{S}^i_{\downarrow\downarrow}$$
$$\boldsymbol{R}^i_{\downarrow\uparrow} = \Big(\boldsymbol{I} - \boldsymbol{S}^i_{\downarrow\uparrow}\boldsymbol{E}_{i+1}\boldsymbol{R}^{i+1}_{\uparrow\downarrow}\boldsymbol{E}_{i+1}\Big)^{-1}\boldsymbol{S}^i_{\downarrow\uparrow}$$
$$\begin{bmatrix} \boldsymbol{t}^i_{\uparrow} \\ \boldsymbol{t}^{i+1}_{\downarrow} \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}^i_{\uparrow\downarrow} & \boldsymbol{R}^i_{\uparrow\uparrow} \\ \boldsymbol{R}^i_{\downarrow\downarrow} & \boldsymbol{R}^i_{\downarrow\uparrow} \end{bmatrix} \begin{bmatrix} \boldsymbol{\Sigma}^i_{\downarrow} \\ \boldsymbol{\Sigma}^{i+1}_{\uparrow} + \boldsymbol{E}_{i+1}\boldsymbol{t}^{i+1}_{\uparrow} \end{bmatrix}$$

- **End**

  (**top of layer stack* *)

- $\boldsymbol{R}^0_{\downarrow\uparrow} = \Big(\boldsymbol{I} - \boldsymbol{S}^{[f,0]}_{\downarrow\uparrow}\boldsymbol{E}_1\boldsymbol{R}^1_{\uparrow\downarrow}\boldsymbol{E}_1\Big)^{-1}\boldsymbol{S}^{[f,0]}_{\downarrow\uparrow}$
- $\boldsymbol{t}^1_{\downarrow} = \boldsymbol{R}^0_{\downarrow\uparrow}(\boldsymbol{\Sigma}^1_{\uparrow} + \boldsymbol{E}_i\boldsymbol{t}^1_{\uparrow})$
- $\Delta\boldsymbol{V}^1_{\downarrow} = \boldsymbol{t}^1_{\downarrow}$

  (**downsweep* *)

- **For** $i = 1, \cdots, L-1$

$$\begin{bmatrix} \Delta\boldsymbol{V}^i_{\uparrow} \\ \Delta\boldsymbol{V}^{i+1}_{\downarrow} \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}^i_{\uparrow\downarrow} \\ \boldsymbol{R}^i_{\downarrow\downarrow} \end{bmatrix} \boldsymbol{E}_i\boldsymbol{V}^i_{\downarrow} + \begin{bmatrix} \boldsymbol{t}^i_{\uparrow} \\ \boldsymbol{t}^{i+1}_{\downarrow} \end{bmatrix}$$

- **End**

  (**bottom of layer stack* *)

- $\Delta\boldsymbol{V}^L_{\uparrow} = \boldsymbol{R}^L_{\uparrow\downarrow}\boldsymbol{E}_L\boldsymbol{V}^L_{\downarrow} + \boldsymbol{t}^L_{\uparrow}$

the right hand side vector $\boldsymbol{\mathcal{S}}_{int}\boldsymbol{\Sigma}$ in (5.52) is defined

$$\boldsymbol{\mathcal{S}}_{int}\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{S}^{[f,0]}_{\downarrow\uparrow} & & & & & \\ & \boldsymbol{S}^{1}_{\uparrow\downarrow} & \boldsymbol{S}^{1}_{\uparrow\uparrow} & & & \\ & \boldsymbol{S}^{1}_{\downarrow\downarrow} & \boldsymbol{S}^{1}_{\downarrow\uparrow} & & & \\ & & & \ddots & & \\ & & & \boldsymbol{S}^{L-1}_{\uparrow\downarrow} & \boldsymbol{S}^{L-1}_{\uparrow\uparrow} & \\ & & & \boldsymbol{S}^{L-1}_{\downarrow\downarrow} & \boldsymbol{S}^{L-1}_{\downarrow\uparrow} & \\ & & & & & \boldsymbol{S}^{[L,\infty)}_{\uparrow\downarrow} \end{bmatrix} \begin{bmatrix} \boldsymbol{\Sigma}^{1}_{\uparrow} \\ \begin{bmatrix} \boldsymbol{\Sigma}^{1}_{\downarrow} \\ \boldsymbol{\Sigma}^{2}_{\uparrow} \end{bmatrix} \\ \vdots \\ \begin{bmatrix} \boldsymbol{\Sigma}^{L-1}_{\downarrow} \\ \boldsymbol{\Sigma}^{L}_{\uparrow} \end{bmatrix} \\ \boldsymbol{\Sigma}^{L}_{\downarrow} \end{bmatrix}$$

$$= \begin{bmatrix} \boldsymbol{S}^{[f,0]}_{\downarrow\uparrow}\boldsymbol{\Sigma}^{1}_{\uparrow} \\ \begin{bmatrix} \boldsymbol{S}^{1}_{\uparrow\downarrow}\boldsymbol{\Sigma}^{1}_{\downarrow} + \boldsymbol{S}^{1}_{\uparrow\uparrow}\boldsymbol{\Sigma}^{2}_{\uparrow} \\ \boldsymbol{S}^{1}_{\downarrow\downarrow}\boldsymbol{\Sigma}^{1}_{\downarrow} + \boldsymbol{S}^{1}_{\downarrow\uparrow}\boldsymbol{\Sigma}^{2}_{\uparrow} \end{bmatrix} \\ \vdots \\ \begin{bmatrix} \boldsymbol{S}^{L-1}_{\uparrow\downarrow}\boldsymbol{\Sigma}^{L-1}_{\downarrow} + \boldsymbol{S}^{L-1}_{\uparrow\uparrow}\boldsymbol{\Sigma}^{L}_{\uparrow} \\ \boldsymbol{S}^{L-1}_{\downarrow\downarrow}\boldsymbol{\Sigma}^{L}_{\downarrow} + \boldsymbol{S}^{L-1}_{\downarrow\uparrow}\boldsymbol{\Sigma}^{L}_{\uparrow} \end{bmatrix} \\ \boldsymbol{S}^{[L,\infty)}_{\uparrow\downarrow}\boldsymbol{\Sigma}^{L}_{\downarrow} \end{bmatrix};$$

we solve the *lower triangular system*

$$\begin{bmatrix} \boldsymbol{D}_0^{-1} & \boldsymbol{A}_{01} & & & \\ & \boldsymbol{D}_1^{-1} & \boldsymbol{A}_{12} & & \\ & & \ddots & & \\ & & & \boldsymbol{D}_{L-1}^{-1} & \boldsymbol{A}_{L-1,L} \\ & & & & \boldsymbol{D}_L^{-1} \end{bmatrix} \begin{bmatrix} \boldsymbol{t}^{1}_{\downarrow} \\ \begin{bmatrix} \boldsymbol{t}^{1}_{\uparrow} \\ \boldsymbol{t}^{2}_{\downarrow} \end{bmatrix} \\ \vdots \\ \begin{bmatrix} \boldsymbol{t}^{L-1}_{\uparrow} \\ \boldsymbol{t}^{L}_{\downarrow} \end{bmatrix} \\ \boldsymbol{t}^{L}_{\uparrow} \end{bmatrix} = \begin{bmatrix} \boldsymbol{S}^{[f,0]}_{\downarrow\uparrow}\boldsymbol{\Sigma}^{1}_{\uparrow} \\ \begin{bmatrix} \boldsymbol{S}^{1}_{\uparrow\downarrow}\boldsymbol{\Sigma}^{1}_{\downarrow} + \boldsymbol{S}^{1}_{\uparrow\uparrow}\boldsymbol{\Sigma}^{2}_{\uparrow} \\ \boldsymbol{S}^{1}_{\downarrow\downarrow}\boldsymbol{\Sigma}^{1}_{\downarrow} + \boldsymbol{S}^{1}_{\downarrow\uparrow}\boldsymbol{\Sigma}^{2}_{\uparrow} \end{bmatrix} \\ \vdots \\ \begin{bmatrix} \boldsymbol{S}^{L-1}_{\uparrow\downarrow}\boldsymbol{\Sigma}^{L-1}_{\downarrow} + \boldsymbol{S}^{L-1}_{\uparrow\uparrow}\boldsymbol{\Sigma}^{L}_{\uparrow} \\ \boldsymbol{S}^{L-1}_{\downarrow\downarrow}\boldsymbol{\Sigma}^{L}_{\downarrow} + \boldsymbol{S}^{L-1}_{\downarrow\uparrow}\boldsymbol{\Sigma}^{L}_{\uparrow} \end{bmatrix} \\ \boldsymbol{S}^{[L,\infty)}_{\uparrow\downarrow}\boldsymbol{\Sigma}^{L}_{\downarrow} \end{bmatrix} \tag{5.53}$$

of (5.52), using *back substitution*, and we solve the *upper triangular* system

$$\begin{bmatrix} \boldsymbol{I}_{3N_k} & & & & \\ \boldsymbol{D}_1\boldsymbol{A}_{10} & \boldsymbol{I}_{6N_k} & & & \\ & & \ddots & & \\ & & & \boldsymbol{I}_{6N_k} & \\ & & & \boldsymbol{D}_L\boldsymbol{A}_{L,L-1} & \boldsymbol{I}_{3N_k} \end{bmatrix} \begin{bmatrix} \Delta\boldsymbol{V}^{1}_{\downarrow} \\ \begin{bmatrix} \Delta\boldsymbol{V}^{1}_{\uparrow} \\ \Delta\boldsymbol{V}^{2}_{\downarrow} \end{bmatrix} \\ \vdots \\ \begin{bmatrix} \Delta\boldsymbol{V}^{L-1}_{\uparrow} \\ \Delta\boldsymbol{V}^{L}_{\downarrow} \end{bmatrix} \\ \Delta\boldsymbol{V}^{L}_{\uparrow} \end{bmatrix} = \begin{bmatrix} \boldsymbol{t}^{1}_{\downarrow} \\ \begin{bmatrix} \boldsymbol{t}^{1}_{\uparrow} \\ \boldsymbol{t}^{2}_{\downarrow} \end{bmatrix} \\ \vdots \\ \begin{bmatrix} \boldsymbol{t}^{L-1}_{\uparrow} \\ \boldsymbol{t}^{L}_{\downarrow} \end{bmatrix} \\ \boldsymbol{t}^{L}_{\uparrow} \end{bmatrix}$$

(5.54)

of (5.52), using *forward substitution*. Also, we have introduced the auxiliary vector

$$\boldsymbol{t} = \left[\ \boldsymbol{t}^1_{\downarrow}, \quad \left[\ \boldsymbol{t}^1_{\uparrow}, \quad \boldsymbol{t}^2_{\downarrow} \ \right], \quad \cdots, \quad \left[\ \boldsymbol{t}^{L-1}_{\uparrow}, \boldsymbol{t}^L_{\downarrow} \ \right], \quad \boldsymbol{t}^L_{\uparrow} \ \right]^T \tag{5.55}$$

to save the result of the upsweep (5.53).

The sweeps (5.53) and (5.54) are summarized in Alg.5.1. This algorithm involves the manipulation of a large number of reference medium reflection and transmission arrays. These calculations may be performed in an efficient manner, by direct manipulation of the component diagonal partitions; a technique that proved highly *vectorizable*, when implemented on a *Fujitsu* VP2200 supercomputer, at the Australian National University.

5.1.2 A forward scattering based splitting and the two way marching algorithm

We now describe an alternative splitting of

$$(\boldsymbol{I} - \boldsymbol{\mathcal{S}}_{int}\boldsymbol{\mathcal{S}}_{layer})\boldsymbol{\mathcal{V}}_{int} = \boldsymbol{b}_{int},$$

from (5.31), that generalizes the *complex screen method* proposed by *Wu* (1994). This splitting is best suited to *smooth* reference media.

Modeling wave propagation in heterogeneous media is expensive. Indeed, although potentially complete, discrete grid modeling schemes for the elastic wavefield such as finite difference-elements and pseudo-spectral methods are now well developed, the sheer volume of computation needed to solve realistic seismic wave problems, precludes them from routine use. Born scattering has been proposed by several authors [e.g. *Kennett* (1986)] as a reasonably cheap wave equation based alternative to the discrete grid methods. In the Born scattering philosophy, the actual medium is cleaved into the sum of a reference (usually homogeneous or plane stratified), and a residual model. We then assume that the wavefield propagates through the reference medium, and the additional material variations of the actual medium that were neglected from the reference model serve as a continuum of independent *point scatterers*, modifying the phase and amplitude of the reference wave field via *single scattering interactions*. The consequence of this modeling strategy is that "*the phases or travel times of both direct and scattered signals are habitually mistimed*" (Coates & Charrette, 1993). The body of

this thesis has concerned itself with re-introducing the neglected *multiple scattering* interactions, to correct for this *mistiming*.

An alternative correction to Born scattering, is the *elastic wave complex screen method* (Wu, 1994), where the least time (or direct) wavefield propagates between successive forward scattering interactions at the correct wave speed. The mistiming, was originally corrected by applying ray theory (Fisk, Charrette, & McCartor, 1992), which is of course a least time theory (being derived from the *least action* principle of Fermat). However, it has been observed by *Wu* (1994), that the least time or direct arrivals may be modeled using wave theory also, by adding *the cumulative forward scattered field to the reference field, to get the wave theory direct field*. We now present a brief critique of the *elastic wave complex screen method*, because the techniques developed in this thesis may be applied to extend it into a *partially interacting wave theory*, i.e. we re-introduce the neglected backscatter. However, in contrast to the multiple scattering formulation of earlier chapters, here we employ the *thin slab Born approximation* of *Wu* (1994), so that all points at a fixed depth are *independent point scatterers, and do not reverberate amongst themselves* (i.e. *side scattering* is neglected).

The *phase screen* method is based on two approximations:

- it exploits the assumed predominance of *forward scattering* over *back scattering* at high frequencies, so that to a first approximation, the wave field may be marched through the medium, by successive forward scattering interactions. The neglect of *back scattering*, avoids the need to invert the large reverberation arrays, such as

  $$\left[\boldsymbol{I} - \boldsymbol{S}_{\downarrow\uparrow}^{(A,B)}\boldsymbol{S}_{\uparrow\downarrow}^{(B,C)}\right]^{-1}$$

  where $\boldsymbol{S}_{\downarrow\uparrow}^{(A,B)}$ is the *net upward incident reflection array* for the depth interval (A, B) above z_B and $\boldsymbol{S}_{\uparrow\downarrow}^{(B,C)}$ is the *net downward incident reflection array* for the depth interval (B, C) below z_B. Various combinations of these reverberation operators occur in the more complete theory based on *coupled plane wave reflection-transmission operator recursions* (Kennett, 1984a; Kennett, 1984b).

- the forward scattering interactions are modelled by passing the wavefield through a series of screens. The key property of the phase screens is that their effect on the wavefield is local in space (i.e. they have diagonal matrix representations in the spatial domain). As a consequence, the forward scattering interactions are efficient to effect.

In the phase screen method of *Fisk, Charrette, & McCartor* (1992), the wave field propagation between successive screens is based on ray theory. The requirement of an accurate model for P-SV wave coupling, has prompted *Wu* (1994) to reformulate the ray theory based method of *Fisk et al.* (1992), by applying a *Born forward scattering method*, where the *cumulative forward scattered field is marched forward.* In this way, the incident field has the correct *least time phase* at each screen, correcting for the *habitual mis-timing* of the traditional Born method. An alternate interpretation of the phase screen method is: *if solving the multiple scattering equations using the Born approximation, is equivalent to a single iteration of the Jacobi method, then solving the same equations with the method proposed by Wu* (1994) *is equivalent to a single iteration of the Gauss-Seidel method.* In the second stage of *Wu* (1994), a *narrow angle* or *parabolic* approximation to the forward scattering operators at each screen is proposed, so that the forward scattering operator may be approximated by a diagonal matrix in the spatial domain.

There are two potential problems with *Wu* (1994):

- the *elastic wave complex screen method* is based on the premise that the net energy scattered forward, is larger than the net energy scattered backward for a medium extending over several wavelengths of the incident field. However, in the Born scattering based phase screen model proposed by *Wu* (1994), backscattering of an isolated thin slab is enhanced with respect to forward scattering. Although back-scatter is enhanced locally, with respect to forward-scatter, it remains an open question whether back-scatter is enhanced relative to forward-scatter on a global scale. From past computational experience, we know that back scatter is more pronounced when the medium changes very rapidly, on the scale length of the incident field (which is certainly the case for a thin slab). When the individual slabs are bonded to form a thicker region, it is hoped that the back scatter will be attenuated through interference.

- by applying the *parabolic or narrow angle* approximation in the derivation of the forward scattering interaction at each screen, some of the gains in accuracy of the wave based theory over the ray based theory have been wasted.

We now show how to correct the forward scattering model of *Wu* (1994), for the neglected backscatter using wave field balances and matrix factorization techniques. We also derive single scattering operators, that are local in

the spatial domain, with out invoking a narrow angle, or *parabolic* approximation. Our approach is based on the *pseudo-spectral approximation*

$$\delta\boldsymbol{\mathcal{S}}^{\rho}(k_y) = \begin{pmatrix} \delta\boldsymbol{S}^{\rho}_{\uparrow\downarrow} & \delta\boldsymbol{S}^{\rho}_{\uparrow\uparrow} \\ \delta\boldsymbol{S}^{\rho}_{\downarrow\downarrow} & \delta\boldsymbol{S}^{\rho}_{\downarrow\uparrow} \end{pmatrix} = -i \begin{pmatrix} \sqrt{\boldsymbol{E}_\rho} & \mathbf{0} \\ \mathbf{0} & \sqrt{\boldsymbol{E}_\rho} \end{pmatrix} \hat{\boldsymbol{D}}^T_{z,\rho}(-\mathbf{k}_x, -k_y) \left(\boldsymbol{I}_8 \otimes \mathbf{DFT}^T(-\mathbf{k}_x)\right)$$
$$\times \hat{\boldsymbol{\Phi}} \Delta\boldsymbol{C}^{\rho}(k_y) \hat{\boldsymbol{\Phi}}^{-1} \left(\boldsymbol{I}_8 \otimes \mathbf{DFT}(\mathbf{k}_x)\right) \hat{\boldsymbol{D}}_{z,\rho}(\mathbf{k}_x, k_y) \begin{pmatrix} \sqrt{\boldsymbol{E}_\rho} & \mathbf{0} \\ \mathbf{0} & \sqrt{\boldsymbol{E}_\rho} \end{pmatrix} \tag{5.56}$$

of the $6N_k \times 6N_k$ array of plane wave reflection/transmission transitions from row_ρ, $1 \le \rho \le L$ of the heterogeneity. The *effective coupling matrix* $\Delta\boldsymbol{\mathcal{C}}^{\rho}_{eff}(k_y)$ for row_ρ was defined by (4.70). The approximation (5.56) neglects the side scattering component of (5.17).

The net result, is an algorithm that is competitive, with the methods developed in the earlier chapters of this thesis, particularly for the subclass of problems where the heterogeneity is embedded in a uniform host, so that the background medium does not support reverberation.

To derive the *forward scattering* splitting, we re-order the interfacial variables $\boldsymbol{\mathcal{V}}_{int}$ in (5.31) into their $\downarrow-$*going* and $\uparrow-$*going* components to get

$$\boldsymbol{\mathcal{V}}_{int} \equiv [\boldsymbol{\mathcal{V}}_{int,\downarrow}, \boldsymbol{\mathcal{V}}_{int,\uparrow}]^T \tag{5.57}$$

When we re-arrange (5.31) consistent with the new ordering (5.57) of the field variable $\boldsymbol{\mathcal{V}}_{int}$, we get

$$\begin{bmatrix} \boldsymbol{A}_{\downarrow\downarrow} & \boldsymbol{A}_{\downarrow\uparrow} \\ \boldsymbol{A}_{\uparrow\downarrow} & \boldsymbol{A}_{\uparrow\uparrow} \end{bmatrix} \begin{bmatrix} \boldsymbol{\mathcal{V}}_{int,\downarrow} \\ \boldsymbol{\mathcal{V}}_{int,\uparrow} \end{bmatrix} = \boldsymbol{b}_{int} \tag{5.58}$$

where the right hand side vector $\boldsymbol{b}_{int}$ was given previously by (5.28) for a source in the surface layer, or (5.29) for a source in the lower halfspace, while the partitions of the coefficient array in (5.58) are defined as:

$$\boldsymbol{A}_{\downarrow\downarrow} = \begin{bmatrix} \left[\boldsymbol{I} - \boldsymbol{S}^{[f,0]}_{\downarrow\uparrow}\delta\boldsymbol{S}^{1}_{\uparrow\downarrow}\right] & & & \\ -\boldsymbol{S}^{1}_{\downarrow\downarrow}\left[\boldsymbol{E}_1 + \delta\boldsymbol{S}^{1}_{\downarrow\downarrow}\right] & \left[\boldsymbol{I} - \boldsymbol{S}^{1}_{\downarrow\uparrow}\delta\boldsymbol{S}^{2}_{\uparrow\downarrow}\right] & & \\ & \ddots & \ddots & \\ & -\boldsymbol{S}^{L-1}_{\downarrow\downarrow}\left[\boldsymbol{E}_{L-1} + \delta\boldsymbol{S}^{L-1}_{\downarrow\downarrow}\right] & \left[\boldsymbol{I} - \boldsymbol{S}^{L-1}_{\downarrow\uparrow}\delta\boldsymbol{S}^{L}_{\uparrow\downarrow}\right] & \end{bmatrix};$$

$$\boldsymbol{A}_{\uparrow\uparrow} = \begin{bmatrix} \left[\boldsymbol{I} - \boldsymbol{S}^{1}_{\uparrow\downarrow}\delta\boldsymbol{S}^{1}_{\downarrow\uparrow}\right] & -\boldsymbol{S}^{1}_{\uparrow\uparrow}\left[\boldsymbol{E}_2 + \delta\boldsymbol{S}^{2}_{\uparrow\uparrow}\right] & & \\ & \left[\boldsymbol{I} - \boldsymbol{S}^{2}_{\uparrow\downarrow}\delta\boldsymbol{S}^{2}_{\downarrow\uparrow}\right] & -\boldsymbol{S}^{2}_{\uparrow\uparrow}\left[\boldsymbol{E}_3 + \delta\boldsymbol{S}^{3}_{\uparrow\uparrow}\right] & \\ & \ddots & \ddots & \\ & & & \left[\boldsymbol{I} - \boldsymbol{S}^{[L,\infty)}_{\uparrow\downarrow}\delta\boldsymbol{S}^{L}_{\downarrow\uparrow}\right] \end{bmatrix};$$

$$\boldsymbol{A}_{\uparrow\downarrow} = -\begin{bmatrix} \boldsymbol{S}^1_{\uparrow\downarrow}\left[\boldsymbol{E}_1 + \delta\boldsymbol{S}^1_{\downarrow\downarrow}\right] & \boldsymbol{S}^1_{\uparrow\uparrow}\delta\boldsymbol{S}^2_{\uparrow\downarrow} & & \\ & \boldsymbol{S}^2_{\uparrow\downarrow}\left[\boldsymbol{E}_2 + \delta\boldsymbol{S}^2_{\downarrow\downarrow}\right] & \boldsymbol{S}^2_{\uparrow\uparrow}\delta\boldsymbol{S}^3_{\uparrow\downarrow} & \\ & & \ddots & \ddots \\ & & & \boldsymbol{S}^{[L,\infty)}_{\uparrow\downarrow}\left[\boldsymbol{E}_L + \delta\boldsymbol{S}^L_{\downarrow\downarrow}\right] \end{bmatrix};$$

$$\boldsymbol{A}_{\downarrow\uparrow} = -\begin{bmatrix} \boldsymbol{S}^{[f,0]}_{\downarrow\uparrow}\left[\boldsymbol{E}_1 + \delta\boldsymbol{S}^1_{\uparrow\uparrow}\right] & & & \\ \boldsymbol{S}^1_{\downarrow\downarrow}\delta\boldsymbol{S}^1_{\downarrow\uparrow} & \boldsymbol{S}^1_{\downarrow\uparrow}\left[\boldsymbol{E}_2 + \delta\boldsymbol{S}^2_{\uparrow\uparrow}\right] & & \\ & \ddots & \ddots & \\ & & \boldsymbol{S}^{L-1}_{\downarrow\downarrow}\delta\boldsymbol{S}^{L-1}_{\downarrow\uparrow} & \boldsymbol{S}^{L-1}_{\downarrow\uparrow}\left[\boldsymbol{E}_L + \delta\boldsymbol{S}^L_{\uparrow\uparrow}\right] \end{bmatrix};$$

We now split (5.58) according to

$$\begin{bmatrix} \boldsymbol{P}^{-1}_{\downarrow\downarrow} & \boldsymbol{0} \\ \boldsymbol{A}_{\uparrow\downarrow} & \boldsymbol{P}^{-1}_{\uparrow\uparrow} \end{bmatrix} \begin{bmatrix} \boldsymbol{\mathcal{V}}_{int,\downarrow} \\ \boldsymbol{\mathcal{V}}_{int,\uparrow} \end{bmatrix} = \boldsymbol{b}_{int} + \begin{bmatrix} \boldsymbol{B}_{\downarrow\downarrow} & \boldsymbol{A}_{\downarrow\uparrow} \\ \boldsymbol{0} & \boldsymbol{B}_{\uparrow\uparrow} \end{bmatrix} \begin{bmatrix} \boldsymbol{\mathcal{V}}_{int,\downarrow} \\ \boldsymbol{\mathcal{V}}_{int,\uparrow} \end{bmatrix} \tag{5.59}$$

where:

$$\boldsymbol{P}^{-1}_{\downarrow\downarrow} = \begin{bmatrix} \boldsymbol{I} & & & \\ -\boldsymbol{S}^1_{\downarrow\downarrow}\left[\boldsymbol{E}_1 + \delta\boldsymbol{S}^1_{\downarrow\downarrow}\right] & \boldsymbol{I} & & \\ & \ddots & \ddots & \\ & & -\boldsymbol{S}^{L-1}_{\downarrow\downarrow}\left[\boldsymbol{E}_{L-1} + \delta\boldsymbol{S}^{L-1}_{\downarrow\downarrow}\right] & \boldsymbol{I} \end{bmatrix};$$

$$\boldsymbol{P}^{-1}_{\uparrow\uparrow} = \begin{bmatrix} \boldsymbol{I} & -\boldsymbol{S}^1_{\uparrow\uparrow}\left[\boldsymbol{E}_2 + \delta\boldsymbol{S}^2_{\uparrow\uparrow}\right] & & \\ & \boldsymbol{I} & -\boldsymbol{S}^2_{\uparrow\uparrow}\left[\boldsymbol{E}_3 + \delta\boldsymbol{S}^3_{\uparrow\uparrow}\right] & \\ & & \ddots & \ddots \\ & & & \boldsymbol{I} \end{bmatrix};$$

$$\boldsymbol{B}_{\downarrow\downarrow} = \begin{bmatrix} \boldsymbol{S}^{[f,0]}_{\downarrow\uparrow}\delta\boldsymbol{S}^1_{\uparrow\downarrow} & & & \\ & \boldsymbol{S}^1_{\downarrow\uparrow}\delta\boldsymbol{S}^2_{\uparrow\downarrow} & & \\ & & \ddots & \\ & & & \boldsymbol{S}^{L-1}_{\downarrow\uparrow}\delta\boldsymbol{S}^L_{\uparrow\downarrow} \end{bmatrix};$$

$$\boldsymbol{B}_{\uparrow\uparrow} = \begin{bmatrix} \boldsymbol{S}^1_{\uparrow\downarrow}\delta\boldsymbol{S}^1_{\downarrow\uparrow} & & & \\ & \boldsymbol{S}^2_{\uparrow\downarrow}\delta\boldsymbol{S}^2_{\downarrow\uparrow} & & \\ & & \ddots & \\ & & & \boldsymbol{S}^{[L,\infty)}_{\uparrow\downarrow}\delta\boldsymbol{S}^L_{\downarrow\uparrow} \end{bmatrix};$$

Multiplying (5.59) through by the inverse of the left hand side array we have

$$\boxed{\begin{bmatrix} \boldsymbol{\mathcal{V}}_{int,\downarrow} \\ \boldsymbol{\mathcal{V}}_{int,\uparrow} \end{bmatrix} = \begin{bmatrix} \boldsymbol{P}_{\downarrow\downarrow}^{-1} & \boldsymbol{0} \\ \boldsymbol{A}_{\uparrow\downarrow} & \boldsymbol{P}_{\uparrow\uparrow}^{-1} \end{bmatrix}^{-1} \boldsymbol{b}_{int} + \begin{bmatrix} \boldsymbol{P}_{\downarrow\downarrow}^{-1} & \boldsymbol{0} \\ \boldsymbol{A}_{\uparrow\downarrow} & \boldsymbol{P}_{\uparrow\uparrow}^{-1} \end{bmatrix}^{-1} \begin{bmatrix} \boldsymbol{B}_{\downarrow\downarrow} & \boldsymbol{A}_{\downarrow\uparrow} \\ \boldsymbol{0} & \boldsymbol{B}_{\uparrow\uparrow} \end{bmatrix} \begin{bmatrix} \boldsymbol{\mathcal{V}}_{int,\downarrow} \\ \boldsymbol{\mathcal{V}}_{int,\uparrow} \end{bmatrix}} \tag{5.60}$$

which is a *forward scattering* analogue of the Lippmann-Schwinger equation (5.36). Using the matrix identity

$$\begin{bmatrix} \boldsymbol{P}_{\downarrow\downarrow}^{-1} & \boldsymbol{0} \\ \boldsymbol{A}_{\uparrow\downarrow} & \boldsymbol{P}_{\uparrow\uparrow}^{-1} \end{bmatrix}^{-1} = \begin{bmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{P}_{\uparrow\uparrow} \end{bmatrix} \begin{bmatrix} \boldsymbol{I} & \boldsymbol{0} \\ -\boldsymbol{A}_{\uparrow\downarrow} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{P}_{\downarrow\downarrow} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{bmatrix}$$

in (5.60) we get

$$\begin{aligned} \begin{bmatrix} \boldsymbol{\mathcal{V}}_{int,\downarrow} \\ \boldsymbol{\mathcal{V}}_{int,\uparrow} \end{bmatrix} &= \begin{bmatrix} \boldsymbol{\mathcal{V}}_{inc,\downarrow} \\ \boldsymbol{\mathcal{V}}_{inc,\uparrow} \end{bmatrix} + \\ & \begin{bmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{P}_{\uparrow\uparrow} \end{bmatrix} \begin{bmatrix} \boldsymbol{I} & \boldsymbol{0} \\ -\boldsymbol{A}_{\uparrow\downarrow} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{P}_{\downarrow\downarrow} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{B}_{\downarrow\downarrow} & \boldsymbol{A}_{\downarrow\uparrow} \\ \boldsymbol{0} & \boldsymbol{B}_{\uparrow\uparrow} \end{bmatrix} \begin{bmatrix} \boldsymbol{\mathcal{V}}_{int,\downarrow} \\ \boldsymbol{\mathcal{V}}_{int,\uparrow} \end{bmatrix} \end{aligned} \tag{5.61}$$

where

$$\begin{bmatrix} \boldsymbol{\mathcal{V}}_{inc,\downarrow} \\ \boldsymbol{\mathcal{V}}_{inc,\uparrow} \end{bmatrix} = \begin{bmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{P}_{\uparrow\uparrow} \end{bmatrix} \begin{bmatrix} \boldsymbol{I} & \boldsymbol{0} \\ -\boldsymbol{A}_{\uparrow\downarrow} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{P}_{\downarrow\downarrow} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{I} \end{bmatrix} \boldsymbol{b}_{int}$$

is the *effective incident field*. Due to the sparsity pattern of $\boldsymbol{P}_{\downarrow\downarrow}^{-1}$ and $\boldsymbol{P}_{\uparrow\uparrow}^{-1}$, we can implement the *down marching* operation $\boldsymbol{P}_{\downarrow\downarrow}$ and the *up marching* operation $\boldsymbol{P}_{\uparrow\uparrow}$ without any matrix inversions.

5.2 A computable representation of the elastic wave Lippmann-Schwinger equation

The solution of the matrix Lippmann-Schwinger equation (5.36), may be written as

$$\boldsymbol{\mathcal{V}}_{int} = \left[\boldsymbol{I} - \boldsymbol{\mathcal{Q}}^{\partial}\Delta\boldsymbol{\mathcal{S}}\right]^{-1} \boldsymbol{\mathcal{V}}_{inc} = \boldsymbol{\mathcal{V}}_{inc} + \boldsymbol{\mathcal{Q}}^{\partial}\Delta\boldsymbol{\mathcal{S}} \left[\boldsymbol{I} - \boldsymbol{\mathcal{Q}}^{\partial}\Delta\boldsymbol{\mathcal{S}}\right]^{-1} \boldsymbol{\mathcal{V}}_{inc} \tag{5.62}$$

where the incident field $\boldsymbol{\mathcal{V}}_{inc}$ was given by (5.38), the planewave propagator for a stratified halfspace $\boldsymbol{\mathcal{Q}}^{\partial}$ was given by (5.37) and finally, the block diagonal plane wave coupling array $\Delta\boldsymbol{\mathcal{S}}$ was given in (5.18). From (5.22) we have

$$\Delta\boldsymbol{\mathcal{S}}(k_y) \;\equiv\; -i\,\boldsymbol{\Theta}_{ou}^{z}\Delta\boldsymbol{\mathcal{T}}(k_y)\boldsymbol{\Theta}_{in}^{z} \tag{5.63}$$

where $\boldsymbol{\Theta}^z_{ou}$, $\Delta\boldsymbol{\mathcal{T}}(k_y)$ and $\boldsymbol{\Theta}^z_{in}$ were defined by (5.19)–(5.21) of Box **5.2**.

Using (5.63) for $\Delta\boldsymbol{\mathcal{S}}(k_y)$ in (5.62), we get

$$\begin{aligned}\boldsymbol{\mathcal{V}}_{int} &= \boldsymbol{\mathcal{V}}_{inc} - i\,\boldsymbol{\mathcal{Q}}^{\partial}\boldsymbol{\Theta}^z_{ou}\Delta\boldsymbol{\mathcal{T}}\boldsymbol{\Theta}^z_{in}\left[\boldsymbol{I} + i\,\boldsymbol{\mathcal{Q}}^{\partial}\boldsymbol{\Theta}^z_{ou}\Delta\boldsymbol{\mathcal{T}}\boldsymbol{\Theta}^z_{in}\right]^{-1}\boldsymbol{\mathcal{V}}_{inc} \\ &= \boldsymbol{\mathcal{V}}_{inc} - i\,\boldsymbol{\mathcal{Q}}^{\partial}\boldsymbol{\Theta}^z_{ou}\Delta\boldsymbol{\mathcal{T}}\left[\boldsymbol{I} + i\,\boldsymbol{\Theta}^z_{in}\boldsymbol{\mathcal{Q}}^{\partial}\boldsymbol{\Theta}^z_{ou}\Delta\boldsymbol{\mathcal{T}}\right]^{-1}\boldsymbol{\Theta}^z_{in}\boldsymbol{\mathcal{V}}_{inc} \qquad (5.64)\end{aligned}$$

Using (5.16) for the diagonal blocks in $\Delta\boldsymbol{\mathcal{T}}$, (5.20) may be written as

$$\Delta\boldsymbol{\mathcal{T}}(k_y) = \Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\left[\boldsymbol{I} - \mathring{\boldsymbol{\mathcal{P}}}^{\,x}_{r,s}\cdot\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\right]^{-1} \qquad (5.65)$$

where $\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)$ is the coupling array for the whole layer stack, that was defined in (4.71), and $\mathring{\boldsymbol{\mathcal{P}}}^{\,x}_{r,s}$ is the block diagonal array

$$\mathring{\boldsymbol{\mathcal{P}}}^{\,x}_{r,s} = \mathrm{diag}\left[\mathring{\boldsymbol{\mathcal{P}}}^{\,x}_{r,s,1},\cdots,\mathring{\boldsymbol{\mathcal{P}}}^{\,x}_{r,s,\rho},\cdots,\mathring{\boldsymbol{\mathcal{P}}}^{\,x}_{r,s,L}\right] \qquad (5.66)$$

comprised from the sideways propagators $\mathring{\boldsymbol{\mathcal{P}}}^{\,x}_{r,s,\rho}$ of (5.15), for each of the L-layers of the reference stratification.

Using (5.65) for $\Delta\boldsymbol{\mathcal{T}}(k_y)$ in (5.64) gives

$$\begin{aligned}\boldsymbol{\mathcal{V}}_{int} = \boldsymbol{\mathcal{V}}_{inc} &- i\,\boldsymbol{\mathcal{Q}}^{\partial}\boldsymbol{\Theta}^z_{ou}\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\left[\boldsymbol{I} - \mathring{\boldsymbol{\mathcal{P}}}^{\,x}_{r,s}\cdot\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\right]^{-1} \\ &\times\left(\boldsymbol{I} + i\,\boldsymbol{\Theta}^z_{in}\boldsymbol{\mathcal{Q}}^{\partial}\boldsymbol{\Theta}^z_{ou}\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\left[\boldsymbol{I} - \mathring{\boldsymbol{\mathcal{P}}}^{\,x}_{r,s}\cdot\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\right]^{-1}\right)^{-1}\boldsymbol{\Theta}^z_{in}\boldsymbol{\mathcal{V}}_{inc} \qquad (5.67)\end{aligned}$$

We introduce the up/down propagator

$$\mathring{\boldsymbol{\mathcal{P}}}^{\,z}_{r,s} \equiv -i\,\boldsymbol{\Theta}^z_{in}\boldsymbol{\mathcal{Q}}^{\partial}\boldsymbol{\Theta}^z_{ou} \qquad (5.68)$$

where $\boldsymbol{\Theta}^z_{ou}$, and $\boldsymbol{\Theta}^z_{in}$ were defined by (5.19) and (5.21) respectively, of Box **5.2**. Using (5.68) for $\mathring{\boldsymbol{\mathcal{P}}}^{\,z}_{r,s}$ in (5.67) we get

$$\begin{aligned}\boldsymbol{\mathcal{V}}_{int} &= \boldsymbol{\mathcal{V}}_{inc} - i\,\boldsymbol{\mathcal{Q}}^{\partial}\boldsymbol{\Theta}^z_{ou}\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y) \\ &\quad\times\left(\boldsymbol{I} - \left[\mathring{\boldsymbol{\mathcal{P}}}^{\,z}_{r,s} + \mathring{\boldsymbol{\mathcal{P}}}^{\,x}_{r,s}\right]\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\right)^{-1}\boldsymbol{\Theta}^z_{in}\boldsymbol{\mathcal{V}}_{inc} \qquad (5.69)\end{aligned}$$

We assume that both the source and receiver positions are excluded from the depth interval $z_0 \le z \le z_L$ of the heterogeneity, so that (5.69) may be written as

$$\begin{aligned}\boldsymbol{\mathcal{V}}_{int,R} &= \boldsymbol{\mathcal{V}}_{inc,R} - i\,\boldsymbol{\mathcal{Q}}^{\partial}_{R}\boldsymbol{\Theta}^z_{ou}\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y) \\ &\quad\times\left(\boldsymbol{I} - \left[\mathring{\boldsymbol{\mathcal{P}}}^{\,z}_{r,s} + \mathring{\boldsymbol{\mathcal{P}}}^{\,x}_{r,s}\right]\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\right)^{-1}\boldsymbol{\Theta}^z_{in}\boldsymbol{\mathcal{V}}_{inc} \qquad (5.70)\end{aligned}$$

where $\boldsymbol{\mathcal{V}}_{int,R}$ is the vector of plane wave amplitudes in the receiver layer, $\boldsymbol{\mathcal{V}}_{inc,R}$ is the corresponding vector when there is no superimposed heterogeneity and $\boldsymbol{\mathcal{Q}}_R^{\partial}$ is the plane wave propagator mapping the secondary source radiation induced by the superimposed lateral heterogeneity from the depth interval $z_0 \leq z \leq z_L$ to the receiver layer. More explicitly, $\boldsymbol{\mathcal{Q}}_R^{\partial}$ may be written as

$$\boldsymbol{\mathcal{Q}}_R^{\partial} \equiv \left[\begin{array}{cccccc} \boldsymbol{\mathcal{Q}}_{R\leftarrow 1}^{\partial} & ,\cdots, & \boldsymbol{\mathcal{Q}}_{R\leftarrow l}^{\partial} & ,\cdots, & \boldsymbol{\mathcal{Q}}_{R\leftarrow L}^{\partial} \end{array} \right]$$

where the partition $\boldsymbol{\mathcal{Q}}_{R\leftarrow l}^{\partial}$ appropriate to layer-l is defined as

$$\begin{aligned} \boldsymbol{\mathcal{Q}}_{R\leftarrow l}^{\partial} &= \left[\begin{array}{c} \boldsymbol{S}_{\downarrow\uparrow}^{(f,R-1+)} \boldsymbol{E}_R \\ \boldsymbol{I}_{3N_k} \end{array} \right] \left(\boldsymbol{I}_{3N_k} - \boldsymbol{S}_{\uparrow\downarrow}^{(R-,L+)} \boldsymbol{S}_{\downarrow\uparrow}^{(f,R-)} \right)^{-1} \boldsymbol{S}_{\uparrow\uparrow}^{(R-,l-1+)} \\ &\quad \times \left(\boldsymbol{I}_{3N_k} - \boldsymbol{S}_{\uparrow\downarrow}^{(l-1+,L+)} \boldsymbol{S}_{\downarrow\uparrow}^{(R-,l-1+)} \right)^{-1} \left[\begin{array}{cc} \boldsymbol{I}_{3N_k} & \boldsymbol{E}_l \boldsymbol{S}_{\uparrow\downarrow}^{(l-,L+)} \end{array} \right] \end{aligned} \quad (5.71)$$

for a receiver at (z_R, x_R) in the middle of layer-R, which is above to the zone of heterogeneity $z_0 \leq z \leq z_L$.

We synthesize the displacement and traction at the receiver location from the plane waves $\boldsymbol{\mathcal{V}}_{int,R}$ at that depth by operating on both sides of (5.70) with the displacement/traction synthesis operator

$$\boldsymbol{\Psi}_{ou,R}^{z}(x_R) \equiv \left(\boldsymbol{I}_6 \otimes \boldsymbol{F}(x_R; \mathbf{k}_x) \right) \boldsymbol{D}_{z,R}(\mathbf{k}_x, k_y) \left(\begin{array}{cc} \sqrt{\boldsymbol{E}_R} & \mathbf{0} \\ \mathbf{0} & \sqrt{\boldsymbol{E}_R} \end{array} \right) \quad (5.72)$$

where $\boldsymbol{E}_R$ is the phase shift array (5.9) appropriate to the receiver layer, $\boldsymbol{F}$ was defined in (3.46) and $\boldsymbol{D}_{z,R}$ is (3.48) evaluating using the reference material properties appropriate to the receiver layer.

Operating on both sides of (5.70) with $\boldsymbol{\Psi}_{in,R}^{z}$ from (5.72) gives

$$\boxed{\boldsymbol{\mathcal{G}} = \mathring{\boldsymbol{\mathcal{G}}} + \mathring{\boldsymbol{\mathcal{P}}}_s^{z,R} \Delta\boldsymbol{\mathcal{C}}_{eff}(k_y) \left(\boldsymbol{I} - \left[\mathring{\boldsymbol{\mathcal{P}}}_{r,s}^{z} + \mathring{\boldsymbol{\mathcal{P}}}_{r,s}^{x} \right] \Delta\boldsymbol{\mathcal{C}}_{eff}(k_y) \right)^{-1} \mathring{\boldsymbol{\mathcal{G}}}_r^{z}} \quad (5.73)$$

We have defined $\mathring{\boldsymbol{\mathcal{G}}}_r^{z}$ in (5.73) as

$$\mathring{\boldsymbol{\mathcal{G}}}_r^{z} \equiv \boldsymbol{\Theta}_{in}^{z} \boldsymbol{\mathcal{V}}_{inc} \equiv \left[\begin{array}{c} \mathring{\boldsymbol{\mathcal{G}}}_r^{z,1} \\ \vdots \\ \mathring{\boldsymbol{\mathcal{G}}}_r^{z,l} \\ \vdots \\ \mathring{\boldsymbol{\mathcal{G}}}_r^{z,L} \end{array} \right]$$

with the partition $\overset{\circ}{\boldsymbol{\mathcal{G}}}{}_r^{z,l}$ for layer-l defined as

$$\overset{\circ}{\boldsymbol{\mathcal{G}}}{}_r^{z,l} = \hat{\boldsymbol{\Phi}}^{-1}\left(\boldsymbol{I}_8 \otimes \mathbf{DFT}(\mathbf{k}_x)\right)\hat{\boldsymbol{D}}_{z,l}(\mathbf{k}_x,k_y)\begin{bmatrix} \sqrt{\boldsymbol{E}_l} \\ \sqrt{\boldsymbol{E}_l}\boldsymbol{S}_{\uparrow\downarrow}^{[l,\infty)}\boldsymbol{E}_l \end{bmatrix} \tag{5.74}$$
$$\times\Big(\boldsymbol{I}_{3N_k} - \boldsymbol{S}_{\downarrow\uparrow}^{(s,l-1]}\boldsymbol{S}_{\uparrow\downarrow}^{(l-1,\infty)}\Big)^{-1}\boldsymbol{S}_{\downarrow\downarrow}^{(s,l-1]}\Big(\boldsymbol{I}_{3N_k} - \boldsymbol{S}_{\downarrow\uparrow}^{[f,s)}\boldsymbol{S}_{\uparrow\downarrow}^{(s,\infty)}\Big)^{-1}\left[\boldsymbol{S}_{\downarrow\uparrow}^{[f,s)}\boldsymbol{\Sigma}_{\uparrow} + \boldsymbol{\Sigma}_{\downarrow}\right]$$

when the source is in the surface zone, or as

$$\overset{\circ}{\boldsymbol{\mathcal{G}}}{}_r^{z,l} = \hat{\boldsymbol{\Phi}}^{-1}\left(\boldsymbol{I}_8 \otimes \mathbf{DFT}(\mathbf{k}_x)\right)\hat{\boldsymbol{D}}_{z,l}(\mathbf{k}_x,k_y)\begin{bmatrix} \sqrt{\boldsymbol{E}_l}\boldsymbol{S}_{\downarrow\uparrow}^{[f,l-1]}\boldsymbol{E}_l \\ \sqrt{\boldsymbol{E}_l} \end{bmatrix}$$
$$\times\Big(\boldsymbol{I}_{3N_k} - \boldsymbol{S}_{\uparrow\downarrow}^{[l,s)}\boldsymbol{S}_{\downarrow\uparrow}^{[f,l)}\Big)^{-1}\boldsymbol{S}_{\uparrow\uparrow}^{[l,z_s)}\Big(\boldsymbol{I}_{3N_k} - \boldsymbol{S}_{\uparrow\downarrow}^{(s,\infty)}\boldsymbol{S}_{\downarrow\uparrow}^{[f,s)}\Big)^{-1}\left[\boldsymbol{\Sigma}_{\uparrow} + \boldsymbol{S}_{\uparrow\downarrow}^{(s,\infty)}\boldsymbol{\Sigma}_{\downarrow}\right] \tag{5.75}$$

when the source is in the lower halfspace.

The $\overset{\circ}{\boldsymbol{\mathcal{P}}}{}_s^{z,R}$ array in (5.73) is defined as

$$\overset{\circ}{\boldsymbol{\mathcal{P}}}{}_s^{z,R} \equiv -i\,\boldsymbol{\Psi}_{in,R}^{z}\boldsymbol{\mathcal{Q}}_R^{\partial}\boldsymbol{\Theta}_{ou}^{z} \equiv \left[\ \left(\overset{\circ}{\boldsymbol{\mathcal{P}}}{}_s^{z,R}\right)_1\ ,\cdots,\ \left(\overset{\circ}{\boldsymbol{\mathcal{P}}}{}_s^{z,R}\right)_l\ ,\cdots,\ \left(\overset{\circ}{\boldsymbol{\mathcal{P}}}{}_s^{z,R}\right)_L\ \right] \tag{5.76}$$

where the partition $\left(\overset{\circ}{\boldsymbol{\mathcal{P}}}{}_s^{z,R}\right)_l$ appropriate to layer-l is defined as

$$\left(\overset{\circ}{\boldsymbol{\mathcal{P}}}{}_s^{z,R}\right)_l = -i\,\left(\boldsymbol{I}_6 \otimes \boldsymbol{F}(x_R;\mathbf{k}_x)\right)\boldsymbol{D}_{z,R}(\mathbf{k}_x,k_y)\begin{bmatrix} \sqrt{\boldsymbol{E}_R}\boldsymbol{S}_{\downarrow\uparrow}^{(f,R-1+)}\boldsymbol{E}_R \\ \sqrt{\boldsymbol{E}_R} \end{bmatrix}$$
$$\times\Big(\boldsymbol{I}_{3N_k} - \boldsymbol{S}_{\uparrow\downarrow}^{(R-,L+)}\boldsymbol{S}_{\downarrow\uparrow}^{(f,R-)}\Big)^{-1}\boldsymbol{S}_{\uparrow\uparrow}^{(R-,l-1+)}\Big(\boldsymbol{I}_{3N_k} - \boldsymbol{S}_{\uparrow\downarrow}^{(l-1+,L+)}\boldsymbol{S}_{\downarrow\uparrow}^{(R-,l-1+)}\Big)^{-1}$$
$$\times\begin{bmatrix} \sqrt{\boldsymbol{E}_l} & \boldsymbol{E}_l\boldsymbol{S}_{\uparrow\downarrow}^{(l-,L+)}\sqrt{\boldsymbol{E}_l} \end{bmatrix}\hat{\boldsymbol{D}}_{z,l}^{T}(-\mathbf{k}_x,-k_y)\left(\boldsymbol{I}_8 \otimes \mathbf{DFT}^T(-\mathbf{k}_x)\right)\hat{\boldsymbol{\Phi}} \tag{5.77}$$

for a receiver in the surface layers.

The array $\overset{\circ}{\boldsymbol{\mathcal{G}}}$ in (5.73) is the displacement-traction Green's tensor for the plane stratified reference medium. To find the corresponding Green's tensor $\boldsymbol{\mathcal{G}}$ for the heterogeneous medium, we must evaluate the *scattering transition*

$$\overset{\circ}{\boldsymbol{\mathcal{P}}}{}_s^{z,R}\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\Big(\boldsymbol{I} - \left[\overset{\circ}{\boldsymbol{\mathcal{P}}}{}_{r,s}^{z} + \overset{\circ}{\boldsymbol{\mathcal{P}}}{}_{r,s}^{x}\right]\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\Big)^{-1}\overset{\circ}{\boldsymbol{\mathcal{G}}}{}_r^{z}$$

in (5.73). This task is considered at length in chapter 6.

The key step in the solution algorithms of chapter 6 entails the evaluation of the matrix/vector product

$$\Big(\boldsymbol{I} - \overset{\circ}{\boldsymbol{\mathcal{P}}}_{r,s}\,\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\Big)\cdot\hat{\boldsymbol{\mathcal{B}}}$$

for an arbitrary $8LN_x-$vector $\hat{\boldsymbol{\mathcal{B}}}$, where the propagator $\overset{\circ}{\boldsymbol{\mathcal{P}}}_{r,s}$ is defined as

$$\begin{aligned}\overset{\circ}{\boldsymbol{\mathcal{P}}}_{r,s} &= \overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{z}_{r,s} + \overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{x}_{r,s} \\ &= -i\,\boldsymbol{\Theta}^{z}_{in}\boldsymbol{\mathcal{Q}}^{\partial}\boldsymbol{\Theta}^{z}_{ou} + \mathrm{diag}\left[\overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{x}_{r,s,1}, \cdots, \overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{x}_{r,s,\rho}, \cdots, \overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{x}_{r,s,L}\right] \quad (5.78)\end{aligned}$$

where we have used the definitions of $\overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{z}_{r,s}$ and $\overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{x}_{r,s}$ from (5.68) and (5.66) respectively. The block diagonal array $\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)$ from (4.71), is comprised from the coupling screens of the individual layers of the heterogeneity. Due to the block diagonal nature of $\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)$ the evaluation of

$$\hat{\boldsymbol{\mathcal{F}}} = \Delta\boldsymbol{\mathcal{C}}_{eff}(k_y) \cdot \hat{\boldsymbol{\mathcal{B}}}$$

is both straightforward to code, and efficient to implement.

In algorithm **5.2** a pseudo-code for the implementation of $\overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{x}_{r,s}\hat{\boldsymbol{\mathcal{F}}}$ is presented, while the pseudo-code for $\overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{z}_{r,s}\hat{\boldsymbol{\mathcal{F}}}$ is given in algorithm **5.3**, where $\hat{\boldsymbol{\mathcal{F}}} = \Delta\boldsymbol{\mathcal{C}}_{eff}(k_y) \cdot \hat{\boldsymbol{\mathcal{B}}}$.

Algorithm 5.2 IMPLEMENT THE SIDEWAYS SWEEPS $\hat{\boldsymbol{\mathcal{B}}} = \overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{x}_{r,s}\hat{\boldsymbol{\mathcal{F}}}$ WITHIN EACH ROW OF THE HETEROGENEITY LATTICE

Given that:

- the $3\bar{N}_k \times 3\bar{N}_k$ diagonal array $\mathbf{E}_{x,\rho}$, is the horizontal phase shift array $\mathbf{E}_x$ of (3.113)
- the planewave propagator $\boldsymbol{\mathcal{Q}}^{x}_{\rho\leftarrow\rho}$ is given by (3.111)
- the $8\bar{N}_k \times 6\bar{N}_k$ array $\hat{\boldsymbol{D}}_{x,\rho}(\mathbf{k}_z, k_y)$ was given by (4.77)
- the $\bar{N}_k-$element row vector $\overline{\boldsymbol{F}}(0)$ was given by (3.72)

The appended subscript ρ indicates that the array should be constructed using the reference material parameters appropriate for layer-ρ.
We evaluate

$$\hat{\boldsymbol{\mathcal{B}}} = \text{diag}\left[\overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{x}_{r,s,1}, \cdots, \overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{x}_{r,s,\rho}, \cdots, \overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{x}_{r,s,L}\right]\hat{\boldsymbol{\mathcal{F}}}$$

where $\hat{\boldsymbol{\mathcal{F}}}$ is an arbitrary $8LN_x-$vector, $\overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{x}_{r,s,\rho}$ was defined in (5.15) as

$$\overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{x}_{r,s,\rho} =$$
$$-i\boldsymbol{I}_{N_x} \otimes \left\{\left(\boldsymbol{I}_8 \otimes \overline{\boldsymbol{F}}(0)\right)\hat{\boldsymbol{D}}_{x,\rho}(\mathbf{k}_z, k_y)\right\}\boldsymbol{\mathcal{Q}}^{x}_{\rho\leftarrow\rho}\boldsymbol{I}_{N_x} \otimes \left\{\hat{\boldsymbol{D}}^{T}_{x,\rho}(-\mathbf{k}_z, -k_y)\ \left(\boldsymbol{I}_8 \otimes \overline{\boldsymbol{F}}^{T}(0)\right)\right\}$$

- (**downsweep* *)
- **For** $\rho = 1, \cdots, L$

  For $\chi = 1, \cdots, N_x$

  $$\begin{bmatrix} \boldsymbol{\Sigma}^{(\rho,\chi)}_{\leftarrow} \\ \boldsymbol{\Sigma}^{(\rho,\chi)}_{\rightarrow} \end{bmatrix} = -i\,\hat{\boldsymbol{D}}^{T}_{x,\rho}(-\mathbf{k}_z, -k_y)\ \left(\boldsymbol{I}_8 \otimes \overline{\boldsymbol{F}}^{T}(0)\right)\hat{\boldsymbol{f}}_{N_x(\rho-1)+\chi}$$

  End $!\chi-$loop

- (* *right sweep* *)

  $\Delta\boldsymbol{V}^{(\rho,1)}_{\rightarrow} = \mathbf{0}$

  For $\chi = 2, \cdots, N_x$

  $\Delta\boldsymbol{V}^{(\rho,\chi)}_{\rightarrow} = \mathbf{E}_{x,\rho}\left(\boldsymbol{\Sigma}^{(\rho,\chi-1)}_{\rightarrow} + \Delta\boldsymbol{V}^{(\rho,\chi-1)}_{\rightarrow}\right)$

  End $!\chi-$loop

- (* *left sweep* *)

  $\Delta\boldsymbol{V}^{(\rho,N_x)}_{\leftarrow} = \mathbf{0}$

  For $\chi = Nx - 1, \cdots, 1$

  $\Delta\boldsymbol{V}^{(\rho,\chi)}_{\leftarrow} = \mathbf{E}_{x,\rho}\left(\boldsymbol{\Sigma}^{(\rho,\chi+1)}_{\leftarrow} + \Delta\boldsymbol{V}^{(\rho,\chi+1)}_{\leftarrow}\right)$

  End $!\chi-$loop

Algorithm 5.3 IMPLEMENT THE UP/DOWN SWEEPS $\hat{\boldsymbol{\mathcal{B}}} = \overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{z}_{r,s}\hat{\boldsymbol{\mathcal{F}}}$ BETWEEN ROWS OF THE HETEROGENEITY LATTICE

Given:

- the $3N_k \times 3N_k$ diagonal, vertical phase shift arrays $\boldsymbol{E}_\rho$, for each $1 \leq \rho \leq L$, that were defined in (5.9)
- the $8N_k \times 6N_k$ arrays $\hat{\boldsymbol{D}}_{z,\rho}(\mathbf{k}_z, k_y)$, for each $1 \leq \rho \leq L$, that were given by (4.73)
- the $N_k \times N_k$, DFT array $\mathbf{DFT}$ that was given by (3.95)
- the upsampling and downsampling operators $\hat{\boldsymbol{\Phi}}[\cdot]$ and $\hat{\boldsymbol{\Phi}}^{-1}[\cdot]$ respectively, that are implemented using Alg.**4.1**
- the interfacial reflection/transmission arrays

$$\begin{bmatrix} \boldsymbol{S}^{\rho}_{\uparrow\downarrow} & \boldsymbol{S}^{\rho}_{\uparrow\uparrow} \\ \boldsymbol{S}^{\rho}_{\downarrow\downarrow} & \boldsymbol{S}^{\rho}_{\downarrow\uparrow} \end{bmatrix}$$

  constructed in BOX **5.1**, for each interface-ρ with $1 \leq \rho \leq L-1$.
- the net downward incident reflection response $\boldsymbol{S}^{[L,\infty)}_{\uparrow\downarrow}$ of the lower halfspace
- the net upward incident reflection response $\boldsymbol{S}^{[f,0]}_{\downarrow\uparrow}$ of the free surface zone

we evaluate

$$\hat{\boldsymbol{\mathcal{B}}} \equiv \overset{\circ}{\boldsymbol{\mathcal{P}}}{}^{z}_{r,s} \cdot \hat{\boldsymbol{\mathcal{F}}} \equiv -i\, \boldsymbol{\Theta}^{z}_{in}\boldsymbol{\mathcal{Q}}^{\partial}\boldsymbol{\Theta}^{z}_{ou} \cdot \hat{\boldsymbol{\mathcal{F}}}$$

where:

- $\hat{\boldsymbol{\mathcal{F}}}$ is an arbitrary $8LN_x-$vector with the $8N_x-$subvector components $\hat{\boldsymbol{\mathcal{F}}}_\rho$ for row$_\rho$ of heterogeneity, for each $1 \leq \rho \leq L$. In turn, $\hat{\boldsymbol{\mathcal{F}}}_\rho$ is comprised from the $8-$vectors, $\boldsymbol{f}_{N_x(\rho-1)+\chi}$ for each $1 \leq \chi \leq N_x$
- the plane wave propagator $\boldsymbol{\mathcal{Q}}^{\partial}$ was given by (5.37) as

$$\boldsymbol{\mathcal{Q}}^{\partial} \equiv (\boldsymbol{I} - \boldsymbol{\mathcal{S}}_{int}\boldsymbol{\mathcal{E}})^{-1}\boldsymbol{\mathcal{S}}_{int};$$

- $\boldsymbol{\Theta}^{z}_{ou}$ was defined in (5.19) as

$$\boldsymbol{\Theta}^{z}_{ou} \equiv \mathrm{diag}\left[\begin{pmatrix} \sqrt{\boldsymbol{E}_1} & \mathbf{0} \\ \mathbf{0} & \sqrt{\boldsymbol{E}_1} \end{pmatrix} \hat{\boldsymbol{D}}^{T}_{z,1}(-\mathbf{k}_x, -k_y)\left(\boldsymbol{I}_8 \otimes \mathbf{DFT}^{T}(-\mathbf{k}_x)\right)\hat{\boldsymbol{\Phi}}, \right.$$
$$\left. \cdots, \begin{pmatrix} \sqrt{\boldsymbol{E}_L} & \mathbf{0} \\ \mathbf{0} & \sqrt{\boldsymbol{E}_L} \end{pmatrix} \hat{\boldsymbol{D}}^{T}_{z,L}(-\mathbf{k}_x, -k_y)\left(\boldsymbol{I}_8 \otimes \mathbf{DFT}^{T}(-\mathbf{k}_x)\right)\hat{\boldsymbol{\Phi}}\right];$$

- $\boldsymbol{\Theta}^{z}_{in}$ was defined in (5.21) as

Algorithm 5.3 *continued*

(**upsweep* *)

- **For** $\rho = L-1, \cdots, 1$

$$\boldsymbol{R}^{\rho}_{\uparrow\downarrow} = \boldsymbol{S}^{\rho}_{\uparrow\downarrow} + \boldsymbol{S}^{\rho}_{\uparrow\uparrow}\boldsymbol{E}_{\rho+1}\boldsymbol{R}^{\rho+1}_{\uparrow\downarrow}\boldsymbol{E}_{\rho+1}\Big(\boldsymbol{I} - \boldsymbol{S}^{\rho}_{\downarrow\uparrow}\boldsymbol{E}_{\rho+1}\boldsymbol{R}^{\rho+1}_{\uparrow\downarrow}\boldsymbol{E}_{\rho+1}\Big)^{-1}\boldsymbol{S}^{\rho}_{\downarrow\downarrow}$$

$$\boldsymbol{R}^{\rho}_{\uparrow\uparrow} = \boldsymbol{S}^{\rho}_{\uparrow\uparrow}\Big(\boldsymbol{I} - \boldsymbol{E}_{\rho+1}\boldsymbol{R}^{\rho+1}_{\uparrow\downarrow}\boldsymbol{E}_{\rho+1}\boldsymbol{S}^{\rho}_{\downarrow\uparrow}\Big)^{-1}$$

$$\boldsymbol{R}^{\rho}_{\downarrow\downarrow} = \Big(\boldsymbol{I} - \boldsymbol{S}^{\rho}_{\downarrow\uparrow}\boldsymbol{E}_{\rho+1}\boldsymbol{R}^{\rho+1}_{\uparrow\downarrow}\boldsymbol{E}_{\rho+1}\Big)^{-1}\boldsymbol{S}^{\rho}_{\downarrow\downarrow}$$

$$\boldsymbol{R}^{\rho}_{\downarrow\uparrow} = \Big(\boldsymbol{I} - \boldsymbol{S}^{\rho}_{\downarrow\uparrow}\boldsymbol{E}_{\rho+1}\boldsymbol{R}^{\rho+1}_{\uparrow\downarrow}\boldsymbol{E}_{\rho+1}\Big)^{-1}\boldsymbol{S}^{\rho}_{\downarrow\uparrow}$$

$$\begin{bmatrix} \boldsymbol{\Sigma}^{\rho}_{\uparrow} \\ \boldsymbol{\Sigma}^{\rho}_{\downarrow} \end{bmatrix} = -i \begin{pmatrix} \sqrt{\boldsymbol{E}_{\rho}} & \boldsymbol{0} \\ \boldsymbol{0} & \sqrt{\boldsymbol{E}_{\rho}} \end{pmatrix} \hat{\boldsymbol{D}}^{T}_{z,\rho}(-\mathbf{k}_x, -k_y)\left(\boldsymbol{I}_8 \otimes \mathbf{DFT}^{T}(-\mathbf{k}_x)\right)\hat{\boldsymbol{\Phi}}\left[\boldsymbol{\mathcal{F}}_{\rho}\right]$$

$$\begin{bmatrix} \boldsymbol{t}^{\rho}_{\uparrow} \\ \boldsymbol{t}^{\rho+1}_{\downarrow} \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}^{\rho}_{\uparrow\downarrow} & \boldsymbol{R}^{\rho}_{\uparrow\uparrow} \\ \boldsymbol{R}^{\rho}_{\downarrow\downarrow} & \boldsymbol{R}^{\rho}_{\downarrow\uparrow} \end{bmatrix} \begin{bmatrix} \boldsymbol{\Sigma}^{\rho}_{\downarrow} \\ \boldsymbol{\Sigma}^{\rho+1}_{\uparrow} + \boldsymbol{E}_{\rho+1}\boldsymbol{t}^{\rho+1}_{\uparrow} \end{bmatrix}$$

- **End**

(**top of layer stack* *)

- $\boldsymbol{R}^{0}_{\downarrow\uparrow} = \Big(\boldsymbol{I} - \boldsymbol{S}^{[f,0]}_{\downarrow\uparrow}\boldsymbol{E}_1\boldsymbol{R}^{1}_{\uparrow\downarrow}\boldsymbol{E}_1\Big)^{-1}\boldsymbol{S}^{[f,0]}_{\downarrow\uparrow}$
- $\boldsymbol{t}^{1}_{\downarrow} = \boldsymbol{R}^{0}_{\downarrow\uparrow}(\boldsymbol{\Sigma}^{1}_{\uparrow} + \boldsymbol{E}_{\rho}\boldsymbol{t}^{1}_{\uparrow})$
- $\Delta\boldsymbol{V}^{1}_{\downarrow} = \boldsymbol{t}^{1}_{\downarrow}$

(**downsweep* *)

- **For** $\rho = 1, \cdots, L-1$

$$\begin{bmatrix} \Delta\boldsymbol{V}^{\rho}_{\uparrow} \\ \Delta\boldsymbol{V}^{\rho+1}_{\downarrow} \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}^{\rho}_{\uparrow\downarrow} \\ \boldsymbol{R}^{\rho}_{\downarrow\downarrow} \end{bmatrix} \boldsymbol{E}_{\rho}\boldsymbol{V}^{\rho}_{\downarrow} + \begin{bmatrix} \boldsymbol{t}^{\rho}_{\uparrow} \\ \boldsymbol{t}^{\rho+1}_{\downarrow} \end{bmatrix}$$

$$\boldsymbol{\mathcal{B}}_{\rho} = \hat{\boldsymbol{\Phi}}^{-1}\left[\left(\boldsymbol{I}_8 \otimes \mathbf{DFT}(\mathbf{k}_x)\right)\hat{\boldsymbol{D}}_{z,\rho}(\mathbf{k}_x, k_y)\begin{pmatrix} \sqrt{\boldsymbol{E}_{\rho}}\Delta\boldsymbol{V}^{\rho}_{\downarrow} \\ \sqrt{\boldsymbol{E}_{\rho}}\Delta\boldsymbol{V}^{\rho}_{\uparrow} \end{pmatrix}\right]$$

- **End**

(**bottom of layer stack* *)

- $\Delta\boldsymbol{V}^{L}_{\uparrow} = \boldsymbol{R}^{L}_{\uparrow\downarrow}\boldsymbol{E}_{L}\boldsymbol{V}^{L}_{\downarrow} + \boldsymbol{t}^{L}_{\uparrow}$
- $\boldsymbol{\mathcal{B}}_{L} = \hat{\boldsymbol{\Phi}}^{-1}\left[\left(\boldsymbol{I}_8 \otimes \mathbf{DFT}(\mathbf{k}_x)\right)\hat{\boldsymbol{D}}_{z,L}(\mathbf{k}_x, k_y)\begin{pmatrix} \sqrt{\boldsymbol{E}_{L}}\Delta\boldsymbol{V}^{L}_{\downarrow} \\ \sqrt{\boldsymbol{E}_{L}}\Delta\boldsymbol{V}^{L}_{\uparrow} \end{pmatrix}\right]$
- **End Algorithm**

Chapter 6

Variational Iterative Methods for scattering problems

6.1 Introduction

We recall from chapter 5, eqn.(5.73), that the 6×6 displacement/traction, Green's tensor field $\boldsymbol{\mathcal{G}}$ of a heterogeneous medium, may be constructed using the formula

$$\boldsymbol{\mathcal{G}} = \overset{\circ}{\boldsymbol{\mathcal{G}}} + \overset{\circ}{\boldsymbol{\mathcal{P}}}{}_{s}^{z,R} \Delta\boldsymbol{\mathcal{C}}_{eff}(k_y) \left(\boldsymbol{I} - \overset{\circ}{\boldsymbol{\mathcal{P}}}_{r,s} \, \Delta\boldsymbol{\mathcal{C}}_{eff}(k_y) \right)^{-1} \overset{\circ}{\boldsymbol{\mathcal{G}}}{}_{r}^{z} \tag{6.1}$$

where the 6×6 array $\overset{\circ}{\boldsymbol{\mathcal{G}}}$ is the corresponding Green's tensor for the stratified reference component of the model. The $8LN_x \times 6$ incident field $\overset{\circ}{\boldsymbol{\mathcal{G}}}{}_{r}^{z}$ was given by either (5.74) or (5.75) depending on whether the the source is above or below the zone of heterogeneity, respectively. The $8LN_x \times 8LN_x$ propagator $\overset{\circ}{\boldsymbol{\mathcal{P}}}_{r,s}$ was given by (5.78). This operator maps the scattered field from one scattering interaction, to the incident field at the subsequent scattering interaction. The scattering interactions are governed by the $LN_x \times LN_x$ diagonal array $\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)$ of (4.71). Each element on the diagonal of $\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)$ is an 8×8 array that models the scattering interaction at a given point in space. Finally, the $6 \times 8LN_x$ propagator $\overset{\circ}{\boldsymbol{\mathcal{P}}}{}_{s}^{z,R}$ carries the scattered field to the receiver. This operator was defined in (5.77). A schematic representation of (6.1) is depicted by Fig.6.1.

The task for this chapter, is to derive some approximate constructions for $\boldsymbol{\mathcal{G}}$ from (6.1), while avoiding a direct construction of the enormous matrix

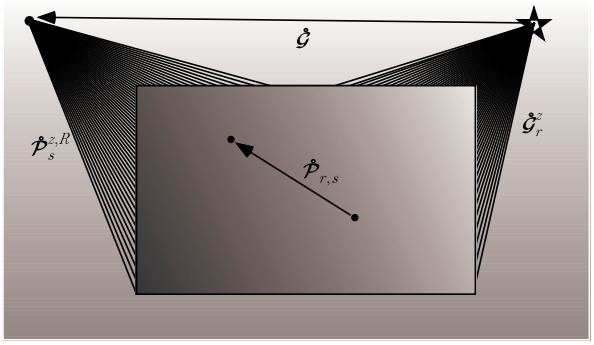

Figure 6.1: The canonical scattering geometry for a source and receiver in the surface zone $z_f \leq z \leq z_0$

inverse

$$\boldsymbol{\Omega} \equiv \left(\boldsymbol{I} - \mathring{\boldsymbol{\mathcal{P}}}_{r,s}\, \Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\right)^{-1} \tag{6.2}$$

in (6.1). We note in passing that $\boldsymbol{\Omega}$ is the time harmonic, elastic wave equation analogue of the *Møller wave operator* of quantum scattering theory [e.g. *Taylor* (1972)].

We will show how to avoid constructing $\boldsymbol{\Omega}$ using expensive matrix inversion, by employing what amounts to a *matrix polynomial approximation* in the Lippmann-Schwinger type operator $\boldsymbol{\mathcal{H}}(k_y)$, defined by

$$\boldsymbol{\mathcal{H}}(k_y) = \left(\boldsymbol{I} - \mathring{\boldsymbol{\mathcal{P}}}_{r,s}\,(k_y)\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\right), \tag{6.3}$$

in terms of the explicitly available matrix operators $\mathring{\boldsymbol{\mathcal{P}}}_{r,s}\,(k_y)$ and $\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)$ that we have met previously. We should point out from the outset, that although the methods to be developed in this chapter are several orders of magnitude faster than direct construction of $\boldsymbol{\Omega}$ in (6.2) using matrix inversion (at least for large problems), the methods are inherently approximate. A nice introduction to *matrix polynomial constructions of a matrix inverse* has been given by *Vorobyev* (1965).

We employ two classes of methods involving matrix polynomials in $\boldsymbol{\mathcal{H}}(k_y)$:

- the first method, is a generalization of the pioneering work of *Lapwood & Hudson* (1975) concerning the *Schwinger-Levine variational formulation of the seismic scattering transition*. We generalize their work in two respects:

1. the method is extended to laterally heterogeneous media
2. the basic method is interpreted as the leading order term in a recursive sequence of additive corrections. The higher order corrections allow us to extend the basic method to *strong scattering environments*

- the second method is a generalization of the *generalized minimal residual method* (GMRES) of *Saad & Schultz* (1986).

Each of these methods constructs the best matrix polynomial approximation in $\boldsymbol{\mathcal{H}}$ to $\boldsymbol{\Omega}$, by choosing the undetermined multipliers in a polynomial of fixed degree, so that a *variational functional* is stationary with respect to small changes in these parameters. The variational formulation of problems governed by integral equations have long been known (Volterra, 1884), and general techniques for the construction of variational principles are now well developed (Gerjouy, Rau & Spruch, 1983).

The symmetry of the field equations (6.3) plays a central role in the variational formulation of the problem. The propagator $\overset{\circ}{\boldsymbol{\mathcal{P}}}_{r,s}(k_y)$ and scattering matrix $\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)$ have the following symmetry properties

$$\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y) \;=\; \Delta\boldsymbol{\mathcal{C}}^T_{eff}(-k_y) \qquad \overset{\circ}{\boldsymbol{\mathcal{P}}}_{r,s}(k_y) \;=\overset{\circ}{\boldsymbol{\mathcal{P}}}^T_{r,s}(-k_y) \tag{6.4}$$

Next we introduce a *bilinear form* $\langle\cdot\ ,\ \cdot\rangle : \mathbb{C}^{8N_xL}\times\mathbb{C}^{8N_xL}\to\mathbb{C}$, defined as

$$\langle \boldsymbol{v}_1, \boldsymbol{v}_2\rangle \equiv \boldsymbol{v}_1^T\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\boldsymbol{v}_2. \tag{6.5}$$

Unfortunately, $\langle \boldsymbol{v}_1, \boldsymbol{v}_1\rangle = 0 \not\Rightarrow \boldsymbol{v}_1 \equiv \boldsymbol{0}$ since $\boldsymbol{v}_1$ is complex valued and the weight matrix $\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)$ used in the definition (6.5) is not *positive definite*. Using (6.4) it follows that

$$\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\Big(\boldsymbol{I}-\overset{\circ}{\boldsymbol{\mathcal{P}}}_{r,s}(k_y)\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)\Big) = \Big(\boldsymbol{I}-\overset{\circ}{\boldsymbol{\mathcal{P}}}_{r,s}(-k_y)\Delta\boldsymbol{\mathcal{C}}_{eff}(-k_y)\Big)^T\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y)$$

so that

$$\langle \boldsymbol{\mathcal{H}}(-k_y)\cdot\boldsymbol{v}_1, \boldsymbol{v}_2\rangle = \langle \boldsymbol{v}_1, \boldsymbol{\mathcal{H}}(k_y)\cdot\boldsymbol{v}_2\rangle \tag{6.6}$$

for arbitrary fields $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$, i.e. the *operator adjoint* $\boldsymbol{\mathcal{H}}^\dagger$ of $\boldsymbol{\mathcal{H}}$ is defined as

$$\boldsymbol{\mathcal{H}}^\dagger(k_y) \equiv \Big(\ \boldsymbol{I}-\overset{\circ}{\boldsymbol{\mathcal{P}}}_{r,s}(-k_y)\Delta\boldsymbol{\mathcal{C}}_{eff}(-k_y)\ \Big) \equiv \boldsymbol{\mathcal{H}}(-k_y) \tag{6.7}$$

relative to the bilinear form of (6.5).

Using the bilinear functional $\langle \cdot\ ,\ \cdot \rangle$ of (6.5), it follows from (6.1) that the (ω, k_y)-response of a 3−D medium with 2−D superimposed heterogeneity driven by a point force $\boldsymbol{f}(\omega, k_y)$ at $\boldsymbol{x}_S$ and recorded by a point receiver at $\boldsymbol{x}_R$ may be written as

$$\begin{aligned} \tilde{\boldsymbol{f}} \cdot \boldsymbol{\mathcal{G}} \cdot \boldsymbol{f} &= \tilde{\boldsymbol{f}} \cdot \mathring{\boldsymbol{\mathcal{G}}} \cdot \boldsymbol{f} + \tilde{\boldsymbol{f}} \cdot \mathring{\boldsymbol{\mathcal{P}}}{}_s^{z,R} \Delta \boldsymbol{\mathcal{C}}_{eff}(k_y) \left(\boldsymbol{I} - \mathring{\boldsymbol{\mathcal{P}}}_{r,s}\, \Delta \boldsymbol{\mathcal{C}}_{eff}(k_y) \right)^{-1} \mathring{\boldsymbol{\mathcal{G}}}{}_r^z \cdot \boldsymbol{f} \\ &= \tilde{\boldsymbol{f}} \cdot \mathring{\boldsymbol{\mathcal{G}}} \cdot \boldsymbol{f} + \left\langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega}(k_y) \cdot \boldsymbol{b} \right\rangle \end{aligned} \tag{6.8}$$

where

$$\boldsymbol{b} = \mathring{\boldsymbol{\mathcal{G}}}{}_r^z \cdot \boldsymbol{f} \tag{6.9}$$

and

$$\tilde{\boldsymbol{b}} = \left[\mathring{\boldsymbol{\mathcal{P}}}{}_s^{z,R}(k_y) \right]^T \cdot \tilde{\boldsymbol{f}}. \tag{6.10}$$

The *receiver directivity vector* $\tilde{\boldsymbol{f}}$ in (6.8) picks out the relevant displacement component to be recorded at $\boldsymbol{x}_R$. Eqn.(6.8) decomposes the *total* response $\tilde{\boldsymbol{f}} \cdot \boldsymbol{\mathcal{G}} \cdot \boldsymbol{f}$ at a fixed receiver position $\boldsymbol{x}_R$ into the sum of a *reference* component

$$\tilde{\boldsymbol{f}} \cdot \mathring{\boldsymbol{\mathcal{G}}} \cdot \boldsymbol{f} \tag{6.11}$$

propagating exclusively through the stratified background medium and a *scattering transition*

$$\left\langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega} \cdot \boldsymbol{b} \right\rangle \tag{6.12}$$

induced by the superimposed heterogeneity. In *Tromp & Dahlen* (1990), an equation for the scattering transition analogous to (6.8) was written down to model the *normal modes* in a heterogeneous earth, although the method of derivation, and the intended application were different from this work.

The calculation of the reference component (6.11) using the *reflectivity method* is now routine (Fuchs & Müller, 1971; Kennett, 1983; Müller, 1985; Mallick & Frazer, 1987). In this chapter we will discuss some methods that are effective for the calculation of scattered field from the superimposed lateral heterogeneity. In particular, we consider the following two strategies to construct the *scattering transition* (6.12):

1. given the incident field (6.9) and the Lippmann-Schwinger type operator (6.3), we solve

$$\boldsymbol{\mathcal{H}}(k_y)\boldsymbol{v} = \boldsymbol{b}, \tag{6.13}$$

for the field vector $\boldsymbol{v}$ using an iterative scheme that is effective for the solution of *complex valued, indefinite* linear systems. Then we form $\langle \tilde{\boldsymbol{b}}, \boldsymbol{v} \rangle$, which is the required scattering transition

2. we construct *high order Schwinger-Levine variational approximations* for a required *scattering transition* $\langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega b} \rangle$ directly. In addition to the *primal* problem (6.13), we introduce the *reciprocal* or *dual* problem

$$\boldsymbol{\mathcal{H}}(-k_y) \cdot \tilde{\boldsymbol{v}} = \tilde{\boldsymbol{b}}, \tag{6.14}$$

where $\tilde{\boldsymbol{b}}(\boldsymbol{x})$ was given by (6.10).

Clearly, once we have solved $\boldsymbol{\mathcal{H}v} = \boldsymbol{b}$, for the field vector $\boldsymbol{v}$ as in class (**1**), we can construct any required scattering transition $\langle \tilde{\boldsymbol{b}}, \boldsymbol{\mathcal{H}}^{-1}\boldsymbol{b} \rangle = \langle \tilde{\boldsymbol{b}}, \boldsymbol{v} \rangle$ in class (**2**) immediately. However, provided the number of required transitions is small (say $1-2$), it is usually more efficient in terms of both computation time and storage, to solve for the transition directly, without first solving for $\boldsymbol{v}$.

All the schemes considered share a common feature:
at iteration n, the current approximation $\boldsymbol{v}_n$ to the solution of $\boldsymbol{\mathcal{H}v} = \boldsymbol{b}$ is chosen to make a *variational functional* $J(\boldsymbol{v})$ stationary (i.e. $\delta J(\boldsymbol{v}_n) = 0$), where the variational trial solution $\boldsymbol{v}_n$ is an element of the *trial space* defined as

$$\boldsymbol{v}_n \in \boldsymbol{v}_0 + \mathcal{K}_n\left(\boldsymbol{r}_0, \boldsymbol{\mathcal{H}}\right), \qquad n = 1, 2, \ldots. \tag{6.15}$$

with $\boldsymbol{v}_0$ the initial approximation to $\boldsymbol{v} = \boldsymbol{\mathcal{H}}^{-1}\boldsymbol{b}$, $\boldsymbol{r}_0 = \boldsymbol{b} - \boldsymbol{\mathcal{H}v}_0$ the corresponding residual vector, and

$$\mathcal{K}_n(\boldsymbol{r}_0, \boldsymbol{\mathcal{H}}) := \operatorname{span}\left\{\boldsymbol{r}_0, \boldsymbol{\mathcal{H}r}_0, \cdots, \boldsymbol{\mathcal{H}}^{n-1}\boldsymbol{r}_0\right\} \tag{6.16}$$

the n-dimensional *Krylov subspace* generated by repeated applications of $\boldsymbol{\mathcal{H}}$ on the initial residual $\boldsymbol{r}_0$. The 2 classes of methods considered here are distinguished by the choice of the variational functional $J(\cdot)$.

6.2 High order Schwinger-Levine approximations of the scattering transition using Padé approximation theory

In this section, we show how to construct the high order approximations to the *scattering transition*

$$\langle \tilde{\boldsymbol{b}}, \boldsymbol{v} \rangle \equiv \langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega b} \rangle \tag{6.17}$$

between a given incident state $\boldsymbol{b}$ and scattered state $\tilde{\boldsymbol{b}}$ using Padé theory. The Møller operator $\boldsymbol{\Omega}$ was given in (6.2), and the bilinear form $< \cdot, \cdot >$ in (6.17) was given by (6.5). The presentation of this thesis was inspired by *Nuttall* (1970). The material now appears to be well known within certain scientific disciplines [e.g. *Garibotti* (1972); *Lucchese, Takatsuka, & McKoy* (1986); *Duneczky & Wyatt* (1988)].

For notational convenience we introduce the *generalized moment* M_n defined by

$$M_n = \left\langle \tilde{\boldsymbol{b}} \ , \ \left(\mathring{\boldsymbol{\mathcal{P}}}_{r,s} \ \Delta\boldsymbol{\mathcal{C}}_{eff} \right)^n \ \boldsymbol{b} \right\rangle \tag{6.18}$$

Then using (6.18), the leading order approximation

$$[0,1]_{<\tilde{\boldsymbol{b}},\boldsymbol{v}>} = \frac{M_0^2}{M_0 - M_1} \tag{6.19}$$

to (6.17) was introduced into computational seismology by *Lapwood & Hudson* (1975). We extend their method handle to heterogeneous media, and show how to compute the higher order corrections needed to secure a reasonable level of convergence to the true transition.

We begin with the simplest case of *weak scattering*. The scattering strength can be quantified by the *spectral radius* $\rho(\mathring{\boldsymbol{\mathcal{P}}}_{r,s} \ \Delta\boldsymbol{\mathcal{C}}_{eff})$ of the *rescattering operator* $\mathring{\boldsymbol{\mathcal{P}}}_{r,s} \ \Delta\boldsymbol{\mathcal{C}}_{eff}$ (MacMillan & Redish, 1986), defined as the maximum complex plane modulus of any eigenvalue $\lambda(\mathring{\boldsymbol{\mathcal{P}}}_{r,s} \ \Delta\boldsymbol{\mathcal{C}}_{eff})$, of $\mathring{\boldsymbol{\mathcal{P}}}_{r,s} \ \Delta\boldsymbol{\mathcal{C}}_{eff}$ i.e.

$$\rho(\mathring{\boldsymbol{\mathcal{P}}}_{r,s} \ \Delta\boldsymbol{\mathcal{C}}_{eff}) = \max|\lambda(\mathring{\boldsymbol{\mathcal{P}}}_{r,s} \ \Delta\boldsymbol{\mathcal{C}}_{eff})|. \tag{6.20}$$

For the very pleasant case of weak scattering

$$\rho(\mathring{\boldsymbol{\mathcal{P}}}_{r,s} \ \Delta\boldsymbol{\mathcal{C}}_{eff}) \ll 1. \tag{6.21}$$

More generally, if $\rho(\mathring{\boldsymbol{\mathcal{P}}}_{r,s} \ \Delta\boldsymbol{\mathcal{C}}_{eff}) < 1$, then the Møller operator of (6.2) has the readily computable *multiple scattering series representation*

$$\boldsymbol{\Omega} = \sum_{n\,=\,0}^{\infty} \left(\mathring{\boldsymbol{\mathcal{P}}}_{r,s} \ \Delta\boldsymbol{\mathcal{C}}_{eff} \right)^n \tag{6.22}$$

so that the $\left\langle \tilde{\boldsymbol{b}}, \boldsymbol{v} \right\rangle$ transition of (6.17) may be written

$$\begin{aligned} \left\langle \tilde{\boldsymbol{b}}, \boldsymbol{v} \right\rangle &= \left\langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega} \boldsymbol{b} \right\rangle \\ &= \sum_{n\,=\,0}^{\infty} \left\langle \tilde{\boldsymbol{b}}\ ,\ \left(\mathring{\boldsymbol{\mathcal{P}}}_{r,s}\ \Delta \boldsymbol{\mathcal{C}}_{eff} \right)^n \boldsymbol{b} \right\rangle \\ &= \sum_{n\,=\,0}^{\infty} M_n \end{aligned} \tag{6.23}$$

with M_n given in (6.18). When $\rho(\mathring{\boldsymbol{\mathcal{P}}}_{r,s}\ \Delta \boldsymbol{\mathcal{C}}_{eff}) \ll 1$, the multiple scattering representation of (6.23) is the most effective means for the construction of $\left\langle \tilde{\boldsymbol{b}}, \boldsymbol{v} \right\rangle$ as only a few terms need be summed in (6.23) to give a highly accurate approximation. Unfortunately, for realistic crustal scattering environments, where the scattering takes place on a length scale spanning several wavelengths of the incident field $\boldsymbol{b}$, the spectral radius $\rho(\mathring{\boldsymbol{\mathcal{P}}}_{r,s}\ \Delta \boldsymbol{\mathcal{C}}(k_y))$ is often not small and (6.23) fails to converge. To circumvent this apparent impasse, we propose to *analytically continue* (6.23) beyond its *domain of convergence* by resumming it as a sequence of *Padé approximants*. An excellent introduction to the Padé resummation of divergent series is given by *Bender & Orszag* (1978), §8.3.

To this end, we introduce a *continuation parameter* θ and consider the multiple scattering series of the hypothetical Møller operator

$$\begin{aligned} \boldsymbol{\Omega}_\theta &= (\boldsymbol{I} - \theta\, \mathring{\boldsymbol{\mathcal{P}}}_{r,s}\ \Delta \boldsymbol{\mathcal{C}}_{eff})^{-1} \\ &= \sum_{n\,=\,0}^{\infty} \theta^n \left(\mathring{\boldsymbol{\mathcal{P}}}_{r,s}\ \Delta \boldsymbol{\mathcal{C}}_{eff} \right)^n \end{aligned} \tag{6.24}$$

where θ is chosen so that $\rho(\theta\ \mathring{\boldsymbol{\mathcal{P}}}_{r,s}\ \Delta \boldsymbol{\mathcal{C}}_{eff}) < 1$. Upon using $\boldsymbol{\Omega}_\theta$ of (6.24) in place of $\boldsymbol{\Omega}$ in (6.23) we get

$$\begin{aligned} \left\langle \tilde{\boldsymbol{b}}, \boldsymbol{v}_\theta \right\rangle &= \left\langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega}_\theta \cdot \boldsymbol{b} \right\rangle \\ &= \sum_{n\,=\,0}^{\infty} \theta^n M_n. \end{aligned} \tag{6.25}$$

We can then investigate the limit

$$\left\langle \tilde{\boldsymbol{b}}, \boldsymbol{v} \right\rangle = \lim_{\theta\,\to\,1} \left\langle \tilde{\boldsymbol{b}}, \boldsymbol{v}_\theta \right\rangle \tag{6.26}$$

as the required analytic continuation of the earlier result (6.23).

Following *Baker & Graves-Morris* (1981), Part 1, we denote the $[L/M]$ rational Padé approximation to an arbitrary analytic function

$$f(z) = \sum_{n\,=\,0}^{\infty} c_n\, z^n$$

as

$$[L/M] \;=\; \frac{a_0 + a_1 z + \cdots + a_L z^L}{b_0 + b_1 z + \cdots + b_M z^M}$$

where the undetermined parameters $\{a_j\}$ and $\{b_j\}$ are chosen so that

$$\sum_{n\,=\,1}^{\infty} c_n z^n \;=\; \frac{a_0 + a_1 z + \cdots + a_L z^L}{b_0 + b_1 z + \cdots + b_M z^M} \;+\; O\left(z^{L+M+1}\right). \tag{6.27}$$

Using a rather elegant result due to Nuttall (*vid. Baker & Graves-Morris* Part.1 (1981)), we construct the $[L/M]$ approximation to (6.27), as an explicit function of the $\{c_j\}$ according to

$$[L/M] = \sum_{n\,=\,0}^{L-M} c_n\, z^n \;+\; \begin{pmatrix} c_{L-M+1} & c_{L-M+2} & \cdots & c_L \end{pmatrix} \times$$
$$\begin{pmatrix} c_{L-M+1} - z\, c_{L-M+2} & c_{L-M+2} - z\, c_{L-M+3} & \cdots & c_L - z\, c_{L+1} \\ c_{L-M+2} - z\, c_{L-M+3} & c_{L-M+3} - z\, c_{L-M+4} & \cdots & c_{L+1} - z\, c_{L+2} \\ \vdots & \vdots & & \vdots \\ c_L - z\, c_{L+1} & c_{L+1} - z\, c_{L+2} & \cdots & c_{L+M-1} - z\, c_{L+M} \end{pmatrix}^{-1} \begin{pmatrix} c_{L-M+1} \\ c_{L-M+2} \\ \vdots \\ c_L \end{pmatrix} \tag{6.28}$$

By setting $L = Q-1$ and $M = Q$ in (6.28), the $[(Q-1)/Q]$ Padé approximate of (6.25), denoted $[(Q-1)/Q]_{<\tilde{\boldsymbol{b}},\boldsymbol{v}>}$, may be written

$$[(Q-1)/Q]_{<\tilde{\boldsymbol{b}},\boldsymbol{v}>} \;=\; \begin{pmatrix} M_0 & M_1 & \cdots & M_{Q-1} \end{pmatrix} \times$$
$$\begin{pmatrix} M_0 - \theta\, M_1 & M_1 - \theta\, M_2 & \cdots & M_{Q-1} - \theta\, M_Q \\ M_1 - \theta\, M_2 & M_2 - \theta\, M_3 & \cdots & M_Q - \theta\, M_{Q+1} \\ \vdots & \vdots & & \vdots \\ M_{Q-1} - \theta\, M_Q & M_Q - \theta\, M_{Q+1} & \cdots & M_{2Q-2} - \theta\, M_{2Q-1} \end{pmatrix}^{-1} \begin{pmatrix} M_0 \\ M_1 \\ \vdots \\ M_{Q-1} \end{pmatrix} \tag{6.29}$$

where the modified moments M_n are given in (6.18). The analytic continuation of (6.26) is now readily accomplished by substituting $\theta = 1$ in (6.29),

for appropriate values of Q. The $Q = 1$ case is given by (6.19). For $Q = 2$ and $\theta = 1$, (6.29) reduces to

$$\begin{aligned}
[1,2]_{<\tilde{\boldsymbol{b}},\boldsymbol{v}>} &= \begin{pmatrix} M_0 & M_1 \end{pmatrix} \begin{pmatrix} M_0 - M_1 & M_1 - M_2 \\ M_1 - M_2 & M_2 - M_3 \end{pmatrix}^{-1} \begin{pmatrix} M_0 \\ M_1 \end{pmatrix} \\
&= \frac{M_0 M_1^2 + M_1^3 - M_0^2 M_2 - 2M_0 M_1 M_2 + M_0^2 M_3}{M_1^2 - M_0 M_2 - M_1 M_2 + M_2^2 + M_0 M_3 - M_1 M_3} \qquad (6.30)
\end{aligned}$$

while for $Q = 3$ and $\theta = 1$, (6.29) becomes

$$\begin{aligned}
[2,3]_{<\tilde{\boldsymbol{b}},\boldsymbol{v}>} &= \begin{pmatrix} M_0 & M_1 & M_2 \end{pmatrix} \begin{pmatrix} M_0 - M_1 & M_1 - M_2 & M_2 - M_3 \\ M_1 - M_2 & M_2 - M_3 & M_3 - M_4 \\ M_2 - M_3 & M_3 - M_4 & M_4 - M_5 \end{pmatrix}^{-1} \begin{pmatrix} M_0 \\ M_1 \\ M_2 \end{pmatrix} \\
&= \frac{N}{D} \qquad (6.31)
\end{aligned}$$

where:

$$N = -M_0 M_2^3 - M_1 M_2^3 - M_2^4 + 2M_0 M_1 M_2 M_3 + 2M_1^2 M_2 M_3 + M_0 M_2^2 M_3 + 3M_1 M_2^2 M_3 - M_0^2 M_3^2 - 2M_0 M_1 M_3^2 - M_1^2 M_3^2 - 2M_0 M_2 M_3^2 - M_0 M_1^2 M_4 - M_1^3 M_4 + M_0^2 M_2 M_4 - 2M_1^2 M_2 M_4 + 2M_0 M_2^2 M_4 + M_0^2 M_3 M_4 + 2M_0 M_1 M_3 M_4 - M_0^2 M_4^2 + M_0 M_1^2 M_5 + M_1^3 M_5 - M_0^2 M_2 M_5 - 2M_0 M_1 M_2 M_5 + M_0^2 M_3 M_5; \quad (6.32)$$

$$D = -M_2^3 + 2M_1 M_2 M_3 + M_2^2 M_3 - M_0 M_3^2 - M_1 M_3^2 - M_2 M_3^2 + M_3^3 - M_1^2 M_4 + M_0 M_2 M_4 - M_1 M_2 M_4 + M_2^2 M_4 + M_0 M_3 M_4 + M_1 M_3 M_4 - 2M_2 M_3 M_4 - M_0 M_4^2 + M_1 M_4^2 + M_1^2 M_5 - M_0 M_2 M_5 - M_1 M_2 M_5 + M_2^2 M_5 + M_0 M_3 M_5 - M_1 M_3 M_5. \quad (6.33)$$

The procedure may be extended to higher values of Q in (6.29), but explicit formulae are too cumbersome to write down. An alternate resummation of the Born series approximation to the scattering transition (6.23) based on the theory of *continued fractions* has been given by *Znojil* (1986). Finally, both Padé and continued fraction theory have matrix generalizations, allowing us to model multiple scattering transitions simultaneously.

In §6.3, we explore a mathematically equivalent but computationally superior algorithm to compute the higher order corrections to the scattering transition (6.17), based on a modified *bi-conjugate gradient recursion*. The relationship between Padé approximation theory and the biconjugate gradient method is now well known [e.g. *Gutknecht* (1990); *Gutknecht* (1992)].

6.3 A variational summation formula for the scattering transition derived from the bi-conjugate gradient method

Given that the *incident* state $\boldsymbol{b}$, and *scattered* state $\tilde{\boldsymbol{b}}$, are each governed by the *Lippmann-Schwinger* type equations

$$\boldsymbol{\mathcal{H}} \cdot \boldsymbol{v} = \boldsymbol{b}, \quad \text{and} \quad \boldsymbol{\mathcal{H}}^{\dagger} \cdot \tilde{\boldsymbol{v}} = \tilde{\boldsymbol{b}} \tag{6.34}$$

where $\boldsymbol{\mathcal{H}}$ was given by (6.3), and the dual operator $\boldsymbol{\mathcal{H}}^{\dagger}$ was given in (6.7). We propose the *trial solutions*

$$\boldsymbol{v}_n = \sum_{j=0}^{n} \alpha_j \boldsymbol{p}_j \tag{6.35}$$

$$\tilde{\boldsymbol{v}}_n = \sum_{j=0}^{n} \tilde{\alpha}_j \tilde{\boldsymbol{p}}_j. \tag{6.36}$$

The scattering transition $\langle \tilde{\boldsymbol{b}}, \boldsymbol{v} \rangle = \langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega} \cdot \boldsymbol{b} \rangle = \langle \boldsymbol{\Omega} \cdot \tilde{\boldsymbol{b}}, \boldsymbol{b} \rangle = \langle \tilde{\boldsymbol{v}}, \boldsymbol{b} \rangle$ may be approximated as

$$\left\langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega} \cdot \boldsymbol{b} \right\rangle \approx \frac{\left\langle \tilde{\boldsymbol{b}}, \boldsymbol{b} \right\rangle^2}{\left\langle \tilde{\boldsymbol{b}}, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{b} \right\rangle} + \sum_{n=1}^{\infty} \frac{\langle \tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{r}_{n-1} \rangle^2}{\langle \tilde{\boldsymbol{p}}_n, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle} \tag{6.37}$$

where $\boldsymbol{\Omega}$ is the *Møller* operator of (6.2), and the *residual fields* $\boldsymbol{r}_n$ and $\tilde{\boldsymbol{r}}_n$ in (6.37) are given by

$$\begin{aligned} \boldsymbol{r}_n &= \boldsymbol{b} - \boldsymbol{\mathcal{H}} \cdot \boldsymbol{v}_n \\ \tilde{\boldsymbol{r}}_n &= \tilde{\boldsymbol{b}} - \boldsymbol{\mathcal{H}}^{\dagger} \cdot \tilde{\boldsymbol{v}}_n. \end{aligned} \tag{6.38}$$

The term $\left\langle \tilde{\boldsymbol{b}}, \boldsymbol{b} \right\rangle^2 \left\langle \tilde{\boldsymbol{b}}, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{b} \right\rangle^{-1}$ in (6.37) was given previously in (6.19) and was originally proposed by *Lapwood & Hudson* (1975). We will now demonstrate how the *high order additive corrections* in (6.37) to the true transition $\langle \tilde{\boldsymbol{b}}, \boldsymbol{v} \rangle$ are calculated in the course of the *Generalized bi-conjugate gradient* (GBCG) scheme. We have used the qualifier "Generalized" in the applellation "Generalized bi-conjugate gradient scheme" to highlight the nonstandard nature of the implementation adopted here. The standard formulation of *Lanczos* (1952), is based around the bilinear form $(\boldsymbol{a}, \boldsymbol{b}) \equiv \boldsymbol{a}^T \cdot \boldsymbol{b}$, for a pair of complex vectors $\boldsymbol{a}$ and $\boldsymbol{b}$. In our formulation, we employ $\langle \boldsymbol{a}, \boldsymbol{b} \rangle \equiv \boldsymbol{a}^T \cdot \Delta \boldsymbol{\mathcal{C}}_{eff} \boldsymbol{b}$,

of (6.5), where the weight matrix has the symmetry $\Delta\boldsymbol{\mathcal{C}}_{eff}(k_y) = \Delta\boldsymbol{\mathcal{C}}^T(-k_y)$. This modified choice of bilinear form is implicit in the work of *Lapwood & Hudson* (1975), although explicit matrix representations are quite different from this thesis.

While the choice of the weighted bilinear form (6.5), adversely affects the stability of traditional interpretations of the BCG method based on vector updates to the solution field $\boldsymbol{v}$, the generalized bilinear form enhances the convergence of the variational scheme presented here, for specific scattering transitions $\left\langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega} \cdot \boldsymbol{b} \right\rangle$. This point is rather subtle, but quite central to the modeling philosophy adopted here. The traditional BCG method, based on vector updates for the solution field $\boldsymbol{v}$ "breaks down" when the term $\langle \tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{r}_{n-1} \rangle$ in (6.37) becomes so small that it is considerably contaminated by roundoff error. However, in the variational representation of the specific transition $\left\langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega} \cdot \boldsymbol{b} \right\rangle$ of (6.37), small values of $\langle \tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{r}_{n-1} \rangle$ drive the correction scheme to convergence. The important point that we wish to make at this point is:"although the scheme of this section bears a superficial resemblance to the standard BCG method, its numerical performance in realistic scattering experiments is substantially superior to the standard method, at the cost of concentrating on *one transition at a time*".

To begin the derivation of (6.37), we introduce the *error functional*

$$\mathcal{E}\left(\tilde{\boldsymbol{v}}_n; \boldsymbol{v}_n\right) = \langle \tilde{\boldsymbol{v}} - \tilde{\boldsymbol{v}}_n, \boldsymbol{\mathcal{H}} \cdot (\boldsymbol{v} - \boldsymbol{v}_n) \rangle \tag{6.39}$$

where the trial solutions $\tilde{\boldsymbol{v}}_n$ and $\boldsymbol{v}_n$ were given in (6.35). We note that the functional (6.39) vanishes in the limit as $\boldsymbol{v}_n \to \boldsymbol{v}$ and $\tilde{\boldsymbol{v}}_n \to \tilde{\boldsymbol{v}}$, where $\boldsymbol{v}$ and $\tilde{\boldsymbol{v}}$ are the exact solutions of (6.34). However, (6.39) does not define a norm since $\mathcal{E}\left(\tilde{\boldsymbol{v}}_n; \boldsymbol{v}_n\right) = 0$ does not imply $\boldsymbol{v}_n = \boldsymbol{v}$ and $\tilde{\boldsymbol{v}}_n = \tilde{\boldsymbol{v}}$. If we expand out the right hand side of (6.39) and rearrange the result, we readily find that

$$\left\langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega} \cdot \boldsymbol{b} \right\rangle \;=\; \mathcal{E}\left(\tilde{\boldsymbol{v}}_n; \boldsymbol{v}_n\right) \;+\; \left\langle \tilde{\boldsymbol{b}}, \boldsymbol{v}_n \right\rangle \;+\; \langle \tilde{\boldsymbol{v}}_n, \boldsymbol{b} \rangle \;-\; \langle \tilde{\boldsymbol{v}}_n, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{v}_n \rangle\,. \tag{6.40}$$

where the left hand side term $\langle \tilde{\boldsymbol{v}}, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{v} \rangle = \left\langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega} \cdot \boldsymbol{b} \right\rangle$ in (6.40) is the required *scattering transition*. When the error functional $\mathcal{E}\left(\tilde{\boldsymbol{v}}_n; \boldsymbol{v}_n\right)$ in (6.40) is negligible, we get

$$\begin{aligned} \left\langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega} \cdot \boldsymbol{b} \right\rangle \;&\approx\; \langle \tilde{\boldsymbol{v}}_n, \boldsymbol{b} \rangle \;+\; \left\langle \tilde{\boldsymbol{b}}, \boldsymbol{v}_n \right\rangle \;-\; \langle \tilde{\boldsymbol{v}}_n, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{v}_n \rangle \\ &\approx\; \mathcal{J}(\tilde{\boldsymbol{v}}_n, \tilde{\boldsymbol{b}}; \boldsymbol{v}_n, \boldsymbol{b}) \end{aligned} \tag{6.41}$$

The *variational functional* $\mathcal{J}(\tilde{\boldsymbol{v}}_n, \tilde{\boldsymbol{b}}; \boldsymbol{v}_n, \boldsymbol{b})$ in (6.41) obeys the *addition*

rule

$$\begin{aligned}
&\mathcal{J}(\tilde{\boldsymbol{v}}_{n-1}+\tilde{\boldsymbol{z}}_n,\tilde{\boldsymbol{b}};\boldsymbol{v}_{n-1}+\boldsymbol{z}_n,\boldsymbol{b})\\
&\quad=\mathcal{J}(\tilde{\boldsymbol{v}}_{n-1},\tilde{\boldsymbol{b}};\boldsymbol{v}_{n-1},\boldsymbol{b})+\left\langle\tilde{\boldsymbol{b}}-\boldsymbol{\mathcal{H}}^{\dagger}\cdot\tilde{\boldsymbol{v}}_{n-1},\boldsymbol{z}_n\right\rangle+\langle\tilde{\boldsymbol{z}}_n,\boldsymbol{b}-\boldsymbol{\mathcal{H}}\cdot\boldsymbol{v}_{n-1}\rangle-\langle\tilde{\boldsymbol{z}}_n,\boldsymbol{\mathcal{H}}\cdot\boldsymbol{z}_n\rangle\\
&\quad=\mathcal{J}(\tilde{\boldsymbol{v}}_{n-1},\tilde{\boldsymbol{b}};\boldsymbol{v}_{n-1},\boldsymbol{b})+\mathcal{J}(\tilde{\boldsymbol{z}}_n,\tilde{\boldsymbol{b}}-\boldsymbol{\mathcal{H}}^{\dagger}\cdot\tilde{\boldsymbol{v}}_{n-1};\boldsymbol{z}_n,\boldsymbol{b}-\boldsymbol{\mathcal{H}}\cdot\boldsymbol{v}_{n-1})
\end{aligned}\tag{6.42}$$

for arbitrary vectors $\tilde{\boldsymbol{z}}_n$ and $\boldsymbol{z}_n$.

Assuming that the trial solutions $\tilde{\boldsymbol{v}}_n$ and $\boldsymbol{v}_n$ in (6.41) are given by (6.35), we can use the addition theorem (6.42) to rewrite (6.41) as

$$\left\langle\tilde{\boldsymbol{b}},\boldsymbol{\Omega}\cdot\boldsymbol{b}\right\rangle\approx\sum_{n\,=\,0}^{\infty}\mathcal{J}(\tilde{\alpha}_n\tilde{\boldsymbol{p}}_n,\tilde{\boldsymbol{r}}_{n-1};\alpha_n\boldsymbol{p}_n,\boldsymbol{r}_{n-1})\tag{6.43}$$

where $\tilde{\boldsymbol{r}}_{n-1}=\tilde{\boldsymbol{b}}-\boldsymbol{\mathcal{H}}^{\dagger}\cdot\tilde{\boldsymbol{v}}_{n-1}$, $\boldsymbol{r}_{n-1}=\boldsymbol{b}-\boldsymbol{\mathcal{H}}\cdot\boldsymbol{v}_{n-1}$ with $\tilde{\boldsymbol{v}}_{-1}=\boldsymbol{v}_{-1}=\boldsymbol{0}$ in (6.43).

We now choose the *variational parameters* $\tilde{\alpha}_j$ and α_j in the trial solutions $\tilde{\boldsymbol{v}}_n$ and $\boldsymbol{v}_n$ so that

$$\sum_{n\,=\,0}^{\infty}\delta\mathcal{J}(\tilde{\alpha}_n\tilde{\boldsymbol{p}}_n,\tilde{\boldsymbol{r}}_{n-1};\alpha_n\boldsymbol{p}_n,\boldsymbol{r}_{n-1})=0\tag{6.44}$$

which defines a *coupled* system for the variational parameters. A *sufficient condition* for (6.44) to hold is:

$$\delta\mathcal{J}(\tilde{\alpha}_n\tilde{\boldsymbol{p}}_n,\tilde{\boldsymbol{r}}_{n-1};\alpha_n\boldsymbol{p}_n,\boldsymbol{r}_{n-1})=0,\qquad n=0,1,\cdots$$

so that the variational parameters may be determined recursively, i.e. α_j defined in terms of $\{\alpha_0,\cdots,\alpha_{j-1}\}$ and similarly for $\tilde{\alpha}_j$.

To begin, at recursion-0 we employ the *Born* approximation

$$\tilde{\boldsymbol{v}}_0\equiv\tilde{\alpha}_0\tilde{\boldsymbol{p}}_0=\tilde{\alpha}_0\tilde{\boldsymbol{b}},\quad\text{and}\quad\boldsymbol{v}_0\equiv\alpha_0\boldsymbol{p}_0=\alpha_0\boldsymbol{b}\tag{6.45}$$

with the variational parameters $\tilde{\alpha}_0$ and α_0 chosen so that

$$\delta\mathcal{J}(\tilde{\alpha}_0\tilde{\boldsymbol{p}}_0,\tilde{\boldsymbol{r}}_{-1};\alpha_0\boldsymbol{p}_0,\boldsymbol{r}_{-1})=\delta\mathcal{J}(\tilde{\alpha}_0\tilde{\boldsymbol{b}},\tilde{\boldsymbol{b}};\alpha_0\boldsymbol{b},\boldsymbol{b})=0\tag{6.46}$$

and

$$\mathcal{J}(\tilde{\alpha}_0\tilde{\boldsymbol{b}},\tilde{\boldsymbol{b}};\alpha_0\boldsymbol{b},\boldsymbol{b})=\tilde{\alpha}_0\left\langle\tilde{\boldsymbol{b}},\boldsymbol{b}\right\rangle\;+\;\alpha_0\left\langle\tilde{\boldsymbol{b}},\boldsymbol{b}\right\rangle\;-\;\tilde{\alpha}_0\alpha_0\left\langle\tilde{\boldsymbol{b}},\boldsymbol{\mathcal{H}}\cdot\boldsymbol{b}\right\rangle.\tag{6.47}$$

To ensure the *stationary condition* (6.46), we impose

$$\frac{\partial}{\partial\,\alpha_0}\mathcal{J}(\tilde{\alpha}_0\tilde{\boldsymbol{b}},\tilde{\boldsymbol{b}};\alpha_0\boldsymbol{b},\boldsymbol{b}) = 0 \quad\Rightarrow\quad \tilde{\alpha}_0 = \frac{\left\langle\tilde{\boldsymbol{b}},\boldsymbol{b}\right\rangle}{\left\langle\tilde{\boldsymbol{b}},\boldsymbol{\mathcal{H}}\cdot\boldsymbol{b}\right\rangle}$$

$$\frac{\partial}{\partial\,\tilde{\alpha}_0}\mathcal{J}(\tilde{\alpha}_0\tilde{\boldsymbol{b}},\tilde{\boldsymbol{b}};\alpha_0\boldsymbol{b},\boldsymbol{b}) = 0 \quad\Rightarrow\quad \alpha_0 = \frac{\left\langle\tilde{\boldsymbol{b}},\boldsymbol{b}\right\rangle}{\left\langle\tilde{\boldsymbol{b}},\boldsymbol{\mathcal{H}}\cdot\boldsymbol{b}\right\rangle} \tag{6.48}$$

so that substituting (6.48) for $\tilde{\alpha}_0$ and α_0 in (6.47) we find

$$\mathcal{J}(\tilde{\alpha}_0\tilde{\boldsymbol{b}},\tilde{\boldsymbol{b}};\alpha_0\boldsymbol{b},\boldsymbol{b}) = \frac{\left\langle\tilde{\boldsymbol{b}},\boldsymbol{b}\right\rangle^2}{\left\langle\tilde{\boldsymbol{b}},\boldsymbol{\mathcal{H}}\cdot\boldsymbol{b}\right\rangle}, \tag{6.49}$$

which is the first term on the right hand side of (6.37). Eqn.(6.49) is essentially eqn.(8.7) of *Lapwood & Hudson* (1975). These authors employed a physical argument based directly on *seismic reciprocity*.

At recursion-n, we choose the variational parameters $\tilde{\alpha}_n$ and α_n in the trial solutions

$$\begin{aligned}\boldsymbol{v}_n &= \sum_{j\,=\,0}^{n}\alpha_j\boldsymbol{p}_j \qquad (6.50)\\ &= \boldsymbol{v}_{n-1}+\alpha_n\boldsymbol{p}_n\\ \tilde{\boldsymbol{v}}_n &= \sum_{j\,=\,0}^{n}\tilde{\alpha}_j\tilde{\boldsymbol{p}}_j\\ &= \tilde{\boldsymbol{v}}_{n-1}+\tilde{\alpha}_n\tilde{\boldsymbol{p}}_n \qquad (6.51)\end{aligned}$$

so that

$$\delta\mathcal{J}(\tilde{\alpha}_n\tilde{\boldsymbol{p}}_n,\tilde{\boldsymbol{r}}_{n-1};\alpha_n\boldsymbol{p}_n,\boldsymbol{r}_{n-1}) = 0 \tag{6.52}$$

where the *residuals* $\tilde{\boldsymbol{r}}_{n-1}$ and $\boldsymbol{r}_{n-1}$ were given by (6.38), so that

$$\begin{aligned}&\mathcal{J}(\tilde{\alpha}_n\tilde{\boldsymbol{p}}_n,\tilde{\boldsymbol{r}}_{n-1};\alpha_n\boldsymbol{p}_n,\boldsymbol{r}_{n-1}) =\\ &\qquad\tilde{\alpha}_n\left\langle\tilde{\boldsymbol{p}}_n,\boldsymbol{r}_{n-1}\right\rangle \;+\; \alpha_n\left\langle\tilde{\boldsymbol{r}}_{n-1},\boldsymbol{p}_n\right\rangle \;-\; \tilde{\alpha}_n\alpha_n\left\langle\tilde{\boldsymbol{p}}_n,\boldsymbol{\mathcal{H}}\cdot\boldsymbol{p}_n\right\rangle\end{aligned} \tag{6.53}$$

To ensure the *stationary condition* (6.52), we impose

$$\frac{\partial}{\partial\,\alpha_n}\mathcal{J}(\tilde{\alpha}_n\tilde{\boldsymbol{p}}_n,\tilde{\boldsymbol{r}}_{n-1};\alpha_n\boldsymbol{p}_n,\boldsymbol{r}_{n-1}) = 0 \quad\Rightarrow\quad \tilde{\alpha}_n = \frac{\langle\tilde{\boldsymbol{r}}_{n-1},\boldsymbol{p}_n\rangle}{\langle\tilde{\boldsymbol{p}}_n,\boldsymbol{\mathcal{H}}\cdot\boldsymbol{p}_n\rangle}$$

$$\frac{\partial}{\partial\,\tilde{\alpha}_n}\mathcal{J}(\tilde{\alpha}_n\tilde{\boldsymbol{p}}_n,\tilde{\boldsymbol{r}}_{n-1};\alpha_n\boldsymbol{p}_n,\boldsymbol{r}_{n-1}) = 0 \quad\Rightarrow\quad \alpha_n = \frac{\langle\tilde{\boldsymbol{p}}_n,\boldsymbol{r}_{n-1}\rangle}{\langle\tilde{\boldsymbol{p}}_n,\boldsymbol{\mathcal{H}}\cdot\boldsymbol{p}_n\rangle} \tag{6.54}$$

so that substituting (6.54) for $\tilde{\alpha}_n$ and α_n in (6.53) we find

$$\mathcal{J}(\tilde{\alpha}_n\tilde{\boldsymbol{p}}_n, \tilde{\boldsymbol{r}}_{n-1}; \alpha_n\boldsymbol{p}_n, \boldsymbol{r}_{n-1}) = \frac{\langle\tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{p}_n\rangle\,\langle\tilde{\boldsymbol{p}}_n, \boldsymbol{r}_{n-1}\rangle}{\langle\tilde{\boldsymbol{p}}_n, \boldsymbol{\mathcal{H}}\cdot\boldsymbol{p}_n\rangle}. \tag{6.55}$$

Finally, substituting (6.49) for

$$\mathcal{J}(\tilde{\alpha}_0\tilde{\boldsymbol{p}}_0, \tilde{\boldsymbol{r}}_{-1}; \alpha_0\boldsymbol{p}_0, \boldsymbol{r}_{-1})$$

and (6.55) for

$$\mathcal{J}(\tilde{\alpha}_n\tilde{\boldsymbol{p}}_n, \tilde{\boldsymbol{r}}_{n-1}; \alpha_n\boldsymbol{p}_n, \boldsymbol{r}_{n-1})$$

in the variational functional (6.42) gives the following *stationary* approximation to the $\langle\tilde{\boldsymbol{b}}, \boldsymbol{\Omega}\cdot\boldsymbol{b}\rangle$ transition

$$\left\langle\tilde{\boldsymbol{b}}, \boldsymbol{\Omega}\cdot\boldsymbol{b}\right\rangle \approx \frac{\left\langle\tilde{\boldsymbol{b}}, \boldsymbol{b}\right\rangle^2}{\left\langle\tilde{\boldsymbol{b}}, \boldsymbol{\mathcal{H}}\cdot\boldsymbol{b}\right\rangle} + \sum_{n\,=\,1}^{\infty}\frac{\langle\tilde{\boldsymbol{p}}_n, \boldsymbol{r}_{n-1}\rangle\,\langle\tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{p}_n\rangle}{\langle\tilde{\boldsymbol{p}}_n, \boldsymbol{\mathcal{H}}\cdot\boldsymbol{p}_n\rangle}. \tag{6.56}$$

The quality of the approximation (6.56) is dependent on the choice of the basis vectors $\tilde{\boldsymbol{p}}_j$ and $\boldsymbol{p}_j$ in (6.35). An obvious choice are the *current steepest descent directions*

$$\boldsymbol{p} = -\nabla_{\boldsymbol{v}}\mathcal{J} = -(\tilde{\boldsymbol{b}} - \boldsymbol{\mathcal{H}}^\dagger\cdot\tilde{\boldsymbol{v}}) = -\tilde{\boldsymbol{r}}$$

and

$$\tilde{\boldsymbol{p}} = -\nabla_{\tilde{\boldsymbol{v}}}\mathcal{J} = -(\boldsymbol{b} - \boldsymbol{\mathcal{H}}\cdot\boldsymbol{v}) = -\boldsymbol{r}$$

so that we might choose $\boldsymbol{p}_n = -\tilde{\boldsymbol{r}}_{n-1}$ and $\tilde{\boldsymbol{p}}_n = -\boldsymbol{r}_{n-1}$. While this choice is reasonable, it does not reproduce the results of §6.2.

Instead, we adopt the *basis* vectors $\tilde{\boldsymbol{p}}_n$ and $\boldsymbol{p}_n$ in the trial solution (6.35) commonly employed with the *bi-conjugate gradient method*, *viz.*

$$\left.\begin{array}{l}\boldsymbol{p}_n = \boldsymbol{r}_{n-1} + \beta_{n-1}\boldsymbol{p}_{n-1}\\ \tilde{\boldsymbol{p}}_n = \tilde{\boldsymbol{r}}_{n-1} + \tilde{\beta}_{n-1}\tilde{\boldsymbol{p}}_{n-1}\end{array}\right\}, \qquad n = 1, 2, \cdots \tag{6.57}$$

with $\tilde{\boldsymbol{p}}_0 = \tilde{\boldsymbol{b}}$ and $\boldsymbol{p}_0 = \boldsymbol{b}$, consistent with the *Born approximation* while the residuals $\tilde{\boldsymbol{r}}_n$ and $\boldsymbol{r}_n$ were given previously in (6.38) and may be updated from one iteration to the next via

$$\begin{aligned}\boldsymbol{r}_n &= \boldsymbol{b} - \boldsymbol{\mathcal{H}}\cdot(\boldsymbol{v}_{n-1} + \alpha_n\boldsymbol{p}_n) = \boldsymbol{r}_{n-1} - \alpha_n\boldsymbol{\mathcal{H}}\cdot\boldsymbol{p}_n\\ \tilde{\boldsymbol{r}}_n &= \tilde{\boldsymbol{b}} - \boldsymbol{\mathcal{H}}^\dagger\cdot(\tilde{\boldsymbol{v}}_{n-1} + \alpha_n\tilde{\boldsymbol{p}}_n) = \tilde{\boldsymbol{r}}_{n-1} - \alpha_n\boldsymbol{\mathcal{H}}^\dagger\cdot\tilde{\boldsymbol{p}}_n.\end{aligned} \tag{6.58}$$

The parameters $\tilde{\beta}_{n-1}$ and β_{n-1} in (6.57) are chosen to enforce the *bi-conjugacy condition*

$$\langle \tilde{\boldsymbol{p}}_m \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle \ = \ 0, \quad m \neq n. \tag{6.59}$$

Exploiting the considerable insight of *Lanczos* (1952), the *bi-conjugacy* condition (6.59) can be achieved by ensuring that $\langle \tilde{\boldsymbol{p}}_{n-1} \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle = \left\langle \tilde{\boldsymbol{p}}_n \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_{n-1} \right\rangle = 0$ or using (6.57), this can be written as

$$\begin{aligned} \left\langle \tilde{\boldsymbol{p}}_{n-1} \ , \ \boldsymbol{\mathcal{H}} \cdot \left(\boldsymbol{r}_{n-1} + \beta_{n-1} \boldsymbol{p}_{n-1} \right) \right\rangle = 0 \quad &\Rightarrow \quad \beta_{n-1} = \ - \ \frac{\langle \tilde{\boldsymbol{p}}_{n-1} \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{r}_{n-1} \rangle}{\left\langle \tilde{\boldsymbol{p}}_{n-1} \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_{n-1} \right\rangle} \\ \left\langle \tilde{\boldsymbol{r}}_{n-1} + \tilde{\beta}_{n-1} \tilde{\boldsymbol{p}}_{n-1} \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_{n-1} \right\rangle = 0 \quad &\Rightarrow \quad \tilde{\beta}_{n-1} = \ - \ \frac{\langle \tilde{\boldsymbol{r}}_{n-1} \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_{n-1} \rangle}{\left\langle \tilde{\boldsymbol{p}}_{n-1} \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_{n-1} \right\rangle} \end{aligned} \tag{6.60}$$

The basic *variational summation* of (6.56) is summarized in Alg.**6.1**. The partial sum ΔS_Q at iteration Q in Alg.**6.1** is identical to the $[(Q-1)/Q]_{<\tilde{\boldsymbol{b}},\boldsymbol{v}>}$ *Padé approximant* of (6.29) with the continuation parameter set at $\theta = 1$.

Next, we derive the more computationally convenient summation algorithm of (6.37) based on alternative formulae for the parameters α_n and β_n, that are identical to those employed in the *bi-conjugate gradient method*. Using the update formula (6.57) in (6.54) we find that

$$\begin{aligned} \alpha_n &= \frac{\langle \tilde{\boldsymbol{p}}_n, \boldsymbol{r}_{n-1} \rangle}{\langle \tilde{\boldsymbol{p}}_n, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle} = \frac{\left\langle \tilde{\boldsymbol{r}}_{n-1} + \tilde{\beta}_{n-1} \tilde{\boldsymbol{p}}_{n-1}, \boldsymbol{r}_{n-1} \right\rangle}{\langle \tilde{\boldsymbol{p}}_n, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle} = \frac{\langle \tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{r}_{n-1} \rangle + \tilde{\beta}_{n-1} \left\langle \tilde{\boldsymbol{p}}_{n-1}, \boldsymbol{r}_{n-1} \right\rangle}{\langle \tilde{\boldsymbol{p}}_n, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle} \\ \tilde{\alpha}_n &= \frac{\langle \tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{p}_n \rangle}{\langle \tilde{\boldsymbol{p}}_n, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle} = \frac{\left\langle \tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{r}_{n-1} + \beta_{n-1} \boldsymbol{p}_{n-1} \right\rangle}{\left\langle \tilde{\boldsymbol{p}}_n, \boldsymbol{\mathcal{H}}^\dagger \cdot \boldsymbol{p}_n \right\rangle} = \frac{\langle \tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{r}_{n-1} \rangle + \beta_{n-1} \left\langle \tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{p}_{n-1} \right\rangle}{\langle \tilde{\boldsymbol{p}}_n, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle} \end{aligned} \tag{6.61}$$

but,

$$\begin{aligned} \left\langle \tilde{\boldsymbol{p}}_{n-1}, \boldsymbol{r}_{n-1} \right\rangle &= \left\langle \tilde{\boldsymbol{p}}_{n-1}, \boldsymbol{r}_{n-2} - \alpha_{n-1} \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_{n-1} \right\rangle \\ &= \left\langle \tilde{\boldsymbol{p}}_{n-1}, \boldsymbol{r}_{n-2} - \frac{\langle \tilde{\boldsymbol{p}}_{n-1}, \boldsymbol{r}_{n-2} \rangle}{\left\langle \tilde{\boldsymbol{p}}_{n-1}, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_{n-1} \right\rangle} \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_{n-1} \right\rangle \\ &= 0 \end{aligned} \tag{6.62}$$

and similarly

$$\begin{aligned} \left\langle \tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{p}_{n-1} \right\rangle &= \left\langle \tilde{\boldsymbol{r}}_{n-2} - \tilde{\alpha}_{n-1} \boldsymbol{\mathcal{H}}^\dagger \cdot \tilde{\boldsymbol{p}}_{n-1}, \boldsymbol{p}_{n-1} \right\rangle \\ &= \left\langle \tilde{\boldsymbol{r}}_{n-2} - \frac{\langle \tilde{\boldsymbol{r}}_{n-2}, \boldsymbol{p}_{n-1} \rangle}{\langle \tilde{\boldsymbol{p}}_{n-1}, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_{n-1} \rangle} \boldsymbol{\mathcal{H}}^\dagger \cdot \tilde{\boldsymbol{p}}_{n-1}, \boldsymbol{p}_{n-1}, \right\rangle \\ &= 0 \end{aligned} \tag{6.63}$$

Algorithm 6.1 THE BASIC VARIATIONAL SUMMATION FOR $< \tilde{\boldsymbol{b}}, \boldsymbol{v} >$

- $\Delta S_{-1} = 0$
- $\boldsymbol{p}_0 = \boldsymbol{r}_{-1} = \boldsymbol{b}$
- $\tilde{\boldsymbol{p}}_0 = \tilde{\boldsymbol{r}}_{-1} = \tilde{\boldsymbol{b}}$
- $\boldsymbol{f}$or $n = 0, 1, \ldots,$ until convergence

$$
\begin{aligned}
\alpha_n &= \frac{\langle \tilde{\boldsymbol{p}}_n \ , \ \boldsymbol{r}_{n-1} \rangle}{\langle \tilde{\boldsymbol{p}}_n \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle} \\
\tilde{\alpha}_n &= \frac{\langle \tilde{\boldsymbol{r}}_{n-1} \ , \ \boldsymbol{p}_n \rangle}{\langle \tilde{\boldsymbol{p}}_n \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle} \\
\Delta S_n &= \Delta S_{n-1} + \alpha_n \langle \tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{p}_n \rangle \\
\boldsymbol{r}_n &= \boldsymbol{r}_{n-1} - \alpha_n \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \\
\tilde{\boldsymbol{r}}_n &= \tilde{\boldsymbol{r}}_{n-1} - \tilde{\alpha}_n \boldsymbol{\mathcal{H}}^{\dagger} \cdot \tilde{\boldsymbol{p}}_n \\
\beta_n &= -\frac{\langle \boldsymbol{p}_n \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{r}_n \rangle}{\langle \boldsymbol{p}_n \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle} \\
\tilde{\beta}_n &= -\frac{\langle \tilde{\boldsymbol{r}}_n \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle}{\langle \tilde{\boldsymbol{p}}_n \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle} \\
\boldsymbol{p}_{n+1} &= \boldsymbol{r}_n + \beta_n \boldsymbol{p}_n \\
\tilde{\boldsymbol{p}}_{n+1} &= \tilde{\boldsymbol{r}}_n + \tilde{\beta}_n \tilde{\boldsymbol{p}}_n
\end{aligned}
$$

- $\boldsymbol{e}$nd
- $\langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega} \cdot \boldsymbol{b} \rangle \approx \Delta S_n$

so that using (6.62) and (6.63) in (6.61) we get

$$\alpha_n = \tilde{\alpha}_n = \frac{\langle \tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{r}_{n-1} \rangle}{\langle \tilde{\boldsymbol{p}}_n, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle} \tag{6.64}$$

Next we derive an alternative to (6.60) for β_n, but first we need to investigate $\langle \tilde{\boldsymbol{r}}_{n-1} \ , \ \boldsymbol{r}_n \rangle$. Performing $\langle \tilde{\boldsymbol{r}}_{n-1} \ , \ (6.58) \rangle$, we have

$$\langle \tilde{\boldsymbol{r}}_{n-1} \ , \ \boldsymbol{r}_n \rangle = \langle \tilde{\boldsymbol{r}}_{n-1} \ , \ \boldsymbol{r}_{n-1} - \alpha_n \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle = \langle \tilde{\boldsymbol{r}}_{n-1} \ , \ \boldsymbol{r}_{n-1} \rangle \left(1 - \frac{\langle \tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle}{\langle \tilde{\boldsymbol{p}}_n, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle} \right) \tag{6.65}$$

where we have used (6.64) for α_n in (6.65), but using (6.57), the $\langle \tilde{\boldsymbol{p}}_n, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle$ term in the denominator of (6.65) may be rewritten as

$$\langle \tilde{\boldsymbol{p}}_n, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle = \left\langle \tilde{\boldsymbol{r}}_{n-1} + \tilde{\beta}_{n-1} \tilde{\boldsymbol{p}}_{n-1}, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \right\rangle = \langle \tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle \tag{6.66}$$

where we have used the *bi-conjugacy* condition $\langle \tilde{\boldsymbol{p}}_{n-1}, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle = 0$ of (6.59). It follows from (6.66), that (6.65) reduces down to

$$\langle \tilde{\boldsymbol{r}}_{n-1} \ , \ \boldsymbol{r}_n \rangle = 0 \tag{6.67}$$

Now, using (6.58), β_n from (6.60) may be rewritten as

$$\beta_n = - \frac{\langle \tilde{\boldsymbol{p}}_n \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{r}_n \rangle}{\langle \tilde{\boldsymbol{p}}_n \ , \ \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n \rangle} = - \frac{\left\langle \boldsymbol{\mathcal{H}}^\dagger \cdot \tilde{\boldsymbol{p}}_n \ , \ \boldsymbol{r}_n \right\rangle}{\left\langle \boldsymbol{\mathcal{H}}^\dagger \cdot \tilde{\boldsymbol{p}}_n \ , \ \boldsymbol{p}_n \right\rangle} = - \frac{\langle \tilde{\boldsymbol{r}}_n - \tilde{\boldsymbol{r}}_{n-1} \ , \ \boldsymbol{r}_n \rangle}{\langle \tilde{\boldsymbol{r}}_n - \tilde{\boldsymbol{r}}_{n-1} \ , \ \boldsymbol{p}_n \rangle} \tag{6.68}$$

but from (6.67) we know that $\langle \tilde{\boldsymbol{r}}_{n-1} \ , \ \boldsymbol{r}_n \rangle = 0$, while in (6.63), we found $\langle \tilde{\boldsymbol{r}}_n, \boldsymbol{p}_n \rangle = 0$ so that (6.68) may be written as

$$\beta_n = \frac{\langle \tilde{\boldsymbol{r}}_n \ , \ \boldsymbol{r}_n \rangle}{\langle \tilde{\boldsymbol{r}}_{n-1} \ , \ \boldsymbol{p}_n \rangle} = \frac{\langle \tilde{\boldsymbol{r}}_n \ , \ \boldsymbol{r}_n \rangle}{\left\langle \tilde{\boldsymbol{r}}_{n-1} \ , \ \boldsymbol{r}_{n-1} + \beta_{n-1} \boldsymbol{p}_{n-1} \right\rangle} = \frac{\langle \tilde{\boldsymbol{r}}_n \ , \ \boldsymbol{r}_n \rangle}{\langle \tilde{\boldsymbol{r}}_{n-1} \ , \ \boldsymbol{r}_{n-1} \rangle} \tag{6.69}$$

where we have used (6.57) and (6.63) again. The arguments used to derive (6.69), are closely duplicated in the derivation of an alternate formula for $\tilde{\beta}_n$ in (6.60). To save space, the result is stated without proof as

$$\tilde{\beta}_n = \frac{\langle \tilde{\boldsymbol{r}}_n \ , \ \boldsymbol{r}_n \rangle}{\langle \tilde{\boldsymbol{r}}_{n-1} \ , \ \boldsymbol{r}_{n-1} \rangle} = \beta_n \tag{6.70}$$

Using the alternative formulae (6.64) and (6.70), in Alg.**6.1**, we get the more efficient scheme of Alg.**6.2**, that effectively sums (6.37).

Algorithm 6.2 THE ENHANCED VARIATIONAL SUMMATION FOR $< \tilde{\boldsymbol{b}}, \boldsymbol{v} >$

- $\Delta S_{-1} = 0$
- $\boldsymbol{r}_{-1} = \boldsymbol{b}$
- $\tilde{\boldsymbol{r}}_{-1} = \tilde{\boldsymbol{b}}$
- ***for*** $n = 0, 1, \ldots,$ until convergence

  $\rho_{n-1} = \langle \tilde{\boldsymbol{r}}_{n-1}, \boldsymbol{r}_{n-1} \rangle$
  if $n = 1$
  $\boldsymbol{p}_n = \boldsymbol{r}_{n-1}$
  $\tilde{\boldsymbol{p}}_n = \tilde{\boldsymbol{r}}_{n-1}$
  else
  $\beta_{n-1} = \rho_{n-1}/\rho_{n-2}$
  $\boldsymbol{p}_n = \boldsymbol{r}_{n-1} + \beta_{n-1}\boldsymbol{p}_{n-1}$
  $\tilde{\boldsymbol{p}}_n = \tilde{\boldsymbol{r}}_{n-1} + \beta_{n-1}\tilde{\boldsymbol{p}}_{n-1}$
  endif
  $\boldsymbol{q}_n = \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_n$
  $\tilde{\boldsymbol{q}}_n = \boldsymbol{\mathcal{H}}^{\dagger} \cdot \tilde{\boldsymbol{p}}_n$
  $\alpha_n = \frac{\rho_{n-1}}{\langle \tilde{\boldsymbol{p}}_{n-1}\ ,\ \boldsymbol{q}_n \rangle}$
  $\Delta S_n = \Delta S_{n-1} + \alpha_n \rho_{n-1}$
  $\boldsymbol{r}_n = \boldsymbol{r}_{n-1} - \alpha_n \boldsymbol{q}_n$
  $\tilde{\boldsymbol{r}}_n = \tilde{\boldsymbol{r}}_{n-1} - \tilde{\alpha}_n \tilde{\boldsymbol{q}}_n$

- ***end***
- $\langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega} \cdot \boldsymbol{b} \rangle \approx \Delta S_n$

6.4 The generalized minimal residual method

In §6.3 we developed a variational scheme to approximate the scattering transition $\left\langle \tilde{\boldsymbol{b}}, \boldsymbol{\Omega} \cdot \boldsymbol{b} \right\rangle$. We will see in §6.5, that the variational method is highly competitive with its computational alternatives, but at the considerable cost of concentrating on one transition at a time. Also, the variational scheme lacks a mathematically rigorous convergence theory. This is due to the choice of the error functional (6.39), which does not define a norm.

During the early stages of the development of the variational method of §6.3, the author implemented a complex arithmetic version of the standard *Generalized minimal residual method* (GMRES) of *Saad & Schultz* (1986), to check the convergence of the variational method. The spinoff from this development was a modified method that could compute all the desired scattering transitions at once, at an acceptable convergence rate. Also, the method tends to be rather robust, even for scattering problems where the spectral radius $\rho(\mathring{\boldsymbol{\mathcal{P}}}_{r,s}\, \Delta\boldsymbol{\mathcal{C}}_{eff}) > 1$, and the error control tends to reflect the true error, rather than some fanciful mathematical notion of that quantity. These qualities suggest that the GMRES method is a robust computational tool in its own right. The author is unaware of any published results on the application of the GMRES method to the solution of acoustic, electromagnetic or elastic wave scattering, with previous computational experience being confined to fluid dynamics and meteorology [e.g. *Kadioğlu & Mudrick* (1992)]. In these published applications, the field variables are *real vectors*.

An elementary introduction to the GMRES method is presented nicely in (*Bruaset* (1995)). An `f77` code for real arithmetic problems, based directly the standard method proposed by *Saad & Schultz* (1986) is available through `Netlib` [see also *Barrett et al.* (1994)]. At the most basic level, the GMRES computes a sequence of vector updates to the approximate solution field, so that the correction introduced at each update minimizes the *least squares measure of the equation residual*. The attendant linear least squares problem is solved via QR factorizations. Each iteration of basic GMRES method requires both an *outer* (*see* Box **6.1**) and an *inner* (*see* Box **6.2**) QR factorization. The outer QR factorization, *orthogonalizes* the trial space basis, so that the least squares problem simplifies to the canonical *upper Hessenberg least squares problem* (*see* Box **6.2**). In *Saad & Schultz* (1986), the outer QR factorization was performed using a *modified Gramm-Schmidt* procedure [e.g. *Watkins* (1992)]. However, we have found that it is preferable to implement the outer QR factorization using a sequence of *unitary Householder reflections*, which are substantially more stable than modified Gramm-

Schmidt factorizations, especially for problems that require several iterations to secure convergence. The second extension of the basic GMRES method concerns the choice of the *trial space basis*. In *Saad & Schultz* (1986), the standard *Krylov* trial space of (6.16) is used. In the outer QR factorization mentioned above, we orthogonalize this basis set. Following *Walker* (1988), we select our trial space basis directly from the orthogonalized Krylov basis, rather than the original basis (6.16). These two extensions of the basic method significantly enhance the observed convergence rate (§6.5).

Unlike the variational treatment of §6.3, the GMRES development does not introduce the *dual* problem $\boldsymbol{\mathcal{H}}^{\dagger}\tilde{\boldsymbol{v}} = \tilde{\boldsymbol{b}}$, but is based solely on the *primal* $N \times N$ matrix/vector equation

$$\boldsymbol{\mathcal{H}}\boldsymbol{v} = \boldsymbol{b}. \tag{6.71}$$

where $N = 8N_xL$, $\boldsymbol{\mathcal{H}}$ was given by (6.3) and $\boldsymbol{b}$ was given by (6.9) Given the starting approximation $\boldsymbol{v}_0 \in C^N$, to the solution of (6.71), we propose to find the *correction* $\boldsymbol{z}_n$, in the current approximate solution $\boldsymbol{v}_n = \boldsymbol{v}_0 + \boldsymbol{z}_n$ at iteration-n so that the *residual* $\boldsymbol{r}_n \equiv \boldsymbol{b} - \boldsymbol{\mathcal{H}}\boldsymbol{v}_n$ is minimized over the n-dimensional *Krylov subspace* $\mathcal{K}_n(\boldsymbol{r}_0, \boldsymbol{\mathcal{H}})$ of (6.16). We recall that $\mathcal{K}_n(\boldsymbol{r}_0, \boldsymbol{\mathcal{H}})$ was generated by repeated applications of $\boldsymbol{\mathcal{H}}$ on the initial residual $\boldsymbol{r}_0 \equiv \boldsymbol{b} - \boldsymbol{\mathcal{H}} \cdot \boldsymbol{v}_0$. The *minimal residual update* $\boldsymbol{z}_n$, is the extremal solution of the *least squares functional for the equation residual*, defined as

$$J(\boldsymbol{z}_n, \boldsymbol{b}) \equiv \min_{\boldsymbol{z}_n \in \mathcal{K}_n(\boldsymbol{r}_0, \boldsymbol{\mathcal{H}})} \|\boldsymbol{b} - \boldsymbol{\mathcal{H}} \cdot (\boldsymbol{v}_0 + \boldsymbol{z}_n)\|_2 \,. \tag{6.72}$$

Since $\boldsymbol{z}_n \in \mathcal{K}_n(\boldsymbol{r}_0, \boldsymbol{\mathcal{H}})$, we also have $\boldsymbol{z}_n \in \mathcal{K}_n(\hat{\boldsymbol{r}}_0, \boldsymbol{\mathcal{H}})$, where where $\hat{\boldsymbol{r}}_0$ is the unit vector defined as $\hat{\boldsymbol{r}}_0 = \boldsymbol{r}_0/\|\boldsymbol{r}_0\|_2$, so that

$$\boldsymbol{z}_n = \left[\hat{\boldsymbol{r}}_0,\ \boldsymbol{\mathcal{H}} \cdot \hat{\boldsymbol{r}}_0, \cdots, \boldsymbol{\mathcal{H}}^{n-1} \cdot \hat{\boldsymbol{r}}_0\right] \cdot \boldsymbol{y}_n$$

where $\boldsymbol{y}_n$ is to be determined. Hence, (6.72) may be rewritten as

$$J(\boldsymbol{y}_n, \boldsymbol{b}) \equiv \min_{\boldsymbol{y}_n \in C^n} \left\| [\hat{\boldsymbol{r}}_0,\ \boldsymbol{\mathcal{H}} \cdot \hat{\boldsymbol{r}}_0, \cdots, \boldsymbol{\mathcal{H}}^n \cdot \hat{\boldsymbol{r}}_0] \cdot \begin{pmatrix} \|\boldsymbol{r}_0\|_2 \\ -\boldsymbol{y}_n \end{pmatrix} \right\|_2 \tag{6.73}$$

Using the *outer* QR factorization

$$[\hat{\boldsymbol{r}}_0,\ \boldsymbol{\mathcal{H}} \cdot \hat{\boldsymbol{r}}_0, \cdots, \boldsymbol{\mathcal{H}}^n \cdot \hat{\boldsymbol{r}}_0] = \boldsymbol{Q}_n \begin{bmatrix} \boldsymbol{R}_n \\ \boldsymbol{0}_{(N-n-1)\times(n+1)} \end{bmatrix}$$

(*see* Box **6.1**) in (6.73), we get

$$\begin{aligned} J(\boldsymbol{y}_n, \boldsymbol{b}) &\equiv \min_{\boldsymbol{y}_n \in C^n} \left\| \boldsymbol{Q}_n \begin{bmatrix} \boldsymbol{R}_n \\ \boldsymbol{0}_{(N-n-1)\times(n+1)} \end{bmatrix} \cdot \begin{pmatrix} \|\boldsymbol{r}_0\|_2 \\ -\boldsymbol{y}_n \end{pmatrix} \right\|_2 \\ &\equiv \min_{\boldsymbol{y}_n \in C^n} \left\| \begin{bmatrix} \boldsymbol{R}_n \\ \boldsymbol{0}_{(N-n-1)\times(n+1)} \end{bmatrix} \cdot \begin{pmatrix} \|\boldsymbol{r}_0\|_2 \\ -\boldsymbol{y}_n \end{pmatrix} \right\|_2 \end{aligned} \tag{6.85}$$

Orthonormalizing the trial space basis using unitary Householder reflections

Box **6.1**

We will show how to construct the *Outer QR* factorization

$$[\hat{\boldsymbol{r}}_0,\ \boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0,\cdots,\boldsymbol{\mathcal{H}}^n\cdot\hat{\boldsymbol{r}}_0]=\boldsymbol{P}_1\boldsymbol{P}_2\cdots\boldsymbol{P}_{n+1}\begin{bmatrix}\boldsymbol{R}_n\\ \boldsymbol{0}_{(N-n-1)\times(n+1)}\end{bmatrix}$$

to orthonormalize the trial space basis $\boldsymbol{\mathcal{H}}^j\cdot\hat{\boldsymbol{r}}_0$, $N-$vectors. The array $\boldsymbol{R}_n$ is an $(n+1)\times(n+1)$ upper triangular matrix, and $\boldsymbol{P}_1\boldsymbol{P}_2\cdots\boldsymbol{P}_{n+1}\equiv\boldsymbol{Q}_n$ is a factorization of the $N\times(n+1)$ array $\boldsymbol{Q}_n$ with orthonormal columns into the product of unitary Householder reflections. Also, in practice $N\gg(n+1)$.

We begin with some basic results concerning unitary Householder reflections (Golub & Van Loan , 1989; Bunch & Grow, 1990; Watkins, 1991). Let us assume that we have an $N-$vector $\boldsymbol{m}=[m_1,m_2,\cdots,m_N]^T$ with complex elements. Then the unitary Householder reflection $\boldsymbol{P}_1$ is given by

$$\boldsymbol{P}_1\equiv\boldsymbol{I}_N-\frac{2\boldsymbol{W}_1\boldsymbol{W}_1^H}{\boldsymbol{W}_1^H\boldsymbol{W}_1}\tag{6.74}$$

where

$$\boldsymbol{W}_1=\begin{bmatrix}m_1\\ m_2\\ \vdots\\ m_N\end{bmatrix}+\begin{bmatrix}\frac{\|[m_1,m_2,\cdots,m_N]^T\|_2 m_1}{|m_1|}\\ 0\\ \vdots\\ 0\end{bmatrix}\begin{matrix}1\\ 2\\ \vdots\\ N\end{matrix}\tag{6.75}$$

so that $\boldsymbol{P}_1$ transforms $\boldsymbol{m}$ into

$$\boldsymbol{P}_1\boldsymbol{m}=-\begin{bmatrix}\frac{\|[m_1,m_2,\cdots,m_N]^T\|_2 m_1}{|m_1|}\\ 0\\ \vdots\\ 0\end{bmatrix}$$

Similarly, the unitary Householder reflection $\boldsymbol{P}_2$ is given by

$$\boldsymbol{P}_2\equiv\boldsymbol{I}_N-\frac{2\boldsymbol{W}_2\boldsymbol{W}_2^H}{\boldsymbol{W}_2^H\boldsymbol{W}_2}\tag{6.76}$$

where

$$\boldsymbol{W}_2=\begin{bmatrix}0\\ m_2\\ \vdots\\ m_N\end{bmatrix}+\begin{bmatrix}0\\ \frac{\|[0,m_2,\cdots,m_N]^T\|_2 m_2}{|m_2|}\\ 0\\ \vdots\end{bmatrix}\begin{matrix}1\\ 2\\ \vdots\\ N\end{matrix}\tag{6.77}$$

so that $\boldsymbol{P}_2$ transforms $\boldsymbol{m}$ into

$$\begin{bmatrix}m_1\\ \frac{\|[0,m_2,\cdots,m_N]^T\|_2 m_2}{|m_2|}\end{bmatrix}$$

Orthonormalizing the trial space basis using unitary Householder reflections

BOX **6.1** *continued*

To orthonormalize the trial space basis

$$[\hat{\boldsymbol{r}}_0,\ \boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0,\cdots,\boldsymbol{\mathcal{H}}^n\cdot\hat{\boldsymbol{r}}_0]$$

we operate with the Householder reflection $\boldsymbol{P}_1$ to get

$$[\boldsymbol{P}_1\hat{\boldsymbol{r}}_0,\ \boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0,\cdots,\boldsymbol{P}_1\boldsymbol{\mathcal{H}}^n\cdot\hat{\boldsymbol{r}}_0]=[-\sigma_1\boldsymbol{e}_1,\ \boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0,\cdots,\boldsymbol{P}_1\boldsymbol{\mathcal{H}}^n\cdot\hat{\boldsymbol{r}}_0] \quad (6.78)$$

where $\sigma_1=\|\hat{\boldsymbol{r}}_0\|_2(\hat{\boldsymbol{r}}_0)_1/|(\hat{\boldsymbol{r}}_0)_1|=(\hat{\boldsymbol{r}}_0)_1/|(\hat{\boldsymbol{r}}_0)_1|$, because $\|\hat{\boldsymbol{r}}_0\|_2=1$, so that

$$-\sigma_1\boldsymbol{e}_1=-\frac{(\hat{\boldsymbol{r}}_0)_1}{|(\hat{\boldsymbol{r}}_0)_1|}\boldsymbol{e}_1=-\begin{bmatrix}\frac{(\hat{\boldsymbol{r}}_0)_1}{|(\hat{\boldsymbol{r}}_0)_1|}\\0\\\vdots\\0\end{bmatrix}$$

and consequently the first coulumn of (6.78) is in its final form. We will now transform the second column of (6.78) in such a way that the first column remains unaltered. To this end, we identify the second column in (6.78) (i.e. $\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0$), with $\boldsymbol{m}$ in (6.77), so that we construct $\boldsymbol{P}_2$ using its definition (6.76) to get

$$\boldsymbol{P}_2\equiv\boldsymbol{I}_N-\frac{2\boldsymbol{W}_2\boldsymbol{W}_2^H}{\boldsymbol{W}_2^H\boldsymbol{W}_2} \quad (6.79)$$

where

$$\boldsymbol{W}_2=\begin{bmatrix}0\\(\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0)_2\\\vdots\\(\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0)_N\end{bmatrix}+\begin{bmatrix}0\\\frac{\|[0,(\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0)_2,\cdots,(\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0)_N]^T\|_2(\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0)_2}{|(\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0)_2|}\\0\\\vdots\end{bmatrix}\begin{matrix}1\\2\\\vdots\\N\end{matrix} \quad (6.80)$$

Operating on (6.78) with $\boldsymbol{P}_2$ from (6.79), we get

$$[\boldsymbol{P}_1\hat{\boldsymbol{r}}_0,\ \boldsymbol{P}_2\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0,\cdots,\boldsymbol{P}_2\boldsymbol{P}_1\boldsymbol{\mathcal{H}}^n\cdot\hat{\boldsymbol{r}}_0]=\left[\begin{bmatrix}-\sigma_1\\0\\0\\\vdots\end{bmatrix},\begin{bmatrix}(\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0)_1\\-\sigma_2\\0\\\vdots\end{bmatrix},\cdots,\boldsymbol{P}_2\boldsymbol{P}_1\boldsymbol{\mathcal{H}}^n\cdot\hat{\boldsymbol{r}}_0\right] \quad (6.81)$$

where $\sigma_2=\frac{\|[0,(\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0)_2,\cdots,(\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0)_N]^T\|_2(\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0)_2}{|(\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0)_2|}$

Continuing in this fashion, we get

$$[\boldsymbol{P}_1\hat{\boldsymbol{r}}_0,\ \boldsymbol{P}_2\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0,\cdots,\boldsymbol{P}_2\boldsymbol{P}_1\boldsymbol{\mathcal{H}}^n\cdot\hat{\boldsymbol{r}}_0]= \quad (6.82)$$

$$\left[\begin{bmatrix}-\sigma_1\\0\\0\end{bmatrix}\ \begin{bmatrix}(\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0)_1\\-\sigma_2\\0\end{bmatrix}\ \begin{bmatrix}(\boldsymbol{P}_n\cdots\boldsymbol{P}_1\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0)_1\\(\boldsymbol{P}_n\cdots\boldsymbol{P}_2\boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0)_2\\\cdot\end{bmatrix}\right]\begin{matrix}1\\2\\\cdot\end{matrix}$$

Orthonormalizing the trial space basis using unitary Householder reflections

Box **6.1** *continued*

The decomposition (6.82) may be written as

$$\boldsymbol{P}_{n+1}\cdots\boldsymbol{P}_2\boldsymbol{P}_1\left[\hat{\boldsymbol{r}}_0,\ \boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0,\cdots,\boldsymbol{\mathcal{H}}^n\cdot\hat{\boldsymbol{r}}_0\right]=\begin{bmatrix}\boldsymbol{R}_n\\ \boldsymbol{0}_{(N-n-1)\times(n+1)}\end{bmatrix} \tag{6.83}$$

but the unitary Householder reflections $\boldsymbol{P}_j, 1\le j\le(n+1))$ are *Hermitian* (i.e. $\boldsymbol{P}_j^H=\boldsymbol{P}_j$), and consequently, they are *involutive* (i.e. $\boldsymbol{P}_j=\boldsymbol{P}_j^{-1}, 1\le j\le(n+1)$), so we multiply both sides of (6.83) by

$$(\boldsymbol{P}_{n+1}\cdots\boldsymbol{P}_2\boldsymbol{P}_1)^{-1}=\boldsymbol{P}_1\boldsymbol{P}_2\cdots\boldsymbol{P}_{n+1}$$

to get

$$\left[\hat{\boldsymbol{r}}_0,\ \boldsymbol{\mathcal{H}}\cdot\hat{\boldsymbol{r}}_0,\cdots,\boldsymbol{\mathcal{H}}^n\cdot\hat{\boldsymbol{r}}_0\right]=\boldsymbol{P}_1\boldsymbol{P}_2\cdots\boldsymbol{P}_{n+1}\begin{bmatrix}\boldsymbol{R}_n\\ \boldsymbol{0}_{(N-n-1)\times(n+1)}\end{bmatrix} \tag{6.84}$$

which is the required QR factorization of the trial space basis.

where the second equation in (6.85) follows from the first equation, since the columns of $N\times(n+1)$ array $\boldsymbol{Q}_n$ are *orthonormal* and consequently, $\|\boldsymbol{Q}_n\cdot\boldsymbol{v}\|_2=\|\boldsymbol{v}\|_2$ for any complex $(n+1)-$vector $\boldsymbol{v}$. Since,

$$\begin{bmatrix}\boldsymbol{R}_n\\ \boldsymbol{0}_{(N-n-1)\times(n+1)}\end{bmatrix}\equiv\left[-\frac{(\hat{\boldsymbol{r}}_0)_1}{|(\hat{\boldsymbol{r}}_0)_1|}\boldsymbol{e}_1,\boldsymbol{H}_n\right]$$

in (6.85) where $\boldsymbol{H}_n$ is an $(n+1)\times n$ upper Hessenberg array, the undetermined vector $\boldsymbol{y}_n$ in (6.85) is the solution of the following *upper Hessenberg least squares problem*

$$J(\boldsymbol{y}_n,\boldsymbol{b})\equiv\min_{\boldsymbol{y}_n\in C^n}\left\|-\frac{\|\boldsymbol{r}_0\|_2(\hat{\boldsymbol{r}}_0)_1}{|(\hat{\boldsymbol{r}}_0)_1|}\boldsymbol{e}_1-\boldsymbol{H}_n\cdot\boldsymbol{y}_n\right\|_2 \tag{6.86}$$

and the $\boldsymbol{e}_1$ is the $(n+1)$-vector $\boldsymbol{e}_1\equiv(1,0,\cdots,0)^T$. To solve the linear least squares problem of (6.86) for $\boldsymbol{y}_n$, we decompose the $(n+1)\times n$ Hessenberg array $\boldsymbol{H}_n$ into its *inner* QR factors so that

$$\boldsymbol{H}_n=\bar{\boldsymbol{Q}}_n\begin{bmatrix}\bar{\boldsymbol{R}}_n\\ \boldsymbol{0}_{1\times n}\end{bmatrix},$$

where $\mathbf{0}_{1\times n}$ is an extra row of zeros, $\bar{\boldsymbol{Q}}_n$ is a $(n+1)\times(n+1)$ array with orthonormal columns, and $\bar{\boldsymbol{R}}_n$ is an $n\times n$ upper triangular array. The *minimal residual solution* $\boldsymbol{y}_n$, of (6.86) , is now written as (*cf.*Box **6.2**)

$$\begin{pmatrix} \boldsymbol{y}_n \\ 0 \end{pmatrix} = -\frac{\|\boldsymbol{r}_0\|_2(\hat{\boldsymbol{r}}_0)_1}{|(\hat{\boldsymbol{r}}_0)_1|} \begin{pmatrix} \bar{\boldsymbol{R}}_n^{-1} \\ \mathbf{0}_{1\times n} \end{pmatrix} \bar{\boldsymbol{Q}}_n^H \boldsymbol{e}_1 \tag{6.87}$$

where $\mathbf{0}_{1\times n}$ is the extra row of zeros, and the *equation residual* is given by $\|\boldsymbol{r}\|_2 \equiv -\frac{\|\boldsymbol{r}_0\|_2(\hat{\boldsymbol{r}}_0)_1}{|(\hat{\boldsymbol{r}}_0)_1|}\boldsymbol{e}_{n+1}\cdot\bar{\boldsymbol{Q}}_n^H\boldsymbol{e}_1$. It is important to note that $\|\boldsymbol{r}\|_2$ is available without the need to construct $\boldsymbol{R}^{-1}$ as in (6.87).

In the standard implementation of *Saad & Schultz* (1986), the

$$\boldsymbol{Q}_n \begin{bmatrix} \boldsymbol{R}_n \\ \mathbf{0}_{(N-n-1)\times(n+1)} \end{bmatrix}$$

factorization is constructed using the *modified Gramm-Schmidt procedure*, while the

$$\bar{\boldsymbol{Q}}_n \begin{bmatrix} \bar{\boldsymbol{R}}_n \\ \mathbf{0}_{1\times n} \end{bmatrix}$$

factorization is built from *unitary Givens rotations*. Unfortunately, the vectors in the Krylov basis set (6.16) become almost linearly dependent as the actual multiple scattering series approaches convergence. The modified Gramm-Schmidt orthogonalization is not robust enough to damp the roundoff inflicted by the increasingly ill-conditioned Krylov basis set, and we have observed a dramatic improvement in the numerical performance of the algorithm by replacing it with an orthogonalization based on *unitary Householder reflections* instead.

Finally, we summarize the basic steps of the modified GMRES algorithm based on Householder reflections in **algorithm 6.3**. We have called the modified method the GMRESH algorithm, to distinguish it from the standard method.

6.5 A numerical comparison of the efficacy and efficiency of the proposed schemes

We consider a stratified halfspace with a 14km deep *monotone increasing velocity gradient zone* comprised of 80 plane layers. We place the source and receiver in the surface layer. In each of the experiments, a *fast and*

Box **6.2** *continued*

or equivalently (because the rotators $\boldsymbol{J}_{i,i-1}$ are *unitary*), we get the *QR factorization* of $\boldsymbol{H}_n$ defined by

$$\boldsymbol{H}_n = \boldsymbol{J}_{2,1}\boldsymbol{J}_{3,2}\cdots\boldsymbol{J}_{n+1,n}\begin{bmatrix}\bar{\boldsymbol{R}}_n\\ \boldsymbol{0}_{1\times n}\end{bmatrix} = \bar{\boldsymbol{Q}}_n\begin{bmatrix}\bar{\boldsymbol{R}}_n\\ \boldsymbol{0}_{1\times n}\end{bmatrix}$$

where $\bar{\boldsymbol{Q}}_n$ is the $(n+1)\times(n+1)$ unitary (so that $\boldsymbol{Q}_n^H\boldsymbol{Q}_n = \boldsymbol{I}_{n+1}$) array

$$\bar{\boldsymbol{Q}}_n \equiv \boldsymbol{J}_{2,1}\boldsymbol{J}_{3,2}\cdots\boldsymbol{J}_{n+1,n} \equiv \begin{bmatrix}(\bar{\boldsymbol{Q}}_n)_1, & (\bar{\boldsymbol{Q}}_n)_2 & ,\cdots, & (\bar{\boldsymbol{Q}}_n)_{n+1}\end{bmatrix}$$

and $\bar{\boldsymbol{R}}_n$ is the $n\times n$ *upper triangular matrix* with zeros below the diagonal, i.e.

$$\begin{bmatrix}\bar{\boldsymbol{R}}_n\\ \boldsymbol{0}_{1\times n}\end{bmatrix} = \begin{bmatrix} * & * & * & \cdots & * & *\\ 0 & * & * & \cdots & * & *\\ 0 & 0 & * & \cdots & * & *\\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots\\ 0 & 0 & 0 & \cdots & * & *\\ 0 & 0 & 0 & \cdots & 0 & *\\ 0 & 0 & 0 & \cdots & 0 & 0\end{bmatrix}.$$

The restriction that $\boldsymbol{H}_n$ is of full column rank is required for the application of a QR factorization method.

Let us suppose that we are required to solve the *full column rank overdetermined linear system*

$$\boldsymbol{H}_n\cdot\boldsymbol{y}_n = \bar{\boldsymbol{Q}}_n\begin{bmatrix}\bar{\boldsymbol{R}}_n\\ \boldsymbol{0}_{1\times n}\end{bmatrix}\cdot\boldsymbol{y}_n = \boldsymbol{b}_n$$

for the unknown $n-$vector $\boldsymbol{y}_n$ given the $(n+1)-$element right hand side vector $\boldsymbol{b}_n$. Using the unitarity of $\bar{\boldsymbol{Q}}_n$ we get

$$\begin{bmatrix}\bar{\boldsymbol{R}}_n\\ \boldsymbol{0}_{1\times n}\end{bmatrix}\cdot\boldsymbol{y}_n = \bar{\boldsymbol{Q}}_n^H\boldsymbol{b}_n = \boldsymbol{J}_{n+1,n}^H\boldsymbol{J}_{n,n-1}^H\cdots\boldsymbol{J}_{2,1}^H\boldsymbol{b}_n = \begin{bmatrix}(\bar{\boldsymbol{Q}}_n^H)_1\cdot\boldsymbol{b}_n\\ (\bar{\boldsymbol{Q}}_n^H)_2\cdot\boldsymbol{b}_n\\ \vdots\\ (\bar{\boldsymbol{Q}}_n^H)_{n+1}\cdot\boldsymbol{b}_n\end{bmatrix}$$

and the corresponding *squared residual* is

$$\|\bar{\boldsymbol{Q}}_n^H\boldsymbol{b}_n - \begin{bmatrix}\bar{\boldsymbol{R}}_n\\ \boldsymbol{0}_{1\times n}\end{bmatrix}\cdot\boldsymbol{y}_n\|_2^2$$

If we choose $\boldsymbol{y}_n$ so that

$$\boldsymbol{y}_n = \bar{\boldsymbol{R}}_n^{-1}\cdot\begin{bmatrix}(\bar{\boldsymbol{Q}}_n^H)_1\cdot\boldsymbol{b}_n\\ (\bar{\boldsymbol{Q}}_n^H)_2\cdot\boldsymbol{b}_n\\ \vdots\\ (\bar{\boldsymbol{Q}}_n^H)_n\cdot\boldsymbol{b}_n\end{bmatrix}$$

Algorithm 6.3 THE GMRESH ALGORITHM

Suppose $\boldsymbol{v}_0$, Tol and Maxit are given.

1. Initialize: Compute $\boldsymbol{r}_0 = \boldsymbol{b} - \boldsymbol{\mathcal{H}} \cdot \boldsymbol{v}_0$, and determine $\boldsymbol{P}_1$ so that $\boldsymbol{P}_1 \boldsymbol{r}_0 = -\frac{\|\boldsymbol{r}_0\|_2 (\hat{\boldsymbol{r}}_0)_1}{|(\hat{\boldsymbol{r}}_0)_1|} \boldsymbol{e}_1 = \|\boldsymbol{r}_0\|_2 \boldsymbol{v}_1$

2. Iterate: For n=1,Maxit, do:

   (a) Update Krylov basis: Set $\boldsymbol{v}_n = \boldsymbol{P}_1 \cdots \boldsymbol{P}_n \boldsymbol{e}_n$ (i.e. $\boldsymbol{v}_n$ is the nth column of $\boldsymbol{Q}_{n-1}$) and compute $\boldsymbol{\mathcal{H}} \cdot \boldsymbol{v}_n$

   (b) Outer QR factorization of Krylov basis: $[\boldsymbol{v}_1, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{v}_1, ..., \boldsymbol{\mathcal{H}} \cdot \boldsymbol{v}_n] = \boldsymbol{Q}_n \begin{bmatrix} \boldsymbol{R}_n \\ \boldsymbol{0}_{(N-n-1)\times(n+1)} \end{bmatrix}$, where $\boldsymbol{R}_n = [-\frac{\|\boldsymbol{r}_0\|_2 (\hat{\boldsymbol{r}}_0)_1}{|(\hat{\boldsymbol{r}}_0)_1|} \boldsymbol{e}_1, \boldsymbol{H}_n]$

   (c) Inner QR factorization: $\boldsymbol{H}_n = \bar{\boldsymbol{Q}}_n \begin{bmatrix} \bar{\boldsymbol{R}}_n \\ \boldsymbol{0}_{1\times n} \end{bmatrix}$

   (d) Find current residual: set $\boldsymbol{w}_n = -\frac{\|\boldsymbol{r}_0\|_2 (\hat{\boldsymbol{r}}_0)_1}{|(\hat{\boldsymbol{r}}_0)_1|} \bar{\boldsymbol{Q}}_n^H \boldsymbol{e}_1$. Set current residual as $\text{res}_n = |\boldsymbol{e}_{n+1} \cdot \boldsymbol{w}_n|$.

   (e) If $\text{res}_n <$ Tol then
   go to **3**

3. Update:

   (a) Set expansion coefficient vector $\boldsymbol{y}_n$ as

$$\begin{pmatrix} \boldsymbol{y}_n \\ 0 \end{pmatrix} = -\frac{\|\boldsymbol{r}_0\|_2 (\hat{\boldsymbol{r}}_0)_1}{|(\hat{\boldsymbol{r}}_0)_1|} \begin{pmatrix} \bar{\boldsymbol{R}}_n^{-1} \\ \boldsymbol{0}_{1\times n} \end{pmatrix} \bar{\boldsymbol{Q}}_n^H \boldsymbol{e}_1$$

   (b) $\boldsymbol{v}_0 \leftarrow \boldsymbol{v}_0 + [\boldsymbol{P}_1 \boldsymbol{e}_1, \boldsymbol{P}_1 \boldsymbol{P}_2 \boldsymbol{e}_2, \cdots, \boldsymbol{P}_1 \cdots \boldsymbol{P}_{n-1} \boldsymbol{P}_n \boldsymbol{e}_n] \cdot \boldsymbol{y}_n$

   (c) If $\text{res}_n <$ Tol then
   Accept $\boldsymbol{v}_0$ as solution
   else
   go to **1**

dense block $14 \times 14\text{km}^2$ is superimposed on the reference medium with a given % perturbation with respect to the reference medium at each point. The dimensions of the block are chosen to span $10\lambda_S \times 10\lambda_S$, where λ_S is the effective SV-wavelength in the stratified reference medium averaged over the gradient zone at 2Hz. We would expect to observe some very strongly scattered coherent phases in this experiment, because the heterogeneity is uniformly dense and fast. This problem is significantly more challenging to solve with an iterative scheme, than a similar sized random medium problem. In our problem, the scattered field from neighbouring regions has similar phase, leading to constructive interference. In random media, the phase from neighbouring points fluctuates, resulting in destructive interference, which essentially reduces the spectral radius of the problem, resulting in faster convergence of the multiple scattering series. Results are only presented for the *in-plane P/SV problem* where $k_y = 0$.

The numerical discretization of the model problem has a matrix dimension of $N = 57600$. To solve a problem of this size by storing the required 3.3×10^9 complex matrix elements (needing ≈ 53 Gbytes) in an array and using some form of *direct method* based on *Gaussian elimination* is well beyond the capabilities of the present generation of super computers.

The iterative schemes tested were:

- the *Born* series approximation of the scattering transition (6.23).
- the analytically continued scattering transition of (6.37), computed via the variational summation method of Alg.**6.2** (labeled BCGVAR in the figures)
- The basic GMRES method of *Saad & Schultz* (1986).
- The GMRESH method
- the standard *bi-conjugate gradient* method of *Lanczos* (1951) and *Lanczos* (1952)
- the *stabilized bi-conjugate gradient* method of *van der Vorst* (1992) [*see also Gutknecht* (1993)], that is essentially a hybrid version of the standard bi-conjugate gradient method and the GMRES(1) method (labeled BCGSTAB in the figures). The standard algorithm has been adapted to handle complex arithmetic.

Some immediate conclusions follow from the experiment:

[Figure: 05 not found]

Figure 6.2: Residual Vs. iteration for 5% heterogeneity

- the fastest methods for the *multi-transition* problem are the GMRESH and BCGSTAB algorithms.
- the variational summation was the fastest method (for a single transition), the 8 iterations took approximately 18 minutes on the DEC-α. This observation is consistent with calculations performed in *Wandzura & Turley* (1992), who considered biconjugate gradient based summations of the *electromagnetic mono-static scattering problem* (i.e. the analysis was restricted to $\tilde{\boldsymbol{b}} = \boldsymbol{b}$ transitions). We performed 8 iterations for a relative error $< 10^{-6}$ at 5% perturbation, and 28 iterations for a relative error $< 10^{-6}$ at 10% perturbation. In the 10% problem, we are effectively performing an *analytic continuation* of the Born series transition beyond its domain of convergence, although this is at the cost of a substantial "blow out" in the required number of iterations.
- the larger the contrast between the plane stratified reference medium and the actual heterogeneous medium (as measured by the % heterogeneity), the slower the convergence of the iterative schemes
- the Born series converged the most slowly of the methods tested at 5% heterogeneity and diverged at 10% heterogeneity. We observe that the first iterate of the Born series (usually referred to as the "Born approximation"), is rather inadequate for the current problems.
- the standard bi-conjugate gradient method stalled, when approaching the converged solution because the $\rho_i =< \tilde{\boldsymbol{r}}_i, \boldsymbol{r}_i >\approx 10^{-14}$, was too small and roundoff would have swamped to true solution.
- the same parameter ρ_i that caused the divergence in the bi-conjugate

[Figure: 10 not found]

Figure 6.3: Residual Vs. iteration for 10% heterogeneity

gradient method actually drives the variational summation method BCGVAR to convergence, since the correction is $\frac{\rho_i^2}{<\tilde{\boldsymbol{p}}_i, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_i>}$, and $<\tilde{\boldsymbol{p}}_i, \boldsymbol{\mathcal{H}} \cdot \boldsymbol{p}_i>$ is not small in all the experiments conducted to date. Indeed, small ρ_i is a necessary, though not sufficient condition for the convergence of the variational method.

- the convergence of GMRES and GMRESH are *monotone*, with the later method markedly superior to the former in both tests. Both methods retain monotone convergence for more difficult problems as well, making them the method of choice for extremely nasty problems.

6.6 The preconditioned GMRESH method

The most direct way to solve

$$\boldsymbol{\mathcal{H}} \cdot \boldsymbol{v} = \boldsymbol{b}. \tag{6.88}$$

for the field vector $\boldsymbol{v}$, is to multiply both sides by the Møller operator

$$\boldsymbol{\Omega} = \boldsymbol{\mathcal{H}}^{-1} = \left(\boldsymbol{I} - \mathring{\boldsymbol{\mathcal{P}}}_{r,s} \cdot \Delta \boldsymbol{\mathcal{C}}_{eff}\right)^{-1} \tag{6.89}$$

to get

$$\boldsymbol{v} = \boldsymbol{\Omega} \cdot \boldsymbol{b}. \tag{6.90}$$

but this method is much too expensive to be of practical use. However, it may be possible to construct an approximation $\tilde{\boldsymbol{\Omega}}$ to $\boldsymbol{\Omega}$, and in instead of (6.89), we solve the *preconditioned equation*

$$\tilde{\boldsymbol{\Omega}} \boldsymbol{\mathcal{H}} \cdot \boldsymbol{v} = \tilde{\boldsymbol{\Omega}} \boldsymbol{b}. \tag{6.91}$$

for the field vector $\boldsymbol{v}$. Provided that $\tilde{\boldsymbol{\Omega}}$ is a reasonable approximation to $\boldsymbol{\Omega}$, solving the preconditioned equation (6.91) using the GMRES method will be more efficient in terms of the number of iterations required to secure convergence than solving the unpreconditioned equation (6.89).

To derive the preconditioning operator $\tilde{\boldsymbol{\Omega}}$ we rewrite (6.89) as

$$\boldsymbol{\Omega} = \left(\boldsymbol{I} - \mathring{\boldsymbol{\mathcal{P}}}_{r,s} \cdot \Delta\boldsymbol{\mathcal{C}}_{eff}\right)^{-1} = \boldsymbol{I} + \mathring{\boldsymbol{\mathcal{P}}}_{r,s} \left(\boldsymbol{I} - \Delta\boldsymbol{\mathcal{C}} \cdot \mathring{\boldsymbol{\mathcal{P}}}_{r,s}\right)^{-1} \Delta\boldsymbol{\mathcal{C}} = \boldsymbol{I} + \tilde{\boldsymbol{\mathcal{P}}}\Delta\boldsymbol{\mathcal{C}} \tag{6.92}$$

however $\tilde{\boldsymbol{\mathcal{P}}} = \mathring{\boldsymbol{\mathcal{P}}}_{r,s} \left(\boldsymbol{I} - \Delta\boldsymbol{\mathcal{C}} \cdot \mathring{\boldsymbol{\mathcal{P}}}_{r,s}\right)^{-1}$ in (6.92) is effectively a propagator for the actual medium because it models all transmission and reflected phases in the reference model, as well as the forward, backward and side scattered phases from the superimposed heterogeneity. This operator could be constructed explicitly using matrix inversion but this would be very expensive. Instead, we will employ a forward scattering modification of the Riccati factorization algorithm Alg.**5.3** to construct the approximate propagator $\tilde{\boldsymbol{\mathcal{P}}}_F$ that includes all up/down transmission and reflected phases in the reference model, and the forward scattered phases from the superimposed heterogeneity, but *ignores backscattering and side scattering*. The preconditioning operator $\tilde{\boldsymbol{\Omega}}$ is then constructed explicitly as

$$\tilde{\boldsymbol{\Omega}} = \boldsymbol{I} + \tilde{\boldsymbol{\mathcal{P}}}_F \Delta\boldsymbol{\mathcal{C}} \tag{6.93}$$

where $\tilde{\boldsymbol{\mathcal{P}}}_F$ is constructed using Alg.**6.4**.

In Fig.6.4 we compare the convergence rates of the preconditioned (PGMRESH) and unpreconditioned GMRESH methods when applied to the 10% model similar to that of §6.5, but with smoother reference properties. It is slightly cheaper to calculate $\tilde{\boldsymbol{\mathcal{P}}}_F$ using Alg.**6.4**, than $\mathring{\boldsymbol{\mathcal{P}}}\ r, s$ from (5.78) using Alg.**5.2** and Alg.**5.3**, because no left/right sweeps are required, although we must calculate the action $\delta\boldsymbol{\mathcal{S}}^{\rho}_{\downarrow\downarrow,\uparrow\uparrow} \cdot \boldsymbol{w}$, $1 \leq \rho \leq L$ [*cf.*(6.94) & (6.95)] on various field vectors $\boldsymbol{w}$. In summary, we have replaced $\boldsymbol{\mathcal{H}} \cdot \boldsymbol{w}$ with the preconditioned operation $\tilde{\boldsymbol{\Omega}}\,(\boldsymbol{\mathcal{H}} \cdot \boldsymbol{w})$ that is slightly less than twice as expensive.

In Fig.6.4 we observe that the preconditioned problem required 18 iterations while the unpreconditioned problem required 38 iterations to secure the same level of convergence. On the basis of this evidence it appears that the preconditioned GMRESH mathod is only slightly cheaper that the unpreconditioned method. However, the principal disadvantage of the GMRES class of methods relative to the BCG based methods of §6.5, is the need

Algorithm 6.4 A Riccati sweep implementation of the up/down preconditioning sweep $\boldsymbol{\mathcal{B}} = \tilde{\boldsymbol{\mathcal{P}}}_F \boldsymbol{\mathcal{F}}$

We introduce the *forward scattering* operator $\delta \boldsymbol{S}^{\rho}_{\downarrow\downarrow}$ defined as

$$\delta \boldsymbol{S}^{\rho}_{\downarrow\downarrow} = -i \begin{pmatrix} \mathbf{0} & \sqrt{\boldsymbol{E}_\rho} \end{pmatrix} \hat{\boldsymbol{D}}^T_{z,\rho}(-\mathbf{k}_x, -k_y) \tag{6.94}$$
$$\times \left(\boldsymbol{I}_8 \otimes \mathbf{DFT}^T(-\mathbf{k}_x)\right) \hat{\boldsymbol{\Phi}} \Delta \boldsymbol{\mathcal{C}}^{\rho}_{eff}(k_y) \hat{\boldsymbol{\Phi}}^{-1} \left(\boldsymbol{I}_8 \otimes \mathbf{DFT}(\mathbf{k}_x)\right) \hat{\boldsymbol{D}}_{z,\rho}(\mathbf{k}_x, k_y) \begin{pmatrix} \sqrt{\boldsymbol{E}_\rho} \\ \mathbf{0} \end{pmatrix}$$

and $\delta \boldsymbol{S}^{\rho}_{\uparrow\uparrow}$ defined as

$$\delta \boldsymbol{S}^{\rho}_{\uparrow\uparrow} = -i \begin{pmatrix} \sqrt{\boldsymbol{E}_\rho} & \mathbf{0} \end{pmatrix} \hat{\boldsymbol{D}}^T_{z,\rho}(-\mathbf{k}_x, -k_y) \tag{6.95}$$
$$\times \left(\boldsymbol{I}_8 \otimes \mathbf{DFT}^T(-\mathbf{k}_x)\right) \hat{\boldsymbol{\Phi}} \Delta \boldsymbol{\mathcal{C}}^{\rho}_{eff}(k_y) \hat{\boldsymbol{\Phi}}^{-1} \left(\boldsymbol{I}_8 \otimes \mathbf{DFT}(\mathbf{k}_x)\right) \hat{\boldsymbol{D}}_{z,\rho}(\mathbf{k}_x, k_y) \begin{pmatrix} \mathbf{0} \\ \sqrt{\boldsymbol{E}_\rho} \end{pmatrix}$$

which are the *single scattering* analogues of $\Delta \boldsymbol{\mathcal{S}}^{\rho}_{\downarrow\downarrow}(k_y)$ and $\Delta \boldsymbol{\mathcal{S}}^{\rho}_{\uparrow\uparrow}(k_y)$ respectively, from (5.17). Also, the *effective coupling matrix* $\Delta \boldsymbol{\mathcal{C}}^{\rho}_{eff}(k_y)$ for row$_\rho$ of the heterogeneity was given by (4.70).

- given the $8LN_x$ *virtual source field* $\boldsymbol{\mathcal{F}}$, compute the *preconditioned field* $\boldsymbol{\mathcal{B}} = \tilde{\boldsymbol{\mathcal{P}}}_F \boldsymbol{\mathcal{F}}$

  (**bottom of layer stack* *)

- $\boldsymbol{R}^L_{\uparrow\downarrow} \equiv \boldsymbol{S}^{[L,\infty)}_{\uparrow\downarrow}$
- $\begin{bmatrix} \boldsymbol{\Sigma}^L_\uparrow \\ \boldsymbol{\Sigma}^L_\downarrow \end{bmatrix} = -i \begin{pmatrix} \sqrt{\boldsymbol{E}_L} & \mathbf{0} \\ \mathbf{0} & \sqrt{\boldsymbol{E}_L} \end{pmatrix} \hat{\boldsymbol{D}}^T_{z,L}(-\mathbf{k}_x, -k_y) \left(\boldsymbol{I}_8 \otimes \mathbf{DFT}^T(-\mathbf{k}_x)\right) \hat{\boldsymbol{\Phi}} \left[\boldsymbol{\mathcal{F}}_L\right]$
- $\boldsymbol{t}^L_\uparrow = \boldsymbol{R}^L_{\uparrow\downarrow} \boldsymbol{\Sigma}^L_\downarrow$

  (**upsweep* *)

- **For** $i = L-1, \cdots, 1$

  $$\boldsymbol{R}^i_{\uparrow\downarrow} = \boldsymbol{S}^i_{\uparrow\downarrow} + \boldsymbol{S}^i_{\uparrow\uparrow} \boldsymbol{E}_{i+1} \boldsymbol{R}^{i+1}_{\uparrow\downarrow} \boldsymbol{E}_{i+1} \left(\boldsymbol{I} - \boldsymbol{S}^i_{\downarrow\uparrow} \boldsymbol{E}_{i+1} \boldsymbol{R}^{i+1}_{\uparrow\downarrow} \boldsymbol{E}_{i+1}\right)^{-1} \boldsymbol{S}^i_{\downarrow\downarrow}$$
  $$\boldsymbol{R}^i_{\uparrow\uparrow} = \boldsymbol{S}^i_{\uparrow\uparrow} \left(\boldsymbol{I} - \boldsymbol{E}_{i+1} \boldsymbol{R}^{i+1}_{\uparrow\downarrow} \boldsymbol{E}_{i+1} \boldsymbol{S}^i_{\downarrow\uparrow}\right)^{-1}$$
  $$\boldsymbol{R}^i_{\downarrow\downarrow} = \left(\boldsymbol{I} - \boldsymbol{S}^i_{\downarrow\uparrow} \boldsymbol{E}_{i+1} \boldsymbol{R}^{i+1}_{\uparrow\downarrow} \boldsymbol{E}_{i+1}\right)^{-1} \boldsymbol{S}^i_{\downarrow\downarrow}$$
  $$\boldsymbol{R}^i_{\downarrow\uparrow} = \left(\boldsymbol{I} - \boldsymbol{S}^i_{\downarrow\uparrow} \boldsymbol{E}_{i+1} \boldsymbol{R}^{i+1}_{\uparrow\downarrow} \boldsymbol{E}_{i+1}\right)^{-1} \boldsymbol{S}^i_{\downarrow\uparrow}$$
  $$\begin{bmatrix} \boldsymbol{\Sigma}^\rho_\uparrow \\ \boldsymbol{\Sigma}^\rho_\downarrow \end{bmatrix} = -i \begin{pmatrix} \sqrt{\boldsymbol{E}_\rho} & \mathbf{0} \\ \mathbf{0} & \sqrt{\boldsymbol{E}_\rho} \end{pmatrix} \hat{\boldsymbol{D}}^T_{z,\rho}(-\mathbf{k}_x, -k_y) \left(\boldsymbol{I}_8 \otimes \mathbf{DFT}^T(-\mathbf{k}_x)\right) \hat{\boldsymbol{\Phi}} \left[\boldsymbol{\mathcal{F}}_\rho\right]$$
  $$\begin{bmatrix} \boldsymbol{t}^i_\uparrow \\ \boldsymbol{t}^{i+1}_\downarrow \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}^i_{\uparrow\downarrow} & \boldsymbol{R}^i_{\uparrow\uparrow} \\ \boldsymbol{R}^i_{\downarrow\downarrow} & \boldsymbol{R}^i_{\downarrow\uparrow} \end{bmatrix} \begin{bmatrix} \boldsymbol{\Sigma}^i_\downarrow \\ \boldsymbol{\Sigma}^{i+1}_\uparrow + \left(\boldsymbol{E}_{i+1} + \delta \boldsymbol{S}^{i+1}_{\uparrow\uparrow}\right) \boldsymbol{t}^{i+1}_\uparrow \end{bmatrix}$$

- **End**

Algorithm 6.4 *continued*

(**downsweep* *)

- **For** $i = 1, \cdots, L-1$

$$\begin{bmatrix} \Delta \boldsymbol{V}^i_{\uparrow} \\ \Delta \boldsymbol{V}^{i+1}_{\downarrow} \end{bmatrix} = \begin{bmatrix} \boldsymbol{R}^i_{\uparrow\downarrow} \\ \boldsymbol{R}^i_{\downarrow\downarrow} \end{bmatrix} \left(\boldsymbol{E}_i + \delta \boldsymbol{S}^i_{\downarrow\downarrow}\right) \boldsymbol{V}^i_{\downarrow} + \begin{bmatrix} \boldsymbol{t}^i_{\uparrow} \\ \boldsymbol{t}^{i+1}_{\downarrow} \end{bmatrix}$$

$$\boldsymbol{\mathcal{B}}_\rho = \hat{\boldsymbol{\Phi}}^{-1} \left[\left(\boldsymbol{I}_8 \otimes \mathbf{DFT}(\mathbf{k}_x)\right) \hat{\boldsymbol{D}}_{z,\rho}(\mathbf{k}_x, k_y) \begin{pmatrix} \sqrt{\boldsymbol{E}_\rho} \Delta \boldsymbol{V}^\rho_{\downarrow} \\ \sqrt{\boldsymbol{E}_\rho} \Delta \boldsymbol{V}^\rho_{\uparrow} \end{pmatrix} \right]$$

- **End**

(**bottom of layer stack* *)

- $\Delta \boldsymbol{V}^L_{\uparrow} = \boldsymbol{R}^L_{\uparrow\downarrow} \left(\boldsymbol{E}_L + \delta \boldsymbol{S}^L_{\downarrow\downarrow}\right) \boldsymbol{V}^L_{\downarrow} + \boldsymbol{t}^L_{\uparrow}$
- $\boldsymbol{\mathcal{B}}_L = \hat{\boldsymbol{\Phi}}^{-1} \left[\left(\boldsymbol{I}_8 \otimes \mathbf{DFT}(\mathbf{k}_x)\right) \hat{\boldsymbol{D}}_{z,L}(\mathbf{k}_x, k_y) \begin{pmatrix} \sqrt{\boldsymbol{E}_L} \Delta \boldsymbol{V}^L_{\downarrow} \\ \sqrt{\boldsymbol{E}_L} \Delta \boldsymbol{V}^L_{\uparrow} \end{pmatrix} \right]$

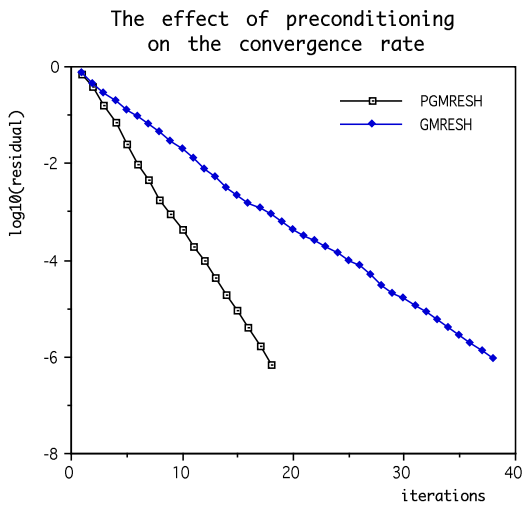

Figure 6.4: Residual Vs. iteration for 10% heterogeneity using a preconditioned generalized minimal residual method

to store all iterates and the requirement that each new iterate be orthogonalized with respect to all the preceding ones. These steps require a large amount of storage and computation. Consequently, halving the number of iterations halves the amount of re-orthogonalization and storage, which is certainly a worthwhile outcome. Also, the BCG methods of §6.5 require the application of both $\boldsymbol{\mathcal{H}}$ and its dual $\boldsymbol{\mathcal{H}}^{\dagger}$ at each iteration, a cost similar to $\tilde{\boldsymbol{\Omega}}\,(\boldsymbol{\mathcal{H}} \cdot \boldsymbol{w})$ in the PGMRES method. In conclusion, on the basis of the numerical experiments conducted to date, the *preconditioned GMRES method is the best overall method for the computation single source/multi-receiver problems with strong heterogeneity superimposed on a reference stratification with large gradients in the material properties*, at least for the in-plane P/SV problem.

Chapter 7

Surface Waves

7.1 The elastodynamic Jost Solutions in a stratified half space

Solving the elastodynamic wave equations is greatly simplified by exploiting their *linearity*. The solution of a linear problem, may expressed in terms of the *basis states* of the solution space. In seismology, the solution space basis is referred to as the *fundamental solution*. When modeling wave propagation in a plane stratified medium, the *plane wave basis* is pre-eminent, because plane stratified media are transversely invariant, and plane waves are the eigen vectors of the translation operator (an eigen basis is always the preferred basis, assuming that it is available). Each of these plane waves is a solution to the wave equation in its own right.

For fixed values of (ω, k_y), the elastodynamic basis states may be classified on the basis of 3 separate physical properties:

- the scattering characteristics of an earth region can be defined by the outcome of a *dual pair of scattering experiments*, initiated by downward incident energy from above, and upward incident energy from below, labeled as experiment "u" and "v" respectively. We shall consider these scattering experiments in detail later, but for now we merely point out that they are the elastodynamic generalization of a related pair of scattering experiments commonly employed in quantum scattering. In the solution of the quantum problem, the outcomes are referred to as the *Jost solutions*, hence we adopt the appellation *elastodynamic Jost solution* to label these outcomes in the elastodynamic problem. When the reference medium is a homogeneous space, rather than a stratified half space, the "u" solution is exclusively comprised of downgoing plane

waves, while the "v" solution is exclusively comprised of upgoing plane waves. But when the reference medium is stratified, "u" and "v" have both up and downgoing components.

- each such elastodynamic Jost solution is further characterized by its incident polarization i.e. either P, SV or SH-wavemotion
- at any depth z the transverse (i.e. x) dependence of the wavefield is expanded in a Fourier basis of plane waves. The incident field is further classified on the basis of the incident horizontal wavenumber k_x

We introduce *multi-indices* $\nu = (J, c, k_x) \in \{\{u, v\} \times \{P, S, H\} \times (-\infty, \infty)\}$ to summarize these three physical properties, i.e. the first index J takes the value $J = u$ for the "u" Jost solution and $J = v$ for the "v" Jost solution. The second index c , takes values P, S and H, denoting incident P SV and SH-polarizations respectively. The final index k_x of the 3-tuple multi-index is the horizontal wavenumber component, so that there are 6 different basis states for each value of k_x.

The solution basis states (or partial wave states) at $\boldsymbol{x} = (z, x)$ and fixed (ω, k_y) may be written in the Dirac notation as

$$|\nu, \boldsymbol{x}\rangle\,, \quad \nu = (J, c, k_x) \in \{\{u, v\} \times \{P, S, H\} \times (-\infty, \infty)\}$$

If we introduce the synthesis operator

$$\sum\!\!\!\!\!\!\int_{\nu} \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathrm{d}k_x \sum_{J \in \{u,v\}} \sum_{c \in \{P,S,H\}}$$

then, at fixed (ω, k_y), we synthesize an arbitrary displacement/ traction field

$$\boldsymbol{b}(\boldsymbol{x}; k_y, \omega) = \tag{7.1}$$
$$[u_z(\boldsymbol{x}; k_y, \omega),\ u_x(\boldsymbol{x}; k_y, \omega),\ u_y(\boldsymbol{x}; k_y, \omega),\ \tau_{zz}(\boldsymbol{x}; k_y, \omega),\ \tau_{xz}(\boldsymbol{x}; k_y, \omega),\ \tau_{yz}(\boldsymbol{x}; k_y, \omega)]^T$$

in a stratified reference medium, from the partial wave states according to

$$\boldsymbol{b}(\boldsymbol{x}; k_y, \omega) = \sum\!\!\!\!\!\!\int_{\nu} |\nu, \boldsymbol{x}\rangle\,. \tag{7.2}$$

The *partial wave synthesis* (7.2) is a rather concise statement of a complex physical process, but has little practical utility unless we know how the partial wave states $|\nu, \boldsymbol{x}\rangle$ evolve over space. We have already met this problem under various guises in the preceding chapters. We recall the spectral representation (3.13), where we constructed the fixed (k_x, k_y, ω) displacement/

traction field $\boldsymbol{b}(k_x, k_y, \omega)$ in free space, using the eigen states of the system matrix (3.7). In this section, we will exploit the plane stratified nature of the reference medium, so that the k_x label of a partial wave state is conserved as the state evolves over depth. Consequently, the spectral representation (3.13) may be used over the depth interval $z_{l-1} \leq z \leq z_l$ of plane reference layer$-l$ for each $1 \leq l \leq L$, but we must allow for coupling between the different $c \in \{P, S, H\}$ labels as we cross the plane interface between adjacent layers. Of course, this problem has been studied in great detail in numerous texts on theoretical seismology.

Due to the plane stratified nature of the reference medium, $\boldsymbol{b}(\boldsymbol{x}; k_y, \omega)$ may be constructed over the depth interval $z_{l-1} \leq z \leq z_l$ of plane reference layer$-l$ for each $1 \leq l \leq L$, according to

$$\begin{aligned} \boldsymbol{b}_l(\boldsymbol{x}; k_y, \omega) &= \sum\!\!\!\!\!\!\int_{\nu_l} |\nu_l, \boldsymbol{x}\rangle \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathrm{d}k\, \boldsymbol{\Theta}_l(\boldsymbol{x}; k_x, k_y) \begin{bmatrix} \boldsymbol{V}_{\downarrow,P_l}(z_{l-1}; k_x, k_y) \\ \boldsymbol{V}_{\downarrow,S_l}(z_{l-1}; k_x, k_y) \\ \boldsymbol{V}_{\downarrow,H_l}(z_{l-1}; k_x, k_y) \\ \boldsymbol{V}_{\uparrow,P_l}(z_l; k_x, k_y) \\ \boldsymbol{V}_{\uparrow,S_l}(z_l; k_x, k_y) \\ \boldsymbol{V}_{\uparrow,S_l}(z_l; k_x, k_y) \end{bmatrix}, \quad z_{l-1} \leq z \leq z_l \end{aligned} \tag{7.3}$$

where:

- the partial wave amplitudes

  $$\boldsymbol{V}_{\downarrow,P_l}(z_{l-1}; k_x, k_y), \quad \boldsymbol{V}_{\downarrow,S_l}(z_{l-1}; k_x, k_y) \quad \text{and} \quad \boldsymbol{V}_{\downarrow,H_l}(z_{l-1}; k_x, k_y)$$

  are the amplitudes of the downgoing P, SV and SH partial waves respectively, phased relative to the top of layer-l

- the partial wave amplitudes

  $$\boldsymbol{V}_{\uparrow,P_l}(z_l; k_x, k_y), \quad \boldsymbol{V}_{\uparrow,S_l}(z_l; k_x, k_y) \quad \text{and} \quad \boldsymbol{V}_{\uparrow,H_l}(z_l; k_x, k_y)$$

  are the amplitudes of the upgoing P, SV and SH partial waves respectively, phased relative to the bottom of layer-l

- the 6×6 array

  $$\begin{aligned} \boldsymbol{\Theta}_l(\boldsymbol{x}; k_x, k_y) &= \mathrm{e}^{ik_x x} \boldsymbol{D}_{z,l}(k_x, k_y) \\ &\times \mathrm{diag}\left[\mathrm{e}^{ik_{z,P_l}(z-z_{l-1})} \quad \mathrm{e}^{ik_{z,S_l}(z-z_{l-1})} \quad \mathrm{e}^{ik_{z,H_l}(z-z_{l-1})} \quad \mathrm{e}^{ik_{z,P_l}(z_l-z)} \quad \mathrm{e}^{ik_{z,S_l}(z_l-z)} \quad \mathrm{e}^{ik_{z,H_l}(z_l-z)} \right] \end{aligned} \tag{7.4}$$

  synthesizes the displacement-traction field at fixed (k_x, k_y, ω) from the partial wave amplitudes

- $\boldsymbol{D}_z(k_x, k_y)$ was given by (3.13)
- $k_{c_l,z}$ was given by (5.11)

Using (7.4), we now construct a 6 element basis $\boldsymbol{B}(\boldsymbol{x}; k_x, k_y)$ of solutions like (7.3) for fixed (k_x, k_y, ω), using the partial wave amplitudes $\boldsymbol{V}_{\downarrow,\uparrow,c_l}$ within each layer. This fundamental solution $\boldsymbol{B}(\boldsymbol{x}; k_x, k_y)$ may be written as

$$\begin{aligned}\boldsymbol{B}(\boldsymbol{x}; k_x, k_y) &= \left[\begin{array}{cc} \boldsymbol{B}_u(\boldsymbol{x}; k_x, k_y) & \boldsymbol{B}_v(\boldsymbol{x}; k_x, k_y) \end{array}\right] \\ &= \boldsymbol{\Theta}_l(\boldsymbol{x}; k_x, k_y) \left[\begin{array}{cc} \boldsymbol{V}_{u,l}(k_x, k_y) & \boldsymbol{V}_{v,l}(k_x, k_y) \end{array}\right], \quad z_{l-1} \le z \le z_l, \quad 1 \le l \le L \\ &= \boldsymbol{\Theta}(\boldsymbol{x}; k_x, k_y) \left[\begin{array}{cc} \boldsymbol{V}_u(k_x, k_y) & \boldsymbol{V}_v(k_x, k_y) \end{array}\right] \end{aligned} \tag{7.5}$$

where:

- the $6L \times 6L$ block diagonal displacement-traction synthesis array $\boldsymbol{\Theta}(\boldsymbol{x}; k_x, k_y)$ at fixed (k_x, k_y, ω), is defined in terms of its layer contributions $\boldsymbol{\Theta}_l(\boldsymbol{x}; k_x, k_y), 1 \le l \le L$, from (7.4) as

$$\boldsymbol{\Theta}(\boldsymbol{x}; k_x, k_y) \equiv \operatorname{diag}\left[\begin{array}{ccc} \boldsymbol{\Theta}_1(\boldsymbol{x}; k_x, k_y) & \cdots & \boldsymbol{\Theta}_L(\boldsymbol{x}; k_x, k_y) \end{array}\right]$$

- the $(6L) \times (3)$ arrays

$$\boldsymbol{V}_u(k_x, k_y) = \begin{pmatrix} \boldsymbol{V}_{u,1}(k_x, k_y) \\ \vdots \\ \boldsymbol{V}_{u,l}(k_x, k_y) \\ \vdots \\ \boldsymbol{V}_{u,L}(k_x, k_y) \end{pmatrix} \text{ and } \boldsymbol{V}_v(k_x, k_y) = \begin{pmatrix} \boldsymbol{V}_{v,1}(k_x, k_y) \\ \vdots \\ \boldsymbol{V}_{v,l}(k_x, k_y) \\ \vdots \\ \boldsymbol{V}_{v,L}(k_x, k_y) \end{pmatrix} \tag{7.6}$$

  are to be constructed from the matrix factorizations of §5.1, after we have chosen the appropriate source radiation

We propose that:

- $\boldsymbol{V}_u(k_x, k_y)$, is initiated by a *free surface source layer* with source radiation $\boldsymbol{\Sigma}_u(k_x, k_y) = [\boldsymbol{\Sigma}_{u,\uparrow}(k_x, k_y), \boldsymbol{\Sigma}_{u,\downarrow}(k_x, k_y)]^T = [\boldsymbol{0}_3, \boldsymbol{I}_3]^T$ radiating P waves, SV waves and SH waves down into the stratified half space. Each such partial wave component of $\boldsymbol{V}_u(k_x, k_y)$ satisfies the radiation condition in the lower half space.

- $\boldsymbol{V}_v(k_x, k_y)$, has its source in the lower half space with source radiation $\boldsymbol{\Sigma}_v(k_x, k_y) = [\boldsymbol{\Sigma}_{v,\uparrow}(k_x, k_y), \boldsymbol{\Sigma}_{v,\uparrow}(k_x, k_y)]^T = [\boldsymbol{I}_3, \boldsymbol{0}_3]^T$ radiating P waves, SV waves and SH waves from the lower half space. Each such partial wave component of $\boldsymbol{V}_v(k_x, k_y)$ satisfies the *traction free condition* at the free surface.

In the downward reflection, and upward transmission problems of this thesis, we are interested in modeling wave propagation on the interval $z_0 \leq \zeta \leq z_L$, so that the source depths z_S for $\boldsymbol{V}_u$ and $\boldsymbol{V}_v$ are chosen outside this interval. Otherwise, if $z_0 \leq z_S \leq z_L$, then $\boldsymbol{V}_u$ and $\boldsymbol{V}_v$ will suffer jump discontinuities across z_S and this will complicate the analysis considerably. Consequently, the source fields for $\boldsymbol{V}_u$ and $\boldsymbol{V}_v$ irradiate the layer stack from above, and below respectively. The fields $\boldsymbol{B}_u(\boldsymbol{x}; k_x, k_y)$ and $\boldsymbol{B}_v(\boldsymbol{x}; k_x, k_y)$ in (7.5), play a role analogous to the matrix *Jost solutions* in mathematically rigorous formulations of quantum scattering [(Taylor, 1972), (Newton, 1982)] and also the theory of solitons (Lamb, 1980). The precise definition of the source radiation $\boldsymbol{\Sigma}_u(k_x, k_y)$ and $\boldsymbol{\Sigma}_v(k_x, k_y)$, initiating the two Jost solutions will not affect the final form of the equations, but may simplify the development.

By setting source depth $z_s = f+$ (i.e. just below the free surface) and source radiation $\boldsymbol{\Sigma}(k_x, k_y) = \boldsymbol{\Sigma}_u(k_x, k_y) = [\boldsymbol{\Sigma}_{u,\uparrow}(k_x, k_y), \boldsymbol{\Sigma}_{u,\downarrow}(k_x, k_y)]^T = [\boldsymbol{0}_3, \boldsymbol{I}_3]^T$ in (5.39), the depth evolution of $\boldsymbol{V}_u(k_x, k_y)$, may be written in terms of its reference layer contributions as

$$\begin{aligned}\boldsymbol{V}_{u,l}(k_x, k_y) &= \begin{bmatrix} \boldsymbol{I}_3 \\ \boldsymbol{S}^{[l,\infty)}_{\uparrow\downarrow} \quad \boldsymbol{E}_l \end{bmatrix} \left[\boldsymbol{I}_3 - \boldsymbol{S}^{(f,l-1]}_{\downarrow\uparrow} \, \boldsymbol{S}^{(l-1,\infty)}_{\uparrow\downarrow} \right]^{-1} \boldsymbol{S}^{(f,l-1]}_{\downarrow\downarrow} \\ &\quad \times \left[\boldsymbol{I}_3 - \boldsymbol{S}^{f}_{\downarrow\uparrow} \, \boldsymbol{S}^{(f,\infty)}_{\uparrow\downarrow} \right]^{-1} (k_x, k_y), \qquad 1 \leq l \leq L \end{aligned} \tag{7.7}$$

and $\boldsymbol{S}^{(f,l-1]}_{\downarrow\uparrow}(k_x, k_y)$ is the net upward incident reflection operator for the half open depth interval $0 < z \leq l-1$ from the top of layer-l, but with the free surface reflection operator $\boldsymbol{S}^{f}_{\downarrow\uparrow}(k_x, k_y)$ set to zero.

The interface variables $\boldsymbol{V}_{u,l}(k_x, k_y)$ of (7.7) characterize the solution field in each of L uniform layers by the amplitudes of $\downarrow-$*going* P, SV and SH plane waves phased relative to the top of layer-l (ie. $z = z^+_{l-1}$), and the amplitudes of $\uparrow-$*going* P, SV and SH waves phased relative to the bottom of layer-l (i.e. $z = z^-_l$) for each $1 \leq l \leq L$ of the reference stratification .

Using $\boldsymbol{V}_{u,l}(k_x, k_y)$ from (7.7), the *Jost solution* $\boldsymbol{B}_u$ is written as

$$\begin{aligned}\boldsymbol{B}_u(\boldsymbol{x}; k_x, k_y) &\equiv \mathrm{e}^{ik_x x} \left[\; (u, P, k_x, k_y, z) \quad (u, S, k_x, k_y, z) \quad (u, H, k_x, k_y, z) \; \right] \\ &\equiv \boldsymbol{\Theta}_l(\boldsymbol{x}; k_x, k_y) \boldsymbol{V}_{u,l}(k_x, k_y), \quad z_{l-1} \leq z \leq z_l \text{ for each } 1 \leq l \leq L \end{aligned} \tag{7.8}$$

where $\boldsymbol{\Theta}_l(\boldsymbol{x}; k_x, k_y)$ was defined in (7.4).

Similarly, from (5.40), with the source depth $z_s = z_L+$ (i.e. at the top of the lower half-space, *see* Fig.5.1), the source radiation $\boldsymbol{\Sigma}(k_x, k_y) = \boldsymbol{\Sigma}_u(k_x, k_y) = [\boldsymbol{\Sigma}_{u,\uparrow}(k_x, k_y), \boldsymbol{\Sigma}_{u,\downarrow}(k_x, k_y)]^T = [\boldsymbol{I}_3, \boldsymbol{0}_3]^T$ and $\boldsymbol{S}_{\uparrow\downarrow}^{(z_s,\infty)}(k_x, k_y) = \boldsymbol{0}_3$ (because we now assume for simplicity, that the medium is homogeneous below the source), the depth evolution of $\boldsymbol{V}_v$, may be written in terms of its reference layer contributions as

$$\boldsymbol{V}_{v,l}(k_x, k_y) = \begin{bmatrix} \boldsymbol{S}_{\downarrow\uparrow}^{[f,l-1]} & \boldsymbol{E}_l \\ & \boldsymbol{I}_3 \end{bmatrix} \left[\boldsymbol{I}_3 - \boldsymbol{S}_{\uparrow\downarrow}^{[l,\infty)} \boldsymbol{S}_{\downarrow\uparrow}^{[f,l)}\right]^{-1} \boldsymbol{S}_{\uparrow\uparrow}^{[l,L]}(k_x, k_y), \qquad 1 \le l \le L. \tag{7.9}$$

Using $\boldsymbol{V}_{v,l}(k_x, k_y)$ from (7.9), the *Jost solution* $\boldsymbol{B}_v$ is written as

$$\begin{aligned} \boldsymbol{B}_v(\boldsymbol{x}; k_x, k_y) &\equiv \mathrm{e}^{ik_x x} \left[\ (v, P, k_x, k_y, z) \quad (v, S, k_x, k_y, z) \quad (v, H, k_x, k_y, z)\ \right] \\ &\equiv \boldsymbol{\Theta}_l(\boldsymbol{x}; k_x, k_y) \boldsymbol{V}_{v,l}(k_x, k_y) \ z_{l-1} \le z \le z_l, \quad \text{for each } 1 \le l \le L \end{aligned} \tag{7.10}$$

where $\boldsymbol{\Theta}_l(\boldsymbol{x}; k_x, k_y)$ was defined in (7.4).

7.1.1 Elementary properties of the Jost solutions

We now introduce the *skew symmetric bilinear functional* $\langle \cdot, \cdot \rangle$, given by

$$\langle \boldsymbol{u}_1, \boldsymbol{u}_2 \rangle \stackrel{\text{def}}{=} (\boldsymbol{u}_1)^T \boldsymbol{J}_6 \boldsymbol{u}_2 \tag{7.11}$$

for a pair of 6−vector fields $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$. If instead, we have a $6 \times l$ array $\boldsymbol{V} = (\boldsymbol{u}_1, \cdots, \boldsymbol{u}_l)$ and a $6 \times m$ array $\boldsymbol{W} = (\boldsymbol{v}_1, \cdots, \boldsymbol{v}_m)$ then

$$\langle \boldsymbol{V}, \boldsymbol{W} \rangle = \begin{pmatrix} \langle \boldsymbol{u}_1, \boldsymbol{v}_1 \rangle & \cdots & \langle \boldsymbol{u}_1, \boldsymbol{v}_m \rangle \\ \cdots & \langle \boldsymbol{u}_i, \boldsymbol{v}_l \rangle & \cdots \\ \langle \boldsymbol{u}_l, \boldsymbol{v}_1 \rangle & \cdots & \langle \boldsymbol{u}_l, \boldsymbol{v}_m \rangle \end{pmatrix} \tag{7.12}$$

is an $l \times m$ array, where $\langle \boldsymbol{u}_i, \boldsymbol{v}_l \rangle$ is defined by (7.11).

In the next section, we will evaluate a number of functionals similar to the prototype

$$\langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{B}_v(\boldsymbol{x}; k_x, k_y) \rangle \tag{7.13}$$

where $\boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y)$ and $\boldsymbol{B}_v(\boldsymbol{x}; k_x, k_y)$ are the *Jost solutions* of (7.8) and (7.10) respectively. The key to the efficient evaluation of (7.13) is to observe that it is depth invariant over the interval $f+ \le z \le L+$ because $\boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y)$ and $\boldsymbol{B}_v(\boldsymbol{x}; k_x, k_y)$ are *homogeneous* solutions of (2.16)

$$\boldsymbol{L}(z, x; \partial_z, \partial_x, k_y) \boldsymbol{B}_v(\boldsymbol{x}; k_x, k_y) = \boldsymbol{L}(z, x; \partial_z, \partial_x, -k_y) \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y) = \boldsymbol{0}$$

(7.14)

where

$$\boldsymbol{L}(z,x;\partial_z,\partial_x,k_y) \equiv \left[\partial_z - \mathring{\boldsymbol{A}}\,(z,x;\partial_x,k_y)\right]$$

was given earlier in (2.17). Using (7.14) we have

$$\begin{aligned}
&\partial_z \left\langle \boldsymbol{B}_u(\boldsymbol{x};-k_x,-k_y), \boldsymbol{B}_v(\boldsymbol{x};k_x,k_y)\right\rangle \\
&\quad = \langle \partial_z \boldsymbol{B}_u(\boldsymbol{x};-k_x,-k_y), \boldsymbol{B}_v(\boldsymbol{x};k_x,k_y)\rangle + \langle \boldsymbol{B}_u(\boldsymbol{x};-k_x,-k_y), \partial_z \boldsymbol{B}_v(\boldsymbol{x};k_x,k_y)\rangle \\
&\quad = \left\langle \mathring{\boldsymbol{A}}\,(z,x;\partial_x,-k_y)\boldsymbol{B}_u(\boldsymbol{x};-k_x,-k_y), \boldsymbol{B}_v(\boldsymbol{x};k_x,k_y)\right\rangle \\
&\qquad\qquad + \left\langle \boldsymbol{B}_u(\boldsymbol{x};-k_x,-k_y), \mathring{\boldsymbol{A}}\,(z,x;\partial_x,k_y)\boldsymbol{B}_v(\boldsymbol{x};k_x,k_y)\right\rangle = \boldsymbol{0}
\end{aligned} \tag{7.15}$$

because

$$\left\langle \mathring{\boldsymbol{A}}\,(z;\partial_x,-k_y)\cdot,\cdot\right\rangle = -\left\langle \cdot, \mathring{\boldsymbol{A}}\,(z;\partial_x,k_y)\cdot\right\rangle \tag{7.16}$$

which follows after two operations of integration by parts.

A final property of the Jost solutions $\boldsymbol{B}_u$ and $\boldsymbol{B}_v$ is the *closure condition* (*vid.* Box **7.1**)

$$\begin{aligned}
\boldsymbol{T}(x;k_y) \equiv \frac{1}{2\pi}\int_{-\infty}^{\infty} \Big[&\boldsymbol{B}_v(\boldsymbol{x};k_x,k_y)\langle \boldsymbol{B}_u(\boldsymbol{x};-k_x,-k_y), \boldsymbol{B}_v(\boldsymbol{x};k_x,k_y)\rangle^{-1} \\
&\qquad \times \int_{-\infty}^{\infty} \langle \boldsymbol{B}_u(\boldsymbol{x};-k_x,-k_y), \boldsymbol{T}(x;k_y)\rangle \mathrm{d}x \\
&-\boldsymbol{B}_u(\boldsymbol{x};k_x,k_y)\langle \boldsymbol{B}_u(\boldsymbol{x};-k_x,-k_y), \boldsymbol{B}_v(\boldsymbol{x};k_x,k_y)\rangle^{-T} \\
&\qquad \times \int_{-\infty}^{\infty} \langle \boldsymbol{B}_v(\boldsymbol{x};-k_x,-k_y), \boldsymbol{T}(x;k_y)\rangle \mathrm{d}x \Big] \mathrm{d}k_x
\end{aligned} \tag{7.17}$$

for $\boldsymbol{T}$ in the space of 6−*component, Fourier transformable vector fields*. The closure condition (7.17), will be needed in the construction of the time harmonic elastic wave Green's tensor for a stratified half space.

Box 7.1 THE CLOSURE PROPERTY OF THE ELASTODYNAMIC JOST SOLUTIONS IN A BASIS OF PLANE WAVES

For convenience, we introduce the short hand notation

$$[\boldsymbol{B}_u(\boldsymbol{x};k_x,k_y,\omega),\boldsymbol{B}_v(\boldsymbol{x};k_x,k_y,\omega)] = \mathrm{e}^{ik_x x}\left[\tilde{\boldsymbol{B}}_u(k_x,k_y),\tilde{\boldsymbol{B}}_v(k_x,k_y)\right] \tag{7.18}$$

where $[\boldsymbol{B}_u(\boldsymbol{x};k_x,k_y,\omega),\boldsymbol{B}_v(\boldsymbol{x};k_x,k_y,\omega)]$ was defined in (7.5).

Using the bilinear form (7.11), we define the *Gramm* or *overlap* matrix $\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\rangle$, of $[\tilde{\boldsymbol{B}}_u,\tilde{\boldsymbol{B}}_v]$ from (7.18) according to

$$\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\right\rangle = \begin{bmatrix} \left\langle\tilde{\boldsymbol{B}}_u(-k_x,-k_y),\tilde{\boldsymbol{B}}_u(k_x,k_y)\right\rangle & \left\langle\tilde{\boldsymbol{B}}_u(-k_x,-k_y),\tilde{\boldsymbol{B}}_v(k_x,k_y)\right\rangle \\ \left\langle\tilde{\boldsymbol{B}}_v(-k_x,-k_y),\tilde{\boldsymbol{B}}_u(k_x,k_y)\right\rangle & \left\langle\tilde{\boldsymbol{B}}_v(-k_x,-k_y),\tilde{\boldsymbol{B}}_v(k_x,k_y)\right\rangle \end{bmatrix}$$

$$= \begin{pmatrix} \mathbf{0}_3 & \left\langle\tilde{\boldsymbol{B}}_u(-k_x,-k_y),\tilde{\boldsymbol{B}}_v(k_x,k_y)\right\rangle \\ -\left\langle\tilde{\boldsymbol{B}}_u(-k_x,-k_y),\tilde{\boldsymbol{B}}_v(k_x,k_y)\right\rangle^T & \mathbf{0}_3 \end{pmatrix}$$

but $\tilde{\boldsymbol{B}}(k_x,k_y)\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\right\rangle^{-1}\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\cdot\right\rangle$ is a *projection* onto the space spanned by $\tilde{\boldsymbol{B}}(k_x,k_y)$, because

$$\begin{aligned}
&\tilde{\boldsymbol{B}}(k_x,k_y)\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\right\rangle^{-1} \\
&\qquad\times\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\right\rangle^{-1}\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\cdot\right\rangle\right\rangle = \\
&\tilde{\boldsymbol{B}}(k_x,k_y)\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\right\rangle^{-1}\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\right\rangle \\
&\qquad\times\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\right\rangle^{-1}\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\cdot\right\rangle = \\
&\tilde{\boldsymbol{B}}(k_x,k_y)\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\right\rangle^{-1}\left\langle\tilde{\boldsymbol{B}}(k_x,k_y),\cdot\right\rangle
\end{aligned}$$

so that

$$\tilde{\boldsymbol{B}}(k_x,k_y)\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\right\rangle^{-1}\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\cdot\right\rangle$$

is *idempotent*.

Now, assuming we are given a field $\boldsymbol{T}(k_x,k_y)=\tilde{\boldsymbol{B}}(k_x,k_y)\cdot\boldsymbol{C}$, with arbitrary expansion coefficients $\boldsymbol{C}$, we have

$$\begin{aligned}
&\tilde{\boldsymbol{B}}(k_x,k_y)\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\right\rangle^{-1}\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\boldsymbol{T}(k_x,k_y)\right\rangle = \\
&\quad\tilde{\boldsymbol{B}}(k_x,k_y)\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\right\rangle^{-1}\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\cdot\boldsymbol{C}\right\rangle = \\
&\quad\tilde{\boldsymbol{B}}(k_x,k_y)\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\right\rangle^{-1}\left\langle\tilde{\boldsymbol{B}}(-k_x,-k_y),\tilde{\boldsymbol{B}}(k_x,k_y)\right\rangle\cdot\boldsymbol{C} = \\
&\quad\tilde{\boldsymbol{B}}(k_x,k_y)\cdot\boldsymbol{C}=\boldsymbol{T}(k_x,k_y)
\end{aligned}$$

$$\langle\boldsymbol{B}(-k_x,-k_y),\boldsymbol{B}(k_x,k_y)\rangle^{-1}$$

Box 7.1 *continued*
and using in the previous equation, we get

$$\tilde{\boldsymbol{B}}_v(k_x,k_y)\langle\tilde{\boldsymbol{B}}_u(-k_x,-k_y),\tilde{\boldsymbol{B}}_v(k_x,k_y)\rangle^{-1}\langle\tilde{\boldsymbol{B}}_u(-k_x,-k_y),\boldsymbol{T}(k_x,k_y)\rangle- \\ \tilde{\boldsymbol{B}}_u(k_x,k_y)\langle\tilde{\boldsymbol{B}}_u(-k_x,-k_y),\tilde{\boldsymbol{B}}_v(k_x,k_y)\rangle^{-T}\langle\tilde{\boldsymbol{B}}_v(-k_x,-k_y),\boldsymbol{T}(k_x,k_y)\rangle \equiv \boldsymbol{T}(k_x,k_y) \tag{7.19}$$

But, using a Fourier transform of x we have

$$\boldsymbol{T}(k_x,k_y)=\int_{-\infty}^{\infty}\mathrm{d}x\,\mathrm{e}^{-ik_xx}\boldsymbol{T}(x,k_y)$$

and substituting this result in (7.19), gives

$$\tilde{\boldsymbol{B}}_v(k_x,k_y)\langle\tilde{\boldsymbol{B}}_u(-k_x,-k_y),\tilde{\boldsymbol{B}}_v(k_x,k_y)\rangle^{-1}\int_{-\infty}^{\infty}\langle\mathrm{e}^{-ik_xx}\tilde{\boldsymbol{B}}_u(-k_x,-k_y),\boldsymbol{T}(x,k_y)\rangle\mathrm{d}x- \\ \tilde{\boldsymbol{B}}_u(k_x,k_y)\langle\tilde{\boldsymbol{B}}_u(-k_x,-k_y),\tilde{\boldsymbol{B}}_v(k_x,k_y)\rangle^{-T}\int_{-\infty}^{\infty}\langle\mathrm{e}^{-ik_xx}\tilde{\boldsymbol{B}}_v(-k_x,-k_y),\boldsymbol{T}(x,k_y)\rangle\mathrm{d}x \\ \equiv \boldsymbol{T}(k_x,k_y) \tag{7.20}$$

However,

$$\boldsymbol{T}(x,k_y)=\frac{1}{2\pi}\int_{-\infty}^{\infty}\mathrm{d}k\,\mathrm{e}^{ik_xx}\boldsymbol{T}(k_x,k_y)$$

so that operating on both sides of (7.20) with $\frac{1}{2\pi}\int_{-\infty}^{\infty}\mathrm{d}k\,\mathrm{e}^{ik_xx}$ gives

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}\Big[\mathrm{e}^{ik_xx}\tilde{\boldsymbol{B}}_v(k_x,k_y)\langle\tilde{\boldsymbol{B}}_u(-k_x,-k_y),\tilde{\boldsymbol{B}}_v(k_x,k_y)\rangle^{-1}\int_{-\infty}^{\infty}\langle\mathrm{e}^{-ik_xx}\tilde{\boldsymbol{B}}_u(-k_x,-k_y),\boldsymbol{T}(x,k_y)\rangle\,\mathrm{d}x \\ -\mathrm{e}^{ik_xx}\tilde{\boldsymbol{B}}_u(k_x,k_y)\langle\tilde{\boldsymbol{B}}_u(-k_x,-k_y),\tilde{\boldsymbol{B}}_v(k_x,k_y)\rangle^{-T}\int_{-\infty}^{\infty}\langle\mathrm{e}^{-ik_xx}\tilde{\boldsymbol{B}}_v(-k_x,-k_y),\boldsymbol{T}(x,k_y)\rangle\mathrm{d}x\Big]\,\mathrm{d}k \\ \equiv \boldsymbol{T}(x,k_y) \tag{7.21}$$

Using the definition (7.18) in (7.21) we get

$$\frac{1}{2\pi}\int_{-\infty}^{\infty}\mathrm{d}k\,\Big[\boldsymbol{B}_v(\boldsymbol{x};k_x,k_y,\omega)\langle\boldsymbol{B}_u(\boldsymbol{x};-k_x,-k_y,\omega),\boldsymbol{B}_v(\boldsymbol{x};k_x,k_y,\omega)\rangle^{-1} \\ \times\int_{-\infty}^{\infty}\langle\boldsymbol{B}_u(\boldsymbol{x};-k_x,-k_y,\omega),\boldsymbol{T}(x,k_y)\rangle\mathrm{d}x \\ -\boldsymbol{B}_u(\boldsymbol{x};k_x,k_y,\omega)\langle\boldsymbol{B}_u(\boldsymbol{x};-k_x,-k_y,\omega),\boldsymbol{B}_v(\boldsymbol{x};k_x,k_y,\omega)\rangle^{-T} \\ \times\int_{-\infty}^{\infty}\langle\boldsymbol{B}_v(\boldsymbol{x};-k_x,-k_y,\omega),\boldsymbol{T}(x,k_y)\rangle\mathrm{d}x\Big] \\ \equiv \boldsymbol{T}(x,k_y) \tag{7.22}$$

7.2 A Jost solution construction for the Lippmann-Schwinger equation

In this section, we re-derive the *Lippmann-Schwinger equation* of (5.36), but this time we employ the *Jost solutions* of (7.5).

The *heterogeneous half space Green's tensor field* $\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)$, driven by a point source at $\boldsymbol{x}_S \equiv (z_S, x_S)$, obeys the following boundary value problem, defined in terms of the bilinear form $\langle\cdot,\cdot\rangle$ of (7.11) as:

$$\left\{\begin{array}{l} \int_{-\infty}^{\infty}\langle \boldsymbol{B}_v((f+,x);-k_x,-k_y),\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\rangle\,\mathrm{d}x = \boldsymbol{0}_{3\times 6} \\ \left[\partial_z - \mathring{\boldsymbol{A}}\,(z,\partial_x,k_y)\right]\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y) = \Delta\boldsymbol{A}(\boldsymbol{x},\partial_x,k_y)\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y) + \delta(\boldsymbol{x}-\boldsymbol{x}_S)\boldsymbol{I}_6 \\ \int_{-\infty}^{\infty}\langle \boldsymbol{B}_u((L+,x);-k_x,-k_y),\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\rangle\,\mathrm{d}x = \boldsymbol{0}_{3\times 6} \end{array}\right. \tag{7.23}$$

where $\mathring{\boldsymbol{A}}\,(z,\partial_x,k_y)$ was given by (2.14) and $\Delta\boldsymbol{A}(\boldsymbol{x},\partial_x,k_y)$ by (2.15). The 3 boundary conditions

$$\int_{-\infty}^{\infty}\langle \boldsymbol{B}_v((f+,x);-k_x,-k_y),\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\rangle\,\mathrm{d}x = \boldsymbol{0}_{3\times 6}$$

ensure that the traction field vanishes at the free surface, while the 3 boundary conditions

$$\int_{-\infty}^{\infty}\langle \boldsymbol{B}_u((L+,x);-k_x,-k_y),\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\rangle\,\mathrm{d}x = \boldsymbol{0}_{3\times 6}$$

ensure that there is no $\uparrow-$*going* incident field at the base of the stratification.

We now construct the depth evolution equations for the bilinear functional

$$\int_{-\infty}^{\infty}\langle \boldsymbol{B}_u(\boldsymbol{x};-k_x,-k_y),\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)\ \rangle\,\mathrm{d}x$$

comprised of $\boldsymbol{B}_u(\boldsymbol{x};-k_x,-k_y)$ from (7.8) and the Greens tensor $\boldsymbol{G}(\boldsymbol{x};\boldsymbol{x}_S;k_y)$

of (7.23) *viz.* ,

$$\begin{aligned}
&\partial_z \int_{-\infty}^{\infty} \langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \, \mathrm{d}x = \\
&\quad \int_{-\infty}^{\infty} \langle \partial_z \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \, \mathrm{d}x \\
&\quad + \int_{-\infty}^{\infty} \langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \partial_z \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \, \mathrm{d}x = \\
&\quad \int_{-\infty}^{\infty} \Big[\langle \mathring{\boldsymbol{A}}\,(z, \partial_x, -k_y) \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \\
&\quad + \langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \mathring{\boldsymbol{A}}\,(z, \partial_x, k_y) \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \Big] \, \mathrm{d}x \\
&\quad + \delta(z - z_S) \boldsymbol{B}_u^T(\boldsymbol{x}_S; -k_x, -k_y) \boldsymbol{J}_6 \\
&\quad + \int_{-\infty}^{\infty} \langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{\Delta A}(\boldsymbol{x}, \partial_x, k_y) \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \, \mathrm{d}x
\end{aligned} \tag{7.24}$$

but

$$\begin{aligned}
&\int_{-\infty}^{\infty} \Big[\langle \mathring{\boldsymbol{A}}\,(z, \partial_x, -k_y) \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \\
&\qquad + \langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \mathring{\boldsymbol{A}}\,(z, \partial_x, k_y) \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \Big] \, \mathrm{d}x = 0,
\end{aligned} \tag{7.25}$$

from (7.16), due to the *quasi-Hamiltonian symmetry* of $\mathring{\boldsymbol{A}}$. Using (7.25) in (7.24) gives

$$\begin{aligned}
\partial_z \int_{-\infty}^{\infty} \langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \, \mathrm{d}x &= \delta(z - z_S) \boldsymbol{B}_u^T(\boldsymbol{x}_S; -k_x, -k_y) \boldsymbol{J}_6 \\
&\quad + \int_{-\infty}^{\infty} \langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{\Delta A}(\boldsymbol{x}, \partial_x, k_y) \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \, \mathrm{d}x
\end{aligned} \tag{7.26}$$

Upon integrating (7.26) over the depth interval $z+ \leq z \leq z_L+$ we get

$$\begin{aligned}
&\overbrace{\int_{-\infty}^{\infty} \langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \, \mathrm{d}x \Big|_{z_{L+}}}^{=\mathbf{0} \text{ by radiation conditions}} \\
&\qquad - \int_{-\infty}^{\infty} \langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \, \mathrm{d}x \Big|_{z_+} = \\
&H(z_S - z) \boldsymbol{B}_u^T(\boldsymbol{x}_S; -k_x, -k_y) \boldsymbol{J}_6 \\
&\qquad + \int_z^{z_L} \int_{-\infty}^{\infty} \langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{\Delta A}(\boldsymbol{x}, \partial_x, k_y) \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \, \mathrm{d}x \, \mathrm{d}z
\end{aligned}$$

so that

$$\int_{-\infty}^{\infty} \langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y) \rangle \, \mathrm{d}x \Big|_{z_+} = -H(z_S - z) \boldsymbol{B}_u^T(\boldsymbol{x}_S; -k_x, -k_y) \boldsymbol{J}_6$$
$$- \int_z^{z_L} \int_{-\infty}^{\infty} \langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \Delta \boldsymbol{A}(\boldsymbol{x}, \partial_x, k_y) \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y) \rangle \, \mathrm{d}x \tag{7.27}$$

Similarly, we construct the evolution equations for

$$\int_{-\infty}^{\infty} \langle \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y) \, \rangle \, \mathrm{d}x$$

comprised of $\boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y)$ from (7.10) and the Greens tensor $\boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)$ of (7.23) *viz.* ,

$$\begin{aligned}
&\partial_z \int_{-\infty}^{\infty} \langle \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y) \rangle \, \mathrm{d}x = \\
&\quad \int_{-\infty}^{\infty} \langle \partial_z \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y) \rangle \, \mathrm{d}x \\
&\quad + \int_{-\infty}^{\infty} \langle \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \partial_z \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y) \rangle \, \mathrm{d}x = \\
&\quad \int_{-\infty}^{\infty} \Big[\langle \mathring{\boldsymbol{A}} \, (z, \partial_x, -k_y) \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y) \rangle \\
&\quad + \langle \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \mathring{\boldsymbol{A}} \, (z, \partial_x, k_y) \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y) \rangle \Big] \, \mathrm{d}x \\
&\quad + \delta(z - z_S) \boldsymbol{B}_v^T(\boldsymbol{x}_S; -k_x, -k_y) \boldsymbol{J}_6 \\
&\quad + \int_{-\infty}^{\infty} \langle \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \Delta \boldsymbol{A}(\boldsymbol{x}, \partial_x, k_y) \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y) \rangle \, \mathrm{d}x
\end{aligned} \tag{7.28}$$

but

$$\begin{aligned}
\int_{-\infty}^{\infty} \Big[&\langle \mathring{\boldsymbol{A}} \, (z, \partial_x, -k_y) \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y) \rangle \\
&+ \langle \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \mathring{\boldsymbol{A}} \, (z, \partial_x, k_y) \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y) \rangle \Big] \, \mathrm{d}x = 0,
\end{aligned} \tag{7.29}$$

from (7.16), due to the *quasi-Hamiltonian symmetry* of $\mathring{\boldsymbol{A}}$. Using (7.29) in (7.28) gives

$$\partial_z \int_{-\infty}^{\infty} \langle \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y) \rangle \, \mathrm{d}x = \delta(z - z_S) \boldsymbol{B}_v^T(\boldsymbol{x}_S; -k_x, -k_y) \boldsymbol{J}_6$$
$$+ \int_{-\infty}^{\infty} \langle \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \Delta \boldsymbol{A}(\boldsymbol{x}, \partial_x, k_y) \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y) \rangle \, \mathrm{d}x \tag{7.30}$$

Upon integrating (7.30) over the depth interval $f+ \leq z \leq z-$ we get

$$\int_{-\infty}^{\infty} \langle \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \, \mathrm{d}x \Big|_{z-}$$
$$- \overbrace{\int_{-\infty}^{\infty} \langle \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \, \mathrm{d}x \Big|_{z_f+}}^{=\mathbf{0} \text{ by free surface conditions}} =$$
$$H(z - z_S)\boldsymbol{B}_v^T(\boldsymbol{x}_S; -k_x, -k_y)\boldsymbol{J}_6$$
$$+ \int_{z_f}^{z} \int_{-\infty}^{\infty} \langle \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \Delta\boldsymbol{A}(\boldsymbol{x}, \partial_x, k_y)\boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \, \mathrm{d}x \, \mathrm{d}z$$

so that

$$\int_{-\infty}^{\infty} \langle \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \, \mathrm{d}x \Big|_{z_-} = H(z - z_S)\boldsymbol{B}_v^T(\boldsymbol{x}_S; -k_x, -k_y)\boldsymbol{J}_6$$
$$+ \int_{z_f}^{z} \int_{-\infty}^{\infty} \langle \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \Delta\boldsymbol{A}(\boldsymbol{x}, \partial_x, k_y)\boldsymbol{G}(\boldsymbol{x}; \boldsymbol{x}_S; k_y)\rangle \, \mathrm{d}x \, \mathrm{d}z \tag{7.31}$$

Next we form the linear combination

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} \Big[\boldsymbol{B}_v(\boldsymbol{x}; k_x, k_y)\langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{B}_v(\boldsymbol{x}; k_x, k_y)\rangle^{-1} \times (7.27)$$
$$- \boldsymbol{B}_u(\boldsymbol{x}; k_x, k_y)\langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{B}_v(\boldsymbol{x}; k_x, k_y)\rangle^{-T} \times (7.31) \Big] \mathrm{d}k_x \tag{7.32}$$

concocted so that the resulting left hand side reduces to

$$\boldsymbol{G}(\boldsymbol{x}, \boldsymbol{x}_S; k_y) \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} \big[\boldsymbol{B}_v(\boldsymbol{x}; k_x, k_y)\langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{B}_v(\boldsymbol{x}; k_x, k_y)\rangle^{-1}$$
$$\times \int_{-\infty}^{\infty} \langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}, \boldsymbol{x}_S; k_y)\rangle \mathrm{d}x$$
$$- \boldsymbol{B}_u(\boldsymbol{x}; k_x, k_y)\langle \boldsymbol{B}_u(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{B}_v(\boldsymbol{x}; k_x, k_y)\rangle^{-T}$$
$$\times \int_{-\infty}^{\infty} \langle \boldsymbol{B}_v(\boldsymbol{x}; -k_x, -k_y), \boldsymbol{G}(\boldsymbol{x}, \boldsymbol{x}_S; k_y)\rangle \mathrm{d}x \Big] \, \mathrm{d}k_x$$

which follows from the *closure* property (7.17), of the Jost solutions $\{\boldsymbol{B}_u, \boldsymbol{B}_v\}$. From (7.32) we get the *Lippmann-Schwinger* integral equation

$$\begin{aligned} \boldsymbol{G}(\boldsymbol{x}, \boldsymbol{x}_S; k_y) &= \mathring{\boldsymbol{G}}(\boldsymbol{x}, \boldsymbol{x}_S; k_y) \\ &+ \int_{f+}^{L+} \int_{-\infty}^{\infty} \mathring{\boldsymbol{G}}(\boldsymbol{x}, \boldsymbol{x}'; k_y)\Delta\boldsymbol{A}(\boldsymbol{x}', \partial_x; k_y)\boldsymbol{G}(\boldsymbol{x}', \boldsymbol{x}_S; k_y) \, \mathrm{d}\boldsymbol{x}' \end{aligned} \tag{7.33}$$

where the *stratified reference Green's tensor* $\mathring{\boldsymbol{G}}\,(\boldsymbol{x}, \boldsymbol{x}_S; k_y)$ in (7.33) is defined as

$$\mathring{\boldsymbol{G}}\,(\boldsymbol{x}, \boldsymbol{x}_S; k_y) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{d}k_x \tag{7.34}$$
$$\times \begin{cases} -\boldsymbol{B}_v(\boldsymbol{x}; \boldsymbol{K}_T)\langle \boldsymbol{B}_u(\boldsymbol{x}'; -\boldsymbol{K}_T), \boldsymbol{B}_v(\boldsymbol{x}'; \boldsymbol{K}_T)\rangle^{-1}\boldsymbol{B}_u^T(\boldsymbol{x}_S; -\boldsymbol{K}_T)\boldsymbol{J}_6 & z < z_S \\ -\boldsymbol{B}_u(\boldsymbol{x}; \boldsymbol{K}_T)\langle \boldsymbol{B}_u(\boldsymbol{x}'; -\boldsymbol{K}_T), \boldsymbol{B}_v(\boldsymbol{x}'; \boldsymbol{K}_T)\rangle^{-T}\boldsymbol{B}_v^T(\boldsymbol{x}_S; -\boldsymbol{K}_T)\boldsymbol{J}_6 & z > z_S \end{cases}$$

for an arbitrary depth point $\boldsymbol{x}'$, a source at $\boldsymbol{x}_S$ a receiver at $\boldsymbol{x}$, and we have introduced $\boldsymbol{K}_T = (k_x, k_y)$ for the horizontal component of the wavenumber.

The *inverse invariant* $\langle\ \boldsymbol{B}_u, \boldsymbol{B}_v\ \rangle^{-1}$ in (7.34) is singular, i.e.

$$\mathrm{Det}\langle \boldsymbol{B}_u(\boldsymbol{x}', -\boldsymbol{K}_T), \boldsymbol{B}_v(\boldsymbol{x}', \boldsymbol{K}_T)\rangle =$$
$$\begin{vmatrix} \langle (u, P, -\boldsymbol{K}_T), (v, P, \boldsymbol{K}_T)\rangle & \langle (u, P, -\boldsymbol{K}_T), (v, S, \boldsymbol{K}_T)\rangle & \langle (u, P, -\boldsymbol{K}_T), (v, H, \boldsymbol{K}_T)\rangle \\ \langle (u, S, -\boldsymbol{K}_T), (v, P, \boldsymbol{K}_T)\rangle & \langle (u, S, -\boldsymbol{K}_T), (v, S, \boldsymbol{K}_T)\rangle & \langle (u, S, -\boldsymbol{K}_T), (v, H, \boldsymbol{K}_T)\rangle \\ \langle (u, H, -\boldsymbol{K}_T), (v, P, \boldsymbol{K}_T)\rangle & \langle (u, H, -\boldsymbol{K}_T), (v, S, \boldsymbol{K}_T)\rangle & \langle (u, H, -\boldsymbol{K}_T), (v, H, \boldsymbol{K}_T)\rangle \end{vmatrix}$$
$$= 0 \tag{7.35}$$

at the surface wave poles $\boldsymbol{K}_T = \boldsymbol{k}_\mu$ on the real axis. In the basic method, we avoid the surface wave poles by moving them off the real wavenumber axis through the agency of the complex frequency. However, surface waves are often an important phase on seismic records at regional offsets and beyond, and it would be very worthwhile to explicitly extract these phases from the companion body wave phases by *evaluating the residues* at the singular points of $\langle\ \boldsymbol{B}_u, \boldsymbol{B}_v\ \rangle^{-1}$.

[Figure: contour not found]

Figure 7.1: The wavenumber contour to separate body wave and surface wave phases. Body wave phases are synthesized from the wavenumber integrals, and the surface wave phases from the residues at the poles, identified as black dots.

7.3 Extracting the surface waves

We deform both ends of the k_x wavenumber contour (*vid.* Fig.7.1) into the $\Im(k_x) \gtrless 0$ plane for $(x - x_S) \gtrless 0$, so that the wave number integrand in (7.34) decays exponentially. Following *Kennett* (1981), we account for the surface waves by calculating the *residues* of the integrand in (7.34) at the values of $\boldsymbol{K}_T = \boldsymbol{k}_\mu$ where the pole condition (7.35) is fulfilled.

We recall that Jost solution $\boldsymbol{B}_u(\boldsymbol{x}'; \pm\boldsymbol{K}_T)$ from (7.8) was constructed to satisfy the *radiation condition* in the lower half space $z > z_L$, while Jost solution $\boldsymbol{B}_v(\boldsymbol{x}'; \pm\boldsymbol{K}_T)$ from (7.10) satisfies the *traction free condition* at the free surface $z = f$. At the poles, we assume "one" of "three" conditions is satisfied:

- the $P-$wave columns $(v, P, \pm\boldsymbol{K}_T, z')$ of $\boldsymbol{B}_v(\boldsymbol{x}'; \pm\boldsymbol{K}_T)$ satisfy the radiation condition and the $P-$wave columns $(u, P, \pm\boldsymbol{K}_T, z')$ of $\boldsymbol{B}_u(\boldsymbol{x}'; \pm\boldsymbol{K}_T)$ satisfy the free surface condition.

- the $SV-$wave columns $(v, S, \pm\boldsymbol{K}_T, z')$ of $\boldsymbol{B}_v(\boldsymbol{x}'; \pm\boldsymbol{K}_T)$ satisfy the radiation condition and the $SV-$wave columns $(u, S, \pm\boldsymbol{K}_T, z')$ of $\boldsymbol{B}_u(\boldsymbol{x}'; \pm\boldsymbol{K}_T)$ satisfy the free surface condition.

- the $SH-$wave columns $(v, H, \pm\boldsymbol{K}_T, z')$ of $\boldsymbol{B}_v(\boldsymbol{x}'; \pm\boldsymbol{K}_T)$ satisfy the radiation condition and the $SH-$wave columns $(u, H, \pm\boldsymbol{K}_T, z')$ of $\boldsymbol{B}_u(\boldsymbol{x}'; \pm\boldsymbol{K}_T)$ satisfy the free surface condition.

By actually locating a surface wave pole, it can be verified that, either

- $(v, P, \boldsymbol{K}_T, z)$ is a linear multiple $(u, P, \boldsymbol{K}_T, z)$
- or $(v, S, \boldsymbol{K}_T, z)$ is a linear multiple $(u, S, \boldsymbol{K}_T, z)$
- or $(v, H, \boldsymbol{K}_T, z)$ is a linear multiple $(u, H, \boldsymbol{K}_T, z)$

The *degenerate* case where any two of the P, S, H columns in $\boldsymbol{B}_u$ and $\boldsymbol{B}_v$ are scalar multiples, leads to a second order pole, and is not considered further, although this possibility cannot be ruled out *a priori*.

Following the method of *Kennett* (1983) Ch.11, we employ Cramer's rule to write

$$\left\langle \boldsymbol{B}_u(\boldsymbol{x}', -\boldsymbol{K}_T), \boldsymbol{B}_v(\boldsymbol{x}', \boldsymbol{K}_T) \right\rangle^{-1} = \frac{\text{Adj}\,\langle \boldsymbol{B}_u(\boldsymbol{x}', -\boldsymbol{K}_T), \boldsymbol{B}_v(\boldsymbol{x}', \boldsymbol{K}_T)\rangle}{\text{Det}\,\langle \boldsymbol{B}_u(\boldsymbol{x}', -\boldsymbol{K}_T), \boldsymbol{B}_v(\boldsymbol{x}', \boldsymbol{K}_T)\rangle} \quad (7.36)$$

where

$$\text{Det}\left(\left\langle \boldsymbol{B}_u(z, -\boldsymbol{K}_T), \boldsymbol{B}_v(\boldsymbol{x}', \boldsymbol{K}_T)\right\rangle\right)$$

is the determinant and

$$\text{Adj}\left(\left\langle \boldsymbol{B}_u(z, -\boldsymbol{K}_T), \boldsymbol{B}_v(\boldsymbol{x}', \boldsymbol{K}_T)\right\rangle\right)$$

is the adjoint matrix (i.e. the transposed matrix of cofactors).

Let's assume $(x - x_S) > 0$, and we have a P-wave pole at $\boldsymbol{K}_T = \boldsymbol{k}_\mu$ in the *first quadrant of the complex k_x plane*, so that the P-wave columns satisfy

$$(u, P, -\boldsymbol{k}_\mu) = \gamma_P (v, P, -\boldsymbol{k}_\mu),$$

for an arbitrary constant γ_P, then using similar arguments to those employed in *Kennett*(1983), Appendix to chapter 11, we get:

$$\begin{aligned}\langle (u, P, -\boldsymbol{K}_T), (v, P, \boldsymbol{K}_T)\rangle &= \langle (u, P, -\boldsymbol{K}_T), (v, S, \boldsymbol{K}_T)\rangle = \langle (u, P, -\boldsymbol{K}_T), (v, H, \boldsymbol{K}_T)\rangle \\ &= \langle (u, S, -\boldsymbol{K}_T), (v, P, \boldsymbol{K}_T)\rangle = \langle (u, H, -\boldsymbol{K}_T), (v, P, \boldsymbol{K}_T)\rangle = 0 \quad (7.37)\end{aligned}$$

because both factors in $\langle \cdot, \cdot \rangle$ satisfy the same homogeneous boundary conditions (i.e. free surface or radiation). Also, the bilinear form $\langle \cdot, \cdot \rangle$ defined on the Jost solutions is depth invariant, for the reasons explained in (7.15), so that if a term like $\langle (u, P, -\boldsymbol{K}_T), (v, P, \boldsymbol{K}_T)\rangle$ vanishes at either the free surface $z = z_f$ or at infinity $z = \infty$ then $\langle (u, P, -\boldsymbol{K}_T), (v, P, \boldsymbol{K}_T)\rangle = 0,\ z_f \leq z \leq \infty$ (which is why we drop the spatial dependence in Jost solutions like $(u, P, -\boldsymbol{K}_T)$ inside $\langle \cdot, \cdot \rangle$, when space on the printed page is at a premium).

Employing (7.37) in (7.36) and canceling common factors, we get

$$\left\langle \boldsymbol{B}_u(\boldsymbol{x}', -\boldsymbol{K}_T), \boldsymbol{B}_v(\boldsymbol{x}', \boldsymbol{K}_T) \right\rangle^{-1} \sim \frac{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}}{\langle (u, P, -\boldsymbol{K}_T), (v, P, \boldsymbol{K}_T) \rangle} \text{ as } \boldsymbol{K}_T \to \boldsymbol{k}_\mu \tag{7.38}$$

where $\boldsymbol{k}_\mu$ is a pole position in the complex wavenumber plane. Substituting (7.38) for $\langle \boldsymbol{B}_u(\boldsymbol{x}', -\boldsymbol{K}_T), \boldsymbol{B}_v(\boldsymbol{x}', \boldsymbol{K}_T) \rangle^{-1}$ in the Green's tensor (7.34), the singular contribution $\mathring{\boldsymbol{G}}_{P,\mu}$ at the $P-$wave pole $\boldsymbol{K}_T = \boldsymbol{k}_\mu$ may be written as

$$\mathring{\boldsymbol{G}}_{P,\mu}(\boldsymbol{x}, \boldsymbol{x}_S; k_y) \sim \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathrm{d}k_x \, \exp\left[ik_x(x - x_S)\right] \tag{7.39}$$
$$\times \begin{cases} -\dfrac{(v, P, \boldsymbol{K}_T, z) \otimes (u, P, -\boldsymbol{K}_T, z_S)\, \boldsymbol{J}_6}{\langle (u, P, -\boldsymbol{K}_T), (v, P, \boldsymbol{K}_T) \rangle} & z < z_S \\ -\dfrac{(u, P, \boldsymbol{K}_T, z) \otimes (v, P, -\boldsymbol{K}_T, z_S)\, \boldsymbol{J}_6}{\langle (u, P, -\boldsymbol{K}_T), (v, P, \boldsymbol{K}_T) \rangle} & z > z_S \end{cases} \quad \text{as} \quad \boldsymbol{K}_T \to \boldsymbol{k}_\mu$$

but employing the linear dependence relation $(u, P, -\boldsymbol{k}_\mu) = \gamma(v, P, -\boldsymbol{k}_\mu)$, in (7.39) and canceling common factors, we get

$$\mathring{\boldsymbol{G}}_{P,\mu}(\boldsymbol{x}, \boldsymbol{x}_S; k_y) \sim \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathrm{d}k_x \, \exp\left[ik_x(x - x_S)\right] \tag{7.40}$$
$$\times \begin{cases} -\dfrac{(v, P, \boldsymbol{K}_T, z) \otimes (v, P, -\boldsymbol{K}_T, z_S)\, \boldsymbol{J}_6}{\langle (v, P, -\boldsymbol{K}_T), (v, P, \boldsymbol{K}_T) \rangle} & 0 \le z, z_S \le \infty \\ -\dfrac{(u, P, \boldsymbol{K}_T, z) \otimes (u, P, -\boldsymbol{K}_T, z_S)\, \boldsymbol{J}_6}{\langle (u, P, -\boldsymbol{K}_T), (u, P, \boldsymbol{K}_T) \rangle} & 0 \le z, z_S \le \infty \end{cases} \quad \text{as} \quad \boldsymbol{K}_T \to \boldsymbol{k}_\mu$$

The surface wave is given by the $2\pi i\times$ residue of (7.40) at $\boldsymbol{k}_\mu = (k_{\mu,x}(k_y, \omega), k_y)$ in first quadrant of the complex k_x-plane, so that

$$\mathring{\boldsymbol{G}}_{P,\mu}(\boldsymbol{x}, \boldsymbol{x}_S; k_y) = -i \frac{\exp\left[ik_{\mu,x}(x - x_S)\right] (u, P, \boldsymbol{k}_\mu, z) \otimes (u, P, -\boldsymbol{k}_\mu, z_S)\, \boldsymbol{J}_6}{\frac{\mathrm{d}}{\mathrm{d}k_x} \langle (u, P, -\boldsymbol{K}_T), (u, P, \boldsymbol{K}_T) \rangle|_{\boldsymbol{K}_T = \boldsymbol{k}_\mu}} \tag{7.41}$$

Similarly, assuming that $(x - x_S) < 0$ we pick up $-2\pi i\times$ the residue at the pole $\boldsymbol{K}_T = -\boldsymbol{k}_\mu = -(k_{\mu,x}(k_y, \omega), k_y)$, in the third quadrant of the complex k_x-plane, so that

$$\mathring{\boldsymbol{G}}_{P,\mu}(\boldsymbol{x}, \boldsymbol{x}_S; k_y) = i \, \frac{\exp\left[ik_{\mu,x}(x - x_S)\right] (u, P, \boldsymbol{k}_\mu, z) \otimes (u, P, -\boldsymbol{k}_\mu, z_S)\, \boldsymbol{J}_6}{\frac{\mathrm{d}}{\mathrm{d}k_x} \langle (u, P, -\boldsymbol{K}_T), \ (u, P, \boldsymbol{K}_T) \rangle|_{\boldsymbol{K}_T = -\boldsymbol{k}_\mu}}$$

(7.42)

but it turns out that

$$\frac{\mathrm{d}}{\mathrm{d}\,k_x}\left\langle (u,P,-k_x)\,,\ (u,P,\boldsymbol{K}_T)\right\rangle\big|_{\boldsymbol{K}_T=-\boldsymbol{k}_\mu} = -\frac{\mathrm{d}}{\mathrm{d}\,k_x}\left\langle (u,P,-\boldsymbol{K}_T)\,,\ (u,P,\boldsymbol{K}_T)\right\rangle\big|_{\boldsymbol{K}_T=\boldsymbol{k}_\mu}$$

so that (7.41) and (7.42) together imply that

$$\mathring{\boldsymbol{G}}_{P,\mu}(\boldsymbol{x},\boldsymbol{x}_S;k_y) = -i\frac{\exp\left[i\boldsymbol{k}_\mu|x-x_S|\right](v,P,\boldsymbol{k}_\mu,z)\otimes(u,P,-\boldsymbol{k}_\mu,z_S)\,\boldsymbol{J}_6}{\frac{\mathrm{d}}{\mathrm{d}\,k_x}\left\langle (u,P,-k_x,\zeta)\,,\ (v,P,k_x,\zeta)\right\rangle\big|_{k_x=\boldsymbol{k}_\mu}} \tag{7.43}$$

We could calculate $\frac{\mathrm{d}}{\mathrm{d}\,k_x}\left\langle (u,P,-k_x,\zeta)\,,\ (v,P,k_x,\zeta)\right\rangle\big|_{k_x=\boldsymbol{k}_\mu}$ in (7.43) using a finite difference approximation to the derivative, but the calculation is rather susceptible to roundoff. Instead, we develop an analytic expression, that is the cartesian analogue of the result in *Kennett* (1983), Appendix to chapter 11.

7.3.1 Calculating the residue at a surface wave pole

Since

$$\partial_z\,(u,P,\pm\boldsymbol{K}_T,z) = \mathring{\boldsymbol{A}}(z,\pm\boldsymbol{K}_T)\,(u,P,\pm\boldsymbol{K}_T,z) \tag{7.44}$$

where $\mathring{\boldsymbol{A}}(z,\boldsymbol{K}_T)$ is the Fourier domain system matrix

$$\mathring{\boldsymbol{A}}(z,\boldsymbol{K}_T) = \tag{7.45}$$

$$\begin{bmatrix} 0 & -ik_x\mathring{\gamma}(z) & -ik_y\mathring{\gamma}(z) & \mathring{a}(z) & 0 & 0 \\ -ik_x & 0 & 0 & 0 & \mathring{b}(z) & 0 \\ -ik_y & 0 & 0 & 0 & 0 & \mathring{b}(z) \\ -\mathring{\rho}(z)\omega^2 & 0 & 0 & 0 & -ik_x & -ik_y \\ 0 & k_x^2\mathring{\zeta}(z)+\mathring{\mu}(z)k_y^2-\mathring{\rho}(z)\omega^2 & k_xk_y(\mathring{\chi}(z)+\mathring{\mu}(z)) & -ik_x\mathring{\gamma}(z) & 0 & 0 \\ 0 & k_xk_y(\mathring{\chi}(z)+\mathring{\mu}(z)) & \mathring{\zeta}(z)k_y^2+k_x^2\mathring{\mu}(z)-\mathring{\rho}(z)\omega^2 & -ik_y\mathring{\gamma}(z) & 0 & 0 \end{bmatrix}$$

it follows that

$$\begin{aligned}\partial_z\frac{\mathrm{d}}{\mathrm{d}\,k_x}\,(u,P,\pm\boldsymbol{K}_T,z) &= \mathring{\boldsymbol{A}}\,(z,\pm\boldsymbol{K}_T)\frac{\mathrm{d}}{\mathrm{d}\,k_x}\,(u,P,\pm\boldsymbol{K}_T,z) \\ &\quad +\frac{\mathrm{d}}{\mathrm{d}\,k_x}\,\mathring{\boldsymbol{A}}\,(z,\pm\boldsymbol{K}_T)\,(u,P,\pm\boldsymbol{K}_T,z)\end{aligned} \tag{7.46}$$

Using (7.44) and (7.46) gives

$$
\begin{aligned}
&\partial_z \left\langle \frac{\mathrm{d}}{\mathrm{d}\, k_x} \left(u, P, -\boldsymbol{K}_T, z\right), \left(u, P, \boldsymbol{K}_T, z\right) \right\rangle \\
&\quad = \left\langle \partial_z \frac{\mathrm{d}}{\mathrm{d}\, k_x} \left(u, P, -\boldsymbol{K}_T, z\right), \left(u, P, \boldsymbol{K}_T, z\right) \right\rangle + \left\langle \frac{\mathrm{d}}{\mathrm{d}\, k_x} \left(u, P, -\boldsymbol{K}_T, z\right), \partial_z \left(u, P, \boldsymbol{K}_T, z\right) \right\rangle \\
&\quad = \left\langle \mathring{\boldsymbol{A}} \left(z, -\boldsymbol{K}_T\right) \frac{\mathrm{d}}{\mathrm{d}\, k_x} \left(u, P, -\boldsymbol{K}_T, z\right), \left(u, P, \boldsymbol{K}_T, z\right) \right\rangle \\
&\qquad + \left\langle \frac{\mathrm{d}}{\mathrm{d}\, k_x} \mathring{\boldsymbol{A}} \left(z, -\boldsymbol{K}_T\right) \left(u, P, -\boldsymbol{K}_T, z\right), \left(u, P, \boldsymbol{K}_T, z\right) \right\rangle \\
&\qquad + \left\langle \frac{\mathrm{d}}{\mathrm{d}\, k_x} \left(u, P, -\boldsymbol{K}_T, z\right), \mathring{\boldsymbol{A}} \left(z, \boldsymbol{K}_T\right) \left(u, P, \boldsymbol{K}_T, z\right) \right\rangle \\
&\quad = - \left\langle \left(u, P, -\boldsymbol{K}_T, z\right), \frac{\mathrm{d}}{\mathrm{d}\, k_x} \mathring{\boldsymbol{A}} \left(z, \boldsymbol{K}_T\right) \left(u, P, \boldsymbol{K}_T, z\right) \right\rangle
\end{aligned}
\tag{7.47}
$$

where we have used the Hamiltonian symmetries

$$
\left\langle \mathring{\boldsymbol{A}} \left(z, -\boldsymbol{K}_T\right) \cdot, \cdot \right\rangle = - \left\langle \cdot, \ \mathring{\boldsymbol{A}} \left(z, \boldsymbol{K}_T\right) \cdot \right\rangle
$$

and

$$
\langle \frac{\mathrm{d}}{\mathrm{d}\, k_x} \mathring{\boldsymbol{A}} \left(z, -\boldsymbol{K}_T\right) \cdot, \cdot \rangle = - \langle \cdot, \ \frac{\mathrm{d}}{\mathrm{d}\, k_x} \mathring{\boldsymbol{A}} \left(z, \boldsymbol{K}_T\right) \cdot \rangle
$$

Upon integrating (7.47) over the depth interval $z \leq \zeta \leq \infty$, and using the *radiation condition* at $\zeta = \infty$, we get

$$
\begin{aligned}
&\left\langle \frac{\mathrm{d}}{\mathrm{d}\, k_x} \left(u, P, -\boldsymbol{K}_T, \zeta\right), \left(u, P, \boldsymbol{K}_T, \zeta\right) \right\rangle \Bigg|_{\zeta = z} = \\
&\qquad \int_z^{\infty} \mathrm{d}\zeta \left\langle \left(u, P, -\boldsymbol{K}_T, \zeta\right), \frac{\mathrm{d}}{\mathrm{d}\, k_x} \mathring{\boldsymbol{A}} \left(\zeta, \boldsymbol{K}_T\right) \left(u, P, \boldsymbol{K}_T, \zeta\right) \right\rangle
\end{aligned}
\tag{7.48}
$$

Similarly, integrating (7.47) over the depth interval $0 \leq \zeta \leq z$, and remembering that $(u, P, \pm \boldsymbol{K}_T, \zeta = 0)$ satisfy the traction free condition at $\zeta = 0$ in the limit as $\pm \boldsymbol{K}_T \to \pm \boldsymbol{k}_\mu$ (which is the only case of interest), we get

$$
\begin{aligned}
&\left\langle \left(u, P, -\boldsymbol{K}_T, \zeta\right), \frac{\mathrm{d}}{\mathrm{d}\, k_x} \left(u, P, \boldsymbol{K}_T, \zeta\right) \right\rangle \Bigg|_{\zeta = z} = \\
&\qquad \int_0^{z} \mathrm{d}\zeta \left\langle \left(u, P, -\boldsymbol{K}_T, \zeta\right), \frac{\mathrm{d}}{\mathrm{d}\, k_x} \mathring{\boldsymbol{A}} \left(\zeta, \boldsymbol{K}_T\right) \left(u, P, \boldsymbol{K}_T, \zeta\right) \right\rangle
\end{aligned}
\tag{7.49}
$$

Now, (7.48) and (7.49) together imply that

$$\frac{\mathrm{d}}{\mathrm{d}\,k_x}\left\langle\left(u,P,-\boldsymbol{K}_T,z\right),\left(u,P,\boldsymbol{K}_T,z\right)\right\rangle = \int_0^\infty \mathrm{d}\zeta\left\langle\left(u,P,-\boldsymbol{K}_T,\zeta\right),\frac{\mathrm{d}}{\mathrm{d}\,k_x}\,\mathring{\boldsymbol{A}}\left(\zeta,\boldsymbol{K}_T\right)\left(u,P,\boldsymbol{K}_T,\zeta\right)\right\rangle \tag{7.50}$$

for an arbitrary depth point z.

7.3.2 The surface wave contributions to the Green's tensor

Finally, using (7.50) in (7.43), and summing over the $N_P(k_y,\omega)$ $P-$wave poles in the first quadrant of the k_x plane at fixed (k_y,ω), we get the surface wave contributions to the Green's tensor as

$$\mathring{\boldsymbol{G}}_P\left(\boldsymbol{x},\boldsymbol{x}_S;k_y\right) = -i\sum_{\mu=1}^{N_P(\omega)}\frac{\exp\left[ik_{\mu,x}\left|x-x_S\right|\right]\left(u,P,\boldsymbol{k}_\mu,z\right)\otimes\left(u,P,-\boldsymbol{k}_\mu,z_S\right)\boldsymbol{J}_6}{\int_0^\infty \mathrm{d}\zeta\left\langle\left(u,P,-\boldsymbol{K}_T,\zeta\right),\frac{\mathrm{d}}{\mathrm{d}\,k_x}\,\mathring{\boldsymbol{A}}\left(\zeta,k_x\right)\left(u,P,\boldsymbol{K}_T,\zeta\right)\right\rangle\Big|_{\boldsymbol{K}_T=\boldsymbol{k}_\mu}} \tag{7.51}$$

The surface wave contributions to $\mathring{\boldsymbol{G}}_S$ and $\mathring{\boldsymbol{G}}_H$ that result from the

$$\left(u,S,-\boldsymbol{k}_\mu\right) = \gamma_S(v,S,-\boldsymbol{k}_\mu),$$

and

$$\left(u,H,-\boldsymbol{k}_\mu\right) = \gamma_H(v,H,-\boldsymbol{k}_\mu),$$

cases respectively, are treated with the same method that we have just outlined for $\mathring{\boldsymbol{G}}_P$ in (7.51), although the pole positions $\boldsymbol{k}_\mu$ will vary in each case.

To summarize, the surface wave Green's tensor may be written as

$$\mathring{\boldsymbol{G}}\left(\boldsymbol{x},\boldsymbol{x}_S;k_y\right) = -i\times \tag{7.52}$$
$$\sum_{c\in\{P,S,H\}}\sum_{\mu_c=1}^{N_c(\omega)}\frac{\exp\left[ik_{\mu_c,x}\left|x-x_S\right|\right]\left(u,P,\boldsymbol{k}_{\mu_c},z\right)\otimes\left(u,P,-\boldsymbol{k}_{\mu_c},z_S\right)\boldsymbol{J}_6}{\int_0^\infty \mathrm{d}\zeta\left\langle\left(u,P,-\boldsymbol{K}_T,\zeta\right),\frac{\mathrm{d}}{\mathrm{d}\,k_x}\,\mathring{\boldsymbol{A}}\left(\zeta,k_x\right)\left(u,P,\boldsymbol{K}_T,\zeta\right)\right\rangle\Big|_{\boldsymbol{K}_T=\boldsymbol{k}_{\mu_c}}}$$

We recall that the specific formula chosen in (7.34) depends on whether $(z-z_S)\gtrless 0$ (the *up/down dichotomy*), while the specific formula for the extracted surface wave components (7.52) depends on whether $(x-x_S)\gtrless 0$ (the *left/right dichotomy*). Note that the complex plane portion wavenumber contour in Fig.7.1 is subject to both the *up/down dichotomy* and the

left/right dichotomy which may be handled by an *alternating direction, Riccati sweep*, i.e. *each offset of the left/right sweep has an embedded up/down sweep*. The possibility of explicitly calculating modal/body wave coupling may be of use in certain studies.

Unfortunately, the stumbling block is the actual calculation of the surface wave pole positions. By employing the complex plane contour of Fig.7.1, the number of plane waves needed to synthesize the body wave phases is greatly reduced, but the number of poles is very large indeed, and it is a major computational problem to locate them to the required accuracy.

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